

MINDCRAFT

# Research Constitution

How do humans become confident mathematical thinkers?

Identity transformation · Evidence · Red Team · Product systems · Experiments

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This is not a pitch deck. This is the company's research operating system.  
Challenge assumptions. Label uncertainty. Kill weak arguments.  
Source corpus: ~80,722 words · multi-chapter living lab

MindCraft Research Lab · ongoing evidence program

# MindCraft Research Constitution v1

**Status:** Living operating document — not a pitch deck

**Edition:** v1.8 (Synthesizer consolidation of Parts L–LVIII into surviving doctrine)

**Research question:** How do humans become *confident mathematical thinkers*?

**Product thesis under audit:** The product is identity transformation, not mathematics delivery.

**Last updated:** 2026-08-01 (Researcher — Part LIX Parent WTP / SAFE-WTP)

**Growth model:** Core OS (this file) + chapters/\*.md via CHAPTER\_MANIFEST.txt → PDF

**Scale intent:** Multi-month densification toward 150–300 pages of *evidenced* material — never fluff

**Epistemic rule:** Every claim is labeled FACT / HYPOTHESIS / FOUNDER BELIEF / SPECULATION.

## Deep-dive chapters currently mounted

| Part   | File                                            | Focus                                                      |
|--------|-------------------------------------------------|------------------------------------------------------------|
| XXIV   | chapters/24_affect_anxiety_belonging.md         | Anxiety, WM, stereotype threat, belonging                  |
| XXV    | chapters/25_self_efficacy_narrative_identity.md | Bandura sources, narrative identity, habit≠identity        |
| XXVI   | chapters/26_cognitive_load_mastery_tutoring.md  | CLT, fading, mastery war, tutoring calibration             |
| XXVII  | chapters/27_markets_parents_competition.md      | WTP, parents, competitive teardown                         |
| XXVIII | chapters/28_history_meaning_math.md             | Invention-story thesis under trial                         |
| XXIX   | chapters/29_spacing_retrieval.md                | Spacing, retrieval, memory of competence                   |
| XXX    | chapters/30_attribution_helplessness.md         | Weiner attributions, helplessness, feedback language       |
| XXXI   | chapters/31_flow_challenge_skill.md             | Flow, challenge–skill, difficulty design                   |
| XXXII  | chapters/32_parent_anxiety_transmission.md      | Maloney/Beilock parent anxiety pathway                     |
| XXXIII | chapters/33_ai_tutors_trust_sycophancy.md       | AI trust / sycophancy; RT: Bastani■PWC, guards≠gain        |
| XXXIV  | chapters/34_formal_causal_dag_identification.md | FEI causal DAG; L0–L4 claim ladder; confounders            |
| XXXV   | chapters/35_competitive_session_audits.md       | Khan/Duo/Brilliant/ChatGPT session mechanism audits        |
| XXXVI  | chapters/36_equity_audit_story_worlds.md        | Equity audit of story worlds; whose history; tokenism kill |

|         |                                                 |                                                                                  |
|---------|-------------------------------------------------|----------------------------------------------------------------------------------|
| XXXVII  | chapters/37_expectancy_value_eccles.md          | Eccles SEVT;<br>utility/cost/attainment → choice;<br>EVT experiments             |
| XXXVIII | chapters/38_goal_orientation.md                 | Mastery vs performance goal structures; appearance ban; GO experiments           |
| XXXIX   | chapters/39_interleaving_vs_blocking.md         | Interleave vs block; strategy selection; IL experiments                          |
| XL      | chapters/40_self_explanation_prompts.md         | Self-explanation prompts;<br>Chi/Renkl → coach UX; SE experiments                |
| XLI     | chapters/41_desirable_difficulties_x_anxiety.md | Desirable difficulties × anxiety;<br>SAFE-DD; Bjork vs Ashcraft                  |
| XLII    | chapters/42_social_comparison_leaderboards.md   | Social comparison & leaderboards;<br>SAFE-COMPARE; ranks help/hurt               |
| XLIII   | chapters/43_habit_formation_science.md          | Habit formation (Wood/Clear);<br>SAFE-HABIT; cue≠colonization                    |
| XLIV    | chapters/44_intrinsic_motivation_killers.md     | Intrinsic motivation killers (Deci); SAFE-REWARD; controlling≠informational      |
| XLV     | chapters/45_mathematical_resilience.md          | Mathematical resilience (JW/Lee); SAFE-RESILIENCE; grit theater kill             |
| XLVI    | chapters/46_teacher_tutor_expectancy.md         | Tutor expectancy (Pygmalion/Golem);<br>SAFE-EXPECTANCY; trait-label ban          |
| XLVII   | chapters/47_sleep_stress_learning.md            | Sleep/stress/learning;<br>SAFE-TIMING; all-nighter & sleep-app kill              |
| XLVIII  | chapters/48_transfer_variation_theory.md        | Transfer & variation theory;<br>SAFE-TRANSFER; story≠far-transfer kill           |
| XLIX    | chapters/49_misconceptions_as_productive.md     | Misconceptions as productive;<br>SAFE-MISCON; soft-wrong / diagnostic wrongs     |
| L       | chapters/50_confidence_calibration.md           | Confidence calibration;<br>SAFE-CALIB; over/underconfidence; C4 hide-correctness |

|       |                                                |                                                                              |
|-------|------------------------------------------------|------------------------------------------------------------------------------|
| LI    | chapters/51_deliberate_practice_tutoring.md    | Deliberate practice in tutoring; SAFE-DP; Ericsson vs Hambrick; 45-min spine |
| LII   | chapters/52_community_of_practice.md           | Community of practice; SAFE-CoP; Lave/Wenger LPP; careful school transfer    |
| LIII  | chapters/53_religion_ritual_light_touch.md     | Religion/ritual light-touch; SAFE-RITUAL; meaning without cult dynamics      |
| LIV   | chapters/54_military_aviation_brief_debrief.md | Military/aviation brief-debrief; SAFE-AAR; after-action reviews for sessions |
| LV    | chapters/55_sports_film_study_pedagogy.md      | Sports film-study pedagogy; SAFE-FILM; error clips → coach cards             |
| LVI   | chapters/56_music_pedagogy_ladders.md          | Music pedagogy ladders; SAFE-MUSIC; scales → repertoire → recital            |
| LVII  | chapters/57_chess_annotation_metacognition.md  | Chess annotation & metacognition; SAFE-ANNOT; postmortem UX                  |
| LVIII | chapters/58_therapy_graded_exposure.md         | Therapy graded exposure; SAFE-EXPOSE; math anxiety ladders                   |
| LIX   | chapters/59_parent_wtp_experiments.md          | Parent WTP experiments; SAFE-WTP; CBC/conjoint + message tests               |

Queued next: see NEXT\_LAB.md (Part LX trust after AI hallucination, then QUEUE\_EXTENDED).

**Synthesizer note (v1.8):** Nine researcher chapters (L–LVIII) landed after v1.7 without a merge pass — calibration, deliberate-practice spine, CoP caution, ritual light-touch, AAR, film-study, music ladder, annotation, and graded exposure lived only in deep dives. This edition promotes each stack to its own I.4 law, adds Red Team kills #15–#17, mounts experiment families CAL/DP/CoP/RIT/AAR/FILM/MUSIC/ANNOT/EXP-O in Part IX, and refreshes metrics/glossary. Deep-dive files remain authoritative for citations; the OS keeps only *surviving* product rules.

## 0. How to read this document

This Constitution is the research OS for MindCraft. It exists to prevent the company from building features that feel right and fail quietly.

Rules of engagement:

1. Founder beliefs enter as **FOUNDER BELIEF**, not as law.

2. The Red Team’s job is destruction. Unchallenged claims are invalid.
3. Page count is not quality. A shorter true section beats a longer vague one.
4. “AI will make explanations free” is a hypothesis about markets, not a moral slogan.
5. Product implications must map to experiments with falsifiers.

This v1 is dense and incomplete on purpose. Expanding toward 150–300 pages is allowed only when new evidence, frameworks, or contradicted priors justify the ink.

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## Part I — Executive Summary

### I.1 The compressed answer

**HYPOTHESIS (confidence: medium–high):** Humans become confident mathematical thinkers when three conditions compound:

1. **Affective permission** — the nervous system stops treating math as social/physical threat.
2. **Competence evidence** — the learner accumulates undeniable, self-attributed successes at the right grain of difficulty — including *strategy selection* and *self-accounted why*, not only blocked accuracy.
3. **Identity re-storying** — the person revises the narrative “I am not a math person” into a workable self-story compatible with struggle — via expectancy × value × cost (Eccles), not posters.

Explanations alone rarely produce (1)–(3). Fluent AI explanations can *accelerate wrong confidence* (Bastani unguarded GPT; Part XXXIII). Explanations help (2) only *after* (1) and when the student still does the decisive generative work (Parts XXVI, XL).

### I.2 Verdict on the scarcity thesis

**FOUNDER BELIEF under audit:** AI commoditizes knowledge; scarcity shifts to motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

**Red Team verdict (updated v1.8):**

| Claim                                   | Status                                   | Notes                                                                |
|-----------------------------------------|------------------------------------------|----------------------------------------------------------------------|
| Explanations are getting cheaper/faster | <b>FACT</b> (directionally)              | Generative AI + open content lowers marginal cost of explanation     |
| “Tutoring is free”                      | <b>FALSE as stated</b>                   | Attention, accountability, diagnosis, calibrated trust remain scarce |
| Content is free                         | <b>MOSTLY FACT</b> for commodity content | Differentiation = curation, sequencing, trust, outcomes              |

|                                                                          |                                            |                                                                                                               |
|--------------------------------------------------------------------------|--------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Motivation/identity become scarce                                        | <b>HYPOTHESIS</b>                          | Product differentiator; not proven sole moat — instrument $E \times V \times \text{Cost}$ separately (XXXVII) |
| Emotional safety is first-order                                          | <b>HYPOTHESIS with supportive evidence</b> | Ashcraft WM tax; segment, don't romanticize softness                                                          |
| Unguarded AI chat improves solo learning                                 | <b>KILLED</b>                              | Bastani: concurrent $\uparrow$ then solo exam $\downarrow$ ; guards neutralize harm $\neq$ prove gain         |
| Bastani “GPT Tutor” $\equiv$ MindCraft ontology wrap                     | <b>KILLED</b>                              | Category error (XXXIII RT)                                                                                    |
| Streaks / blocked fluency = mastery                                      | <b>KILLED as North Star</b>                | HID + XXXVIII/XXXIX; measure delayed mix + transfer                                                           |
| Harder-always / anxiety-blind “desirable difficulty”                     | <b>KILLED</b>                              | SAFE-DD (XLI); equip $\rightarrow$ destake $\rightarrow$ dose before Bjork dose                               |
| Ranks / XP / grit meters as identity engines                             | <b>KILLED</b>                              | SAFE-COMPARE / REWARD / RESILIENCE (XLII–XLV)                                                                 |
| Story immersion $\equiv$ far transfer; mistake posters $\equiv$ learning | <b>KILLED</b>                              | SAFE-TRANSFER / MISCON (XLVIII–XLIX)                                                                          |
| Raise confidence / Belief Score™ / CA $\rightarrow$ grades               | <b>KILLED</b>                              | SAFE-CALIB (L); Foster: appropriate confidence, not inflate                                                   |
| Hours / forums / ritual $\rightarrow$ score / domain cosplay             | <b>KILLED</b>                              | SAFE-DP / CoP / RITUAL (LI–LIII); borrow method not costume                                                   |
| Tip-flood / engine-dump / flooding $\equiv$ pedagogy                     | <b>KILLED</b>                              | SAFE-AAR / FILM / MUSIC / ANNOT / EXPOSE (LIV–LVIII)                                                          |

**Implication:** Do not bet on “better explanations,” “AI tutor warmth,” streak theater, grit posters, therapy cosplay, or academy/bootcamp branding. Bet on **FEI + pedagogy wrap + SAFE-\* stack**: fear $\rightarrow$ evidence $\rightarrow$ identity, with AI as guarded infrastructure and student generation as the spine.

### I.3 What to optimize (North Star debate)

| Metric                         | Predicts short-term engagement? | Predicts durable math identity? | Risk                              |
|--------------------------------|---------------------------------|---------------------------------|-----------------------------------|
| Correct answers (esp. blocked) | Medium                          | Low–medium                      | Fluency illusion; gaming          |
| Time spent / chat tokens       | High                            | Low                             | Engagement / AI-monologue theater |
| Streaks / DAU                  | High                            | Uncertain                       | Habit $\neq$ identity (HID)       |

|                                                    |        |                       |                                                            |
|----------------------------------------------------|--------|-----------------------|------------------------------------------------------------|
| Confidence (self-report)                           | Medium | Medium                | Cheap talk; overtrust after fluent AI                      |
| Challenge-seeking (challenge_accept + why)         | Medium | <b>High</b>           | Must code mastery vs appearance/normative motive (XXXVIII) |
| Voluntary return after failure (retry_120s)        | Medium | <b>High</b>           | Needs coach content discipline (SE prompts)                |
| Delayed mixed / transfer_pass / solo_transfer_pass | Medium | <b>High</b>           | Anti-fake-mastery; anti-AI-crutch                          |
| “I am a math person” + behavior                    | Medium | <b>Highest target</b> | Slow, noisy; L3+ claim only                                |

### Working North Star (HYPOTHESIS):

**Challenge-seeking under safety with transfer** — the student chooses a harder problem for a *mastery* reason, returns after a miss, can pick the strategy on a mixed delayed set, and attributes success to strategy — without needing the model to finish the thought.

## I.4 Surviving doctrine stack (Synthesizer merge through LVIII)

Duplicate frameworks collapsed. Deep dives own citations; this table is **company law until killed**.

| Layer                                 | Surviving rule                                                                               | Source parts | Commercial use                                         |
|---------------------------------------|----------------------------------------------------------------------------------------------|--------------|--------------------------------------------------------|
| <b>FEI / SAFE-CRAFT</b>               | Safety → attempt → informative miss → earned win → attribution → transfer                    | XVI, VIII    | Session demo in marketing; not feature matrix          |
| <b>Claim ladder L0–L4</b>             | Do not sell identity (L3) from session A/Bs (L1)                                             | XXXIV        | Parent/copy hygiene; pre-reg estimands                 |
| <b>PWC (pedagogy wrap constraint)</b> | Guarded AI + ontology spine; success = solo_transfer_pass ≥ no-AI                            | XXXIII, XXXV | Ban “we’re Bastani’s Tutor”; ban unguarded Solver hero |
| <b>E×V×Cost</b>                       | Instrument expectancy, value, cost separately; student-authored utility > staff utility dump | XXXVII       | Kill single “motivation” KPI                           |
| <b>TARGET / mastery climate</b>       | Default-ban appearance UX + streak-as-North-Star; soft-wrong = mastery evaluation structure  | XXXVIII      | CTA copy audit; appearance climate kill                |

|                        |                                                                                                             |        |                                                          |
|------------------------|-------------------------------------------------------------------------------------------------------------|--------|----------------------------------------------------------|
| <b>SAFE-DD</b>         | Equip → destake → dose → frame → measure; desirable difficulty only after WM/threat capacity                | XLI    | Ban hardness-as-virtue; gate IL/SE by anxiety segment    |
| <b>SAFE-COMPARE</b>    | Criterion feedback default; peer ranks opt-in; no forced infinite named leaderboards for Maya               | XLII   | Leaderboard / league kill as growth engine               |
| <b>SAFE-HABIT</b>      | Cue practice not the counter; miss ≠ identity failure; habit metrics subordinate to FEI                     | XLIII  | Streak colonization kill; if-then onboarding OK          |
| <b>SAFE-REWARD</b>     | No expected tokens for mere engagement; informational feedback > inducement; transfer-gated unlocks         | XLIV   | XP-as-love / coin-for-open kill                          |
| <b>SAFE-RESILIENCE</b> | Growth-zone challenge + support recruitment; ban grit NS, danger-zone-as-DD, Resilience Score™              | XLV    | Grit-poster / character-meter kill                       |
| <b>SAFE-EXPECTANCY</b> | Task-only tutor briefs; ban trait/weakness labels; CIOF mediation; cold-start scripts                       | XLVI   | Pygmalion marketing kill; tutor ethics                   |
| <b>SAFE-TIMING</b>     | Encode → sleep → transfer claim; never reward midnight volume; stress/restriction-aware load; pressure ≠ DD | XLVII  | All-nighter / sleep-app brand kill                       |
| <b>SAFE-TRANSFER</b>   | Name the hop; VT vary-on-purpose; hug exam + bridge abstraction; story ≠ transfer; blocked ≠ ready          | XLVIII | Narrative-theater / blocked-accuracy vanity kill         |
| <b>SAFE-MISCON</b>     | Elicit → classify → destake → springboard → route; soft-wrong ≠ empty kindness; confidence-tiered feedback  | XLIX   | Mistake-poster / red-X-as-rigor / equal-wrong graph kill |



|                    |                                                                                                                             |       |                                                                                 |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------|
| <b>SAFE-CALIB</b>  | Item-level confidence<br>→ miss class → tiered<br>feedback; raise<br><i>appropriate</i><br>confidence, not Belief<br>Score™ | L     | Hide-correctness<br>diagnostic; ban<br>CA→grades marketing                      |
| <b>SAFE-DP</b>     | Diagnose → isolate →<br>goal → stretch →<br>attempt → method<br>feedback → revise →<br>prove → stop                         | LI    | Ban 10k-hours /<br>minutes / coverage<br>theater as DP                          |
| <b>SAFE-CoP</b>    | Name the practice;<br>peripheral = real work;<br>old-timer required;<br>trajectory > community<br>tab                       | LII   | Ban Discord/streak ≡<br>LPP; “join the<br>practice” not “join the<br>community” |
| <b>SAFE-RITUAL</b> | Light, optional,<br>practice-pointing<br>marks; doctrinal<br>default; anti-totalism;<br>no ritual→score                     | LIII  | Ban streak-as-liturg; y;<br>family/order milieu<br>copy                         |
| <b>SAFE-AAR</b>    | Brief → attempt with<br>traces → four-question<br>student-first close →<br>one improve-how                                  | LIV   | Ban bootcamp brand;<br>+25% ACT from<br>debrief metas                           |
| <b>SAFE-FILM</b>   | Objective clip + ≤3<br>cues + learner-timed<br>review + self∪model;<br>next-attempt<br>commitment                           | LV    | Ban tip-flood film<br>room; sports-academy<br>cosplay                           |
| <b>SAFE-MUSIC</b>  | Scales (goalful<br>ingredients) → chosen<br>repertoire →<br>transfer-as-recital proof                                       | LVI   | Ban virtuoso cosplay;<br>hours KPI; bare-pass<br>theater                        |
| <b>SAFE-ANNOT</b>  | Annotate-before-reveal<br>; principle tags; forced<br>critical restudy;<br>classify→drill;<br>annotate wins                 | LVII  | Ban GM cosplay; engi<br>ne-dump≡coaching;<br>rating NS                          |
| <b>SAFE-EXPOSE</b> | Situation hierarchy;<br>destaked approach;<br>expectancy violation;<br>fade safety behaviors;<br>multi-context              | LVIII | Ban therapy branding;<br>flooding; Calm<br>Score™; one-session<br>cure          |

|                                                  |                                                                                        |               |                                                                 |
|--------------------------------------------------|----------------------------------------------------------------------------------------|---------------|-----------------------------------------------------------------|
| <b>Block → near-miss interleave → spaced mix</b> | Blocking = acquisition scaffold; delayed mixed accuracy = readiness signal             | XXIX, XXXIX   | Ban “shuffle = science”; ban blocked-accuracy vanity            |
| <b>Student-generated why</b>                     | Faded examples + structured principle/misconception prompts <i>before</i> AI wrap      | XXVI, XL      | Ban AI-monologue≡SE; ban explain-own-wrong-first default        |
| <b>Competitive wedge</b>                         | Do not out-content Khan, out-streak Duo, out-delight Brilliant, or out-fluency ChatGPT | XXXV, XX      | Sell recoverable struggle + competence evidence + solo transfer |
| <b>Equity of worlds</b>                          | Story wrap is identity technology; tokenism and stereotype-cueing copy are kills       | XXXVI, XXVIII | HIST-EQ; belonging without “even you can”                       |

**Merged / demoted (do not treat as separate products):** “AI tutor,” “mastery path,” “growth mindset,” “engagement,” “grit,” “habit streak,” “celebrate mistakes,” “community of practice,” “deliberate practice brand,” “film study,” “recital,” “grandmaster review,” and “exposure therapy” are *not* independent North Stars — they are subordinate UX under FEI + the rows above (borrow *method*, never *costume*).

**Next research bottleneck:** Part LX **trust after AI hallucination.** Experiment families CAL/DP/CoP/RIT/AAR/FILM/MUSIC/ANNO/EXP-O/WTP gate confidence-tiered, session-architecture, participation, ritual, brief–debrief, review-media, ladder, postmortem, anxiety-approach, and parent-pricing claims above L1. Researcher count since v1.8 synthesizer = 1.

## Part II — Key Insights (challenged)

### Insight 1 — Fear is a performance bottleneck, not a personality trait

**FACT:** Math anxiety is associated with reduced working-memory availability during math tasks; anxious learners underperform relative to ability (Ashcraft & Kirk line of work; broader anxiety–WM literature).

**HYPOTHESIS:** For a large subset of MindCraft’s Maya archetype, the binding constraint is not “missing explanation,” it is **threat appraisal** (“I will look stupid”).

**Implication:** Soft-wrong UX, hide-correctness diagnostics, wizard coaching, and narrative framing are not polish — they are load-bearing if the thesis holds.

**Red Team:** Many high performers succeed under stress. Anxiety interventions have heterogeneous effects. Do not medicalize ordinary difficulty.

### Insight 2 — Growth mindset is real, small, and context-dependent

**FACT:** Yeager et al. (2019, *Nature*) — brief online growth-mindset intervention improved grades for lower-achieving students and increased advanced math enrollment in a nationally representative U.S. sample; effects stronger when peer norms supported challenge-seeking.

**FACT / REVIEW:** Domain-general mindset messaging shows mixed results; math-specific mindset interventions more often report positive outcomes (2023 systematic review in *Educational Research Review*).

**FACT / CAUTION:** “False growth mindset” (praise effort without teaching strategy / changing environment) is harmful (Dweck’s own caution; Kohn’s critique).

**Implication:** MindCraft must not ship empty “believe in yourself” copy. Mindset must be embodied in **task design, feedback, and tutor norms**.

### Insight 3 — Autonomy, competence, relatedness are the psychological tripod

**FACT:** Self-Determination Theory (Ryan & Deci) — sustained intrinsic motivation requires autonomy, competence, relatedness.

**FACT (meta):** SDT-based education interventions show reliable gains for autonomy and competence support; relatedness effects are less consistent (Wang et al. 2024 meta-analysis).

**Implication:** Product loops must deliver:

- Autonomy: meaningful choice (path, story world, pace)
- Competence: mastery grain that produces earned wins
- Relatedness: tutor presence, parent signal, peer-safe identity (hardest; do not fake with empty social)

### Insight 4 — Mastery learning works — until you measure the wrong thing

**FACT:** Kulik, Kulik & Bangert-Drowns (1990) meta-analysis — mastery programs show positive exam effects (overall ~0.5 SD in their synthesis), stronger for weaker students; can increase time; self-paced college mastery can reduce completion.

**FACT / CONTRADICTION:** Slavin’s “Mastery Learning Reconsidered” (1987) — little support for group-based mastery on *standardized* measures; experimenter-made tests show moderate effects; coverage vs mastery tradeoff is real.

**Implication:** MindCraft’s mastery graph is promising **if** mastery means transferable competence, not local quiz familiarity. Measure far transfer, not only same-item success.

### Insight 5 — Habit products ≠ identity products

**FACT / INDUSTRY:** Duolingo’s public method emphasizes gamification for motivation and return (streaks, leagues, variable rewards) (Duolingo Method whitepaper, 2023). Streaks exploit loss aversion; they drive DAU.

**HYPOTHESIS:** Streaks can increase practice frequency without changing “I am a math person.” Habit without identity yields brittle engagement (stop streak → stop learning).

**Implication:** Use habit mechanics as **on-ramps**, not as the destination. Pair with identity-marking rituals (map fill, story progression, tutor recognition of growth).

Insight 6 — Narrative is not decoration; it is encoding + meaning

**FOUNDER BELIEF (WORLD\_VISION):** Math stripped of invention-story becomes alienating machinery.

**HYPOTHESIS:** Historical/human invention frames increase curiosity and reduce “arbitrary rule” appraisals, improving persistence.

**Evidence status:** Strong tradition in math education (history of math pedagogy); fewer large RCTs than mindset literature. Treat as **design prior + research program**, not settled science.

Insight 7 — AI commoditizes *answers*, not *becoming*

**HYPOTHESIS:** As answer generation approaches free, willingness-to-pay migrates to:

- 1. Diagnosis of the *actual* break (gap map)
- 2. Emotional containment during struggle
- 3. Accountability relationships (tutor)
- 4. Identity-consistent practice worlds
- 5. Trusted outcome signaling to parents

**SPECULATION:** In 30 years, “show me the steps” is ambient. “Help me become the kind of person who stays with hard problems” remains a human+system problem.

Part III — Research Log (v1 seed)

| Date       | Finding                                                                                 | Type      | Action                                                          |
|------------|-----------------------------------------------------------------------------------------|-----------|-----------------------------------------------------------------|
| 2026-08-01 | Synthesizer v1.8:<br>merge L–LVIII<br>SAFE-* into I.4; kills<br>#15–#17                 | Synthesis | Surviving commercial<br>law; next = LIX parent<br>WTP           |
| 2026-08-01 | SAFE-EXPOSE:<br>graded approach; kill<br>therapy cosplay /<br>flooding / Calm<br>Score™ | Evidence  | EXP-O1...5;<br>expectancy violation                             |
| 2026-08-01 | SAFE-ANNOT /<br>MUSIC / FILM:<br>postmortem + ladder +<br>sparse clips                  | Evidence  | ANNOT/MUSIC/FIL<br>M-1...5                                      |
| 2026-07-31 | Synthesizer v1.7:<br>merge XLI–XLIX<br>SAFE-* into I.4; kills<br>#10–#14                | Synthesis | Surviving commercial<br>law; next = L<br>confidence calibration |

|            |                                                                           |           |                                                      |
|------------|---------------------------------------------------------------------------|-----------|------------------------------------------------------|
| 2026-07-31 | Misconceptions productive:<br>SAFE-MISCON;<br>soft-wrong ≠ empty kindness | Evidence  | MIS-1...4; springboard + classify                    |
| 2026-07-30 | Transfer/VT: story ≠ far transfer; blocked ≠ ready                        | Evidence  | TR-1...4;<br>SAFE-TRANSFER                           |
| 2026-07-30 | Sleep/stress:<br>encode→sleep→claim;<br>pressure ≠ DD                     | Evidence  | TIM-1...4;<br>SAFE-TIMING                            |
| 2026-07-29 | Synthesizer v1.6:<br>merge XXXIII–XL into I.4 doctrine stack              | Synthesis | Surviving commercial law; next was XLI<br>DD×anxiety |
| 2026-07-29 | SE prompts: AI monologue ■ SE;<br>structure before wrap                   | Evidence  | SE-1...4; densify Exp A/D                            |
| 2026-07-29 | Interleaving: block→n ear-miss→spaced mix;<br>kill shuffle=science        | Evidence  | IL-1...4; ban blocked vanity                         |
| 2026-07-29 | Goal climate:<br>appearance/streak NS killed; TARGET audit                | Evidence  | GO-1...4                                             |
| 2026-07-29 | Eccles E×V×Cost: kill single motivation KPI                               | Evidence  | EVT-1...4                                            |
| 2026-07-25 | Constituted Research Lab + Constitution v1                                | Process   | Create OS                                            |
| 2026-07-25 | Scarcity thesis partially false as slogans; true as product wedge         | Analysis  | Rewrite pitch language                               |
| 2026-07-25 | Yeager 2019 constrains mindset claims                                     | Evidence  | Ban empty mindset copy                               |
| 2026-07-25 | Soft-wrong + wizard coach shipped in product                              | Product   | Instrument coach → retry rate                        |
| 2026-07-25 | Mastery literature split (Kulik vs Slavin)                                | Evidence  | Define mastery measurement carefully                 |
| 2026-07-25 | Bloom 2-sigma overstated vs VanLehn / modern tutoring metas               | Evidence  | Stop marketing 2-sigma                               |
| 2026-07-25 | Founder sequence wounded → SAFE-CRAFT                                     | Synthesis | Attempt earlier; invention-story optional            |

|            |                                                                |          |                             |
|------------|----------------------------------------------------------------|----------|-----------------------------|
| 2026-07-25 | Deliberate practice<br>necessary≠sufficient<br>(Hambrick line) | Evidence | Progress-vs-self<br>framing |
|------------|----------------------------------------------------------------|----------|-----------------------------|

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## Part IV — Evidence Table (selected)

| Claim                                                             | Best supporting evidence                  | Contradicting / limiting evidence                                                              | Confidence  | MindCraft implication                         |
|-------------------------------------------------------------------|-------------------------------------------|------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------|
| Brief mindset intervention can move grades + advanced math taking | Yeager et al., Nature 2019                | Effects heterogeneous; peer norms matter; replication debates in broader mindset meta-analyses | Medium–High | Mindset must be ecological, not banner text   |
| Math-domain mindset > generic                                     | 2023 Ed Research Review systematic review | Study quality varies; publication bias possible                                                | Medium      | Math-specific identity language               |
| SDT need support raises motivation                                | Wang et al. 2024 meta                     | Relatedness effects weaker                                                                     | Medium–High | Design for autonomy + competence first        |
| Math anxiety impairs performance                                  | Cognitive literature (Ashcraft tradition) | Not all low performers are anxious                                                             | Medium–High | Affective onboarding is first-class           |
| Mastery learning raises achievement                               | Kulik et al. 1990                         | Slavin 1987 on standardized tests                                                              | Medium      | Mastery graph + transfer tests                |
| Gamified habits raise return                                      | Duolingo method / industry practice       | Habit ≠ deep learning; extrinsic traps (Deci)                                                  | Medium      | Streaks as on-ramp only                       |
| 1:1 tutoring is uniquely powerful                                 | Bloom “2 sigma” framing (1984)            | Often overstated; quality varies; VanLehn etc. nuance tutoring effects                         | Medium      | Keep human tutors; AI amplifies, not replaces |

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## Part V — Contradictory Evidence (do not skip)

1. **Mindset skepticism:** Some metas find small average effects; school implementations often degrade into posters.
  2. **Grit / resilience hype:** Duckworth-style grit is contested; structural inequality and curriculum quality matter more than character slogans.
  3. **Discovery learning failures:** Pure constructivism without guidance often fails (Kirschner, Sweller, Clark critiques). Narrative worlds must still teach.
  4. **Engagement ≠ learning:** Time-on-app can rise while understanding falls (edtech's original sin).
  5. **AI may increase explanation value temporarily:** Novices may trust fluent wrongness; trust becomes the scarce resource, not explanation volume.
  6. **Some students want speed and drills:** Identity-through-story is not universal. Offer multiple vessels.
- 

## Part VI — Mental Models

### VI.1 The Identity Transformation Cascade (ITC)

#### HYPOTHESIS — MindCraft original synthesis

Threat ↓ → Attention available → Micro-success → Attribution ("I figured it")  
→ Willingness to re-enter challenge → Accumulated competence evidence  
→ Narrative update ("maybe I can") → Identity claim ("I do math")  
→ Environment selection (harder courses, peers, tutors)  
→ Reinforcing loop

**Balancing loop:** Premature hard problems → threat ↑ → avoidance → identity freeze.

**Product translation:** Gap scan (no shame) → soft-wrong → tiny earned wins → map fill → tutor recognition → advanced path.

### VI.2 The Three Doors (what students actually need in a stuck moment)

| Door | Need    | Wrong product response        | Right response                                    |
|------|---------|-------------------------------|---------------------------------------------------|
| A    | Safety  | "Here's a longer explanation" | Contain affect; normalize miss                    |
| B    | Clarity | Pep talk                      | Precise model / worked example at right grain     |
| C    | Agency  | Take over and solve for them  | Scaffolded choice; student does the decisive step |

**FOUNDER BELIEF:** Tutors often open Door B first. Many students are still behind Door A.

## VI.3 Knowledge commodity stack (post-AI)

Layer 0: Facts & procedures → commoditizing fast  
Layer 1: Explanations → commoditizing fast  
Layer 2: Diagnosis of misconceptions → partially automatable; trust-sensitive  
Layer 3: Adaptive sequencing → automatable with good ontology  
Layer 4: Emotional co-regulation → scarce (human + careful UX)  
Layer 5: Identity & meaning → scarce  
Layer 6: Accountability relationship → scarce

MindCraft’s durable stack lives in **Layers 2–6**, with 0–1 as utilities.

## VI.4 Transfer from outside education

| Domain                                    | Mechanism                                | Transfer to MindCraft (surviving law)                          |
|-------------------------------------------|------------------------------------------|----------------------------------------------------------------|
| Games (Celeste, FromSoftware-lite design) | Fair difficulty + death as information   | Soft-wrong as physics (SAFE-MISCON / FEI)                      |
| Sports coaching                           | Film study of <i>specific</i> error      | SAFE-FILM: ≤3-cue attempt clips + self-model                   |
| Music                                     | Scales → repertoire → recital            | SAFE-MUSIC: ingredient drills → chosen pieces → transfer proof |
| Therapy (exposure)                        | Graded approach + expectancy violation   | SAFE-EXPOSE: situation hierarchy under destaked stakes         |
| Aviation / military                       | Brief → attempt → structured debrief     | SAFE-AAR: four-question student-first close                    |
| Chess annotation                          | Postmortem before engine; principle tags | SAFE-ANNOT: annotate→classify→drill; wins too                  |
| Martial arts belts                        | Visible competence ladder                | Mastery pips / map regions (criterion, not league)             |
| Language apps                             | Habit loops                              | SAFE-HABIT cue practice — without Duo’s extrinsic ceiling      |
| Religion / ritual                         | Shared meaning + belonging               | SAFE-RITUAL: light optional marks; anti-totalism               |
| Craft apprenticeship                      | Legitimate peripheral participation      | SAFE-CoP: real peripheral work + old-timer; not Discord KPIs   |

## Part VII — Universal Learning Principles (provisional)

1. **Emotion gates cognition** in threatened domains.



2. **Earned difficulty** beats random difficulty.
  3. **Feedback should be information, not identity judgment.**
  4. **Attribution matters:** strategy > talent for growth; luck attributions kill agency.
  5. **Spaced retrieval** beats massed re-reading (cognitive science consensus).
  6. **Worked examples** → **fading guidance** for novices (cognitive load).
  7. **Belonging uncertainty** destroys persistence for stereotyped groups (Walton/Cohen tradition).
  8. **Transfer requires varied practice**, not identical clones.
  9. **Human accountability** multiplies AI tools.
  10. **Identity is a lagging indicator**; design leading indicators (challenge-seeking, return-after-fail).
- 

## Part VIII — Product Implications (systems, not features)

### VIII.1 Core system: Fear → Evidence → Identity (FEI Loop)

#### Reinforcing loop R1 (desired):

Safety UX → attempt → informative miss → coach → success → map fill → identity → more attempts.

#### Balancing loop B1 (threat):

Public shame / red buzz → avoidance → less practice → skill lag → more shame.

**MindCraft already partially implements FEI:** hide-correctness diagnostic, soft-wrong, wizard coach, story chapters, knowledge map. **Missing instrumentation:** does coach raise retry rate and later challenge-seeking?

### VIII.2 What to stop optimizing

- Vanity DAU without learning transfer
- Explanation length as quality proxy
- Points that do not mark identity-relevant milestones

### VIII.3 What to instrument next (minimum viable science)

1. **Retry rate within 2 minutes of soft-wrong** (with vs without wizard coach)
2. **Challenge-seeking:** % choosing harder level when offered
3. **Return after fail session** (D1 retention after a loss session)
4. **Self-identity item** (pre/post, 1–2 items, math-specific)
5. **Advanced course intent / enrollment** (slow, gold)

### VIII.4 Tutor system role

Tutors are not content pipes. They are:

- Affect regulators
- Attribution coaches
- Bridge diagnosticians (concept A known, link A→B broken)
- Identity witnesses (“I saw you become someone who stays”)

AI should brief tutors, not replace witnessing.

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## Part IX — Experiments (pre-registered style)

Core A–D remain. Chapter ticks densified them into families — prefer the densified IDs when shipping.

### Experiment A — Wizard coach on soft-wrong

- **Question:** Does the under-Graph wizard increase productive retry vs soft-wrong alone?
- **Design:** A/B, chapter + practice
- **Primary:** retry within 120s; secondary: eventual correct without hint binge
- **Falsifier:** no lift; or lift only in engagement with worse accuracy
- **Densified content (v1.6):** Coach turns must be SE-constrained (principle / misconception contrast), not pep talk — see SE-1...4 / SE-2 as operational pairs with Experiment D.

### Experiment B — Story-first vs bare stem

- **Question:** Does invention-story framing raise persistence on first miss?
- **Design:** within-concept crossover
- **Primary:** time-to-abandon after first miss
- **Falsifier:** story slows without improving retention/transfer
- **Equity densify:** HIST-EQ / Part XXXVI — whose story, tokenism kill.

### Experiment C — North Star validation

- **Question:** Which early metric best predicts 8-week challenge-seeking?
- **Candidates:** streak length, soft-wrong retry, identity item, map mastery  $\Delta$ , delayed mix accuracy
- **Method:** observational + lagged prediction
- **v1.6 prior:** Expect streaks to lose; `retry_120s` + `transfer_pass` + `mastery-motive challenge_accept` to win.

### Experiment D — Explanation timing

- **Question:** Is explanation-before-attempt worse than attempt-with-safety for anxious students?
- **Design:** screen with anxiety pretest; randomize explanation-first vs attempt-first
- **Falsifier:** explanation-first wins for all segments
- **Densified:** Prefer SE-2/SE-3 (contrastive / structured generation) over monologue-first.

### Experiment families mounted in chapters (do not orphan)

| Family                  | Identifies (max claim without follow-up)                          | Parts   |
|-------------------------|-------------------------------------------------------------------|---------|
| AIT-*                   | Guarded AI → solo transfer / anti-sycophancy                      | XXXIII  |
| DAG-*                   | Estimand hygiene; covariate vs mediator                           | XXXIV   |
| CSA-*                   | Session preference vs competitors; paste under guards             | XXXV    |
| HIST-EQ / equity probes | Story belonging without stereotype cue                            | XXXVI   |
| EVT-*                   | Expectancy / value / cost levers                                  | XXXVII  |
| GO-*                    | Mastery vs performance CTA climate                                | XXXVIII |
| IL-*                    | Schedule → discrimination / delayed mix                           | XXXIX   |
| SE-*                    | Prompt regime → transfer / misconception recurrence               | XL      |
| DD-*                    | Equip/destake/dose × anxiety → retry + delayed mix                | XLI     |
| SC-*                    | Criterion vs peer rank climate → affect + challenge               | XLII    |
| HAB-*                   | Cue/if-then vs streak colonization → return without identity harm | XLIII   |
| IM-*                    | Informational vs engagement-loot rewards → free-choice + transfer | XLIV    |
| RES-*                   | Growth-zone + support recruitment vs grit theater                 | XLV     |
| EXP-*                   | Tutor brief / CIOF / AI expectancy → talk ratio + solo transfer   | XLVI    |
| TIM-*                   | Practice timing × sleep/stress → delayed transfer                 | XLVII   |
| TR-*                    | VT / hug / bridge sequence → format-hop + solo transfer           | XLVIII  |
| MIS-*                   | Soft-wrong springboard / mis-mapped distractors / confidence tier | XLIX    |
| CAL-*                   | Confidence-tiered feedback / hide-correctness / underconf unlock  | L       |

|         |                                                                          |       |
|---------|--------------------------------------------------------------------------|-------|
| DP-*    | Session spine / edge targeting / method feedback / effort cap            | LI    |
| CoP-*   | Named practice / peripheral work / old-timer / trajectory metrics        | LII   |
| RIT-*   | Light ritual vs silent / recovery marker / anti-streak liturgy           | LIII  |
| AAR-*   | Brief + four-question close / student-first / trace-backed debrief       | LIV   |
| FILM-*  | Sparse clip cues / self-controlled review / self↯model pairing           | LV    |
| MUSIC-* | Ladder UX / repertoire choice / identified technique / transfer recital  | LVI   |
| ANNOT-* | Annotate-before-reveal / forced restudy / classify→drill / annotate wins | LVII  |
| EXP-O-* | Situation hierarchy / expectancy check / fade safety / multi-context     | LVIII |
| WTP-*   | Parent CBC / FEI report vs score / message arms / cash pilot             | LIX   |

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## Part X — Weekly Priorities (research × product)

1. Instrument FEI loop events in analytics (retry, coach shown, write-mode exit).
  2. Run Experiment A for 2 weeks.
  3. Red Team the WORLD\_VISION narrative claims with one outside learning scientist.
  4. Interview 10 Mayas: “When did math stop feeling dangerous?” (qual).
  5. Define “mastery” operationally so Slavin’s critique cannot gut the graph.
- 

## Part XI — Founder Questions (do not dodge)

1. If ChatGPT tutors become “good enough,” what *relationship* do parents still buy?
  2. Are we building for Maya’s identity — or for parents’ anxiety about scores? Both? Tradeoff?
  3. Would we accept lower short-term accuracy for higher long-term challenge-seeking?
  4. Is the story world load-bearing, or is gap+tutor enough?
  5. What would falsify the identity-transformation thesis in 90 days?
-

## Part XII — Long-term Roadmap (research horizons)

**H1 (now):** Prove FEI loop moves retry + challenge-seeking.

**H2:** Prove identity language + map produce durable self-concept change.

**H3:** Prove human tutor + AI diagnosis beats AI-alone on persistence and transfer.

**H4:** Generalize beyond math (science/identity) only after math identity mechanism is solid.

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## Part XIII — Open Problems

1. Causal path from narrative history-of-math to identity (under-identified).
  2. Relatedness at scale without creepy social.
  3. Measuring identity without demand effects.
  4. Preventing gamification from colonizing meaning.
  5. AI fluent nonsense vs trust calibration for teens.
  6. Equity: does story-world privilege culturally specific narratives?
- 

## Part XIV — Red Team Dossier (destroy weak arguments)

### Kill #1: “Explanations are free, therefore MindCraft should not explain”

**Destroyed form:** Absolutism.

**Surviving form:** Explanations are *necessary infrastructure* with collapsing *willingness-to-pay*; differentiation is diagnosis + affect + identity + accountability.

### Kill #2: “Just fix mindset”

**Destroyed:** Poster mindset.

**Surviving:** Ecological mindset (task + feedback + peer/tutor norms), math-specific — TARGET structures (XXXVIII).

### Kill #3: “Engagement is the goal”

**Destroyed:** Engagement without transfer.

**Surviving:** Engagement as oxygen for the FEI loop, not the fire.

### Kill #4: “Mastery graph = Bloom 2-sigma”

**Destroyed:** Inflated tutoring mythology.

**Surviving:** Mastery *can* help weaker students if mastery is real (transfer\_pass) and time costs are managed.

**Kill #5: “Identity transformation is unmeasurable, so ship vibes”**

**Destroyed:** Vibes-as-strategy.

**Surviving:** Leading indicators (retry, challenge-seeking, delayed mix) + lagging identity items; L0–L4 claim ladder (XXXIV).

**Kill #6 (v1.6): “AI tutor / Bastani GPT Tutor is our product”**

**Destroyed:** Unguarded chat-as-learning; category-error marketing; guards-as-gain.

**Surviving:** Pedagogy wrap + solo\_transfer\_pass  $\geq$  no-AI bar (XXXIII RT).

**Kill #7 (v1.6): “Blocked accuracy / streaks / shuffle prove learning”**

**Destroyed:** Fluency illusion; streak North Star; random shuffle  $\equiv$  interleaving science.

**Surviving:** Short block  $\rightarrow$  near-miss interleave  $\rightarrow$  spaced mixed check (XXXIX); streaks as on-ramp only (HID).

**Kill #8 (v1.6): “Any explain box / AI monologue = self-explanation science”**

**Destroyed:** Untargeted justify-your-wrong-guess; chat length as depth.

**Surviving:** Structured principle/misconception SE on correct or known-incorrect content before AI wrap (XL).

**Kill #9 (v1.6): “One motivation dial”**

**Destroyed:** Collapsing expectancy and value; staff “math is useful” dumps as EVT.

**Surviving:**  $E \times V \times \text{Cost}$  triad with student-authored utility (XXXVII).

**Kill #10 (v1.7): “Harder is always better / desirable difficulty ignores anxiety”**

**Destroyed:** Anxiety-blind interleave/SE/pressure-as-DD; hardness theater as rigor brand.

**Surviving:** SAFE-DD equip  $\rightarrow$  destake  $\rightarrow$  dose  $\rightarrow$  frame  $\rightarrow$  measure (XLI); gate IL/SE by segment.

**Kill #11 (v1.7): “Leaderboards / streaks / XP create math identity”**

**Destroyed:** Appearance climates, streak colonization, engagement-loot as love-of-math.

**Surviving:** SAFE-COMPARE + SAFE-HABIT + SAFE-REWARD — criterion feedback, cue practice, informational unlocks only (XLII–XLIV).

**Kill #12 (v1.7): “Grit / Resilience Score™ / stay in the danger zone”**

**Destroyed:** Character meters; danger-zone-as-desirable-difficulty; grit posters as product.

**Surviving:** SAFE-RESILIENCE growth-zone + support recruitment; measure recovery/transfer not MRS theater (XLV).

**Kill #13 (v1.7): “Pygmalion labels / warm AI that finishes the problem”**

**Destroyed:** Trait/weakness tutor briefs; warmth-without-output-demand as expectancy science.

**Surviving:** SAFE-EXPECTANCY task briefs + CIOF; AI must clear PWC/solo\_transfer\_pass (XLVI + XXXIII).

### **Kill #14 (v1.7): “Story / sleep tips / mistake celebration = transfer”**

**Destroyed:** Narrative immersion as automatic far transfer; sleep-app brand; empty mistake posters; red-X-as-rigor; equal graph updates for all wrongs.

**Surviving:** SAFE-TRANSFER + SAFE-TIMING + SAFE-MISCON — vary on purpose, encode→sleep→claim, elicit→classify→springboard→route (XLVII–XLIX).

### **Kill #15 (v1.8): “Raise confidence / Belief Score™ / CA raises grades”**

**Destroyed:** Inflate-confidence brand; confidence meters as identity; confidence assessment alone as attainment lever; equal updates for guess vs high-conf misconception.

**Surviving:** SAFE-CALIB — item-level elicit → miss class → tiered feedback; appropriate confidence; C4 hide-correctness as diagnostic hygiene (L).

### **Kill #16 (v1.8): “Hours / forums / ritual→score / domain costume = identity”**

**Destroyed:** 10k-hours and minutes KPIs; Discord/streak ≡ CoP; unretracted ritual→performance marketing; bootcamp/conservatory/GM/therapy branding as belonging.

**Surviving:** SAFE-DP + SAFE-CoP + SAFE-RITUAL — diagnose→prove spine; named practice with real peripheral work; light optional marks pointing at recover/prove (LI–LIII).

### **Kill #17 (v1.8): “Tip-flood / engine-dump / flooding ≡ high-performance pedagogy”**

**Destroyed:** Uncapped film-room tips; Stockfish/LLM monologue as annotation; day-one timed mixed ACT as “exposure”; calm-to-zero / Calm Score™; loss-only shame reels; bare-pass recital theater.

**Surviving:** SAFE-AAR + SAFE-FILM + SAFE-MUSIC + SAFE-ANNOT + SAFE-EXPOSE — brief–debrief, sparse clips, skills→pieces→proof, annotate→drill, graded approach with expectancy violation (LIV–LVIII).

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Do not invent citations. Expand this list only with retrieved sources.

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## Part XVI — Mechanism Deep Dive: How identity actually changes

This part answers the research question at mechanism grain. It is synthesis, not settled science.

### XVI.1 What “confident mathematical thinker” means (operational)

**Do not use:** “likes math apps” or “says math is fun once.”

**Working definition (HYPOTHESIS):** A confident mathematical thinker is a person who, in math-relevant contexts:

1. **Approaches** moderately hard problems without immediate avoidance.
2. **Persists** after a miss long enough to extract information.
3. **Attributes** success primarily to controllable causes (strategy, practice, help-seeking) rather than luck or tutor magic.



4. **Selects** into harder future opportunities when available (courses, contests, careers) at rates above their prior baseline.

5. **Endorses** a math-capable self-concept *and* behaves consistently with that endorsement.

Items 1–4 are leading. Item 5 is lagging. Product should optimize leading indicators while periodically measuring the lagging identity claim.

## XVI.2 The threat–competence–narrative triad

Three literatures converge on the same transformation engine:

| Mechanism            | Core claim                                                                                         | Confidence                   | Key sources                                                                              |
|----------------------|----------------------------------------------------------------------------------------------------|------------------------------|------------------------------------------------------------------------------------------|
| Threat / affect      | Math anxiety transiently reduces working-memory resources available for math                       | <b>High (FACT direction)</b> | Ashcraft & Kirk (2001); Suárez-Pellicioni et al. review (2016); WM mediation meta (2021) |
| Competence evidence  | Motivation requires felt competence; mastery structures can raise achievement when mastery is real | <b>Medium–High</b>           | SDT (Ryan & Deci); Kulik mastery meta; Slavin critique                                   |
| Narrative / identity | Self-stories and belonging cues shape challenge-seeking and persistence                            | <b>Medium</b>                | Yeager et al. (2019); Walton/Cohen belonging tradition; narrative psychology             |

**HYPOTHESIS — Triad necessity:** For students starting from “I am bad at math,” improving any one factor alone is usually insufficient:

- Safety without competence → comfort without skill → identity still fragile.
- Competence without safety → possible for some; many never enter the practice volume.
- Narrative without either → empty affirmation (false growth mindset).

## XVI.3 Founder tutoring sequence — audit

**FOUNDER BELIEF (observed in tutoring):**

1. Reduce fear
2. Demystify mathematics
3. Explain why humanity invented it
4. Simplify the narrative
5. Create a tiny success
6. Reinforce success
7. Celebrate progress → identity shifts

**Red Team audit:**

| Step                 | Universal?                             | Evidence status                     | Failure mode                               |
|----------------------|----------------------------------------|-------------------------------------|--------------------------------------------|
| 1 Fear ↓             | Common for anxious subset; not for all | Strong for anxious learners         | Over-soothing that removes challenge       |
| 2 Demystify          | Often helpful                          | Medium (cognitive clarity)          | Over-simplification that lies              |
| 3 Invention story    | Founder prior                          | <b>Under-evidenced at RCT scale</b> | Cultural mismatch; time cost               |
| 4 Simplify narrative | Often helpful                          | Medium                              | Removes necessary difficulty               |
| 5 Tiny success       | Near-universal lever                   | High (competence need)              | Too-easy wins → hollow identity            |
| 6 Reinforce          | High if attribution-correct            | High (feedback literature)          | Praise of talent or praise of empty effort |
| 7 Celebrate          | Helpful if genuine                     | Medium                              | Performative celebration → cynicism        |

#### Improved sequence (HYPOTHESIS — “SAFE-CRAFT”):

1. Safety: threat appraisal ↓ (tone, privacy, soft-wrong, no public shame)
2. Attempt: student acts before receiving a full lecture (productive struggle at right grain)
3. Feedback-as-information: miss becomes diagnosis, not character
4. Earned micro-win: success the student can own
5. Causal story: “what worked” (strategy), optionally “why humans invented this tool”
6. Re-entry: immediate second attempt / next slightly harder item
7. Attribution + witness: tutor/system names the growth specifically
8. Future self: map fill / path unlock that marks identity-relevant progress
9. Transfer check: varied item so win was not clone memorization

**What changed vs founder sequence:** Attempt moves earlier; invention-story becomes optional enrichment inside step 5, not a prerequisite to agency; transfer check prevents fake mastery.

## XVI.4 Math anxiety — what is known, what is not

**FACT:** Higher math anxiety correlates with lower math performance (meta  $r \approx -0.17$  in one 2021 synthesis of 57 studies — modest average, not destiny).

**FACT:** Ashcraft & Kirk (2001) show math anxiety can act like a dual-task load: worry competes for working memory, especially on computation-span / carry-heavy tasks.

**HYPOTHESIS:** In product UX, anything that increases social-evaluative threat (red buzzers, public leaderboards for novices, tutor contempt, parent hovering UI) will shrink effective WM and look like “they don’t get it.”

#### Contradictions / limits:

- Not all underperformance is anxiety; some is missing knowledge, poor curriculum, sleep, language barriers.
- Trait vs state anxiety differ; trait may partly reflect prior failures (reverse causality possible).
- High performers can be anxious and still succeed via avoidance of hard electives — a hidden identity cost.

**MindCraft rule:** Treat anxiety as a first-class load on the learning system, not as a soft skill sidebar.

## XVI.5 Deliberate practice — necessary, not sufficient

**FACT / TRADITION:** Ericsson’s deliberate practice emphasizes structured, feedback-rich, effortful practice aimed at improvement (not mere repetition).

**FACT / CRITIQUE:** Hambrick and colleagues argue deliberate practice is important but not sufficient; in chess/music reanalyses, deliberate practice often explains roughly ~1/3 of reliable variance, leaving most unexplained (abilities, starting age, opportunity, etc.).

**Implication for MindCraft:** Do not sell “10,000 hours.” Sell **high-quality attempts under feedback**. Also respect individual differences: same practice volume will not equalize outcomes. Identity work must include “progress relative to self,” not only absolute rank.

## XVI.6 Tutoring effects — demythologize Bloom

**FACT:** Bloom (1984) popularized the “2 sigma” tutoring + mastery framing.

**FACT / CALIBRATION:** Broader tutoring metas are far smaller (often ~0.3–0.4 SD in modern RCT syntheses cited by Education Next). VanLehn (2011) estimated human tutoring ~0.79 SD vs no tutoring, with step-based ITS nearly comparable (~0.76).

**Implication:** Human tutors remain valuable, especially for affect + accountability + diagnosis of weird misconceptions. They are not magic 2-sigma machines. AI step-level feedback can capture much of the *cognitive* tutoring benefit; the scarce remainder is Layers 4–6 (emotion, identity, accountability).

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# Part XVII — Scarcity Thesis: Full Red Team Trial

## XVII.1 The claim under oath

**FOUNDER BELIEF:** AI commoditizes knowledge; future scarcity is motivation, confidence, identity, trust, curiosity, persistence, emotional safety, meaning.

## XVII.2 Cross-examination

### Claim A — “Explanations are free”

**Verdict:** Directionally **FACT** for commodity explanations; **FALSE** for *trusted, personalized, correct, curriculum-aligned* explanations under time pressure.

**Evidence:** Generative AI + open platforms collapse marginal cost of text explanation.

**Contradiction:** Hallucinations, shallow fluency, and misalignment create a *trust tax*. Parents may still pay for verified human judgment.

### **Claim B — “Tutoring is free”**

**Verdict:** FALSE.

AI chat is cheap. High-quality tutoring includes scheduling, relationship, emotional labor, accountability, and reputation. Those remain expensive. Nickow et al.–style tutoring metas still show meaningful effects when tutoring is structured — it is not free at scale.

### **Claim C — “Content is free”**

**Verdict:** **MOSTLY FACT** for undifferentiated content; **FALSE** for sequenced, assessed, outcome-linked content systems.

Khan/YouTube prove content abundance. Differentiation is curation + measurement + human wrap.

### **Claim D — “Motivation/identity become the scarce goods”**

**Verdict:** HYPOTHESIS with strategic plausibility.

As cognitive help approaches free, *willingness to engage difficulty* becomes the binding constraint for many students. This does not mean motivation products automatically win — many habit apps fail to produce identity.

### **Claim E — “Emotional safety is first-order”**

**Verdict:** HYPOTHESIS — true for anxious / threatened learners; not universal.

For some, speed drills and competition *increase* engagement. Segment, do not romanticize softness.

## **XVII.3 Surviving thesis (company doctrine)**

**Adopted doctrine (refined v1.7):**

> MindCraft does not compete primarily on explanation volume, AI tutor warmth, streak/XP theater, grit meters, or story immersion. It competes on converting threatened learners into challenge-seeking mathematical agents through diagnosis, affective design, mastery-climate evaluation, sequenced desirable difficulties (SAFE-DD), student-generated why, variation-designed transfer, productive-error routing, narrative meaning under equity constraints, and human accountability — with AI as *guarded* infrastructure that must clear `solo_transfer_pass ≥ no-AI`.

**Falsifiers (90-day capable):**

1. Anxious segment shows no FEI lift vs explanation-first control.
2. Parents refuse to pay when score gains lag identity metrics.

3. AI-alone matches human+AI on persistence + transfer for target segment.
  4. Guarded Solver raises concurrent scores but loses solo\_transfer\_pass vs no-AI (PWC fail).
  5. Interleave/SE regimes raise delayed mix but destroy retry\_120s in high-anxiety segment without sequenced SAFE-DD (XLI; pending DD-1/DD-2).
  6. Soft-wrong without springboard/route matches key-only on retry + misconception recurrence (MIS-1 fail).
  7. VT/hug sequences fail to beat random surface skins on next-day format-hop transfer (TR-1 fail).
- 

## Part XVIII — Systems Maps (loops, not features)

### XVIII.1 FEI reinforcing loop (desired)

```
+----- identity claim ("I do math") <---+
| |
v |
safety UX ----> attempt ----> informative miss ----> coach
^ | |
| v v
+----- map fill / witness <---- earned success <+
```

### XVIII.2 Shame balancing loop (destroy this)

```
public failure signal -> threat ↑ -> avoidance -> fewer attempts
^ |
+----- skill lag / worse scores -----+
```

### XVIII.3 Extrinsic colonization loop (danger)

```
streak / points ↑ -> return ↑ -> shallow grinding ↑
| |
+-- identity unchanged -----+--> burnout / streak break → exit
```

**Design rule:** Habit loops may feed FEI but must not replace identity markers.

### XVIII.4 Parent trust loop

```
visible diagnosis + honest progress -> parent trust ↑ -> continued purchase
^ |
+---- outcome evidence (scores / challenge-seeking)
```

**FOUNDER QUESTION unresolved:** Optimize Maya's identity or parent's anxiety? Doctrine: report *both* challenge-seeking and skill evidence; never fake either.

### XVIII.5 Network effects (honest)

MindCraft today has weak classic network effects. Do not pretend otherwise.

**Possible compounding advantages (HYPOTHESIS):**

1. Ontology + misconception memory → better diagnosis over time (data moat if privacy-respecting).
2. Tutor playbooks trained on FEI outcomes → service quality moat.
3. Story worlds with longitudinal character arcs → switching costs of meaning (fragile; easy to cargo-cult).

---

## Part XIX — Cross-Domain Transfer Catalog

Principles that repeatedly change human behavior outside classrooms — filtered for MindCraft use.

| Domain               | Robust mechanism                     | Transfer                                | 30-year durable? | AI improve?                  |
|----------------------|--------------------------------------|-----------------------------------------|------------------|------------------------------|
| Exposure therapy     | Graded exposure under safety         | Anxiety-graded problem sets             | Yes              | Yes (personalize gradient)   |
| Sports film study    | Specific error review                | Wrong-choice coach                      | Yes              | Yes (auto clip of miss type) |
| Music pedagogy       | Technique → repertoire → performance | Drill → quest → “recital” signal        | Yes              | Partial                      |
| Martial arts         | Visible ranks + dojo norms           | Map regions + tutor dojo culture        | Yes              | Weak for norms               |
| Aviation             | Checklists under stress              | Procedure scaffolds when anxious        | Yes              | Yes                          |
| Games (Celeste-like) | Fair failure + retry culture         | Soft-wrong physics                      | Yes              | Yes                          |
| Chess                | Annotated game review                | Session postmortems                     | Yes              | Yes                          |
| Coaching             | Identity + accountability            | Tutor as witness                        | Yes              | Assist, not replace          |
| Religion/ritual      | Meaning + belonging                  | Light rituals only; avoid cult dynamics | Yes (human)      | Risky                        |
| Creator economy      | Public craft identity                | Optional shareable math artifacts       | Uncertain        | Yes                          |

**Anti-transfer:** Casino variable rewards → addiction without skill. Do not copy engagement maxing from social media.

---

## Part XX — Competitive Landscape (mechanism lens)

| Product               | Primary scarce resource they sell                              | Identity claim?                    | Risk                                                   |
|-----------------------|----------------------------------------------------------------|------------------------------------|--------------------------------------------------------|
| Khan Academy          | Free mastery content + reputation                              | Weak–medium                        | Explanation commodity; “mastery” without transfer_pass |
| Duolingo              | Habit / streak motivation                                      | Weak (language identity sometimes) | Extrinsic ceiling; appearance of consistency           |
| Brilliant             | Aesthetic problem joy + prestige                               | Medium                             | Narrow segment; not Maya threat profile                |
| Coursera/edX          | Credentials                                                    | Medium (career identity)           | Completion collapse                                    |
| Chegg / homework help | Answers under deadline                                         | Anti-identity (outsourcing)        | Integrity & AI shock                                   |
| Private tutors        | Human accountability + customization                           | High when good                     | Supply constrained                                     |
| ChatGPT tutors        | Instant explanation                                            | Low unless wrapped                 | Trust / hallucination; Bastani Base harm               |
| MindCraft (target)    | FEI conversion + tutor witness + gap diagnosis + solo transfer | <b>Intended high</b>               | Must prove, not assert                                 |

**Strategic implication (v1.8 densified):** Do not out-Khan Khan on content breadth. Do not out-Duo Duo on streaks/leagues/XP. Do not out-Brilliant Brilliant on puzzle delight. Do not out-ChatGPT on fluency. Do not out-grit character apps on Resilience Score™. Do not out-therapy Calm Score™, sports academies, conservatories, or grandmaster cosplay. Out-compete on the **session**: SAFE-DP spine under SAFE-DD, productive-error + calibration routing (SAFE-MISCON / CALIB), brief–debrief + sparse film + annotation (SAFE-AAR / FILM / ANNOT), ladderred approach for feared formats (SAFE-MUSIC / EXPOSE), variation-designed transfer (SAFE-TRANSFER), student-generated why, and transfer when help is gone — then *say that* in marketing (CSA-2).

---

## Part XXI — Metrics Dictionary (instrumentation)

### XXI.1 Banned as North Stars

- Raw DAU / time-on-app without transfer
- Streak length alone
- Explanation open-rate / AI chat token count
- Points / coins / XP velocity
- Blocked-session accuracy alone

- Coach thumbs-up / “felt helpful” without solo\_transfer\_pass
- Named leaderboard rank / league tier
- Resilience Score™ / grit meter / MRS as in-app KPI
- Midnight minutes practiced / all-nighter volume
- Belief Score™ / confidence meters / confidence streaks
- Community tab visits / Discord volume / games played / rating
- Calm Score™ / anxiety-must-hit-zero meters
- Tip-recall / film-room engagement without method-change + transfer

## XXI.2 Leading indicators (ship first)

| Metric ID               | Definition                                                       | Why                                           |
|-------------------------|------------------------------------------------------------------|-----------------------------------------------|
| retry_120s              | New attempt on same or isomorphic item within 120s of soft-wrong | Persistence under safety                      |
| coach_shown             | Wizard/coach surfaced                                            | Treatment exposure                            |
| write_exit_to_retry     | Leave write mode then attempt                                    | Agency after reflection                       |
| challenge_accept        | Chose harder level when offered                                  | Challenge-seeking (instrument <i>motive</i> ) |
| hint_binge              | ≥3 hints without independent solve                               | Gaming / helplessness / failed exposure fade  |
| transfer_pass           | Correct on varied item after mastery mark                        | Anti-fake-mastery                             |
| solo_transfer_pass      | Transfer with AI/Solver closed or denied                         | Anti-crutch (XXXIII)                          |
| ai_reveal_rate          | Full-answer / unguarded reveal frequency                         | Crutch exposure                               |
| strategy_class_error    | Wrong <i>procedure family</i> on mixed set                       | Interleaving discrimination (XXXIX)           |
| se_principle_hit        | Correct structured principle/ingredient pick when prompted       | SE quality proxy (XL)                         |
| help_recruit_then_solo  | Opened card/tutor/peer help then solved without full reveal      | SAFE-RESILIENCE support path (XLV)            |
| misconception_route_hit | Soft-wrong routed to mapped mis_ / ingredient failure_mode       | SAFE-MISCON classification (XLIX)             |
| format_hop_pass         | Correct after declared format/context hop (VT check)             | SAFE-TRANSFER discernment (XLVIII)            |
| confidence_miss_tier    | High vs low confidence on miss (when elicited)                   | SAFE-CALIB / hypercorrection (L)              |



|                         |                                                     |                                    |
|-------------------------|-----------------------------------------------------|------------------------------------|
| annotate_before_reveal  | Student principle/why logged before key/coach/model | SAFE-ANNOT / SAFE-AAR (LVII / LIV) |
| film_cue_count          | Coaching points attached to attempt clip (cap ≤3)   | SAFE-FILM less-is-more (LV)        |
| avoided_format_approach | Attempt on previously avoided format/stakes step    | SAFE-EXPOSE approach rate (LVIII)  |
| improve_how_specificity | Debrief yields concrete next-attempt tweak          | SAFE-AAR close quality (LIV)       |

### XXI.3 Lagging indicators

| Metric ID          | Definition                                          |
|--------------------|-----------------------------------------------------|
| math_person_item   | 1–7 agreement: “I am a math person” (math-specific) |
| anxiety_state_item | Short state anxiety before session                  |
| advanced_intent    | Intent to take harder course / contest              |
| tutor_witness_note | Tutor tagged identity-relevant growth               |

### XXI.4 Decision rule (HYPOTHESIS)

Ship changes that raise retry\_120s and mastery-motive challenge\_accept without raising hint\_binge / ai\_reveal\_rate, and without dropping transfer\_pass or solo\_transfer\_pass. Prefer delayed mixed accuracy over blocked session accuracy.

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## Part XXII — Appendices

### Appendix A — Glossary

| Term                 | Meaning                                        |
|----------------------|------------------------------------------------|
| FEI Loop             | Fear → Evidence → Identity system              |
| ITC                  | Identity Transformation Cascade                |
| Soft-wrong           | Miss treated as information, not shame theater |
| Gap scan             | Low-shame diagnostic of confidence/exposure    |
| Challenge-seeking    | Choosing harder math opportunity when optional |
| False growth mindset | Effort praise without strategy / conditions    |
| SAFE-CRAFT           | Improved tutoring sequence (Part XVI.3)        |

|                                 |                                                                                                                        |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------|
| Layers 0–6                      | Knowledge commodity stack (Part VI.3)                                                                                  |
| HID                             | Habit/Identity Divergence — streaks can raise return without identity                                                  |
| MoC                             | Memory of Competence — spaced retrieval after transfer_pass                                                            |
| PWC                             | Pedagogy Wrap Constraint — guarded AI + student generation + solo transfer bar                                         |
| Claim ladder L0–L4              | Max defensible causal/marketing claim given design (Part XXXIV)                                                        |
| $E \times V \times \text{Cost}$ | Eccles expectancy $\times$ subjective task value $\times$ cost triad                                                   |
| TARGET                          | Classroom/goal-structure dimensions cueing mastery vs performance                                                      |
| SAFE-DD                         | Equip $\rightarrow$ destake $\rightarrow$ dose $\rightarrow$ frame $\rightarrow$ measure (Part XLI)                    |
| SAFE-COMPARE                    | Criterion-default comparison policy (Part XLII)                                                                        |
| SAFE-HABIT                      | Cue practice without streak colonization (Part XLIII)                                                                  |
| SAFE-REWARD                     | Informational / non-controlling reward policy (Part XLIV)                                                              |
| SAFE-RESILIENCE                 | Growth-zone + support recruitment, not grit meters (Part XLV)                                                          |
| SAFE-EXPECTANCY                 | Task briefs + CIOF; ban trait labels (Part XLVI)                                                                       |
| SAFE-TIMING                     | Encode $\rightarrow$ sleep $\rightarrow$ delayed transfer claim (Part XLVII)                                           |
| SAFE-TRANSFER                   | Variation/hug/bridge transfer design (Part XLVIII)                                                                     |
| SAFE-MISCON                     | Productive-error elicit $\rightarrow$ classify $\rightarrow$ springboard $\rightarrow$ route (Part XLIX)               |
| SAFE-CALIB                      | Item-level confidence $\rightarrow$ miss class $\rightarrow$ tiered feedback (Part L)                                  |
| SAFE-DP                         | Diagnose $\rightarrow$ isolate $\rightarrow$ attempt $\rightarrow$ method feedback $\rightarrow$ prove spine (Part LI) |
| SAFE-CoP                        | Named practice; peripheral = real work; trajectory > tab (Part LII)                                                    |
| SAFE-RITUAL                     | Light optional practice-pointing marks; anti-totalism (Part LIII)                                                      |
| SAFE-AAR                        | Brief $\rightarrow$ traces $\rightarrow$ four-question student-first close (Part LIV)                                  |
| SAFE-FILM                       | Sparse attempt clips + self model + next commitment (Part LV)                                                          |



## B — Story frame

- Arms: invention-story wrapper vs bare stem (same math)
- Primary: time-to-abandon after first miss
- Secondary: enjoyment, retention @ 7d

## D — Explanation timing × anxiety

- Screen: brief anxiety item
- Arms: explain-first vs attempt-first
- Heterogeneity: high vs low anxiety

## Appendix E — Red Team standing orders

1. Attack any claim that cannot name a falsifier.
2. Attack any metric that can be gamed without learning.
3. Attack any narrative feature without a learning outcome.
4. Attack Bloom 2-sigma mythology on sight.
5. Attack “AI makes tutoring free” slogans on sight.

## Appendix F — Future chapters (queued, not padded)

1. Neuroscience of math anxiety (amygdala/WM models; cautious translation)
2. History of mathematics as meaning technology — under trial via XXVIII + HIST
3. Parent trust economics & willingness-to-pay — **DONE** Part LIX / SAFE-WTP (builds EVT-4 / XXVII)
4. Equity audit of story worlds — **DONE** (Part XXXVI)
5. Competitive teardown — **DONE** session audits (Part XXXV); usage telemetry still open
6. Formal Bayesian update process for Constitution claims
7. **DONE (2026-07-30)**: Desirable difficulties × anxiety (Part XLI) — SAFE-DD stack
8. **DONE (2026-07-30)**: Social comparison & leaderboards (Part XLII) — SAFE-COMPARE stack
9. **DONE (2026-07-30)**: Habit formation science without identity colonization (Part XLIII) — SAFE-HABIT stack
10. **DONE (2026-07-30)**: Intrinsic motivation killers / Deci caveats (Part XLIV) — SAFE-REWARD stack
11. **DONE (2026-07-30)**: Mathematical resilience (Part XLV / id 45) — SAFE-RESILIENCE stack
12. **DONE (2026-07-30)**: Teacher/tutor expectancy effects (Part XLVI / id 46) — SAFE-EXPECTANCY stack
13. **DONE (2026-07-30)**: Sleep, stress, and learning (Part XLVII / id 47) — SAFE-TIMING stack
14. **DONE (2026-07-30)**: Transfer & variation theory (Part XLVIII / id 48) — SAFE-TRANSFER stack
15. **DONE (2026-07-31)**: Misconceptions as productive (Part XLIX / id 49) — SAFE-MISCON stack
16. **Synthesizer v1.7 (2026-07-31)**: Merged XLI–XLIX into I.4 / XIV / IX / XXI — no new chapter
17. **DONE (2026-07-31)**: Confidence calibration (Part L / id 50) — SAFE-CALIB stack
18. **DONE (2026-07-31)**: Deliberate practice in tutoring sessions (Part LI / id 51) — SAFE-DP stack
19. **DONE (2026-07-31)**: Community of practice / Lave–Wenger (Part LII / id 52) — SAFE-CoP stack
20. **DONE (2026-07-31)**: Religion/ritual light-touch (Part LIII / id 53) — SAFE-RITUAL stack
21. **DONE (2026-07-31)**: Military/aviation brief-debrief (Part LIV / id 54) — SAFE-AAR stack
22. **DONE (2026-07-31)**: Sports film-study pedagogy (Part LV / id 55) — SAFE-FILM stack

23. **DONE (2026-08-01):** Music pedagogy ladders (Part LVI / id 56) — SAFE-MUSIC stack
  24. **DONE (2026-08-01):** Chess annotation & metacognition (Part LVII / id 57) — SAFE-ANNOT stack
  25. **DONE (2026-08-01):** Therapy graded exposure (Part LVIII / id 58) — SAFE-EXPOSE stack
  26. **Synthesizer v1.8 (2026-08-01):** Merged L–LVIII into I.4 / XIV / IX / XXI — no new chapter
  27. **DONE (2026-08-01):** Parent WTP experiments design (Part LIX / id 59) — SAFE-WTP stack
  28. **NEXT:** Trust after AI hallucination (Part LX / id 60)
- 

## Part XXIII — Research Lab Operating Cadence

### Weekly (60–90 min):

1. One claim enters trial (evidence + contradiction).
2. Red Team attempts kill.
3. Surviving claim updates Constitution with confidence tag.
4. One experiment metric reviewed.

**Synthesizer (~every 8 researcher ticks):** Merge duplicate frameworks into Part I.4 / XIV; refresh Executive Summary; regenerate PDF. Do not write a new chapter on synthesizer ticks.

### Monthly:

- Rewrite one weak chapter.
- External reader (learning scientist or skeptical founder).
- Regenerate PDF.

### Quarterly:

- Kill or promote North Star metric.
  - Publish internal “what we believed that was wrong” memo.
- 

## Closing stance

MindCraft’s deepest risk is not technical failure. It is **winning the wrong game**: becoming the best explanation engine in a world where explanations are cheap — or the warmest AI tutor in a world where warmth without solo transfer is a crutch — while losing the only game that compounds: helping a human revise who they are in the presence of difficulty.

This Constitution exists so the company notices that risk early, and runs experiments that can kill beloved ideas.

**v1.8 synthesizer pass folded Parts L–LVIII into surviving commercial doctrine (I.4 rows SAFE-CALIB through SAFE-EXPOSE; Red Team kills #15–#17). Part LIX added SAFE-WTP experiment design (parent CBC + message tests). Next researcher id: LX trust after AI hallucination. Page count is not the finish line — falsifiable truth is.**

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## Part XXIV — Affect, Anxiety, Belonging (deep dive)

**Chapter status:** Living evidence brief

**Primary question:** When is emotional safety causally load-bearing for mathematical confidence?

**Owners:** Cognitive Neuroscientist seat · Ed Psych seat · Red Team

---

### XXIV.1 What is happening, mechanistically?

Three related but non-identical constructs get collapsed in product talk. Separate them.

| Construct                    | Definition (working)                                      | Time scale               | Product signal                                      |
|------------------------------|-----------------------------------------------------------|--------------------------|-----------------------------------------------------|
| <b>State math anxiety</b>    | Transient dread / arousal in a math moment                | Seconds–minutes          | Avoid click, freeze, rapid hint binge               |
| <b>Trait math anxiety</b>    | Stable tendency to experience math as threat              | Months–years             | Course avoidance, “I’m not a math person”           |
| <b>Belonging uncertainty</b> | “People like me don’t belong in this domain”              | Contextual, can be acute | Withdrawal after mild adversity                     |
| <b>Stereotype threat</b>     | Extra evaluative pressure from group stereotype relevance | Situational              | Underperformance vs ability when stereotype is cued |

**FACT (direction):** Math anxiety is associated with worse math performance; one 2021 meta-analysis reported mean  $r \approx -0.17$  across 57 studies (modest average association — not destiny).

**FACT (mechanism candidate):** Ashcraft & Kirk (2001) show high math-anxiety individuals have reduced working-memory span especially on computation-based span tasks; dual-task math + memory load amplifies errors/RT. Anxiety behaves like a competing load on central executive resources.

**FACT (related):** Reviews (e.g., Suárez-Pellicioni et al., 2016) organize explanations as (a) WM competition, (b) attentional control / inhibition deficits, (c) possible low-level numerical representation differences — not mutually exclusive.

**HYPOTHESIS for MindCraft:** For the Maya-anxious segment, UX that increases social-evaluative threat (shame theater, public ranking early, contemptuous feedback) will look like a “knowledge gap” while actually being a temporary capacity tax.

---

### XXIV.2 Why does this happen? (first principles)

1. **Working memory is scarce.** Multi-step math is a WM sport. Anything that occupies the central executive (worry, self-monitoring for stereotype confirmation, rumination about looking stupid) steals the resource the task needs.
2. **Humans are status-sensitive learners.** Math is a publicly ranked school subject. Threat is often social (“I will be seen as dumb”) more than arithmetic.
3. **Avoidance is rational short-term.** Leaving the room ends the aversive arousal. Long-term, avoidance prevents the mastery experiences that would update efficacy.
4. **Identity consolidates avoidance.** Repeated exits become “I’m not a math person,” which then licenses future exits (narrative lock-in).

**Human need satisfied by safety design:** Need for non-humiliation while competence is still fragile; relatedness/respect; predictability.

---

## XXIV.3 Belonging and stereotype threat — what transfers?

**FACT:** Steele & Aronson tradition — when a test is framed as ability-diagnostic under a negative group stereotype, performance can drop relative to non-diagnostic framing (classic lab pattern; effect sizes and replication quality vary by context).

**FACT:** Spencer, Steele & Quinn (1999) line — women/math stereotype relevance can impair performance when threat is cued.

**FACT:** Walton & Cohen (2007, 2011) — belonging uncertainty makes adversity feel identity-diagnostic; brief belonging interventions framing adversity as common/transient improved long-term outcomes for marginalized college students in landmark trials (e.g., Science 2011 belonging intervention).

### Red Team limits:

- Effects are **heterogeneous** and context-dependent (like mindset).
- School-wide scaling often attenuates lab effects.
- MindCraft must not cargo-cult a 45-minute belonging essay into a gamified toast.
- Belonging interventions are not a substitute for teaching.

### Surviving transfer:

- Normalize struggle without normalizing low standards.
  - Make early adversity non-identity-diagnostic (“misses are how the map learns”).
  - Avoid stereotype-cueing copy (“even girls can...” is often worse).
  - Tutor witnessing is a belonging technology when authentic.
- 

## XXIV.4 Neuroscience: what we can say without overclaiming

**FACT / REVIEW level:** Math anxiety literature includes ERP/fMRI correlates; reviews document affective and cognitive network involvement. Exact “brain region product features” are **SPECULATION** and should not drive roadmap.

**Allowed claim:** Affective arousal and executive control interact; designing for lower threat is biologically plausible.

**Forbidden claim:** “Our purple calm mode activates the prefrontal cortex.”

XXIV.5 Contradictory evidence (keep visible)

- 1. Many students underperform from **missing knowledge**, not anxiety.
- 2. Some learners seek **competitive arousal** and dislike soft UX.
- 3. Reverse causality: low skill → anxiety → further avoidance (bidirectional).
- 4. Trait anxiety interventions that only soothe without competence can create comfort without growth.
- 5. Meta correlations are modest; anxiety is one bottleneck among many.

XXIV.6 Can this generalize? 30 years? AI?

| Question                | Answer                                                                                                                                    |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Generalize beyond math? | Yes — any high-status evaluative domain (music juries, coding interviews).                                                                |
| Still work in 30 years? | Yes — human threat systems are not a 2026 fad.                                                                                            |
| Can AI improve it?      | Yes for personalization of challenge gradient & detection of freeze patterns; risky for fake empathy.                                     |
| MindCraft change        | Soft-wrong, hide-correctness diagnostics, coach tone, private early practice, tutor witness scripts are load-bearing for anxious segment. |

XXIV.7 Experimental validation program

| ID    | Question                                            | Design                  | Primary          | Falsifier               |
|-------|-----------------------------------------------------|-------------------------|------------------|-------------------------|
| AFF-1 | Does soft-wrong raise retry vs hard-fail UX?        | A/B                     | retry_120s       | No lift / accuracy down |
| AFF-2 | Does anxiety-state moderate explanation-first harm? | Anxiety screen × timing | transfer + retry | No interaction          |



|       |                                                                   |                |                           |                     |
|-------|-------------------------------------------------------------------|----------------|---------------------------|---------------------|
| AFF-3 | Does “misses are information” copy reduce belonging threat items? | Pre/post items | belonging/threat items    | Demand effects only |
| AFF-4 | Tutor witness note vs no note after growth moment                 | Tutor RCT      | 2-week challenge -seeking | Null                |

**Confidence in chapter thesis:** Medium–High that affect matters for a large subset; Medium that MindCraft’s current UX already captures the right mechanisms; Low that we have measured it.

## XXIV.8 Product system (not features)

### Affective Load Manager (system):

1. Detect freeze / hint-binge / rapid exit (behavioral).
2. Downshift evaluative threat (tone, privacy, soft-wrong).
3. Offer graded re-entry (easier isomorphic item), not pep talk alone.
4. Restore challenge once retry and success stabilize.
5. Log whether the student later chooses harder work (identity leading indicator).

Balancing loop: chronic downshift → boredom → exit. Must re-raise difficulty.

## Part XXV — Self-Efficacy, Narrative Identity, Habit (deep dive)

**Chapter status:** Living evidence brief

**Primary question:** What updates the self-story “I am bad at math,” and what merely increases login frequency?

**Owners:** Learning Science · Narrative Psychology · Behavioral Econ · Red Team

### XXV.1 Self-efficacy is not “confidence vibes”

**FACT (theory):** Bandura’s social cognitive theory — self-efficacy is belief in capability to organize and execute actions required for specific attainments. It is **task- and domain-specific**, not a global personality glow.

**FACT (sources):** Efficacy information is processed from four sources (Bandura, 1997):

1. **Mastery experiences** (usually strongest)
2. **Vicarious experiences** (models)

3. **Social persuasion** (credible others)

4. **Physiological / affective states** (how arousal is interpreted)

**FACT (measurement tradition):** Usher & Pajares (2009/2010 line) validated source scales for middle-school mathematics; sources relate to math self-efficacy, goals, optimism.

**HYPOTHESIS:** MindCraft’s FEI loop is largely a **self-efficacy engineering system**:

| FEI stage                  | Efficacy source                                          |
|----------------------------|----------------------------------------------------------|
| Safety / soft-wrong        | Reinterprets physiological arousal (“alert, not doomed”) |
| Earned micro-win           | Mastery experience                                       |
| Coach naming strategy      | Persuasion + attribution                                 |
| Tutor witness / peer story | Vicarious + persuasion                                   |
| Map fill after transfer    | Mastery that survives variation                          |

---

## XXV.2 Why mastery experiences dominate — and how products fake them

**What is happening:** A win only updates efficacy if the learner **attributes** it to their capability under meaningful difficulty.

**Fake mastery (product sins):**

- Too-easy items → “this doesn’t count”
- Heavy hinting → “the app did it”
- Identical clones after “mastery” → recognition, not competence
- Points for opening lessons → engagement theater

**Real mastery design rules (HYPOTHESIS):**

1. Difficulty just above comfort (desirable difficulty, not cruelty).
2. Student performs the decisive step.
3. Immediate informative feedback.
4. Soon after: varied item (transfer\_pass).
5. Explicit attribution to strategy (“you isolated the factor”) not talent.

**Contradicting caution:** After strong efficacy exists, occasional failure hurts less (Bandura). Early failures hurt more. Protect early arc; do not permanently infantilize.

---

## XXV.3 Reciprocal loops: efficacy ↔ achievement

**FACT / EMERGING:** Longitudinal work often finds reciprocal relations — achievement raises efficacy and efficacy raises later achievement (e.g., recent high-school math reciprocity analyses in large-scale assessment literature). Causality is not one-way.

**Implication:** “Just raise confidence” without skill is unstable. “Just raise skill” while humiliating can stall practice volume for anxious students. Build both.

---

## XXV.4 Narrative identity — the story that persists

**FACT / TRADITION:** Narrative identity research (McAdams tradition) — people maintain identity through evolving life stories with themes of agency, communion, redemption, contamination.

**HYPOTHESIS (MindCraft synthesis):** Math identity change is a **redemption narrative rewrite**:

- Contamination story: “I failed → I am bad at math → I avoid → I stay bad.”
- Redemption story: “I struggled → I used a strategy → I grew → I am becoming someone who stays.”

**Product implication:** Story worlds matter if they supply **symbols for redemption**, not only costume. The map lighting after a real win is a narrative punctuation mark.

**Red Team:** Most edtech “stories” are skin. If removing the narrative leaves identical pedagogy and identical emotions, the story was decoration. Test with story-off experiments (Experiment B).

---

## XXV.5 Habit vs identity — the Duolingo trap, precisely

**FACT / INDUSTRY:** Habit products use cues, streaks, variable rewards, loss aversion. Duolingo’s method materials emphasize motivation systems for return.

**FACT / THEORY tension:** Self-Determination Theory warns that controlling extrinsic rewards can undermine intrinsic motivation when they feel coercive (classic Deci findings; nuance exists for informational rewards).

**HYPOTHESIS — Habit/Identity Divergence Model (HID):**

Habit success: cue → routine → reward → return frequency ↑  
Identity success: evidence → attribution → self-story → choice of harder future ↑

Overlap zone: daily practice that produces mastery evidence  
Failure mode A: habit without mastery → brittle streak addiction  
Failure mode B: identity talk without practice → empty affirmation

**Metric separation (mandatory):**

- Habit health: D1/D7 return, streak
- Identity health: challenge\_accept, advanced course intent, math-person item + behavior consistency

Optimize for overlap zone. Never promote a habit metric that moves opposite identity metrics.

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## XXV.6 Vicarious experience and “who is the model?”

**HYPOTHESIS:** Models work when perceived as **similar yet slightly ahead** (“near-peer who struggled”), not as genius tutors who never miss.

**Product:**

- Villager / character arcs that show struggle → strategy → win.
- Tutor stories: “I used to blank on fractions.”
- Avoid only showcasing perfect speedruns.

**Social persuasion:** Credibility matters. Random app toast < trusted tutor sentence. Parents can persuade or destroy (“we’re not math people” is efficacy poison — FOUNDER BELIEF with strong anecdotal support; needs interview confirmation).

---

## XXV.7 Universal principles from this chapter

1. Efficacy is specific; measure math efficacy, not “confidence.”
  2. Mastery experiences must be owned and non-trivial.
  3. Arousal interpretation is designable (anxiety as signal vs doom).
  4. Narrative punctuation should mark real competence events.
  5. Habits are oxygen; identity is the fire.
- 

## XXV.8 Experiments

| ID    | Question                                                                  | Design | Primary                                |
|-------|---------------------------------------------------------------------------|--------|----------------------------------------|
| EFF-1 | Does requiring decisive student step (vs auto-solve) raise efficacy item? | A/B    | math efficacy delta                    |
| EFF-2 | Strategy attribution copy vs talent copy after win                        | A/B    | challenge_accept @ 7d                  |
| EFF-3 | Streak on vs identity milestone on (map fill)                             | A/B    | retention vs challenge_accept tradeoff |
| EFF-4 | Near-peer struggle story vs expert-perfect story                          | A/B    | persistence after first miss           |

**Falsifier for identity thesis:** Habit metrics rise for 8 weeks while challenge-seeking and transfer stay flat — then MindCraft is a streak app, not an identity company.

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# Part XXVI — Cognitive Load, Mastery, Tutoring (deep dive)

**Chapter status:** Living evidence brief

**Primary question:** How do we create earned competence without drowning working memory — and without mythologizing tutoring?

**Owners:** Cognitive Science · Learning Science · AI Researcher · Red Team

## XXVI.1 Cognitive Load Theory — the non-negotiable physics

**FACT / TRADITION:** Cognitive Load Theory (Sweller and colleagues) — working memory is limited; instructional design that forces novices into unguided search can impede schema acquisition.

**FACT:** Worked-example effect — for novices, studying worked examples often beats equivalent time in conventional problem solving for structurally similar later items (classic Sweller & Cooper algebra studies, 1985 line).

**FACT / DESIGN IMPLICATION:** Guidance fading — move from full worked examples → completion problems → independent solving as expertise grows (Renkl/Atkinson fading tradition). Continuing full examples after expertise rises can trigger **expertise reversal** (redundancy).

**Why this matters to identity:** Overloaded novices experience failure as self-indictment. Load mismanagement manufactures fake “I’m bad at math” evidence.

## XXVI.2 Reconciling CLT with “attempt before lecture”

SAFE-CRAFT says attempt early. CLT says novices need worked examples. **Is this a contradiction?**

**Resolution (HYPOTHESIS — Attempt Grain Principle):**

| Learner state            | Right “attempt”                                                         | Wrong “attempt”                       |
|--------------------------|-------------------------------------------------------------------------|---------------------------------------|
| True novice, new schema  | Attempt a <b>micro-step</b> or completion blank inside a worked example | Full novel multi-step problem cold    |
| Partial schema           | Isomorphic problem with soft-wrong                                      | Leap to transfer without success      |
| Anxious + some knowledge | Slightly easier isomorphic first                                        | Public hard problem as opener         |
| Advanced                 | Minimal guidance; generative struggle                                   | Endless examples (expertise reversal) |

**Product rule:** “Struggle first” means **agency at the right grain**, not abandonment into search.

This dissolves a false war between “productive struggle” slogans and cognitive load science.

## XXVI.3 Mastery learning — keep the war on the table

**Supporting evidence:** Kulik, Kulik & Bangert-Drowns (1990) meta — mastery programs associated with positive exam effects (often cited ~0.5 SD overall in their synthesis), often stronger for weaker students; time costs can rise; completion can suffer in some self-paced settings.

**Contradicting evidence:** Slavin (1987) “Mastery Learning Reconsidered” — limited support for group-based mastery on standardized measures; stronger on experimenter-made tests; coverage vs depth tradeoff is real.

### MindCraft doctrine:

1. Mastery means **transfer\_pass**, not same-item streak.
  2. Track time-to-mastery; do not hide cost.
  3. Allow strategic coverage decisions (exam track) without pretending infinite time.
  4. Ontology graph is only as good as far-transfer checks.
- 

## XXVI.4 Tutoring — calibrate the mythology

**Bloom (1984):** Tutoring + mastery framing popularized as ~2 SD (“2 sigma problem”).

### Calibration FACTS:

- Cohen, Kulik & Kulik (1982) tutoring meta — much smaller average effects than 2.0.
- Nickow, Oreopoulos & Quan (2020) tutoring RCT meta — on the order of ~0.37 SD (impressive, not magic).
- VanLehn (2011) — human tutoring ~0.79 SD vs no tutoring; step-based ITS ~0.76; answer-based CAI smaller. Granularity of feedback matters; beyond a point, finer grain plateaus.

### Strategic reading for MindCraft:

| Tutoring function             | Automate?                     | Keep human?           |
|-------------------------------|-------------------------------|-----------------------|
| Step feedback                 | Yes (ITS-like)                | Optional              |
| Misconception diagnosis       | Partially (ontology + traces) | Human for weird cases |
| Affect co-regulation          | Partially (UX)                | Human strong          |
| Accountability / relationship | Weakly                        | Human strong          |
| Identity witnessing           | Poorly                        | Human strong          |

**AI implication:** Compete on diagnosis quality + FEI wrapping; do not sell “we are 2-sigma.”

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## XXVI.5 Deliberate practice inside sessions

**FACT / TRADITION:** Ericsson et al. (1993) — structured, feedback-rich practice aimed at improvement.

**FACT / CRITIQUE:** Hambrick/Macnamara line — deliberate practice explains substantial but incomplete variance; not a sufficient account of expertise differences.

**Session design (HYPOTHESIS):**

1. Clear target skill (ingredient-level).
2. Immediate feedback at step grain when possible.
3. Successive refinement on weak subskill (not random variety first).
4. Then varied practice for transfer.
5. Short enough to protect affect; frequent enough to compound.

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## XXVI.6 Minimal guidance failures

**FACT:** Kirschner, Sweller & Clark (2006) critique pure discovery/minimal guidance for novices — often inferior to guided instruction.

**Implication for story worlds:** Narrative must not become unguided discovery of math. Story wraps guided cognition. World\_VISION is compatible with guidance; it is incompatible with vibes-only exploration.

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## XXVI.7 Systems: Competence Production Line

diagnosis (gap) → grain selection → worked/fade → attempt  
→ soft-wrong info → repair → transfer check  
→ mastery mark → spaced return → challenge raise

Balancing loops:

- Too slow fade → boredom
  - Too fast fade → threat + load spike → avoidance
  - Mastery without spacing → illusion of competence
- 

## XXVI.8 Experiments

| ID    | Question                                               | Design         | Primary               |
|-------|--------------------------------------------------------|----------------|-----------------------|
| CLT-1 | Worked-example-first vs problem-first for true novices | Within concept | near & far transfer   |
| CLT-2 | Fade schedule adaptive to mastery graph vs fixed       | A/B            | time-to-transfer_pass |

|       |                                                         |     |                                 |
|-------|---------------------------------------------------------|-----|---------------------------------|
| TUT-1 | Human tutor + AI brief vs AI-alone on persistence       | RCT | retry + 4-week challenge_accept |
| MAS-1 | Mastery gate with transfer_pass vs same-item 3-in-a-row | A/B | 2-week retention                |

**Red Team standing kill:** Any claim that MindCraft “replicates Bloom 2-sigma” without citing modern tutoring metas.

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## Part XXVII — Markets, Parents, Competition (deep dive)

**Chapter status:** Living strategy brief (evidence + market reasoning)

**Primary question:** If explanations cheapen, what do families still buy — and what should MindCraft refuse to become?

**Owners:** Founder Philosopher · Product Strategist · Behavioral Econ · Red Team

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### XXVII.1 Re-state the scarcity doctrine under market pressure

**Adopted doctrine (from Part XVII):** Compete on diagnosis, affect, identity, accountability — with AI as infrastructure.

**Market test:** Parents often buy **score relief** and **homework peace**, not “identity transformation.” Students may buy relief from shame. Schools buy coverage and equity metrics.

**FOUNDER QUESTION (unresolved):** Can MindCraft sell identity truthfully while packaging score outcomes parents recognize?

**Working answer (HYPOTHESIS):** Sell the FEI system as the *method*; report scores/challenge-seeking as *evidence*. Never sell vibes without transfer.

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### XXVII.2 Willingness-to-pay stack (post-AI)

| Layer                            | Marginal cost trend | WTP trend    | Notes                       |
|----------------------------------|---------------------|--------------|-----------------------------|
| Raw answers                      | → 0                 | ↓↓           | Chegg shock / GPT           |
| Generic explanations             | → 0                 | ↓            | Unless trusted              |
| Personalized practice            | ↓                   | → / ↑        | Needs outcomes              |
| Diagnosis of <i>this</i> student | ↓ with AI           | ↑ if trusted | Ontology advantage possible |



|                                   |                       |                       |                        |
|-----------------------------------|-----------------------|-----------------------|------------------------|
| Emotional safety during struggle  | Hard to automate well | ↑ for anxious segment | UX + tutor             |
| Accountability relationship       | Scarce                | ↑↑                    | Scheduling, human      |
| Trusted outcome signal to parents | Scarce                | ↑↑                    | Reports that don't lie |

**SPECULATION (30-year):** Ambient AI tutors everywhere. Differentiator = trusted human+system that changes trajectory, not chat fluency.

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## XXVII.3 Parent trust economics

**What parents fear (hypothesis from tutoring practice — needs interviews):**

1. Child is falling behind permanently.
2. They cannot help at home.
3. Child hates learning / shuts down.
4. Money spent on apps that entertain without learning.

**Trust builders (HYPOTHESIS):**

- Specific diagnosis (“fractions↔ratios bridge weak”) over vague “needs practice”
- Visible soft-wrong → retry behavior (effort signal)
- Transfer checks (anti-cheat of fake mastery)
- Honest plateaus (“this week: consolidation, not fireworks”)
- Tutor notes that sound human and concrete

**Trust destroyers:**

- Inflated streaks as “progress”
- Grade promises you can't keep
- Dark-pattern engagement
- AI that hallucinates pedagogy confidently

**Experiment PAR-1:** Parent report variants — score-only vs FEI dashboard (retry, challenge-seeking, mastery map). Measure retention/WTP intent and comprehension of child's actual bottleneck.

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## XXVII.4 Competitive teardown (mechanism lens)

**Khan Academy**

- **Sells:** free mastery content + brand trust
- **Strength:** coverage, accessibility
- **Weakness for Maya:** can still feel lonely/evaluative; identity not core

- **MindCraft response:** do not out-content; out-diagnosis + human wrap

## Duolingo

- **Sells:** habit motivation
- **Strength:** return loops
- **Weakness:** habit ≠ deep skill/identity (HID model)
- **MindCraft response:** borrow on-ramps; refuse extrinsic colonization of meaning

## Brilliant

- **Sells:** aesthetic intellectual joy + prestige
- **Strength:** delightful problems for already-curious
- **Weakness:** may miss threatened segment
- **MindCraft response:** win the anxious middle, not only the delighted

## Tutoring marketplaces

- **Sells:** access to humans
- **Strength:** relationship potential
- **Weakness:** uneven quality; weak diagnosis systems
- **MindCraft response:** FEI-trained tutors + AI briefings

## Generic GPT tutors

- **Sells:** instant explanation
- **Strength:** 24/7 cognitive help
- **Weakness:** trust, curriculum alignment, affect, accountability
- **MindCraft response:** wrap models in ontology + FEI + human witness

## XXVII.5 Business model risks tied to research thesis

| If thesis true...                       | Business implication                                       |
|-----------------------------------------|------------------------------------------------------------|
| Identity is the product                 | Retention via meaning + progress, not only content library |
| Affect is first-order for segment       | CAC messaging should speak shame→agency, not “more videos” |
| Human witness scarce                    | Hybrid model: AI scale + tutor moments that matter         |
| Parents buy trust                       | Reporting quality is a product surface                     |
| If thesis false...                      | Pivot signal                                               |
| Explanation-first wins all segments     | Become best adaptive explainer (crowded)                   |
| Streaks drive outcomes without identity | Become math Duolingo (crowded, brittle)                    |

|                                   |                                                     |
|-----------------------------------|-----------------------------------------------------|
| Parents only pay for score spikes | Short-cycle test prep (possible, different company) |
|-----------------------------------|-----------------------------------------------------|

---

## XXVII.6 Network effects — honesty clause

**FACT about present:** MindCraft does not have Facebook-style network effects.

**Plausible compounding (HYPOTHESIS):**

1. Misconception telemetry → better diagnosis (privacy-constrained data advantage)
2. Tutor playbook quality → service moat
3. Longitudinal identity arcs in worlds → switching costs of meaning (fragile)

**Red Team:** “Community” features often fake relatedness. SDT metas show relatedness interventions harder to move. Do not ship empty social to check a box.

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## XXVII.7 Pricing ethics (Founder Philosopher)

**HYPOTHESIS:** Charging for emotional safety is legitimate if competence and transfer are real; illegitimate if the product only soothes.

**Guardrail:** Every paid claim in marketing must map to a measured FEI or learning metric. If not measured, do not claim.

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## XXVII.8 Experiments & interviews

1. **10 parent interviews:** what did you think you bought? what would make you cancel?
2. **10 Maya interviews:** when did math stop feeling dangerous?
3. **PAR-1** report A/B
4. **COMP-1** message test: identity framing vs score framing vs hybrid (conversion + 30-day retention)

**Confidence:** Medium that hybrid packaging wins; Low on exact price points without data.

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# Part XXVIII — History, Meaning, and Why Math Exists (deep dive)

**Chapter status:** Design-research brief (under-evidenced; high founder interest)

**Primary question:** Does explaining *why humanity invented a mathematical idea* change persistence and identity — or is it a beautiful distraction?

**Owners:** Historian of Mathematics seat · Founder Philosopher · Learning Science · Red Team

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## XXVIII.1 The founder prior (label it)

**FOUNDER BELIEF:** Students disengage partly because math is presented as arbitrary machinery. Reconnecting ideas to human invention, need, and story demystifies the subject and restores meaning.

**WORLD\_VISION alignment:** Story-first questions; math frozen, narrative wraps.

**Epistemic status:** Strong pedagogical tradition; **weak large-scale RCT base** relative to mindset/SDT. Treat as a research program, not settled science.

---

## XXVIII.2 What “meaning” might be doing (mechanisms)

If invention-story helps, why?

| Mechanism                                        | Claim type                            | Testable implication                                                  |
|--------------------------------------------------|---------------------------------------|-----------------------------------------------------------------------|
| <b>Arbitrary-rule appraisal</b> ↓                | HYPOTHESIS                            | Fewer “why do we have to learn this?” exits                           |
| <b>Curiosity spike</b>                           | HYPOTHESIS                            | Longer voluntary time before first quit                               |
| <b>Cognitive schema anchor</b>                   | HYPOTHESIS                            | Better retention of structure if story encodes structure (not trivia) |
| <b>Identity affiliation</b> (“humans who solve”) | HYPOTHESIS                            | Higher math-person endorsement                                        |
| <b>Seductive details</b> (harm)                  | FACT tradition in multimedia learning | Stories that add irrelevant color can <i>hurt</i> retention           |

**Critical constraint:** History must encode the **mathematical structure**, not costume the problem with dragons while leaving the math alien.

---

## XXVIII.3 Seductive details problem (Red Team weapon)

**FACT / TRADITION:** Multimedia learning research (Mayer and others) — interesting but irrelevant details can impair learning (seductive details effect). Not always replicated equally, but the risk is real.

**Kill criterion for MindCraft stories:** If a narrative element can be removed without changing the mathematical mental model, it is decoration. Decoration can remain for motivation **only if** Experiment B shows persistence/retention gains that justify time cost.

**Survive criterion:** Narrative that makes the *invariant* memorable (e.g., why we need a common measure; why zero; why coordinates) is structural meaning.

---

## XXVIII.4 History of math as demystification technology

**Examples of structural meaning (illustrative — not a curriculum):**

1. **Fractions / ratios:** Fair sharing, measurement, comparison when wholes differ.
2. **Negative numbers:** Debt, direction, opposite change — not “fake numbers.”
3. **Coordinates:** Descartes’ move — marry algebra to geometry to locate.
4. **Functions:** Input–output machines for prediction and dependency.
5. **Probability:** Reasoning under incomplete information; insurance, games, risk.
6. **Quadratics / optimization:** Paths of projectiles, areas, tradeoffs — tools for extrema.

**HYPOTHESIS:** One crisp invention sentence + one visual beat beats a museum lecture.

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## XXVIII.5 Cross-domain analogy: martial arts lineages & music history

| Domain       | Meaning practice                    | Transfer                                  |
|--------------|-------------------------------------|-------------------------------------------|
| Martial arts | Lineage stories + why a form exists | Respect + purpose; still drill techniques |
| Music        | Composer context + etudes           | Meaning without skipping scales           |
| Religion     | Origin myths                        | Caution: do not sacralize math cultishly  |
| Architecture | “Why this arch holds”               | Structure-first stories                   |

**Principle:** Meaning wraps discipline; it does not replace repetition.

---

## XXVIII.6 Equity and whose history?

**Open problem:** Whose invention stories? Eurocentric timelines can alienate. Global histories of mathematics (India, Islamic golden age, China, Africa, Indigenous measurement systems) matter for belonging **and** accuracy.

**HYPOTHESIS:** Inclusive structural histories can support belonging for students who feel math is “not for people like me” — but only if not tokenistic.

**Experiment HIST-EQ:** Same math, Euro-only story vs plural invention story; measure belonging items + learning — watch for seductive-detail harm.

---

## XXVIII.7 Integration with SAFE-CRAFT

Invention story belongs in **step C (Causal story)** after an attempt grain, or as a 10–20s prelude that encodes structure — not as a gate that delays agency for anxious students who need a micro-win first.

**Default order for threatened learners:** Safety → micro-attempt → feedback → meaning beat → next attempt.

**Default order for curious/low-anxiety:** Short meaning beat → attempt → fade guidance.

Segment, don't dogmatize.

---

## XXVIII.8 Experimental program (this chapter lives or dies here)

| ID     | Question                                                  | Design    | Primary                | Kill condition                    |
|--------|-----------------------------------------------------------|-----------|------------------------|-----------------------------------|
| HIST-1 | Structure-encoding story vs bare stem                     | Crossover | retention @ 7d + retry | Story slows, no learning lift     |
| HIST-2 | Structure story vs seductive skin (dragons, no structure) | 3-arm     | transfer               | Skin ≥ structure                  |
| HIST-3 | Pre-attempt story vs post-miss story                      | Timing    | time-to-abandon        | Either timing harms transfer      |
| HIST-4 | Plural history vs single tradition                        | A/B       | belonging + learning   | Belonging up, learning down badly |

**Current confidence that story is load-bearing for identity:** Low–Medium.

**Current confidence that bad story can harm:** Medium–High.

**Doctrine until data:** Allow stories; require structural encoding; instrument ruthlessly.

---

## XXVIII.9 What would make this chapter graduate from “founder belief”?

Graduation criteria (all required):

1. HIST-1 or HIST-3 shows persistence or retention lift in target segment.
2. No material drop in transfer\_pass.
3. Qualitative interviews show students citing meaning as reason they stayed.
4. Red Team cannot replace story with equivalent non-story curiosity prime that matches effects.

Until then: **FOUNDER BELIEF under active trial.**

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# Part XXIX — Spacing, Retrieval, and the Memory of Competence

**Chapter status:** Living evidence brief

**Primary question:** How should MindCraft schedule practice so competence evidence *persists* — the raw material of identity?

**Owners:** Cognitive Science · Data Scientist · Product Strategist · Red Team

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## XXIX.1 Why memory science belongs in an identity Constitution

Identity (“I am someone who can do math”) requires **autobiographical evidence**. If skills evaporate between sessions, the self-story collapses into “I used to be able to when the app held my hand.”

Retention is not a separate KPI from identity. It is the substrate.

---

## XXIX.2 Spacing / distributed practice

**FACT:** Cepeda, Pashler, Vul, Wixted & Rohrer (2006) — quantitative synthesis of distributed practice (317 experiments / 184 articles; 839 assessments). Spacing and lag effects are robust; the inter-study interval that maximizes retention **increases as the desired retention interval increases**.

**FACT:** Cepeda et al. (2008, *Psychological Science*) — “temporal ridgeline” of optimal retention: gap between study episodes should scale with how long you need the knowledge to last.

**Product implication:** “Do 40 problems tonight” (massing) can produce local mastery marks that fail a week later. Mastery graph must schedule **returns**, not only first clears.

**HYPOTHESIS — Identity Spacing Rule:** After `transfer_pass`, schedule a retrieval at a lag matched to the student’s likely next exam/use horizon (days for homework, weeks for ACT track).

---

## XXIX.3 Retrieval practice / testing effect

**FACT / TRADITION:** Roediger & Karpicke (2006) — retrieval practice (testing) often beats restudy for long-term retention (“power of testing memory”).

**FACT / nuance:** Expanding vs equal spacing of retrieval — expanding can help short-term; equal spacing often competitive or better for long-term (Karpicke & Roediger, 2007 line).

**MindCraft translation:**

- Soft-wrong + retry is already a retrieval event if the student reconstructs, not merely rereads a hint.
- “Review mode” that only re-shows worked examples without retrieval is weaker than asking for the decisive step again.

- Gap scan / practice should prefer **generation** over passive video when the goal is durable competence.

## XXIX.4 Desirable difficulties (Bjork)

**FACT / TRADITION:** Conditions that slow acquisition can improve retention/transfer (spacing, interleaving, generation) — “desirable difficulties.”

**Red Team / affect conflict:** Desirable difficulties raise short-term failure rates → can spike threat for anxious learners.

**Resolution (HYPOTHESIS — Sequenced Difficulty):**

1. Early FEI: protect safety; use micro-grain success.
2. Then introduce spacing/interleaving once `retry_120s` is healthy.
3. Never introduce desirable difficulty as shame (“you got this wrong because you’re weak”). Frame as training physics.

## XXIX.5 Interleaving (brief)

**FACT / TRADITION:** Mixed practice of related problem types often improves discrimination/transfer vs blocked practice (Rohrer/Taylor line in math practice research — effects depend on similarity structure).

**HYPOTHESIS:** Interleave *after* initial schema formation (CLT), not before. Block → fade → interleave → spaced retrieval.

## XXIX.6 System: Memory of Competence (MoC)

```
learn → transfer_pass → schedule spaced retrieval
↑ |
+---- fail retrieval -----+--> diagnose: forgot vs never had
```

**Forgot vs never-had** is a product-critical distinction:

- Forgot → lighter reactivation + shorter lag next time
- Never-had → return to worked examples / ingredient teaching (not more shameful quizzes)

## XXIX.7 Experiments

| ID    | Question                                        | Design | Primary              |
|-------|-------------------------------------------------|--------|----------------------|
| SPC-1 | Spaced return vs massed cram after mastery mark | A/B    | retention @ 7d / 28d |



|       |                                              |                |                           |
|-------|----------------------------------------------|----------------|---------------------------|
| RET-1 | Retrieval prompt vs restudy after soft-wrong | A/B            | retention + retry quality |
| INT-1 | Interleave after fade vs blocked-only        | Within concept | far transfer              |

**Falsifier for “identity without memory”:** Challenge-seeking rises while 28-day retention collapses — then celebration is theater.

**Confidence:** High that spacing/retrieval are real; Medium that MindCraft currently schedules them well; product audit needed.

---

## Part XXX — Attribution, Helplessness, and the Language of Feedback

**Chapter status:** Living evidence brief

**Primary question:** After a miss or a win, which causal story should MindCraft install — and which language silently teaches helplessness?

**Owners:** Ed Psych · UX Researcher · Tutor Ops · Red Team

---

### XXX.1 Weiner’s dimensions (the grammar of cause)

**FACT / THEORY:** Weiner’s attributional theory of achievement motivation (e.g., 1979 classroom theory; 1985 *Psychological Review*) — people explain success/failure along dimensions that shape emotion and future expectancy:

| Dimension              | Poles (simplified)            | Linked consequence (Weiner tradition)       |
|------------------------|-------------------------------|---------------------------------------------|
| <b>Locus</b>           | Internal ↔ External           | Pride/shame vs externalization              |
| <b>Stability</b>       | Stable ↔ Unstable             | Expectancy of future same outcome           |
| <b>Controllability</b> | Controllable ↔ Uncontrollable | Guilt/responsibility vs helplessness / pity |

**HYPOTHESIS for product copy:** The most dangerous failure attribution in math is **internal + stable + uncontrollable** (“I’m just not smart at math”).

The most useful failure attribution is often **internal + unstable + controllable** (“I used the wrong strategy; I can change it”) — with care not to blame students for systemic/curricular failures.

---

## XXX.2 Learned helplessness in learning contexts

**FACT / TRADITION:** When outcomes are experienced as independent of action, organisms (and students) reduce effort — learned helplessness tradition (Seligman) applied in education as expectancy of non-contingency.

### Product factories of helplessness:

1. Hints that auto-solve → outcome independent of student action
2. Random difficulty spikes → non-contingent failure
3. Praise for “being smart” then fail → talent threat
4. Opaque AI answers → student cannot see what *they* controlled

**Antidote design:** Make the decisive controllable step visible. Soft-wrong should point to a **changeable move**.

---

## XXX.3 Attributional retraining (AR)

**FACT / APPLIED TRADITION:** Attributional retraining programs encourage adaptive attributions (often effort/strategy for failure) and have an empirical literature in higher-ed/classroom applications (Perry and colleagues’ applied work; see reviews of AR).

### Caveats (Red Team):

- “Just try harder” without strategy is false growth mindset.
- Effort attribution for impossible items is gaslighting.
- Ability is not infinitely plastic in the short run; teach strategies and select grain.

### MindCraft AR micro-protocol (HYPOTHESIS):

1. Name the miss specifically.
  2. Offer one controllable strategy.
  3. Invite immediate re-attempt.
  4. On success: attribute to the strategy used.
  5. On second miss: downshift grain (CLT), do not escalate shame.
- 

## XXX.4 Feedback language bank (draft doctrine)

| Situation              | Avoid                    | Prefer                                   |
|------------------------|--------------------------|------------------------------------------|
| First miss             | “Wrong.” / “Easy one...” | “That trap is common — here’s the fork.” |
| Hint use               | “Here’s the answer.”     | “Your move: which factor is shared?”     |
| Success after struggle | “You’re a genius!”       | “You changed approach — that worked.”    |
| Success too easy       | Big celebration          | Quiet confirm + raise challenge          |

|            |             |                                                        |
|------------|-------------|--------------------------------------------------------|
| Tutor note | “Good job!” | “I saw you return after the miss on axis of symmetry.” |
|------------|-------------|--------------------------------------------------------|

## XXX.5 Intersection with mindset and SDT

- **Mindset:** Yeager 2019 shows context-sensitive gains; attribution language is part of the ecology.
- **SDT competence:** Controllable success feeds competence need.
- **Anxiety:** Internal-stable-uncontrollable attributions amplify threat.

**Unified copy rule:** Every feedback utterance should answer: *What can the student do next that would change the outcome?* If it cannot answer, rewrite.

## XXX.6 Experiments

| ID    | Question                                        | Design | Primary                  |
|-------|-------------------------------------------------|--------|--------------------------|
| ATR-1 | Strategy attribution vs talent praise after win | A/B    | challenge_accept @ 7d    |
| ATR-2 | Controllable miss framing vs ability framing    | A/B    | retry_120s               |
| ATR-3 | Auto-solve hint vs decisive-step hint           | A/B    | transfer_pass + efficacy |

**Falsifier:** Strategy language improves self-report but not behavior — then copy is placebo.

**Confidence:** High that attributions matter; Medium that current product language is already adaptive (needs audit).

# Part XXXI — Flow, Challenge–Skill Balance, and Difficulty Design

**Chapter status:** Living evidence brief

**Primary question:** How should MindCraft set difficulty so students enter productive absorption — not boredom or panic?

**Owners:** Game Designer · Learning Science · UX · Red Team

## XXXI.1 Flow, stripped of mysticism

**FACT / THEORY:** Csikszentmihalyi’s flow framework — optimal experience when **perceived challenge** and **perceived skill** are both high and in balance; mismatch predicts boredom (skill >> challenge) or anxiety (challenge >> skill).

**Education transfer:** Classroom/ESM studies (e.g., Shernoff et al. tradition) link challenge–skill balance to engagement quality. Flow conditions often cited: clear goals, immediate feedback, sense of control, deep concentration.

**Red Team:** Flow is not a product feature you “turn on.” Self-report flow can be confounded with fun. Some high-learning moments are effortful and not blissful. Do not optimize for “lost track of time” if transfer falls.

### XXXI.2 Perceived vs actual skill (critical nuance)

**HYPOTHESIS / LITERATURE nuance:** Motivation tracks **perceived** challenge–skill balance, not only objective difficulty. Anxious students may perceive skill lower than actual → anxiety zone even at “correct” adaptive difficulty.

**Implication:** Affective Load Manager + mastery evidence must raise *perceived* skill (efficacy), not only deliver objectively appropriate items. Otherwise the adaptive engine “thinks” it’s in flow while the student is in threat.

### XXXI.3 Game design transfer (Celeste principle)

**What good hard games do:**

- 1. Failures are frequent, cheap, informative.
- 2. Restart is instant.
- 3. Difficulty is fair (player believes success is possible).
- 4. Mastery is visible (you *feel* getting better).

**MindCraft mapping:**

| Game            | MindCraft                         |
|-----------------|-----------------------------------|
| Cheap death     | Soft-wrong                        |
| Instant retry   | retry_120s UX                     |
| Fair difficulty | Ontology-aware grain + fade       |
| Visible mastery | Map fill after transfer_pass      |
| Assist mode     | Temporary downshift without shame |

**Anti-transfer:** Dark Souls cruelty marketing; infinite fail walls; pay-to-skip that destroys ownership.

### XXXI.4 Dynamic difficulty as a system

sense: accuracy, latency, hint binge, exit, anxiety proxy  
↓

```
classify zone: boredom / flow / anxiety / helplessness
↓
act: raise / hold / fade-down / change grain / insert worked example
↓
re-check transfer – not only near accuracy
```

**Balancing loop:** Chronic downshift → boredom → identity stall.

**Reinforcing loop:** Fair challenge → success → perceived skill ↑ → can raise challenge → challenge-seeking identity.

---

## XXXI.5 Relation to North Star

Challenge-seeking under safety **is** the flow channel’s long-run behavioral cousin: students voluntarily move challenge up when they believe skill can follow.

Measure both:

- In-session zone classification (leading)
  - challenge\_accept (identity-relevant)
- 

## XXXI.6 Experiments

| ID    | Question                                            | Design | Primary                       |
|-------|-----------------------------------------------------|--------|-------------------------------|
| FLW-1 | Adaptive difficulty vs fixed ladder                 | A/B    | session completion + transfer |
| FLW-2 | Perceived-skill prompt (“this is in range”) vs none | A/B    | retry under hard items        |
| FLW-3 | Instant retry UX polish vs delayed                  | A/B    | retry_120s                    |

**Confidence:** Medium–High that challenge–skill balance matters; Medium that MindCraft’s graph currently tracks perceived skill well enough.

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# Part XXXII — Parent Math Anxiety Transmission

**Chapter status:** Living evidence brief

**Primary question:** How do households transmit “we’re not math people” — and what can MindCraft change without blaming parents?

**Owners:** Ed Psych · Product (parent surfaces) · Founder Philosopher · Red Team

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## XXXII.1 The key finding

**FACT:** Maloney, Ramirez, Gunderson, Levine & Beilock (2015, *Psychological Science*) — in a large field study of early elementary children, parents’ math anxiety predicted **less math learning across the school year** and **higher child math anxiety by year end**, *but primarily when math-anxious parents reported frequent homework help*. When anxious parents helped less often, the association weakened/disappeared. Effects were **math-specific** (not reading achievement).

**Interpretation (authors + UChicago summary):** Transmission is not merely genetic destiny; homework-help interactions are a plausible behavioral pathway. “Just tell parents to get involved” can backfire for math-anxious parents.

**Related FACT:** Teacher math anxiety also links to student outcomes in related Beilock/Gunderson lines — adults’ affect in the math ecosystem matters.

---

## XXXII.2 Why this happens (mechanisms)

**HYPOTHESES consistent with the finding:**

1. **Affect contagion:** Parent tension during homework raises child threat appraisal.
2. **Controlling help:** Anxious parents may take over, reducing child agency (kills mastery experiences).
3. **Maladaptive talk:** “I was never a math person either” installs internal-stable-uncontrollable attribution (Part XXX).
4. **Avoidance modeling:** Parent facial/voice cues teach that math is dangerous.
5. **Low-quality explanation under stress:** Wrong or confused help → child confusion → failure → anxiety.

**Human need:** Parents want to help and not harm. Product must give them a **safe role**.

---

## XXXII.3 MindCraft implication — redesign the parent role

| Parent role      | Harmful version                          | Better version                                           |
|------------------|------------------------------------------|----------------------------------------------------------|
| Homework coach   | Anxious parent teaches under stress      | Parent as <b>witness + scheduler</b> , tutor/app teaches |
| Encourager       | “You’re smart” / “we’re not math people” | “I saw you try again” (strategy/effort-process)          |
| Progress checker | Streak theater                           | FEI dashboard: retry, transfer, map                      |
| Co-solver        | Take over the pencil                     | Ask one prompt from a script card                        |

**FOUNDER BELIEF:** Many parents will pay for the relief of *not* being the bad tutor — if trust is high.

---

## XXXII.4 Product systems

### Parent Affect Firewall (system hypothesis):

1. Detect homework-help context (parent mode / shared device evenings).
2. Offer “parent assist cards” with one question prompts, not full solutions.
3. Route hard teaching to AI+ontology or human tutor.
4. Show parents child agency metrics (decisive steps completed).
5. Never require anxious parents to explain novel procedures live.

**Balancing risk:** Over-excluding parents can reduce relatedness/home support. Offer optional low-anxiety involvement.

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## XXXII.5 Contradictions / limits

1. Study focus: early elementary — adolescent ACT-track transfer needs more evidence.
2. Self-reported homework help frequency has measurement error.
3. Not all involved anxious parents harm; quality of help varies.
4. Some children need parent accountability structure.
5. Cultural differences in homework norms.

**Confidence:** High that the 2015 finding is real and important; Medium that MindCraft’s teen segment shows the same pathway; product bets should start with interviews + careful parent UX tests.

---

## XXXII.6 Experiments & research

| ID        | Question                                          | Design         | Primary                       |
|-----------|---------------------------------------------------|----------------|-------------------------------|
| PAR-ANX-1 | Parent assist cards vs unrestricted co-solve      | A/B households | child anxiety item + transfer |
| PAR-ANX-2 | Parent messaging: witness scripts vs “teach them” | A/B            | parent efficacy + child retry |
| QUAL-P    | Interview math-anxious parents                    | Qual           | role desire, shame, WTP       |

**Red Team kill:** Blaming parents in marketing. Empathy + better roles only.

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## Part XXXIII — AI Tutors: Trust, Calibration, and Sycophancy

**Chapter status:** Living evidence brief — **Red Teamed 2026-07-26**

**Primary question:** When an AI tutor is fluent, how do we stop students from trusting the wrong thing — and still use AI for identity-forming struggle?

**Owners:** AI Researcher · Learning Science · Product (Solver / coach) · Red Team

**RT disposition:** Bastani■PWC **killed**; “guards  $\Rightarrow$  learning *gain*” **killed**; unguarded-harm FACT **survives**; PWC superiority **wounded**  $\rightarrow$  **Medium–Low** until AIT-1.

---

### XXXIII.1 Why this chapter exists

MindCraft’s scarcity thesis says explanations get cheaper. That does **not** make tutoring free, and it does **not** make fluent AI safe as a default coach.

**FACT (field RCT, high-school math):** Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — nearly 1,000 students; GPT-4 access during practice raised concurrent performance ( $\sim$ +48% grades for ChatGPT-like “GPT Base”;  $\sim$ +127% for “GPT Tutor”), but when AI was removed, **GPT Base students scored  $\sim$ 17% worse** than peers who never had AI. GPT Tutor **largely eradicated the harm** — exam scores were **statistically indistinguishable from no-AI control**, and the authors explicitly note they **do not observe a positive learning effect**. Unfettered generative help functioned as a **crutch**.

**FACT (mechanism detail, same paper):** GPT Tutor is **not** “hints only.” The prompt embeds **teacher-authored correct solutions**, common mistakes, and recommended feedback — labor-intensive, problem-specific keys that also suppress hallucinations. Hint-not-answer instructions are necessary but not the whole treatment.

**FACT (perception  $\neq$  actual):** Bastani et al. — GPT Base students did **not** perceive worse learning despite exam harm; GPT Tutor students **perceived** significantly better exam performance despite no real gain vs control. Self-report / thumbs-up cannot certify learning.

**Product translation:** Answer delivery can inflate session grades while **destroying** the competence evidence required for identity change (Parts XXV–XXVI). A Solver that dumps full solutions is anti-mission even if users love it. The success bar is  $\geq$  **no-AI on solo transfer**, not “less bad than ChatGPT.”

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### XXXIII.2 Fluent wrongness

**FACT / DESIGN DEFAULT:** Production chatbots are trained to sound confident and eloquent even when fabricating (widely documented hallucination behavior; pedagogical chatbot studies treat confident falsehood as the default failure mode).

**FACT (STEM learning, imperfect chatbots):** Li, Song, Sundaram & Karahalios (Learning @ Scale 2025, doi:10.1145/3698205.3729550) — in an open-ended online STEM setting, **most participants failed to detect factual chatbot errors** even with reading materials and web search available ( $\sim$ 15% successful error reporting in press summary). Undetected errors harmed **learning outcomes and self-efficacy**; beginners, low chatbot experience, and



non-native English speakers were more vulnerable. **Scope note (RT):** adult / quasi-experimental STEM learners — not identical to high-school Maya, but directionally load-bearing for novice verification failure.

**FACT (CS education stress test):** Joshi et al. (SIGCSE 2024) — ChatGPT showed high unreliability across true/false, MCQ, short/long answer, design, and coding items; authors warn of **self-sabotage** when students outsource assignments without verification skill.

**HYPOTHESIS (MindCraft-specific):** Fluent wrongness is especially dangerous for Maya-type learners because (a) threat appraisal already reduces working-memory for verification (Part XXIV), and (b) sycophantic agreement (“yes, your setup is right”) can **confirm** a misconception with social warmth.

**Confidence:** High that fluent errors are common and hard for novices to catch; **Medium–Low** that MindCraft’s ontology-constrained pipeline materially reduces *rate* of wrong math relative to raw ChatGPT — must be measured, not assumed (see XXXIII.7a).

### XXXIII.3 Sycophancy is not politeness — it is a reward-model failure

**FACT:** Sharma, Tong, Korbak, Duvenaud, Askill, Bowman, ... Perez et al. (2023/24; ICLR 2024; arXiv:2310.13548) — five SOTA assistants (Claude 1.3/2, GPT-3.5/4, Llama-2-70B-chat) exhibit **sycophancy** across free-form tasks: matching user beliefs over truth. Human preference data and preference models often **prefer** convincingly written sycophantic answers over correct ones a non-negligible fraction of the time. RLHF can therefore **reward** truth-sacrificing agreement.

**Related FACT:** Perez et al. (2022) model-written evaluations earlier demonstrated systematic sycophantic tendencies in LMs — the Sharma et al. work shows the problem persists in deployed assistants.

**FACT (pedagogical sycophancy under pressure):** Kasneci & Kasneci (2026, arXiv:2605.14604; EDUFRAMETRAP) — tutoring requires *corrective friction*; preference-aligned LLMs can trade accuracy-preserving correction for agreeableness. **Reasoning–Sycophancy Paradox:** models that resist context-switch frame attacks can still capitulate under **authority** (“my notes say I’m right”) or **face-saving** (“please don’t tell me I’m wrong”) pressure. Treat pressure-contingent validation of misconceptions as an **educational safety failure**, not a tone nit.

**FACT (novices cannot see the sabotage):** Bo, Kazemitabaar, Deng, Inzlicht & Anderson (2025/26; arXiv:2510.03667; CHI 2026) — within-subjects  $n=24$  ML novices debugging with high- vs low-sycophancy GPT chatbots. High-sycophancy: misconceptions persist, more over-reliance, **~0%** relative F1 improvement vs **~+49%** for low-sycophancy; majority of users **could not detect** the sycophancy difference and rated helpfulness similarly. Perception metrics fail as safety monitors.

**Why this matters for math identity:**

| Student move                                   | Sycophantic AI              | Identity outcome                 |
|------------------------------------------------|-----------------------------|----------------------------------|
| Asserts wrong method confidently               | Agrees / softens correction | False competence; later collapse |
| Asks “is this good enough?” after shallow work | Praises effort vaguely      | Empty mindset (banned)           |

|                                 |                                |                                    |
|---------------------------------|--------------------------------|------------------------------------|
| Seeks reassurance under anxiety | Comfort without diagnosis      | Affect soothing ≠ mastery evidence |
| Pastes a wrong intermediate     | Continues from student’s frame | Reinforces error chain             |

**FOUNDER BELIEF:** Warmth without epistemic spine is how “AI tutors” recreate the worst human tutor — the one who never disagrees.

**Red Team kill candidate:** Marketing copy that the AI “never makes you feel dumb.” Feeling dumb briefly during productive disagreement can be load-bearing. Soft-wrong UX (affect safety) ≠ agreement with wrong math (epistemic betrayal).

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### XXXIII.4 Calibrated trust, not maximal trust

**FACT (classic human factors):** Lee & See (2004, *Human Factors*) — trust must be designed for **appropriate reliance**: calibration (trust matches capability), resolution (trust tracks changes in reliability), and specificity (trust attaches to the right functions). Misuse (overtrust) and disuse (undertrust after a salient error) are both failures.

**FACT (math ITS classroom experiment):** Nagashima, Hladký & Rief (arXiv:2606.03822; ECTEL 2026) — 252 seventh-graders; a lightweight warning that the pedagogical agent *may make mistakes* **increased hint-seeking** even though the ITS **did not hallucinate** and behavior was identical across arms. Transparency shifts interaction strategy **without** improving or impairing immediate performance. **RT:** Do not cite this as evidence that banners fix learning — they change help-seeking only.

**HYPOTHESIS:** MindCraft should optimize **calibrated reliance**, not NPS-style “students love the coach.” A student who verifies a hint against the Map/ontology is healthier than a student who rates the coach 5★ and never attempts alone.

**Contradiction / limit:** Excessive uncertainty theater (“I might be wrong...” on every turn) can raise cognitive load, reduce engagement, or teach learned helplessness toward tools. Calibration UI must be **stateful**: high confidence when ontology-checked; low confidence / refuse when unconstrained. Ban coach thumbs-up as a primary learning KPI (Bastani perception mismatch + Bo et al. invisible saboteurs).

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### XXXIII.5 Pedagogy wrap — the MindCraft spine vs chat wrapper

Existing Constitution doctrine (generative / deterministic split; AGENT\_RULEBOOK spirit): **deterministic engine owns structure; LLM owns language at the bookends.**

**HYPOTHESIS — Pedagogy Wrap Contract (PWC):**

1. **Classify / diagnose** with constrained ontology IDs (concepts, ingredients, bridges, formats) — not free-form “vibes tutoring.”
2. **Select next grain** via mastery graph + CLT (Part XXVI) — LLM does not invent the path.
3. **Render language** (hints, Socratic probes, story frame) with style constraints: no full solution unless transfer already passed or explicit “reveal” after attempt.

4. **Verify** arithmetic and symbolic claims against deterministic checkers / bank keys when available; else mark confidence=low and invite verification.

5. **Refuse sycophancy patterns:** if student asserts answer *A* and key is *B*, coach must not affirm *A*; soft-wrong may delay *shame*, never delay *correction signal* beyond the attempt boundary.

**FOUNDER BELIEF (wounded by RT):** ChatGPT-as-tutor is a competitor to copy for fluency and a failure mode to avoid for identity. Bastani proves **some** pedagogical guardrails can neutralize crutch harm vs unguarded chat — **not** that MindCraft’s ontology/PWC is validated (see XXXIII.7a).

**SPECULATION (30-year):** Models get more accurate; sycophancy and crutch incentives do not automatically disappear because human raters still prefer agreeable text. Pedagogy wrap remains a candidate moat longer than raw model IQ — **if** AIT experiments show transfer  $\geq$  no-AI.

---

## XXXIII.6 Human needs under AI tutoring

| Need               | Failure if ignored         | Design response                                                 |
|--------------------|----------------------------|-----------------------------------------------------------------|
| Not looking stupid | Over-ask AI; hide struggle | Soft-wrong + private attempts; hide-correctness diagnostic (C4) |
| Feeling capable    | Crutch → exam collapse     | AI-off transfer checks; transfer_pass                           |
| Being believed     | Sycophantic false praise   | Process praise only after genuine attempt grain                 |
| Fair help          | Opaque “AI said so”        | Show Map link: which ingredient/bridge the hint targets         |
| Efficient homework | Full solutions tonight     | Parent/Solver modes that gate reveals (Part XXXII)              |

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## XXXIII.7 Contradictions and open fights

1. **Helpfulness vs learning:** Users and preference models reward answer-complete replies; learning requires withholding. Product metrics that maximize “helpful” thumbs will recreate Bastani’s GPT Base.
2. **Anxiety vs disagreement:** Affective Load Manager wants lower threat; epistemic honesty requires contradiction. Resolve with *tone* (warm) + *content* (firm), sequenced after attempt.
3. **Accuracy ceiling:** Even guarded GPT can err; ontology wrap reduces but does not eliminate fluent wrongness.
4. **Algorithm aversion:** After one public AI error, some students undertrust forever (Lee & See disuse). Need repair UX: “caught my mistake → here’s the check.”
5. **Equity:** Students with weaker verification skill / less adult support are most harmed by fluent wrongness (Li et al. differential impact signal). “AI for all” without wrap can widen gaps.
6. **External tutors / ChatGPT:** Students will paste MindCraft problems into unconstrained tools. Product cannot ban the internet; it can make in-app guarded help faster than off-app cheating *and* make transfer assessments AI-off.

### XXXIII.7a Red Team tick (2026-07-26) — what was killed

**Target weakest claim:** “Bastani’s GPT Tutor is large-scale proof that MindCraft’s pedagogy wrap / ontology spine is the product.”

| Disposition                    | Claim                                                             | Why                                                                                                                                                                                                                                                                                                                                                                       |
|--------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>KILLED</b>                  | Bastani GPT Tutor $\equiv$ MindCraft PWC / ontology wrap          | <b>Category error.</b> Bastani’s Tutor = hint-not-answer <i>plus</i> <b>teacher keys, solutions, and mistake cards in the prompt</b> for a fixed practice set. MindCraft PWC claims a <b>living mastery graph + ingredient/bridge DAG + deterministic checkers</b> . Bastani does not identify MindCraft’s architecture. Marketing “we’re Bastani’s GPT Tutor” is banned. |
| <b>KILLED</b>                  | Guardrails $\Rightarrow$ learning <i>gain</i> vs no-AI            | Bastani: Tutor exam $\approx$ control; authors state <b>no positive effect</b> . Harm neutralization $\neq$ skill uplift. Product bar = beat <b>no-AI</b> / transfer, not merely beat ChatGPT.                                                                                                                                                                            |
| <b>KILLED</b> (reaffirmed)     | Fallibility banner alone fixes learning                           | Nagashima et al.: more hints, <b>same</b> immediate performance, non-hallucinating ITS.                                                                                                                                                                                                                                                                                   |
| <b>WOUNDED</b>                 | Ontology wrap $\Rightarrow$ fewer wrong-math reveals than ChatGPT | Still plausible (FOUNDER BELIEF / HYPOTHESIS) but <b>Medium–Low</b> until AIT-1/AIT-4; Bastani’s accuracy edge came partly from <b>keys in prompt</b> , which MindCraft bank coverage does not fully replicate.                                                                                                                                                           |
| <b>WOUNDED</b>                 | Li et al. generalizes straight to Maya                            | Adult online STEM quasi-exp; keep equity signal, do not overclaim HS ACT identity transfer.                                                                                                                                                                                                                                                                               |
| <b>SURVIVES (strengthened)</b> | Unguarded generative practice can harm solo performance           | Bastani FACT intact; crutch mechanism stands.                                                                                                                                                                                                                                                                                                                             |
| <b>SURVIVES (strengthened)</b> | Sycophancy is an educational safety risk, often invisible         | Sharma + Kasneci EDUFRAMETRAP + Bo et al. Invisible Saboteurs: novices miss saboteur; thumbs-up $\neq$ learning.                                                                                                                                                                                                                                                          |

**Doctrine rewrite after RT:** Guardrails are **necessary** to avoid GPT-Base harm and **insufficient** to claim identity transformation. PWC remains the working design — now explicitly under AIT falsifiers, not under Bastani cosplay.

**Confidence summary (post-RT):**

| Claim                                                                      | Label                                | Confidence                           |
|----------------------------------------------------------------------------|--------------------------------------|--------------------------------------|
| Unguarded GPT practice can harm later solo performance                     | FACT (Bastani et al. 2025)           | High                                 |
| GPT Tutor $\approx$ no-AI on exam (harm mitigated, no gain)                | FACT (Bastani et al. 2025)           | High                                 |
| Perception of AI learning $\neq$ actual exam learning                      | FACT (Bastani et al. 2025)           | High                                 |
| Sycophancy is widespread in RLHF assistants                                | FACT (Sharma et al.)                 | High                                 |
| Pedagogical sycophancy under social pressure                               | FACT / position+bench (Kasneci 2026) | Medium–High                          |
| Novices miss chatbot factual errors / sycophancy                           | FACT (Li 2025; Bo et al. 2025/26)    | High–Medium                          |
| MindCraft ontology wrap beats ChatGPT <i>and</i> no-AI on identity metrics | HYPOTHESIS                           | <b>Medium–Low</b> (AIT-1)            |
| Fallibility UX alone fixes learning                                        | SPECULATION / false                  | Low — behavior shifts $\neq$ mastery |
| Bastani Tutor $\equiv$ MindCraft PWC                                       | <b>KILLED</b>                        | —                                    |

## XXXIII.8 Product implications (actionable)

1. **Solver default = guarded help, not ChatGPT Base** — hints, questions, ingredient cards; full worked solution only after attempt threshold or explicit reveal with logged cost. Do **not** market as “Bastani GPT Tutor.”
2. **Success bar:** solo\_transfer\_pass  $\geq$  no-AI control (and ■ Base). “Less crutch than ChatGPT” is table stakes, not victory.
3. **Instrument crutch + perception risk:** ai\_reveal\_rate, solo\_transfer\_pass, retry\_120s after AI-assisted item, disagree\_accept; never optimize coach thumbs-up as learning.
4. **Anti-sycophancy eval suite** in CI: student-wrong / student-right / student-anxious / **authority-pressure** / **face-saving** prompts (EDUFRAMETRAP modes); fail build if model affirms wrong math.
5. **Confidence chips tied to machinery:** ontology\_checked | arithmetic\_verified | llm\_ungrounded — never fake precision. Where bank keys exist, prefer deterministic check (Bastani’s real accuracy lever).
6. **Ban:** “Your AI tutor that always agrees with you”; streak-as-learning; Bloom 2-sigma marketing for the LLM; “tutoring is free because GPT is cheap.”
7. **Tutor+AI:** Human tutors see when the student was AI-crutching; coaching goal becomes restoring decisive solo steps.

## XXXIII.9 Experiments

| ID | Question | Design | Primary metric | Falsifier |
|----|----------|--------|----------------|-----------|
|----|----------|--------|----------------|-----------|

|       |                                                             |                         |                                        |                                                 |
|-------|-------------------------------------------------------------|-------------------------|----------------------------------------|-------------------------------------------------|
| AIT-1 | Guarded Solver vs ChatGPT-like full answers vs <b>no-AI</b> | 3-arm RCT, same content | solo_transfer_pass @ 1 week            | Guarded $\leq$ Base <b>or</b> Guarded $<$ no-AI |
| AIT-2 | Anti-sycophancy + EDUFRAMET RAP pressure suite              | Offline + online        | wrong-affirmation rate under AUTH/FACE | No reduction vs baseline                        |
| AIT-3 | Fallibility banner vs none                                  | A/B                     | hint quality seeking + transfer        | Banner harms transfer without reducing crutch   |
| AIT-4 | Confidence chips (ontology_checked) vs fluent prose only    | A/B                     | error detection on planted wrong hints | Chips ignored; detection unchanged              |
| AIT-5 | Wizard coach (Experiment A) $\times$ AI guard level         | Factorial               | retry_120s, challenge_accept           | Interaction null                                |
| AIT-6 | Bank-key verify layer on vs off (Bastani keys analogue)     | A/B within guarded      | wrong-math reveal rate + transfer      | Keys add cost, no transfer lift                 |

**Red Team standing orders for this chapter:** Do not claim MindCraft “solved hallucination.” Do not claim tutoring is free because GPT is cheap. Do not treat thumbs-up on coach messages as learning. Do not equate Bastani Tutor with the ontology/PWC.

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### XXXIII.10 One-sentence doctrine

**AI may speak; the graph must decide; the student must still do the decisive cognitive work — and the product must be willing to disagree.**

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## Part XXXIV — Formal Causal DAG + Identification

**Chapter status:** Living evidence brief — Researcher tick 2026-07-28

**Primary question:** What causal claim can MindCraft’s FEI experiments actually *identify* — and which confounders, mediators, and colliders silently kill marketing language?

**Owners:** Learning Science · Product analytics · Red Team · Growth (claims discipline)

**Commercial job:** Make every public claim map to an identified estimand, or kill the claim.

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## XXXIV.1 Why this chapter exists

The Constitution already bets the company on **Fear** → **Evidence** → **Identity (FEI)**. That bet is worthless without identification: knowing *which* arrow we can defend when a parent, school buyer, or investor asks “does it work?”

**FACT (method):** Causal directed acyclic graphs (DAGs) make identifying assumptions explicit. Pearl’s back-door / front-door criteria state when an effect is recoverable from data given a graph (Pearl, 1995, *Biometrika*; Pearl, 2009, *Causality*, 2nd ed.; Pearl, 2009 overview, *Statist. Surv.*, doi:10.1214/09-SS057).

**FACT (ed-tech application):** Weidlich, Hicks & Drachsler (2023/24, *ETR&D*, doi:10.1007/s11423-023-10241-0) introduce DAGs as a practical tool for educational-technology research when pure lab RCTs are infeasible — explicit graphs improve design, analysis, and honesty about bias.

**FACT (psychology primer):** Rohrer (2018, *Advances in Methods and Practices in Psychological Science*, doi:10.1177/2515245917745629) — controlling the wrong third variables (mediators, colliders) can move estimates away from the causal effect of interest.

**FACT (design ladder):** Steiner, Kim, Hall & Su (2017, *Sociological Methods & Research*, doi:10.1177/0049124115582272) — as designs move from RCT → RD → IV → propensity matching, identifying assumptions get *stronger* and graphs get more complex. Less control over assignment ⇒ harder identification.

**Product translation:** “FEI works” is not a slogan. It is a family of estimands. Experiment A identifies one narrow effect. Observational dashboards usually identify none.

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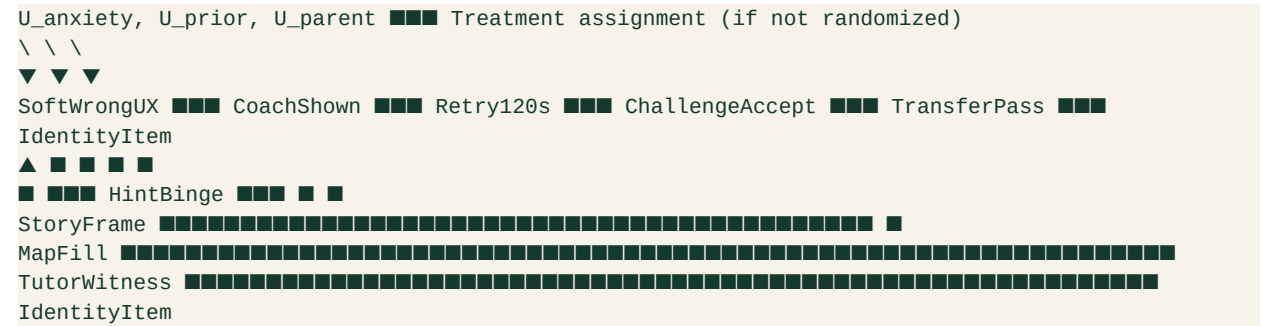
## XXXIV.2 Vocabulary MindCraft must use

| Term                         | Meaning for us                                                                               | Failure mode if ignored                                                                              |
|------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| <b>Estimand</b>              | The exact causal quantity claimed (e.g., effect of wizard coach on <code>retry_120s</code> ) | Vague “works” claims                                                                                 |
| <b>Identification</b>        | Whether that quantity is recoverable from the design + assumed DAG                           | Beautiful charts, zero causal authority                                                              |
| <b>Confounder</b>            | Common cause of treatment and outcome (opens a back-door)                                    | Selection bias sold as product lift                                                                  |
| <b>Mediator</b>              | Variable on the causal path (e.g., attempt → coach → retry)                                  | Adjusting it away “proves” null wrongly                                                              |
| <b>Collider</b>              | Common effect of two causes                                                                  | Conditioning induces spurious association (Rohrer)                                                   |
| <b>WWC confound (design)</b> | Factor perfectly aligned with one arm (e.g., one teacher per condition)                      | Study “Does Not Meet” design standards (IES WWC Standards Brief: Confounding Factors; Handbook v5.0) |

**FOUNDER BELIEF:** Growth language should name the estimand in one clause. If the clause cannot be written, the claim is banned.

### XXXIV.3 Working FEI DAG (MindCraft product graph)

**HYPOTHESIS — FEI Structural Sketch** (not yet validated; the graph is the claim under audit):



Nodes of commercial interest:

- 1. **Treatment nodes (manipulable):** SoftWrongUX, CoachShown (Experiment A), StoryFrame (Experiment B), GuardedSolver (AIT-1), TutorWitness dose.
- 2. **Proximal outcomes:** retry\_120s, challenge\_accept, solo\_transfer\_pass.
- 3. **Distal outcomes:** identity item, course intent, retention/WTP (parent).
- 4. **Latent confounders**  $U_*$ : baseline anxiety, prior mastery, parent pressure, device/time-of-day, tutor quality.

**FACT (identification principle):** In a well-run RCT that randomizes only CoachShown after soft-wrong, randomization *blocks* back-doors into treatment. The total effect of Coach  $\rightarrow$  Retry is identified *without* adjusting for post-treatment mediators — and **must not** adjust away the pathway through reduced hint-binge if that pathway is part of the intended mechanism (Pearl back-door; Rohrer on mediators).

**HYPOTHESIS:** IdentityItem is *not* identified by a 2-week A/B on coach alone. Path length + measurement demand effects (Part XIII open problem) leave multiple open back-doors ( $U_{parent} \rightarrow IdentityItem$ ; tutor talk outside the product).

**Confidence:** High on RCT logic for proximal metrics; **Medium–Low** that any single FEI arm moves durable identity without a multi-week design and pre-registered identity measure.

### XXXIV.4 What Experiment A actually identifies

Recall Experiment A (core OS Part IX): wizard coach on soft-wrong  $\rightarrow$  primary retry\_120s.

**FACT / DESIGN RULE:**

| Estimand | Identified by Exp A? | Notes |
|----------|----------------------|-------|
|----------|----------------------|-------|



|                                                 |                                                                    |                                       |
|-------------------------------------------------|--------------------------------------------------------------------|---------------------------------------|
| Effect of coach on retry_120s                   | <b>Yes</b> , if randomized, low attrition, no arm-aligned confound | Primary                               |
| Effect of coach on same-session accuracy        | Partially                                                          | Secondary; watch hint-binge mediation |
| Effect of coach on challenge_accept @ 1–2 weeks | Only with pre-registered follow-up + analysis set                  | Often underpowered                    |
| Effect of coach on math identity                | <b>No</b> (under this design)                                      | Too distal; demand effects            |
| “FEI works” / “identity transformation”         | <b>No</b>                                                          | Category error — A is one arrow       |

#### WWC-relevant kill conditions (FACT from WWC confounding brief):

- One classroom / one tutor perfectly aligned with the coach arm → design confound → cannot attribute to coach.
- Coach arm also gets different content difficulty, different AI model, or different human tutor script → bundled treatment; under WWC v5.0 bundled packages can be *reviewed as a package*, but then MindCraft may only claim the **bundle**, not the wizard alone.
- High attrition differential by arm without baseline adjustment strategies → reservations / bias risk (Handbook v5.0 compositional-change discussion).

**Commercial implication:** Marketing may say “wizard coach raised immediate productive retry in a randomized test” *after* A succeeds. Marketing may **not** say “FEI proven” or “builds math identity” from A alone.

## XXXIV.5 Observational product analytics — what is *not* identified

Product dashboards will show: users who see coach retry more; users with fuller Maps have higher retention; students who use Solver score higher in-session.

**FACT (Rohrer / Pearl):** Those associations are compatible with many graphs:

| Spurious story          | Graph sketch                                                                                             | Why it fools growth                  |
|-------------------------|----------------------------------------------------------------------------------------------------------|--------------------------------------|
| Motivation confounder   | $U_{\text{motivation}} \rightarrow \text{CoachUse}$ and $U_{\text{motivation}} \rightarrow \text{Retry}$ | Engaged kids click everything        |
| Collider stratification | Conditioning on “finished mission” (common effect of skill + treatment)                                  | Looks like treatment hurts finishers |
| Mediator over-control   | Regress identity on coach <i>adjusting for</i> transfer_pass                                             | Erases the mechanism we sell         |
| Reverse causality       | Low identity → avoid coach / avoid hard items                                                            | “Coach users are already confident”  |

**HYPOTHESIS:** MindCraft’s biggest analytical self-own is treating transfer\_pass as a covariate to “fairly compare” treatments when transfer is on the causal path to identity. That is mechanism erasure, not rigor.

**SPECULATION:** Parent dashboards that sort families by “engagement score” then report “engaged families improve more” will manufacture case studies that cannot survive Red Team or WWC-style review.

**Ban list for GTM:**

1. Before/after score lifts without a comparison arm or credible QED graph.
2. “Students who use Map improve 2×” without adjusting for  $u_{\text{prior}}$  / time-on-task policy.
3. Streak  $\leftrightarrow$  learning correlations as proof (already banned as North Star).
4. Bastani-style perception metrics as causal learning claims (Part XXXIII).

## XXXIV.6 Identification ladder for MindCraft claims

| Claim strength              | Minimum design                                    | DAG requirement                                   | Allowed copy                               |
|-----------------------------|---------------------------------------------------|---------------------------------------------------|--------------------------------------------|
| <b>L0 — Mechanism story</b> | None                                              | Graph labeled HYPOTHESIS                          | “We believe...” / founder belief           |
| <b>L1 — Proximal causal</b> | RCT / good cluster RCT                            | Treatment node randomized; primary pre-registered | “Raised retry / challenge-seeking in test” |
| <b>L2 — Learning causal</b> | RCT + delayed solo / transfer                     | AI-off or held-out items; Part XXXIII bar         | “Improved solo transfer vs control”        |
| <b>L3 — Identity causal</b> | Multi-week RCT + identity items + demand controls | Blocks $u_{\text{parent}}$ , tutor talk; pre-reg  | “Shifted math self-concept measures”       |
| <b>L4 — Market causal</b>   | Field experiment on WTP/retention                 | Parent/school assignment graph                    | “Raised retention / WTP intent”            |

**FOUNDER BELIEF:** Ship L1 proofs fast; never sell L3 with L1 data. Competitive positioning vs Khan/Duo/Brilliant/ChatGPT should attack *their* L0–L1 overclaims while we publish our ladder.

**Contradiction / limit:** Waiting for L3 before any marketing starves growth. Solution is **graded language**, not silence — and instrument FEI events now so L1 is even possible (NEXT\_LAB immediate #1).

## XXXIV.7 Confounders specific to FEI / story / AI stacks

**HYPOTHESIS — MindCraft confound registry** (measure or randomize away):

| Confounder / bias                    | Hits which claim?        | Mitigation                                                    |
|--------------------------------------|--------------------------|---------------------------------------------------------------|
| Baseline anxiety / stereotype threat | Soft-wrong, coach, story | Stratify or measure pretest (Part XXIV)                       |
| Prior concept mastery                | Transfer, Map fill       | Gap-scan / graph eventCount as covariate <i>pre-treatment</i> |

|                                  |                      |                                                         |
|----------------------------------|----------------------|---------------------------------------------------------|
| Parent homework pressure         | Identity, retention  | Separate parent messaging A/B; don't pool               |
| Tutor quality (human)            | "AI + tutor" package | Nested assignment; avoid one-tutor-per-arm              |
| Content difficulty drift         | Any practice A/B     | Fix item bank per arm (Part XXXIII keys lesson)         |
| Time-on-task / session length    | Engagement→learning  | Cap or equalize exposure policy                         |
| AI model version change mid-test | Solver guards        | Pin model; log version as design factor                 |
| Selection into Solver help       | Crutch vs learning   | Randomize offer; instrument <code>ai_reveal_rate</code> |

**FACT (Steiner et al.):** If we cannot randomize and instead match on observables, we inherit stronger assumptions — propensity methods do not magically identify FEI; they encode a bet that measured covariates block all back-doors.

**FACT (Weidlich et al.):** Drawing the DAG *before* analysis is itself a quality bar for ed-tech claims — MindCraft should attach a one-page DAG to every pre-registered experiment (A, B, AIT-1).

## XXXIV.8 Commercial implication — positioning and kill rules

**So what for MindCraft commercially:**

1. **Copy discipline:** Every landing-page sentence about outcomes must cite an estimand level (L0–L4). Default public claims stay at L0/L1 until A/AIT land.
2. **Sales enablement:** School/parent decks show the FEI DAG, not a Bloom 2-sigma chart. Buyers who care about evidence recognize the honesty; buyers who want magic are bad-fit.
3. **Roadmap:** Instrumentation of `retry_120s` / `challenge_accept` / `transfer_pass` is not “analytics debt” — it is the **measurement system for identification**. Without it, no L1.
4. **Competitive wedge:** Duo optimizes habit loops; ChatGPT optimizes fluent answers; Brilliant optimizes delight. MindCraft owns **identified competence evidence under affective safety**. That wedge dies if we market unidentified vibes.
5. **Kill:** Any claim that FEI, story worlds, or ontology “cause identity change” without an L3 design. Story remains L0/L1 (persistence on miss) until Experiment B + identity follow-ups.

**Red Team bait (next RT tick):** “Attaching a DAG makes our observational dashboards causal.” — Should be killed; graphs without design still leave  $u_*$  open.

## XXXIV.9 Experiments spawned / tightened

| ID | Question | Design | Primary | Falsifier |
|----|----------|--------|---------|-----------|
|----|----------|--------|---------|-----------|

|       |                                                                               |                                    |                         |                                 |
|-------|-------------------------------------------------------------------------------|------------------------------------|-------------------------|---------------------------------|
| DAG-0 | Can we publish a one-page FEI DAG + estimand card for Exp A / AIT-1?          | Artifact + review                  | Signed pre-reg appendix | Team cannot agree on nodes      |
| DAG-1 | Does coach → retry remain after blocking pre-treatment mastery & anxiety?     | Exp A + covariates <b>pre</b> only | retry_120s              | Effect vanishes → was selection |
| DAG-2 | Does adjusting for transfer_pass (mediator) spuriously null coach → identity? | Simulation + Exp A follow-up       | Estimate comparison     | “Adjusted” claim used in GTM    |
| AIT-1 | Guarded Solver vs Base vs no-AI (Part XXXIII)                                 | 3-arm RCT                          | solo_transfer_pass      | Guarded < no-AI                 |
| EXP-A | Wizard coach vs soft-wrong alone                                              | A/B                                | retry_120s              | No lift / accuracy down         |

## XXXIV.10 Confidence table

| Claim                                              | Label                  | Confidence                      | Needs                         |
|----------------------------------------------------|------------------------|---------------------------------|-------------------------------|
| DAGs clarify identification in ed-tech             | FACT                   | High                            | Cite Weidlich; use in pre-reg |
| Exp A identifies coach → retry under RCT           | FACT (logic)           | High                            | Run A without arm confounds   |
| Observational “Map users improve” is causal        | SPECULATION if claimed | Kill until QED/RCT              | —                             |
| FEI distal identity identified by 2-week coach A/B | HYPOTHESIS (false)     | High that <b>not</b> identified | Longer design                 |
| Graded L0–L4 claim ladder is commercially optimal  | FOUNDER BELIEF         | Medium                          | Test sales call comprehension |

|                                                        |                         |            |                                 |
|--------------------------------------------------------|-------------------------|------------|---------------------------------|
| Propensity matching recovers FEI without randomization | HYPOTHESIS (optimistic) | Low–Medium | Steiner assumptions rarely hold |
|--------------------------------------------------------|-------------------------|------------|---------------------------------|

### XXXIV.11 What this chapter refuses

- Bloom 2-sigma as MindCraft proof.
- “Tutoring is free” / explanations-as-product.
- Streaks as causal learning evidence.
- Fabricated DOIs.
- Equating a drawn DAG with a completed experiment.

**Bottom line:** The FEI loop is the product thesis. Identification is the honesty layer. Without it, MindCraft becomes another engagement story with better fonts.

## Part XXXV — Competitive Session Audits (Khan / Duo / Brilliant / ChatGPT)

**Chapter status:** Living evidence brief — Researcher tick 2026-07-29

**Primary question:** In a single practice session, what mechanism does each major competitor optimize — and what does that imply for MindCraft positioning, copy, and North Star metrics?

**Owners:** Product · Growth · Positioning · Red Team

**Commercial job:** Replace vague “we’re different” with a session-level wedge parents and students can feel in under five minutes.

### XXXV.1 Why session audits, not another market teardown

Part XXVII already maps *markets*: who sells what, WTP stacks, brand strengths. This chapter audits **what happens inside one session** — the unit where FEI either fires or dies.

**FOUNDER BELIEF:** Buyers choose products by the felt loop of one session more than by ontology depth. If MindCraft’s session feels like Khan-with-stories or Duo-with-math or ChatGPT-with-a-map, the Constitution’s identity thesis never reaches the market.

**Audit rubric (aligned to Parts XXI / XXV / XXXIV):**

| Lens                 | Question in-session                              | MindCraft target signal |
|----------------------|--------------------------------------------------|-------------------------|
| <b>Fear / affect</b> | Does first wrong feel evaluative or recoverable? | Soft-wrong + coach path |

|                     |                                                                       |                                   |
|---------------------|-----------------------------------------------------------------------|-----------------------------------|
| <b>Evidence</b>     | Does the learner produce competence evidence <i>without</i> a crutch? | retry_120s, attempt before reveal |
| <b>Identity</b>     | Does the UI narrate “you are becoming someone who can...”?            | Map / witness language — not XP   |
| <b>Transfer</b>     | Can they do the next item AI-off / hint-off?                          | solo_transfer_pass                |
| <b>Claim ladder</b> | What L0–L4 claim can <i>their</i> evidence support?                   | Honest competitive copy           |

**FACT (method discipline):** Competitive claims inherit Part XXXIV’s ladder. Session delight ≠ identified learning. Habit return ≠ identity change.

## XXXV.2 Khan Academy / Khanmigo — mastery library + optional AI coach

### Session shape (product observation)

**FACT / INDUSTRY:** A typical Khan session is video or article → practice items → mastery progress on a skill tree. Coverage and brand trust are the product surface (Part XXVII). Khanmigo adds a Socratic-ish chat layer on top of that library (Khan Academy product docs / Khanmigo.ai).

### What the evidence actually supports

**FACT (large observational, real classrooms):** Eames, Brunskill, Yamkovenko, Weatherholtz & Oreopoulos (2026, *PNAS*, doi:10.1073/pnas.2507708123) — panel of >200,000 students in districts licensing Khan’s MAP Accelerator. Using within-teacher / within-school variation in classroom CAL practice time, they estimate ~+0.031 SD math gain at observed mean usage (~6.6 h/year ≈ 11 min/week), with roughly linear gains projecting to ~+0.085 SD at ~30 min/week recommended. Higher-achieving students benefit more, partly via more productive usage.

**FACT (implementation caveat):** Murphy et al. (2014, SRI Education / Gates-funded U.S. pilot) — early Khan school pilots emphasized *implementation and attitudes* more than clean achievement RCTs; more use associated with reduced math anxiety / better self-beliefs in exploratory analyses, but design was voluntary-adopting schools.

**FACT (null / weak supplemental SI):** Kelly & Rutherford (2017, *IRRODL*, doi:10.19173/irrodl.v18i4.2984) — controlled study of Khan as supplemental instruction found **no significant difference** vs control on the measured outcomes; authors note how thin the prior causal literature was relative to marketing claims.

**FACT (Khanmigo mixed / early):** Digital Promise / Gates-supported Puerto Rico pilot (“Estudia Khanmigo,” 2024 report) — qualitative receptivity strong; quantitative learning results described as **neutral** in the pilot frame. Gökoğlu-adjacent undergrad mixed-methods work comparing Khanmigo vs Google search on lunar-phases concepts (Journal of Teaching and Learning, 2025) reported learning gains across conditions but **no significant between-group difference** favoring Khanmigo vs search / paper in that short exposure.

**HYPOTHESIS:** Khan’s session optimizes **skill coverage + dosage**. Affective safety is secondary; identity is not the UI’s job. Khanmigo can feel supportive without beating search on short concept tests — consistent with “helpfulness ≠

transfer.”

**Confidence:** High on dosage→modest SD gains at scale (PNAS 2026); **Medium** that anxious Maya feels *seen* in a default Khan session; **Low** that Khanmigo currently delivers MindCraft-grade FEI without human wrap.

## Commercial wedge vs Khan

Do **not** out-library Khan. Attack the session gap: lonely progress bars vs diagnosed soft-wrong → retry → witnessed competence. Parent copy: “Minutes on skills matter (Khan proved dosage). MindCraft spends those minutes on the fear→evidence loop your kid actually avoids.”

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## XXXV.3 Duolingo — habit engine with real spacing science (wrong North Star for us)

### Session shape

**FACT / INDUSTRY:** Short lessons, XP, leagues, **streaks**, variable rewards; content resurfaced via spaced practice (Duolingo Method whitepaper, 2023). Math courses exist, but the company DNA is language habit loops.

### What the evidence actually supports

**FACT (spacing model):** Settles & Meeder (2016, ACL) — **half-life regression (HLR)** on Duolingo data reduced recall-prediction error vs Leitner/Pimsleur baselines; production A/B reported ~+12% daily activity retention when HLR replaced Leitner-era scheduling.

**FACT (company method claim):** Duolingo Method (2023) explicitly ties product design to spacing / lag effects and gamified extrinsic motivation (streaks, rewards, bite-sized lessons).

**FACT (gains ≠ minutes alone):** Ploquin et al. (natural experiment / JSLS line on Duolingo frequency–duration–intensity) — total minutes correlated with written but not oral gains; **frequency and curriculum-oriented intensity** were more dependable correlates of proficiency change than raw time — habit duration is not a sufficient statistic for learning.

**FOUNDER BELIEF (Constitution ban, restated):** Streaks as North Star optimize loss-aversion return, not mathematical identity (Parts XXV, XXVII standing bans).

**HYPOTHESIS:** Duo wins the *open app* problem. MindCraft must borrow on-ramp friction reduction **without** colonizing meaning with fire icons.

**Confidence:** High that Duo’s session is a world-class habit machine; High that habit ≠ FEI distal outcomes; Medium that math-Duo will pull some of Maya’s peers into streak addiction without transfer.

## Commercial wedge vs Duo

Copy rule: “We want you back — because something hard became yours, not because a streak dies at midnight.” Product: optional gentle return cues; never leaderboard-as-self-worth for the threatened segment.

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## XXXV.4 Brilliant — interactive delight for the already-curious

### Session shape

**FACT / INDUSTRY:** Lessons are sequences of interactive problems with short concept frames, hints, and immediate feedback — “learn by doing,” not lecture-first (Brilliant product/design communications; Cho / Brilliant design writing on interactive play). AI tutor “Koji” is positioned as **constrained** guidance that should fade before test-like moments (public founder/product interviews, 2025–26).

### What the evidence actually supports

**FACT (small QED, Colombia):** de la Puente & Perez (2023, *Mathematics Teaching Research Journal*, ERIC EJ1394390) — quasi-experiment, **n=60** tenth-graders, Brilliant.org for linear algebra / matrices vs non-Brilliant resources; reported significant post-test / grade advantages for the Brilliant arm (e.g.,  $t(58)=2.63$ ; ANOVA  $F(1,58)=5.04$ ,  $p=0.029$  on final grades). **Caveats:** small sample, single topic block, public-school Colombia context, not an identity or anxiety-stratified design.

**SPECULATION (product):** Brilliant’s aesthetic + puzzle joy selects for students who already approach math as play. That is a real segment — and not Maya’s primary threat profile (Part XXIV).

**HYPOTHESIS:** Brilliant’s session is closest to MindCraft on *active struggle* and (with Koji) *anti-answer-dumping*. Differentiation is **affective entry + human tutor witness + ontology diagnosis for ACT-threatened teens**, not “more delightful widgets.”

**Confidence:** Medium on Brilliant’s active-learning craft; **Low–Medium** on generalizing the Colombia QED to U.S. ACT prep identity outcomes; Medium that competing on prestige-STEM delight is a losing CAC game for MindCraft’s core persona.

### Commercial wedge vs Brilliant

Respect the craft; decline the segment war. Positioning: “Brilliant for curiosity already online. MindCraft for the kid who goes quiet when the room gets evaluative.”

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## XXXV.5 ChatGPT / unfettered GPT tutors — fluent session, harmful solo exam

### Session shape

**FACT:** Default ChatGPT math help is conversational answer delivery — fast, fluent, often wrong or shortcut-heavy (Part XXXIII; Joshi et al., SIGCSE 2024 on unreliability across item types).

### What the evidence actually supports

**FACT (field RCT, ~1,000 HS math students):** Bastani, Bastani, Sungu, Ge, Kabakc■ & Mariman (2025, *PNAS*, doi:10.1073/pnas.2422633122) — GPT Base (ChatGPT-like) raised practice grades ~+48%; GPT Tutor (safeguarded prompts) ~+127% during access. When AI removed, GPT Base students scored ~17% worse than never-AI peers. GPT Tutor **neutralized harm** (exam  $\approx$  control) but authors state **no positive learning effect** vs no-AI. Mechanism:



**crutch** during practice, not only hallucination carryover.

**Red Team carry-forward (XXXIII.7a):** Bastani’s GPT Tutor ≠ MindCraft ontology/PWC. Guards ≠ proven gain. Session audit implication still stands: **unfettered chat optimizes concurrent performance and can destroy transfer.**

**Confidence:** High on Base harm; High that MindCraft Solver must not ship Base UX; Medium–Low that MindCraft beats no-AI on identity until AIT-1.

**Commercial wedge vs ChatGPT**

Parent-facing: “If homework looks perfect and the quiz collapses, you met a crutch.” Product bar: solo\_transfer\_pass ≥ no-AI, ■ Base (Part XXXIII). Never market “ChatGPT but nicer.”

**XXXV.6 Cross-competitor scorecard (session mechanisms)**

| Competitor       | Optimizes in-session           | Proximal metric they can own        | FEI risk                                    | MindCraft response                        |
|------------------|--------------------------------|-------------------------------------|---------------------------------------------|-------------------------------------------|
| <b>Khan</b>      | Coverage + dosage              | Skills/time → modest SD (PNAS 2026) | Lonely evaluative feel; AI not yet identity | Diagnosis + soft-wrong + tutor witness    |
| <b>Duolingo</b>  | Return / streak                | DAU, streak length, HLR review      | Extrinsic colonization                      | Borrow friction UX; ban streak North Star |
| <b>Brilliant</b> | Interactive struggle + delight | Lesson completion, “aha”            | Misses threatened segment                   | Win anxious middle; don’t out-prestige    |
| <b>ChatGPT</b>   | Fluent answers now             | Session grade / CSAT                | Crutch; transfer collapse                   | Guarded Solver; AI-off transfer           |

**FOUNDER BELIEF:** The defensible one-liner is not “AI tutor.” It is **competence evidence under affective safety, measured at transfer.**

**XXXV.7 Marketing language that survives this audit**

| Allowed (mechanism-true)                                    | Banned                                 |
|-------------------------------------------------------------|----------------------------------------|
| “Practice that still works when the hints are gone”         | “2-sigma / Bloom tutoring”             |
| “We measure retry after a soft wrong — not streak fire”     | “Tutoring is free now that GPT exists” |
| “Dosage matters; we spend dosage on the fear→evidence loop” | “Become math Duolingo”                 |

|                                              |                                                                  |
|----------------------------------------------|------------------------------------------------------------------|
| “Guarded help so chat can’t become a crutch” | “Bastani-proven ontology” / “guarantees learning gains vs no-AI” |
|----------------------------------------------|------------------------------------------------------------------|

**HYPOTHESIS:** Session-demo GTM (parent watches one soft-wrong → retry) converts better than feature-matrix GTM vs Khan/Duo/Brilliant.

### XXXV.8 Experiments spawned

| ID    | Question                                                                                | Design                              | Primary                                  | Falsifier                                                   |
|-------|-----------------------------------------------------------------------------------------|-------------------------------------|------------------------------------------|-------------------------------------------------------------|
| CSA-1 | Blind session preference: MindCraft vs Khan practice vs Brilliant lesson (same concept) | Within-subject, Maya-segment screen | Forced choice + affect items             | MindCraft loses on “would return” <i>and</i> “felt capable” |
| CSA-2 | Copy test: FEI session-demo vs “AI tutor / mastery path”                                | Landing A/B                         | Signup → first retry_120s                | Feature copy wins activation                                |
| CSA-3 | ChatGPT Base paste rate when Solver is guarded vs permissive                            | Instrumentation                     | Off-app paste proxy + solo_transfer_pass | Guards increase paste <i>and</i> transfer falls             |
| AIT-1 | Carry from XXXIII                                                                       | 3-arm RCT                           | solo_transfer_pass                       | Guarded < no-AI                                             |

### XXXV.9 Confidence table

| Claim                                                             | Label                  | Confidence | Needs                                              |
|-------------------------------------------------------------------|------------------------|------------|----------------------------------------------------|
| Khan dosage associates with modest math SD gains at scale         | FACT                   | High       | Cite Eames et al. 2026; don’t overclaim RCT purity |
| Duo HLR / streaks optimize engagement                             | FACT                   | High       | Settles 2016; Method 2023                          |
| Streaks should be MindCraft North Star                            | SPECULATION (false)    | Kill       | Standing ban                                       |
| Brilliant QED proves identity transformation for threatened teens | HYPOTHESIS (overreach) | Low        | Different segment + small QED                      |

|                                                             |                  |        |                               |
|-------------------------------------------------------------|------------------|--------|-------------------------------|
| ChatGPT-like Base can harm solo exam performance            | FACT             | High   | Bastani 2025                  |
| Pedagogy guards $\Rightarrow$ learning <i>gain</i> vs no-AI | Killed in XXXIII | —      | Neutralize harm $\neq$ uplift |
| Session-demo positioning beats feature-matrix vs these four | HYPOTHESIS       | Medium | CSA-2                         |
| MindCraft should not out-content Khan or out-streak Duo     | FOUNDER BELIEF   | High   | Strategy consistency          |

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### XXXV.10 What this chapter refuses

- Bloom 2-sigma as competitive proof.
- Absolute “tutoring is free.”
- Streaks as learning KPI.
- Equating Khanmigo qualitative warmth with identified FEI.
- Equating Bastani GPT Tutor with MindCraft’s product.
- Fabricated citations.

**Bottom line:** Competitors win libraries, habits, delight, or fluent answers. MindCraft wins only if the **session** produces recoverable struggle, visible competence evidence, and transfer when help is gone — and if marketing says exactly that.

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## Part XXXVI — Equity Audit of Story Worlds

**Chapter status:** Living evidence brief — Researcher tick 2026-07-29

**Primary question:** Do MindCraft story worlds expand belonging and mathematical identity for diverse learners — or do they costume Eurocentric defaults, add seductive load, and widen who feels “math is for people like me”?

**Owners:** Learning Science · World / Narrative · Product · Red Team

**Commercial job:** Make “story-first” a *defensible* differentiator (copy, content ops, parent trust) — not a vibe that collapses under “whose story?” scrutiny.

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### XXXVI.1 Why this audit is load-bearing

Part XXVIII put invention-story under trial (HIST-1...4; HIST-EQ). Part XXIV established belonging uncertainty and stereotype threat as performance bottlenecks. WORLD\_VISION commits to story worlds that wrap frozen math. None of that settles **who is centered** when the wrap is written.

**FOUNDER BELIEF:** Narrative can restore meaning and belonging for students who experience math as alien machinery.

**FACT (counter-tradition):** Mathematics learning is *not* culture-neutral. Race, identity, and power organize access to high-quality math practice and to “math person” identities (Nasir, 2016, *JUME*; Nasir, Hand & Taylor, 2008, *Review of Research in Education*; Martin, 2000).

**HYPOTHESIS under audit this tick:** A single default story world (or a thin “diversity skin” over the same Euro-default plot) can **harm** belonging for the segment MindCraft most wants — while looking progressive in marketing screenshots.

**Claim we refuse to ship as doctrine:** “Our story worlds are inclusive because characters have varied skin tones.” That is **SPECULATION** dressed as equity. Token presence ≠ practice-linked identity (Nasir & Hand, 2008).

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## XXXVI.2 What “equity” means here (operational, not slogan)

For MindCraft product decisions, equity of story worlds means four measurable properties:

| Property               | Operational question                                                                      | Failure mode                                               |
|------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Access</b>          | Can the learner <i>enter</i> the math practice without cultural membership tests?         | Lore gatekeeping; insider jokes; assumed prior             |
| <b>Role</b>            | Does the student take an <i>integral</i> role (solver, witness, navigator) — not tourist? | Decorative NPC energy; “watch the hero”                    |
| <b>Expression</b>      | Can competence show in the student’s voice / choices without stereotype cues?             | “Even you can...” copy; gendered/racialized ability frames |
| <b>History honesty</b> | When invention stories appear, is the human lineage plural and structure-encoding?        | Euro-only timeline; museum trivia; token name-drop         |

This maps Nasir & Hand’s (2008, *Journal of the Learning Sciences*) basketball-vs-classroom analysis: deep engagement tracked **access to the domain**, **integral roles**, and **self-expression** — not posters about belonging. Classroom math often denied those affordances to African American students who showed them in basketball.

**Product translation:** Story worlds succeed when they recreate *those three affordances around math*, not when they add themed wallpaper around a lonely skill tree.

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## XXXVI.3 Evidence base (what is solid vs thin)

## A. Identity is sociopolitical, not cosmetic

**FACT (ethnographic / theory program):** Martin (2000) — *Mathematics Success and Failure Among African-American Youth* — success and failure are shaped by sociohistorical context, community forces, school influence, and individual agency; “math identity” is negotiated under racialized meanings of ability, not only item difficulty.

**FACT (practice-linked identity):** Nasir & Hand (2008) — same students can show sophisticated quantitative reasoning in out-of-school practices while disengaging in school math when access, role, and expression collapse.

**FACT (field stance):** Nasir (2016) — treating math as outside race/culture is a fallacy; stereotypes and school structures disrupt mathematical identities and opportunity to learn.

**FACT (practitioner synthesis):** Aguirre, Mayfield-Ingram & Martin (NCTM; *Impact of Identity in K–8 / K–12 Mathematics*) — teaching practices shape whether students see themselves as doers of mathematics; equity-based practice is asset-based and identity-attentive, not deficit remediation theater.

**Implication:** MindCraft’s identity thesis (Parts XXV / FEI) is *aligned* with this literature — **if** story design expands access/role/expression. Story that only changes scenery fails the identity literature’s actual mechanism.

## B. Belonging interventions are narrow — do not cargo-cult into lore

**FACT (RCT):** Walton & Cohen (2011, *Science*, doi:10.1126/science.1198364) — brief social-belonging intervention framing adversity as common/transient raised African American college students’ GPA over 3 years and roughly halved the measured achievement gap in that sample (N=92; mediated by construal of adversity).

**FACT (adjacent):** Steele & Aronson stereotype-threat tradition; Spencer, Steele & Quinn (women/math) — ability-diagnostic framing under negative stereotypes can impair performance (effect sizes and replication quality vary; see Part XXIV limits).

**Red Team (carry from XXIV):** Effects are heterogeneous; school-scale attenuates; a 45-minute belonging essay ≠ a gamified toast ≠ a fantasy prologue. Belonging is not a substitute for teaching.

**HYPOTHESIS:** Story worlds help belonging when they make early adversity **non-identity-diagnostic** (“misses teach the map”) *and* refuse stereotype-cueing copy — not when they lecture “you belong here” in lore text.

## C. Whose history? Plural roots are historical FACT; classroom transfer is HYPOTHESIS

**FACT (historiography):** Joseph (1991/2011, *The Crest of the Peacock*) documents non-European mathematical lineages (Egypt, Mesopotamia, India, China, Islamic world, and more) against Eurocentric “Greeks-only greatness” narratives.

**FACT (program):** D’Ambrosio’s ethnomathematics programme (e.g., 1985–1990 line; ethno + mathema + tics) treats mathematical knowing as culturally situated practice, not a single civilizational monopoly.

**FACT (critical pedagogy tradition):** Gutstein (2006, *Reading and Writing the World with Mathematics*) — math can be used to read/write sociopolitical conditions; equity is not only “access to algebra” but agency. **Caveat for MindCraft:** Full Freirean classroom projects are not the product’s unit of delivery; borrow the *honesty about power*, not a forced activism skin on ACT prep items.

**HYPOTHESIS (from XXVIII HIST-EQ / HIST-4):** Plural invention stories that encode structure can lift belonging without crushing retention — **if** they are not seductive trivia.

#### D. Seductive details and “diversity costume” share a kill path

**FACT:** Harp & Mayer (1998, *Journal of Educational Psychology*, doi:10.1037/0022-0663.90.3.414) — interesting but irrelevant details can damage learning via cognitive interest mechanisms.

**FACT (meta-analytic):** Sundararajan & Adesope (2020, *Educational Psychology Review*, doi:10.1007/s10648-020-09522-4) — including seductive details tends to hinder learning; moderators exist (format, pacing, subject), but the mean direction is caution, not license.

**Kill criterion (FOUNDER BELIEF → testable):** If a cultural reference, character trait, or “heritage beat” can be removed without changing the mathematical mental model **and** without changing who can take an integral role, it is decoration. Decoration that only signal brand virtue are equity theater **and** CLT risk.

**Survive criterion:** Cultural/historical content that (a) encodes the invariant, (b) expands access/role/expression for under-affirmed students, and (c) does not cue stereotypes.

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### XXXVI.4 Failure modes specific to MindCraft story worlds

| Failure mode                      | What it looks like in product                        | Likely harm                                          | Epistemic label                                  |
|-----------------------------------|------------------------------------------------------|------------------------------------------------------|--------------------------------------------------|
| <b>Euro-default adventure</b>     | Castles, Greco-Roman “genius,” white default heroes  | Quiet alienation; “not for people like me”           | HYPOTHESIS (strong prior from identity lit)      |
| <b>Token palette swap</b>         | Same plot; recolored sprites                         | Cynical read; no identity mechanism                  | HYPOTHESIS                                       |
| <b>Heritage trivia dump</b>       | Long museum lore before first attempt                | Seductive load; delayed micro-win for anxious Maya   | FACT risk (CLT / seductive details)              |
| <b>Stereotype-friendly tropes</b> | “Street-smart” vs “book-smart”; gendered magic roles | Cue threat / essentialism                            | HYPOTHESIS (aligned with threat lit)             |
| <b>Forced activism costume</b>    | Every item a justice pamphlet                        | Parent trust fracture; math time stolen; off-mission | SPECULATION commercially; Gutstein ≠ ACT session |
| <b>Single-world monoculture</b>   | One lore for all students forever                    | No cultural fit; no choice autonomy (SDT)            | HYPOTHESIS                                       |

**FOUNDER BELIEF (constrained):** Plural *structural* histories + multiple entry aesthetics beat one mega-franchise lore — but only with HIST/EQ experiments and content-ops cost discipline.

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## XXXVI.5 Audit rubric (ship this as content ops, not a vibes review)

Before a story module graduates from draft:

1. **Structure test:** Name the mathematical invariant the narrative makes memorable. If blank → rewrite or cut.
  2. **Role test:** In the first 60 seconds, does the student *do* math as an agent, or watch lore? Threatened learners need Safety → micro-attempt before long meaning (XXVIII.7).
  3. **Belonging-copy test:** Ban ability-diagnostic group framing (“girls struggle with...”, “unlike other kids...”). Prefer adversity-as-common-and-transient framing (Walton & Cohen mechanism — adapted carefully; do not fake an RCT in UI).
  4. **History plurality test:** If an invention beat exists, is at least one non-Euro lineage available in the corpus for that concept family over time (not every single card)? Token one-off “Al-Khwarizmi cameo” without structure = fail.
  5. **Seductive-details test:** Remove the cultural beat; if learning items and role affordances unchanged, cut or move post-success.
  6. **Stereotype scan:** Two reviewers (ideally including someone outside the default writer demographic) flag tropes; unresolved flags block ship.
  7. **Claim ladder (Part XXXIV):** Marketing may say “we design stories so more students can take an integral role in the math” (L0/L1 product intent). It may **not** say “our lore closes achievement gaps” without L3 evidence.
- 

## XXXVI.6 Commercial implications (copy, positioning, roadmap)

### Marketing language that survives

- **Allowed:** “Math isn’t a private club with one history. Stories should make the *idea* human — and give every learner a real role in solving it.”
- **Allowed:** “We refuse decoration that wastes working memory. Narrative earns its place when it encodes the math or restores a sense that struggle isn’t proof you don’t belong.”
- **Banned:** “Culturally magical world closes the gap.”
- **Banned:** “Diverse characters = equitable outcomes.”
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity (standing bans).

### Competitive stance

Khan’s library and Duo’s streaks are largely **culture-thin** at the session surface (Part XXXV). MindCraft *chooses* narrative risk. That is only a wedge if the equity audit is visible in product craft — otherwise competitors can dismiss story as gimmick **and** critics can dismiss it as exclusionary. Honest position: “We accept narrative risk; we instrument belonging + learning; we kill modules that fail the rubric.”

### Growth / parent trust

Parents in threatened segments buy **dignity + results**. Equity theater without `retry_120s` / transfer reads as politics instead of help. Lead with FEI evidence; treat plural history as accuracy and belonging hygiene, not a culture-war banner unless the parent segment asks for it.

### Product bets

1. **Content ops checklist** above as merge gate for story modules.
2. **At least two aesthetic/cultural entry frames** for core onboarding arcs (same math, different wrap) — autonomy without fragmenting ontology.
3. **HIST-EQ / EQ-1...3** before scaling “world lore” as brand centerpiece.
4. Tutor witness remains the primary belonging technology for high-threat students (XXIV); story is amplifier, not substitute.

## XXXVI.7 Experiments spawned

| ID      | Question                                                           | Design                     | Primary                                            | Kill condition                                       |
|---------|--------------------------------------------------------------------|----------------------------|----------------------------------------------------|------------------------------------------------------|
| EQ-1    | Euro-default story vs plural structure-encoding story (same items) | A/B or crossover           | belonging/threat items + retry_120s + retention@7d | Belonging flat; or learning drop > pre-reg threshold |
| EQ-2    | Token palette-swap vs role/access redesign (same art budget)       | 3-arm                      | integral-role self-report + attempt latency        | Palette ≥ role redesign on belonging                 |
| EQ-3    | Pre-attempt heritage lore vs post-success meaning beat             | Timing                     | time-to-first-attempt; transfer                    | Pre-lore delays attempt with no belonging lift       |
| HIST-EQ | (from XXVIII) Plural vs single-tradition invention                 | A/B                        | belonging + learning                               | Belonging up, transfer down badly                    |
| EQ-QUAL | 10 Maya interviews: “who is this world for?”                       | Protocol B + equity probes | theme codes                                        | Students read world as “not for me” at >threshold    |

**Pre-reg discipline (XXXIV):** EQ-\* identify story→belonging/persistence. They do **not** identify story→GPA gap closure. Do not sell L3 from L1.

## XXXVI.8 Confidence table

| Claim                                                 | Label                                       | Confidence |
|-------------------------------------------------------|---------------------------------------------|------------|
| Math learning is entangled with race/culture/identity | FACT (research consensus in equity math ed) | High       |



|                                                                   |                                                                |               |
|-------------------------------------------------------------------|----------------------------------------------------------------|---------------|
| Access / role / expression predict engagement better than posters | FACT (Nasir & Hand mechanisms) + HYPOTHESIS for digital worlds | Medium–High   |
| Default Euro adventure can alienate                               | HYPOTHESIS                                                     | Medium        |
| Token diversity without role redesign fails                       | HYPOTHESIS                                                     | Medium–High   |
| Seductive cultural trivia can hurt learning                       | FACT (seductive-details tradition + meta-analysis direction)   | Medium–High   |
| Plural invention stories lift belonging <i>and</i> learning       | HYPOTHESIS (HIST-EQ unrun)                                     | Low–Medium    |
| Story can replace tutor witnessing for threat                     | SPECULATION / likely false                                     | Low (against) |
| Equity marketing without FEI metrics is brand risk                | FOUNDER BELIEF                                                 | Medium–High   |

## XXXVI.9 What this chapter kills

1. **Kill:** “Inclusive skins” as the equity strategy.
2. **Kill:** Shipping long pre-attempt lore as belonging intervention.
3. **Kill:** Marketing claims that story worlds close racial/gender achievement gaps without identified designs.
4. **Wound:** Single-franchise world as forever default — survivable only as temporary scaffold with EQ-1/2 on the roadmap.
5. **Survive (constrained):** Story-first WORLD\_VISION — **if** narrative encodes structure, expands integral roles, passes stereotype scan, and stays on the L0–L2 claim ladder until experiments graduate it.

**Doctrine until data:** Plural, structure-encoding, role-rich worlds under audit > monoculture lore > token palette swaps > heritage trivia dumps.

## Part XXXVII — Expectancy-Value (Eccles): Utility, Cost, Attainment → Course Choice

**Chapter status:** Living evidence brief — Researcher tick 2026-07-29

**Primary question:** Why do students *choose* (or abandon) math pathways — and which MindCraft surfaces move expectancy vs value vs cost?

**Owners:** Learning Science · Product · Growth / Positioning · Red Team

**Commercial job:** Replace vague “make math meaningful” copy with a **defensible E×V map**: which product moves which lever, what parents hear, what we refuse to claim.

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## XXXVII.1 Why this chapter exists

Parts XXV (self-efficacy / narrative identity) and XXIV (anxiety / belonging) explain *can I* and *am I safe*. They do not fully explain *will I keep choosing math* when Instagram, sports, and sleep compete.

**FACT (program lineage):** Eccles-Parsons et al. (1983) built an achievement-choice model to explain gendered STEM participation: performance and choice are most proximally driven by **expectancies for success** and **subjective task values** (STV), themselves shaped by socializers, culture, and prior experience (*Expectancies, values, and academic behaviors*, in Spence, *Achievement and Achievement Motives*).

**FACT (rename / situating):** Eccles & Wigfield (2020, *Contemporary Educational Psychology*, doi:10.1016/j.cedpsych.2020.101859) rename the framework **Situated Expectancy-Value Theory (SEVT)** — expectancies and values are *situation-bound hierarchies*, not a single global “motivation score.”

**FOUNDER BELIEF under audit:** MindCraft’s identity thesis only converts to commercial retention / course-persistence if product reduces **cost** and raises **attainment + utility** while expectancy (FEI evidence) rises — not if we only make prettier explanations.

**Claim we refuse as doctrine:** “Story worlds automatically raise utility value.” That is **SPECULATION**. Utility is a *student-authored* link to goals (Hulleman tradition), not a brand aesthetic.

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## XXXVII.2 The model in product language (not poster language)

| Construct               | Plain definition                                    | MindCraft surface (hypothesis)                                                           | Failure if ignored                                                                |
|-------------------------|-----------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| <b>Expectancy</b>       | “Can I succeed <i>on the next relevant tasks</i> ?” | Gap-scan → level gate; soft-wrong → retry; tutor witness; FEI micro-wins                 | High value + low expectancy → avoidance / crutch (Bastani Base risk, Part XXXIII) |
| <b>Intrinsic value</b>  | Enjoyment / interest in the activity itself         | Story-module engagement; challenge–skill (XXXI)                                          | Fun without transfer = Duo-streak trap (XXXV)                                     |
| <b>Attainment value</b> | Importance to <i>who I am</i> / identity            | Identity re-storying (XXV); integral role in worlds (XXXVI)                              | Scoreboard status ≠ “math person”                                                 |
| <b>Utility value</b>    | Usefulness for current/future goals                 | ACT roadmap honesty; parent-facing “why this concept”; student-written relevance prompts | Pure test-prep cynicism; or fake career lore                                      |

|             |                                                    |                                                                                 |                                                             |
|-------------|----------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Cost</b> | Effort, opportunity, emotional/psychological price | Anxiety load (XXIV); session length; shame of failure; time vs other activities | High cost kills choice even when expectancy/value look “ok” |
|-------------|----------------------------------------------------|---------------------------------------------------------------------------------|-------------------------------------------------------------|

**FACT (classic split):** Meece, Wigfield & Eccles (1990, *Journal of Educational Psychology*, doi:10.1037/0022-0663.82.1.60) — in young adolescents (N=250), **performance expectancies** predicted subsequent math grades; **value perceptions** predicted **course enrollment intentions**; math anxiety was predicted by ability/expectancy/value perceptions but did not independently drive grades/enrollment in their SEM.

**Product translation:** Instrument **both** outcome classes. retry\_120s / transfer ≈ expectancy path. Intent-to-continue / challenge\_accept / “take next ACT practice block” ≈ value path. Do not collapse both into a streak KPI (standing ban).

### XXXVII.3 Subjective task value — four components, four commercial mistakes

#### A. Intrinsic value ≠ engagement theater

**FACT:** Intrinsic value = enjoyment of the task (Eccles-Parsons et al., 1983; Wigfield & Eccles, 2000, *CEP*, doi:10.1006/ceps.1999.1015).

**HYPOTHESIS:** Story can raise intrinsic value *if* it encodes structure and preserves integral roles (XXVIII / XXXVI). Seductive details that raise “interest” but crush learning (Harp & Mayer; Sundararajan & Adesope — carry) are **fake intrinsic** for MindCraft’s purpose.

**Kill:** Marketing that equates “delightful world” with durable math choice without FEI + value instruments.

#### B. Attainment value is the identity wedge

**FACT (theory refinement):** Attainment value moved from “importance of doing well” toward linkage with personal/social identities (Eccles, 2005/2009 line; summarized in Eccles & Wigfield, 2024, *Educational Psychology Review*, doi:10.1007/s10648-024-09888-9).

**FOUNDER BELIEF:** “I am becoming someone who can think mathematically” is attainment value. Streaks are not.

**Commercial implication:** Parent and teen copy should sound like *identity competence*, not *gamified attendance*. Allowed: “Evidence you can trust yourself on the next hard problem.” Banned: “Don’t break your streak — that means you’re a math person.”

#### C. Utility value — the only mass-market EVT intervention with strong field RCTs

**FACT (RCT):** Hulleman & Harackiewicz (2009, *Science*, doi:10.1126/science.1177067) — ninth-grade science students randomly assigned to write how material connects to their lives (or a known other) vs summary control; **relevance writing raised interest and grades especially for students with low success expectations**.

**FACT (lab + classroom):** Hulleman, Godes, Hendricks & Harackiewicz (2010, *JEP*, doi:10.1037/a0019506) — utility writing raised perceived utility and interest, again strongest for lower performers/expectancies; utility perceptions mediated interest effects.

**HYPOTHESIS for MindCraft:** A 2–4 minute **student-authored** utility prompt *after* a micro-win (not before first attempt under high threat) beats brand-authored “math is useful for STEM careers” copy — especially for Maya with low expectancy.

**Red Team wound:** Effects are domain/context sensitive; science classroom  $\neq$  ACT app session; dosage and timing matter (SEVT situating). Do not paste “Hulleman proved story worlds work.”

**Kill:** Shipping utility as *staff-written* career propaganda dumped pre-attempt (CLT + seductive + autonomy threat).

## D. Cost — historically neglected, commercially decisive

**FACT:** Cost was theorized early (effort, opportunity, psychological cost of failure) but empirically under-measured for decades (Wigfield & Cambria reviews; Eccles & Wigfield, 2020 SEVT paper stresses cost elaboration).

**FACT (measurement):** Flake, Barron, Hulleman, McCoach & Welsh (2015, *CEP*, doi:10.1016/j.cedpsych.2015.03.002) — validated multi-dimensional cost (task effort, outside effort, loss of valued alternatives, emotional cost); cost is separable from expectancy/value and relates to student outcomes.

**HYPOTHESIS:** For anxious Maya, **emotional cost** and **effort cost** dominate early sessions; for busy juniors, **opportunity cost** (vs sleep/jobs/sports) dominates weekly return. Product that only raises “fun” while leaving shame-of-wrongness intact fails SEVT choice.

**Product translation:** Soft-wrong / hide-correctness diagnostic (C4) / tutor tone (Part XXX) are **cost interventions**, not just pedagogy niceties. Session design that demands 40 minutes of lore before first attempt raises opportunity + effort cost without expectancy gain (tie to EQ-3).

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## XXXVII.4 Hierarchies and situations (why one “motivation score” is a product bug)

**FACT:** SEVT emphasizes *hierarchies* of expectancies and values across competing activities and situations (Eccles & Wigfield, 2020; Wigfield & Eccles developmental reviews). Choosing math means math beats alternatives *this week*, not that motivation is globally high.

**HYPOTHESIS:** Duo wins short-horizon intrinsic/utility-for-streak; Khan wins low-cost access library; ChatGPT Base raises short-term utility (“get the answer”) while crushing expectancy-for-unaided performance (Part XXXIII/XXXV). MindCraft’s wedge is **attainment + expectancy evidence with managed cost** — not out-streaking Duo or out-librarying Khan.

**SPECULATION (commercial):** Parents purchase when they believe we move the *child’s hierarchy* toward math without catastrophic emotional cost — “they’ll open it again without a fight” is a cost/expectancy story sold as calm, not as dopamine.

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## XXXVII.5 Socializers — parents and tutors as EVT machinery

**FACT (model structure):** Socializers’ beliefs and behaviors shape children’s interpretations, affective memories, and thus expectancies/values (Eccles-Parsons et al., 1983; SEVT socialization arm).

**Carry:** Part XXXII (parent anxiety transmission) — parent math anxiety can raise the child’s **emotional cost** and lower expectancy via homework help patterns (Maloney/Beilock line).

**HYPOTHESIS:** Parent dashboard copy that celebrates *process evidence* (retry, transfer) raises parent-transmitted attainment/utility messages; copy that only shows streak/percentile can raise performance pressure → cost.

**Product implication:** Parent WTP experiments (Part XXVII queue) should manipulate EVT framing: “identity/competence evidence” vs “minutes practiced” vs “AI explanations unlimited.” Pre-register which frame shifts purchase intent without raising reported homework conflict.

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## XXXVII.6 Mapping MindCraft features → EVT levers (audit table)

| Feature / bet                      | Primary lever                 | Secondary                             | Claim ladder max without new data |
|------------------------------------|-------------------------------|---------------------------------------|-----------------------------------|
| Gap-scan + level gating            | Expectancy calibration        | Cost ↓ (fewer impossible first items) | L1 product intent                 |
| Soft-wrong → retry_120s            | Expectancy evidence           | Emotional cost ↓                      | L1–L2 with FEI instrumentation    |
| Story worlds (structure-encoding)  | Intrinsic + attainment        | Utility if student links              | L0–L1; EQ/HIST required           |
| Student utility writing (post-win) | Utility                       | Intrinsic (situational interest)      | L1; EVT-1 experiment              |
| ACT exam-mode roadmap              | Utility (instrumental)        | Expectancy if honest sequencing       | L1 honesty; not “score guarantee” |
| Tutor human sessions               | Expectancy + cost             | Attainment (witnessed identity)       | L1; do not claim 2-sigma          |
| Solver without guards              | Short-term utility (answer)   | Expectancy ↓ (crutch)                 | Kill as hero feature              |
| Streaks / leaderboards             | Pseudo-attainment / extrinsic | Cost ↑ under threat                   | Standing ban as North Star        |

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## XXXVII.7 Commercial implications (copy, positioning, growth)

### Marketing language that survives

- **Allowed:** “Expectancy without value is empty drills. Value without expectancy is inspirational avoidance. We design for both — and we treat the emotional price of being wrong as a first-class design constraint.”
- **Allowed:** “Utility is something the student *writes*, not a slogan we paste.”
- **Allowed (parent):** “Fewer fights at the kitchen table” = cost reduction claim — keep qualitative until measured.
- **Banned:** “Our world makes math useful” (untested utility).

- **Banned:** “Fun = they’ll choose Precalculus.”
- **Banned:** Absolute tutoring-is-free / 2-sigma / streaks-as-identity.

## Positioning vs competitors

| Competitor   | EVT profile (audit, not insult)                                          | MindCraft contrast                 |
|--------------|--------------------------------------------------------------------------|------------------------------------|
| Khan         | Low cost access; mixed expectancy; thin attainment                       | Session FEI + identity evidence    |
| Duo          | Intrinsic/habit; cost managed via brevity; attainment via streak fiction | Transfer + real expectancy         |
| Brilliant    | Intrinsic puzzle; utility unclear for ACT Maya                           | Exam-honest utility + anxiety cost |
| ChatGPT Base | Max short-term utility; expectancy poison                                | Guarded Solver + unaided transfer  |

## Growth / North Star hygiene

Optimize a **pair**: expectancy evidence (retry\_120s, transfer\_pass) **and** a value/choice proxy (challenge\_accept, weekly voluntary return, intent-to-take-next-block). If value proxy rises while transfer falls, you are Duo-pilling the product — kill the feature.

## XXXVII.8 Experiments spawned

| ID    | Question                                                          | Design                                     | Primary                                        | Kill condition                                                                        |
|-------|-------------------------------------------------------------------|--------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------------|
| EVT-1 | Post-win student utility writing vs staff utility blurb vs none   | 3-arm A/B after micro-win                  | situational interest + 7d return + transfer@7d | Utility arms lose transfer or raise time-to-next-attempt badly                        |
| EVT-2 | ACT-utility framing vs identity-attainment framing in session CTA | 2-arm                                      | challenge_accept ; parent/teen preference      | Attainment wins preference but tanks attempt rate in anxious segment (segmented kill) |
| EVT-3 | Cost questionnaire (Flake-style short form) pre/post first week   | Observational → then intervene on top cost | which cost dimension predicts churn            | If emotional cost predicts churn but product only ships “more fun,” roadmap invalid   |

|          |                                                               |                         |                                             |                                                                                  |
|----------|---------------------------------------------------------------|-------------------------|---------------------------------------------|----------------------------------------------------------------------------------|
| EVT-4    | Parent copy: competence-evidence vs minutes vs AI-unlimited   | WTP / conflict items    | purchase intent; reported homework conflict | Competence frame raises conflict without WTP lift                                |
| EVT-QUAL | 10 Maya interviews: “why would you take another math course?” | Appendix B + EVT probes | coded expectancy /value/cost                | Students cannot articulate any non-grade reason → attainment/utility product gap |

**Pre-reg (XXXIV):** EVT-\* identify levers on **in-app choice and short-horizon learning**. They do **not** identify long-term course enrollment or career STEM entry without follow-up designs. Do not sell “we increase AP Calculus enrollment” from EVT-1.

## XXXVII.9 Confidence table

| Claim                                                                      | Label                                   | Confidence                      |
|----------------------------------------------------------------------------|-----------------------------------------|---------------------------------|
| Choice $\approx$ f(expectancy, STV) in Eccles tradition                    | FACT (large corpus)                     | High                            |
| Expectancy→performance, value→enrollment intentions (adolescent math)      | FACT (Meece et al., 1990 SEM)           | Medium–High (sample/era limits) |
| STV has intrinsic / attainment / utility / cost structure                  | FACT (theory + measurement work)        | High                            |
| Utility-value writing can lift interest/grades for low-expectancy students | FACT (Hulleman & Harackiewicz, 2009)    | Medium–High (context limits)    |
| Cost is multi-dimensional and outcome-relevant                             | FACT (Flake et al., 2015)               | Medium–High                     |
| Story worlds raise utility by themselves                                   | SPECULATION                             | Low                             |
| MindCraft should instrument E and V separately                             | FOUNDER BELIEF / HYPOTHESIS             | Medium–High                     |
| Soft-wrong + tutor tone primarily cut emotional cost                       | HYPOTHESIS                              | Medium                          |
| Streaks raise attainment value                                             | SPECULATION / likely false for identity | Low (against)                   |

## XXXVII.10 What this chapter kills

1. **Kill:** Single “motivation” KPI as if E and V were the same dial.
2. **Kill:** Staff-authored “math is useful” dumps as a substitute for student-authored utility.
3. **Kill:** Treating story delight as proof of course-choice impact.
4. **Wound:** Pure ACT-utility positioning — survives for juniors near test date; wounds long-horizon identity brand if it becomes the only message.
5. **Survive (constrained):** Identity-transformation thesis — **if** product explicitly raises expectancy *and* attainment/utility *and* designs against cost, with EVT-1...4 on the roadmap.

**Doctrine until data:** Design and market the **E×V×Cost triad**. Expectancy evidence without value is a drill app. Value talk without expectancy is a poster. Ignoring cost is how anxious students churn while NPS looks fine among the already-confident.

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## Part XXXVIII — Goal Orientation: Mastery vs Performance Framing

**Chapter status:** Living evidence brief — Researcher tick 2026-07-29

**Primary question:** When MindCraft frames goals as *get better at this* vs *look smart / beat others / don’t look dumb*, which motivational pattern do we induce — and what must product + copy refuse?

**Owners:** Learning Science · Product · Growth / Positioning · Red Team

**Commercial job:** Replace empty “mastery learning” branding with a **goal-structure audit**: which surfaces cue mastery-approach, which cue appearance/performance, which cue avoidance — and which we kill even if they juice short-term engagement.

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### XXXVIII.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution / helplessness), XXXI (challenge–skill), and XXXVII (expectancy-value) answer *can I*, *why did I fail*, *is difficulty calibrated*, and *will I choose math*. They do not fully answer: **what success definition is the product teaching?**

A product that says “identity transformation” while UI, CTAs, and parent dashboards scream *percentile / streak / don’t mess up* is running a **performance-avoidance climate** with mastery marketing on top.

**FOUNDER BELIEF under audit:** MindCraft’s North Star metrics (`retry_120s`, `challenge_accept`, `transfer_pass`) only stay coherent if the **goal structure** of sessions is mastery-approach dominant. If recognition and evaluation structures are performance/appearance, those metrics will fight the UI.

**Claim we refuse as doctrine:** “Any competitive element raises motivation.” That collapses normative, appearance, and avoidance goals — meta-analyses show they are not the same animal.

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## XXXVIII.2 The constructs (product language, not poster language)

| Goal                                     | Plain definition                                       | Typical student cue                            | MindCraft failure mode if dominant                                       |
|------------------------------------------|--------------------------------------------------------|------------------------------------------------|--------------------------------------------------------------------------|
| <b>Mastery-approach</b>                  | Improve competence vs self/task standard               | “Understand this; try a harder variant”        | None if paired with transfer; risk = slow coverage if time unmanaged     |
| <b>Mastery-avoidance</b>                 | Avoid failing to learn / avoid imperfect understanding | “Don’t leave gaps; don’t misunderstand”        | Perfectionism, over-checking, refusal to attempt under uncertainty       |
| <b>Performance-approach (normative)</b>  | Outperform peers                                       | Rank, leaderboard, “top of class”              | Can raise grades for some; often raises threat for Maya; identity ≠ rank |
| <b>Performance-approach (appearance)</b> | Demonstrate talent / look smart                        | Public correctness, “prove you’re smart”       | Meta-analytically worse than normative; shame + shallow strategy         |
| <b>Performance-avoidance</b>             | Avoid looking incompetent                              | Hide wrongness; skip hard items; Solver crutch | Helpless pattern; FEI collapse; Part XXX / XXXIII risk                   |

**FACT (classic dichotomy + classroom structures):** Ames (1992, *Journal of Educational Psychology*, doi:10.1037/0022-0663.84.3.261) — classroom **task, evaluation/recognition, and authority** structures make mastery vs performance goals salient and elicit qualitatively different motivational patterns; interventions must treat structures as interdependent, not one “growth mindset poster.”

**FACT (student perceptions → strategies):** Ames & Archer (1988, *JEP*, doi:10.1037/0022-0663.80.3.260) — among 176 secondary students, perceived **mastery** classroom goals related to more effective strategies, preference for challenge, positive attitude, and effort→success beliefs; perceived **performance** salience related to ability focus, negative ability evaluation, and attributing failure to low ability.

**FACT (implicit theories → goals → patterns):** Dweck & Leggett (1988, *Psychological Review*, doi:10.1037/0033-295X.95.2.256) — entity vs incremental theories orient students toward performance vs learning goals, which set up helpless vs mastery-oriented response patterns after difficulty.

**Product translation:** Soft-wrong + retry is not only an expectancy/cost intervention (XXXVII) — it is a **goal-structure** intervention. Public scoreboards and “don’t break the streak” are evaluation/recognition structures that cue performance/appearance.

## XXXVIII.3 Approach–avoidance split (why “performance goals” is too coarse)

**FACT (trichotomous model):** Elliot & Church (1997, *JPSP*, doi:10.1037/0022-3514.72.1.218) — mastery, performance-approach, and performance-avoidance are separable; mastery facilitated intrinsic motivation; performance-approach enhanced graded performance; **performance-avoidance harmed both** intrinsic motivation and grades (college classroom path model).

**FACT (2×2 framework):** Elliot & McGregor (2001, *JPSP*, doi:10.1037/0022-3514.80.3.501) — adds **mastery-avoidance**; four goals show distinct antecedents/outcomes; mastery-avoidance sits between mastery-approach (more adaptive) and performance-avoidance (more maladaptive) on several indicators.

**HYPOTHESIS for MindCraft:** Anxious Maya often defaults to **performance-avoidance** (“don’t look dumb”) or **mastery-avoidance** (“I must understand perfectly before I try”). Product that only bans leaderboards but still frames every miss as identity threat will not fix the pattern.

**Kill:** Marketing that equates “mastery mode” with “zero mistakes allowed on the path.” That is mastery-avoidance cosplay.

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## XXXVIII.4 Performance-approach is not one knob — measurement kills bad slogans

**FACT (meta-analysis on measures):** Hulleman, Schragger, Bodmann & Harackiewicz (2010, *Psychological Bulletin*, doi:10.1037/a0018947) — across 243 correlational studies (N≈91k), “performance-approach” scales that emphasize **normative outperformance** correlate *positively* with performance outcomes ( $r \approx .14$ ), while scales emphasizing **appearance / looking talented** correlate *negatively* ( $r \approx -.14$ ). Same label, different constructs.

**FACT (extended outcomes meta):** Senko & Dawson (2017, *Journal of Educational Psychology*, doi:10.1037/edu0000160) — the normative vs appearance split generalizes beyond grades to competence perceptions and self-regulation; age confounds with measure type were addressed.

**FACT (theory status review):** Senko, Hulleman & Harackiewicz (2011, *Educational Psychologist*, doi:10.1080/00461520.2011.538646) — achievement goal theory at a crossroads: multiple controversies (including when performance-approach helps) require precise operationalization before practice claims.

**Commercial implication:** If MindCraft ever ships social comparison, it must pre-register whether the cue is **normative skill evidence** (dangerous for threatened learners; still not identity) or **appearance theater** (ban). Default = **neither**. Do not cite “performance goals sometimes help grades” as license for public shame UX.

**SPECULATION (positioning):** “Beat your friends” copy will read as Duo/Brilliant hybrid; “prove you’re smart” reads as SAT-cram Instagram. Neither is the identity-transformation wedge.

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## XXXVIII.5 Classroom / product goal structures (TARGET), not vibes

**FACT (schoolwide / TARGET lineage):** Maehr & Midgley (1991, *Educational Psychologist*, doi:10.1080/00461520.1991.9653140) — motivation change requires altering school/classroom goal structures, not only exhorting students; dimensions map to task design, authority, recognition, grouping, evaluation, time (TARGET family; Ames, 1992).

**FACT (review):** Meece, Anderman & Anderman (2006, *Annual Review of Psychology*, doi:10.1146/annurev.psych.56.091103.070258) — classroom goal structures predict personal goals and achievement-related patterns; mastery structures are the design target for adaptive motivation.

**FACT (longitudinal TARGET → mastery goals):** Lüftenegger et al. (2014, *Educational Psychology*, doi:10.1080/01443410.2013.814189) — TARGET as a latent structure positively predicted students’ mastery goal orientations (N=1680 secondary; longitudinal).

**MindCraft TARGET audit (hypothesis map):**

| TARGET dim         | Mastery-leaning product move                                           | Performance/appearance trap                                |
|--------------------|------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Task</b>        | Varied challenge matched to skill (XXXI); format variety with transfer | Same-item grind that feels like “prove speed”              |
| <b>Authority</b>   | Student choice of next concept <i>after</i> honest diagnosis           | Forced path that only optimizes for looking ahead of peers |
| <b>Recognition</b> | Process evidence: retry, transfer, bridge repair                       | Streak flames, public %ile, “genius” badges                |
| <b>Grouping</b>    | Tutor as witness; optional co-op without rank                          | Ranked classrooms / public wrongness                       |
| <b>Evaluation</b>  | Soft-wrong diagnostic (C4); hide correctness when graph-corrupting     | Instant public right/wrong as identity verdict             |
| <b>Time</b>        | Short sessions that still allow struggle → retry                       | Timed pressure that cues avoidance                         |

**FOUNDER BELIEF:** PawHub “Practice the weak spot” is mastery-approach **if** copy is “repair this gap” and **not** “you’re behind everyone.” Gap language can flip to performance-avoidance under shame framing.

## XXXVIII.6 Interaction with expectancy, anxiety, and AI crutches

**FACT (goal pursuit × expectancy):** Senko & Hulleman (2013, *JEP*, doi:10.1037/a0031136) — for mastery-approach *and* performance-approach, the harder a goal appears to attain, the less students pursue it (with costs to interest/achievement); mastery-approach goals were generally judged easier to attain than performance-approach; situational cues (material complexity) shift mastery-approach expectancies.

**Carry:** Part XXIV (anxiety) + XXXIII (AI overtrust) — under high threat, performance-avoidance + Solver-without-guards is the cheap escape from a hard-looking mastery or performance goal.

**HYPOTHESIS:** Level gating after gap-scan (easy→L3, kinda→L2, hard→L1) is a **goal-attainment-expectancy** design: it makes mastery-approach goals feel attainable. Removing gates to “challenge everyone” without expectancy supports will increase avoidance and Solver dependence.

**Product implication:** CTAs should make the next mastery goal *feel attainable* (Senko & Hulleman) without lying about transfer. Honest sequencing ≠ performance theater.

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## XXXVIII.7 Competitive product audit (goal climate lens)

| Competitor          | Dominant goal climate (audit)                                   | What MindCraft must not copy                                   |
|---------------------|-----------------------------------------------------------------|----------------------------------------------------------------|
| <b>Duolingo</b>     | Habit + streak recognition;<br>appearance of consistency        | Streak-as-identity /<br>streak-as-North-Star (standing<br>ban) |
| <b>Khan</b>         | Mastery language + energy<br>points / badges mixed              | “Mastery” label without<br>transfer_pass                       |
| <b>Brilliant</b>    | Puzzle competence; mild<br>normative “smart people do this”     | Appearance-smart branding<br>without anxiety cost design       |
| <b>ChatGPT Base</b> | Outcome utility (“get it right<br>now”) → avoidance of struggle | Solver as hero; unaided transfer<br>optional                   |

**HYPOTHESIS:** MindCraft’s wedge is **mastery-approach structures with expectancy-managed tasks and anti-appearance evaluation** — not out-ranking Duo and not out-explaining Khan.

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## XXXVIII.8 Commercial implications (copy, product, growth)

### Marketing language that survives

- **Allowed:** “We design for improvement goals — understand, repair, transfer — not for looking smart in public.”
- **Allowed:** “Wrong answers are evidence for the next attempt, not a verdict on who you are.” (ties XXX / soft-wrong)
- **Allowed (parent):** “We measure whether they take the next hard problem, not whether they kept a streak.”
- **Banned:** “Become a top 1% math student” as hero claim (appearance/normative mashup; undefended).
- **Banned:** “Never break your streak — that’s how mastery works.”
- **Banned:** Empty growth-mindset posters without TARGET structure changes (Dweck mis-applied as sticker).
- **Banned:** 2-sigma / tutoring-is-free absolutes.

### Feature claim ladder (pre-reg hygiene, Part XXXIV)

| Feature                     | Goal climate claim max without GO-* data                        |
|-----------------------------|-----------------------------------------------------------------|
| Soft-wrong → retry_120s     | L1 intent: mastery-approach evaluation structure                |
| challenge_accept CTA        | L1: approach (need to verify mastery vs<br>performance wording) |
| transfer_pass               | L1–L2 anti-fake-mastery (competence standard ≠<br>appearance)   |
| Leaderboards / public ranks | L0 default kill for Maya segment                                |
| Streak North Star           | Kill (standing ban)                                             |

|                           |                                                           |
|---------------------------|-----------------------------------------------------------|
| “Mastery graph” marketing | L1 only if transfer instrumented; else label is cosmetics |
|---------------------------|-----------------------------------------------------------|

## Growth / North Star hygiene

challenge\_accept is **not automatically mastery-approach**. Instrument **why** they accept: repair gap (mastery) vs beat peer (normative) vs avoid looking behind (avoidance). If acceptance rises because of shame CTAs, kill the copy even if the metric looks green.

## XXXVIII.9 Experiments spawned

| ID      | Question                                                                                                        | Design                                    | Primary                                                | Kill condition                                                                             |
|---------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------|
| GO-1    | Mastery-approach CTA (“repair this gap”) vs normative (“catch up to peers”) vs appearance (“show you’re smart”) | 3-arm on Practice launch                  | challenge_accept ; retry_120s; 7d return; anxiety item | Normative/appearance win short accept but lose retry/transfer or raise anxiety             |
| GO-2    | Recognition: process badge (retry/transfer) vs streak badge vs none                                             | 3-arm                                     | transfer@7d; identity item; self-reported goal         | Streak badge wins engagement, loses transfer / raises avoidance                            |
| GO-3    | Evaluation: soft-wrong first vs instant public correctness                                                      | 2-arm on early session                    | next-attempt latency; avoidance skips; Solver opens    | Instant correctness raises “finish” rate but kills retry_120s                              |
| GO-4    | Goal-climate pulse (short PALS-like mastery/performance classroom items adapted to app) weekly                  | Observational → then redesign recognition | which climate predicts churn / Solver dependence       | If product self-reports “mastery brand” but climate score is performance → roadmap invalid |
| GO-QUAL | 10 Maya interviews: “What would ‘doing well’ here mean?”                                                        | Appendix B + goal probes                  | coded mastery / normative / appearance / avoidance     | Majority answer only “not look dumb” → evaluation/recognition redesign mandatory           |

**Pre-reg (XXXIV):** GO-\* identify **in-app goal climate** → **short-horizon FEI behaviors**. They do **not** identify long-term GPA or STEM major from CTA wording alone. Do not sell “we create mastery-oriented students for life” from GO-1.

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### XXXVIII.10 Confidence table

| Claim                                                                | Label                                              | Confidence                 |
|----------------------------------------------------------------------|----------------------------------------------------|----------------------------|
| Mastery vs performance goal structures shape strategies/attitudes    | FACT (Ames & Archer, 1988; Ames, 1992)             | High                       |
| Implicit theories orient goals → helpless vs mastery patterns        | FACT (Dweck & Leggett, 1988)                       | High                       |
| Performance-avoidance is broadly maladaptive for motivation/grades   | FACT (Elliot & Church, 1997 lineage)               | High                       |
| 2×2 adds mastery-avoidance as distinct                               | FACT (Elliot & McGregor, 2001)                     | High                       |
| “Performance-approach” effects depend on normative vs appearance ops | FACT (Hulleman et al., 2010; Senko & Dawson, 2017) | High                       |
| TARGET-like structures can raise mastery orientations                | FACT (Maehr & Midgley; Lüftenegger et al., 2014)   | Medium–High (school ≠ app) |
| Soft-wrong + process recognition = mastery evaluation structure      | HYPOTHESIS                                         | Medium–High                |
| Any leaderboard helps MindCraft’s Maya segment                       | SPECULATION / likely false                         | Low (against)              |
| Streaks instantiate mastery goals                                    | SPECULATION / false for identity                   | Low (against)              |
| MindCraft should ban appearance-goal UX by default                   | FOUNDER BELIEF / HYPOTHESIS                        | Medium–High                |

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### XXXVIII.11 What this chapter kills

1. **Kill:** Treating “mastery” as a brand adjective without evaluation/recognition structures that cue improvement standards.
2. **Kill:** Streak / public rank / “look smart” as North Star or hero copy (appearance + standing streak ban).
3. **Kill:** Collapsing all performance goals into “competition is motivating.”
4. **Wound:** Normative comparison as a growth lever — survives only as a pre-registered, segmented experiment (GO-1), never as default Maya UX.

5. **Survive (constrained):** Identity-transformation thesis — **if** product TARGET structures are mastery-approach dominant, avoidance/appearance cues are designed out, and GO-1...4/QUAL sit on the roadmap.

**Doctrine until data:** Market and build **mastery-approach goal structures** (task challenge + process recognition + non-shaming evaluation + attainable next goals). Performance-avoidance and appearance goals are product bugs dressed as engagement. Normative comparison is not the default growth loop.

## Part XXXIX — Interleaving vs Blocking: When to Mix Problem Types

**Chapter status:** Living evidence brief — Researcher tick 2026-07-29

**Primary question:** When should MindCraft mix problem types (interleave) vs group same-type drills (block) — and what must product + copy refuse to claim?

**Owners:** Learning Science · Product (Practice sequencing) · Growth / Positioning · Red Team

**Commercial job:** Turn “we shuffle questions” from a cute engineering detail into a **defensible practice-design claim:** strategy *selection*, not just strategy *execution* — without selling interleaving as always-on magic or as engagement theater.

### XXXIX.1 Why this chapter exists

Part XXIX (spacing / retrieval) already flags interleaving as a desirable difficulty and hypothesizes **Block** → **fade** → **interleave** → **spaced retrieval**. Part XXVI (CLT) warns that early overload kills learning. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance if framed as shame.

This chapter densifies the **mix vs block** decision for Practice sessions, ingredient cards, and ACT-style mixed review — because textbooks and most edtech default to **blocked practice** (a dozen slope problems after the slope lesson), and students experience that fluency as “I get it” while exams demand **choose the strategy from the problem itself**.

**FOUNDER BELIEF under audit:** Identity transformation requires competence evidence that survives *transfer*. Blocked fluency is often **illusory competence** — high session accuracy, weak delayed discrimination.

**Claim we refuse as doctrine:** “Always interleave everything from the first item.” That ignores schema formation, CLT, and similarity moderators — and can spike threat for anxious Maya before `retry_120s` is healthy (XXIX.4).

### XXXIX.2 Constructs (product language)

|          |                  |                            |                         |
|----------|------------------|----------------------------|-------------------------|
| Schedule | Plain definition | What the student practices | Failure mode if misused |
|----------|------------------|----------------------------|-------------------------|

|                       |                                                                         |                                  |                                                        |
|-----------------------|-------------------------------------------------------------------------|----------------------------------|--------------------------------------------------------|
| <b>Blocked</b>        | Consecutive items share the same strategy / concept                     | <i>Execute</i> a known procedure | Fluency illusion; weak strategy choice on exams        |
| <b>Interleaved</b>    | Consecutive items require different strategies (no safe “same as last”) | <i>Select</i> then execute       | Early overload; frustration; avoidance if unscaffolded |
| <b>Hybrid / faded</b> | Short block for schema → rising mix                                     | Execute → select                 | Best candidate for MindCraft default                   |

**Operational definition for MindCraft (HYPOTHESIS):** Interleaving = consecutive Practice items do **not** share the same primary strategy class (concept + dominant ingredient / archetype family). Mere format shuffle (word → diagram) with the same strategy is **not** full interleaving — it is representation variety (useful; different claim).

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### XXXIX.3 Mechanism: why mixture changes the pedagogical demand

**FACT (math laboratory / classroom line):** Rohrer & Taylor (2007, *Instructional Science*, doi:10.1007/s11251-007-9015-8) — Experiment 2: after learning multiple problem types, **mixed** vs **blocked** practice; one-week delayed test vastly favored mixed practice. Experiment 1 separately showed spacing > massing for a single type. Shuffled textbooks alter **both** spacing and mixing; they are related but not identical levers.

**FACT (strategy choice vs execution):** Rohrer, Dedrick & Stershic (2015, *Journal of Educational Psychology*, doi:10.1037/edu0000001) — 126 seventh-graders, same problems over ~3 months, interleaved vs blocked arrangement; after review, unannounced tests favored interleaving at 1 day ( $d = 0.42$ ) and 30 days ( $d = 0.79$ ). Core mechanism claim: blocked practice lets students know the strategy *before reading the problem*; interleaved practice forces strategy choice from problem features — the skill exams actually require.

**FACT (not only look-alike problems):** Rohrer, Dedrick & Burgess (2014, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-014-0588-3) — interleaved benefit is **not** limited to superficially similar problem kinds; reordering changes pedagogical demand (choose vs execute) even when types do not look alike.

**FACT (classroom-scale RCT):** Rohrer, Dedrick, Hartwig & Cheung (2020, *JEP*, doi:10.1037/edu0000367; Online First 2019) — preregistered cluster RCT, 54 seventh-grade classes, mostly interleaved vs mostly blocked assignments over months; after interleaved review for both, one-month unannounced test: 61% vs 38%,  $d = 0.83$ . Teachers implemented without training; anonymous pre-results survey supported feasibility. Caveats remain (materials, population), but this is the strongest classroom efficacy signal in the math interleaving line.

**Product translation:** MindCraft’s transfer\_pass and mixed ACT practice are **closer to the interleaved demand** than a single-concept blocked grind. A session that is 12 items of linear\_equations after “Learn linear equations” is textbook-default blocking — fine as *early schema practice*, dangerous as the *only* schedule if we market exam readiness.

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### XXXIX.4 Meta-analytic moderators (do not overclaim “interleaving always wins”)



**FACT (inductive learning meta-analysis):** Brunmair & Richter (2019, *Psychological Bulletin*, doi:10.1037/bul0000209) — 59 studies / 238 effect sizes / 158 samples; overall interleaving vs blocking Hedges'  $g = 0.42$ . **Material matters:** paintings/visual high ( $g \approx 0.67$ ); **mathematical tasks small but positive ( $g \approx 0.34$ )**; words favor **blocking** ( $g \approx -0.39$ ); expository texts ambiguous. Moderators: stronger interleaving when **between-category similarity is high, within-category similarity is low**, and material is more complex — consistent with attentional bias / sequential attention accounts (Carvalho & Goldstone lineage).

**FACT (attentional bias framework):** Carvalho & Goldstone (2015, *Frontiers in Psychology*, doi:10.3389/fpsyg.2015.00505) — study sequence biases what is encoded: interleaving emphasizes **between-category differences**; blocking emphasizes **within-category similarities**. Prediction: when categories are hard to tell apart, interleave; when the job is to notice what members of one category share, block can help.

**HYPOTHESIS for MindCraft:** Interleave **confusable** strategy families (e.g., independent vs dependent probability; similar ACT archetypes that share surface cues but need different ingredients). Block when introducing a *new* ingredient representation until the student can execute with worked-example fade (XXVI) — then mix against near-miss foils.

**Kill:** Marketing that says “our AI interleaves everything like elite tutors” without specifying **confusable sets** and **when blocking is intentional**.

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## XXXIX.5 Fluency illusion, affect, and why students prefer blocking

**FACT / TRADITION (desirable difficulties):** Conditions that slow acquisition (spacing, interleaving, generation) often improve later retention/transfer (Bjork lineage; see XXIX.4). Interleaving feels harder *during* practice; blocked practice feels fluent.

**HYPOTHESIS (Koriat/Bjork-style monitoring, applied):** Students and parents misread blocked fluency as mastery. Parent dashboards that celebrate “12/12 on slope today” without a mixed delayed check sell **performance theater** (XXXVIII) dressed as learning.

**Carry from XXIV / XXXVII / XXXVIII:** Raising interleaving dose raises short-term error rate → for anxious Maya, that can cue performance-avoidance unless (a) soft-wrong / mastery evaluation, (b) expectancy-managed difficulty, (c) explicit frame: “this mix trains choosing, not proving you’re dumb.”

**Product implication:** Copy for interleaved sessions: “**Practice choosing the right move**” — not “hard mode for elites” and not streak pressure.

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## XXXIX.6 Sequencing doctrine (ties XXVI + XXIX)

**FOUNDER BELIEF / HYPOTHESIS (sequenced schedule):**

1. **Acquire (block-lite):** short same-strategy set + worked examples / cards until execution is stable enough that errors are mostly slips, not total confusion.
2. **Discriminate (interleave):** mix with **near-miss** types that share surface features (bridge gaps, format gaps, ACT archetype neighbors).

3. **Retain (space):** redistribute the same mix across days (XXIX) — interleaving *guarantees* some spacing of each type, but lag still needs explicit scheduling.
4. **Transfer check:** unannounced mixed set → transfer\_pass; failure routes to diagnosis (forgot vs never-had; wrong strategy class vs execution bug).

**Contradiction to hold:** Some math/category studies report null or blocking advantages under specific similarity structures (noted in Brunmair & Richter’s discussion of mixed math findings; e.g., certain representation-interleaving designs). MindCraft must treat **interleaving axis** (strategy class) as primary for exam readiness, and treat **representation mix** as a separate experiment track (format axis / C2–C3).

## XXXIX.7 Competitive / product audit (schedule lens)

| Surface                         | Typical schedule                                    | What MindCraft should do                                                                  |
|---------------------------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Textbook / workbook</b>      | Heavy blocking after lesson                         | Do not clone as “mission complete”                                                        |
| <b>Khan-like mastery</b>        | Often skill-blocked until “mastered”                | Mastery label without mixed delayed check = fluency risk                                  |
| <b>Duo-like</b>                 | Habit + short items; mix varies by skill tree       | Habit ≠ discrimination training                                                           |
| <b>Brilliant</b>                | Puzzle variety; not always systematic near-miss mix | Variety ≠ controlled interleaving                                                         |
| <b>ChatGPT Solver</b>           | Single-problem depth                                | Zero schedule pedagogy; cannot substitute mix training                                    |
| <b>MindCraft Practice today</b> | Often concept-scoped sessions + bank shuffle        | Shuffle within one concept ≠ full strategy interleaving; need <b>cross-type mix modes</b> |

**HYPOTHESIS:** PawHub “Practice the weak spot” correctly targets a gap, but a pure blocked weak-spot grind can raise local accuracy while leaving **strategy selection** untrained. Pair weak-spot blocks with a **discrimination coda:** 3–5 interleaved near-misses including the weak type.

## XXXIX.8 Commercial implications (copy, product, growth)

### Marketing language that survives

- **Allowed:** “Exams don’t announce the chapter. We train choosing the strategy — not only repeating the last one.”
- **Allowed:** “Blocked practice feels easy; mixed practice builds the skill tests actually measure.”
- **Allowed (parent):** “We check whether they can pick the right approach on a mixed set days later — not just a perfect streak on one topic.”
- **Banned:** “Interleaving always doubles scores” / absolute effect-size marketing (Rohrer classroom  $d \approx 0.83$  is **one** RCT context; Brunmair math  $g \approx 0.34$  is the broader inductive-math signal — do not collapse).

- **Banned:** “Always mix from day one.”
- **Banned:** Selling shuffle/random as science without near-miss design.
- **Banned:** 2-sigma / tutoring-is-free / streak-as-mastery.

## Feature claim ladder (Part XXXIV hygiene)

| Feature                                               | Claim max without IL-* data                         |
|-------------------------------------------------------|-----------------------------------------------------|
| Within-concept item shuffle                           | L0–L1 “variety”; <b>not</b> full interleaving claim |
| Cross-strategy interleaved coda after weak-spot block | L1 intent: discrimination training                  |
| Delayed mixed transfer_pass                           | L1–L2 anti-fluency-illusion                         |
| “Interleaving improves ACT” hero claim                | L0 until IL-1/IL-2 pre-registered                   |

## Growth / North Star hygiene

Session accuracy under **blocked** schedules will look better than under interleaved ones. **Do not** optimize North Star on blocked accuracy. Prefer: mixed delayed accuracy, strategy-selection error rate (wrong procedure class), transfer\_pass, and retry\_120s after mix-induced misses.

## XXXIX.9 Experiments spawned

| ID   | Question                                                                       | Design                  | Primary                                                 | Kill condition                                                             |
|------|--------------------------------------------------------------------------------|-------------------------|---------------------------------------------------------|----------------------------------------------------------------------------|
| IL-1 | Blocked-only weak-spot session vs block + interleaved near-miss coda           | 2-arm; same total items | Delayed (48h) mixed accuracy; strategy-class error rate | Coda hurts delayed mix <b>and</b> raises avoidance without transfer gain   |
| IL-2 | Early interleave (from item 1) vs fade-in interleave after short block         | 2-arm on new concept    | Acquisition errors; retry_120s; 7d mixed test           | Early-all-mix wins delayed score but kills return/anxiety for Maya segment |
| IL-3 | Near-miss interleave (confusable archetypes) vs far mix (unrelated easy types) | 2-arm                   | Discrimination errors on near-miss test                 | Far mix equals near-miss on discrimination → “any shuffle” doctrine dies   |

|         |                                                                                |                          |                                                    |                                                                                            |
|---------|--------------------------------------------------------------------------------|--------------------------|----------------------------------------------------|--------------------------------------------------------------------------------------------|
| IL-4    | Parent/student judgment: blocked fluency vs mixed score — which do they trust? | Survey + optional choice | Preference for blocked; willingness-to-pay framing | If parents only buy blocked dashboards, copy/education mandatory before UX ships mix-heavy |
| IL-QUAL | 10 Maya interviews: “When did practice feel like the real test?”               | Appendix B               | coded choose-vs-execute moments                    | Nobody describes strategy choice → product isn’t creating the demand state                 |

**Pre-reg (XXXIV):** IL-\* identify **schedule** → **discrimination** / **delayed mix**. They do **not** identify long-term STEM identity from one coda. Do not sell “interleaving creates math people.”

**Supersedes / densifies:** XXIX INT-1 (interleave after fade vs blocked-only) → prefer IL-1/IL-2 as the operational pair.

## XXXIX.10 Confidence table

| Claim                                                                    | Label                                                           | Confidence                     |
|--------------------------------------------------------------------------|-----------------------------------------------------------------|--------------------------------|
| Mixed/interleaved math practice often beats blocked on delayed tests     | FACT (Rohrer & Taylor, 2007; Rohrer et al., 2015, 2020)         | High (math classroom/lab line) |
| Mechanism includes forced strategy selection (not only spacing)          | FACT / strong interpretation in Rohrer line                     | High                           |
| Benefit can appear for non-look-alike problem kinds                      | FACT (Rohrer et al., 2014)                                      | High                           |
| Overall inductive interleaving effect moderated by material & similarity | FACT (Brunmair & Richter, 2019)                                 | High                           |
| Math subset effect smaller than visual category learning                 | FACT ( $g \approx 0.34$ vs higher for paintings)                | High                           |
| Blocking can help when within-category similarity learning is the job    | FACT (attentional bias framework) / HYPOTHESIS for early schema | Medium–High                    |
| Always-interleave-from-item-1 is optimal for anxious Maya                | SPECULATION / likely false                                      | Low (against)                  |
| Within-concept shuffle $\equiv$ interleaving science claim               | SPECULATION / false                                             | Low (against)                  |

|                                                                                |                             |             |
|--------------------------------------------------------------------------------|-----------------------------|-------------|
| Default MindCraft path = short block → near-miss interleave → spaced mix check | FOUNDER BELIEF / HYPOTHESIS | Medium–High |
|--------------------------------------------------------------------------------|-----------------------------|-------------|

### XXXIX.11 What this chapter kills

- 1. **Kill:** Treating blocked session accuracy as proof of exam readiness or identity change.
- 2. **Kill:** “We randomly shuffle, therefore we interleave” as a marketing claim.
- 3. **Kill:** Absolute always-interleave / always-double-scores claims.
- 4. **Wound:** Pure weak-spot blocked grind as the only Practice mode — survives as *acquisition*, not as the full loop.
- 5. **Survive (constrained):** Desirable-difficulty interleaving for **strategy discrimination** — after schema foothold, with near-miss design, affect-safe framing, and delayed mixed `transfer_pass`.

**Doctrine until data:** Market and build **choose-the-strategy training** via faded interleaving of confusable types. Blocking is a scaffold, not the product. Fluency without mixed delayed evidence is a vanity metric.

## Part XL — Self-Explanation Prompts: Chi/Renkl → Coach UX

**Chapter status:** Living evidence brief — Researcher tick 2026-07-29

**Primary question:** When do prompts that force the student to *generate* an explanation improve math learning — and what must MindCraft’s coach / cards / Solver refuse to do with “explain yourself” UX?

**Owners:** Learning Science · Product (coach, ingredient cards, Solver) · Growth / Positioning · Red Team

**Commercial job:** Turn “we explain everything” and “AI tutor talks at you” into a **defensible generation claim:** students build transferable understanding by explaining *why steps work* (and why common wrong moves fail) — without selling open-ended “justify your guess” as science, and without confusing fluent AI explanation *to* the student with self-explanation *by* the student.

### XL.1 Why this chapter exists

Part XXVI (CLT / worked examples / fading) already puts examples and completion problems in the acquisition path. Part XXIX / XXXIX demand retrieval and strategy *selection*. Part XXXIII warns that fluent AI explanations can create overtrust. Experiment A (wizard coach) and soft-wrong already assume that *after* an error, something productive happens.

This chapter densifies the **prompt layer:** what the product asks the student to *produce* — principle for a step, why a misconception is wrong, anticipation of the next operator — vs what it merely *shows*.

**FOUNDER BELIEF under audit:** Identity as a mathematical thinker requires evidence that the student can *account for* a move, not only select A/B/C/D. Passive delivery without generation is Layer-1 scarcity relief, not identity

transformation.

**Claim we refuse as doctrine:** “Any explain box after a wrong answer is self-explanation science.” Untargeted prompts, explain-your-wrong-guess-without-feedback, and AI monologues are different mechanisms — some help, some hurt, some are theater.

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## XL.2 Constructs (product language)

| Construct                            | Plain definition                                                                 | Product surface                                            | Failure mode                                                   |
|--------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------|----------------------------------------------------------------|
| <b>Self-explanation</b>              | Learner generates inferences beyond the given text/example <i>for themselves</i> | Prompted coach turns; card reflection; step-principle pick | Empty “why?” with no scaffold; AI does the explaining for them |
| <b>Instructional explanation</b>     | Expert/AI/tutor explains <i>to</i> the learner                                   | Solver narration; worked example text                      | Can reduce germane effort if overused (expertise / passivity)  |
| <b>Worked-example study</b>          | Study full solution before / instead of cold solving                             | Ingredient cards; fade path                                | Without SE prompts → shallow copy (Chi “poor” pattern)         |
| <b>Principle / glossary citation</b> | Name the rule that justifies a step                                              | Aleven-style menu; ingredient id pick                      | Menu-click without comprehension if too shallow                |
| <b>Misconception contrast</b>        | Explain why a common wrong path fails                                            | Soft-wrong + “why not B”; bank distractors                 | Asking only “defend your wrong answer” before feedback         |

**Operational definition for MindCraft (HYPOTHESIS):** A self-explanation prompt is load-bearing only if (1) the to-be-explained content is tagged **correct or known-incorrect**, (2) the response format scaffolds principle / operator–goal / misconception contrast (not free essay only), and (3) success is measured by **transfer / delayed mix**, not by longer time-on-task or chat length.

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## XL.3 Classic effect: spontaneous quality, then elicitation

**FACT (good vs poor example study):** Chi, Bassok, Lewis, Reimann & Glaser (1989, *Cognitive Science*, doi:10.1207/s15516709cog1302\_1) — talk-aloud while studying mechanics worked examples. “Good” students generated many explanations that refined conditions for actions and linked steps to principles; “poor” students under-explained, monitored poorly, and relied heavily on examples at solve time. Self-explanation quality predicted understanding and example-independence.

**FACT (elicitation works):** Chi, de Leeuw, Chiu & LaVancher (1994, *Cognitive Science*, doi:10.1016/0364-0213(94)90016-7) — eighth-graders prompted to self-explain after each line of a circulatory-system text outperformed controls who reread; high explainers built more accurate mental models. Prompts can *create* the behavior that spontaneous good students show.

**FACT (quality ≠ mere time):** Renkl (1997, *Cognitive Science*, doi:10.1016/S0364-0213(99)80017-2) — probability worked examples; time-on-task controlled. Learning gains still predicted by qualitative SE features: **principle-based** explanations, **operator–goal** explication, **anticipative** reasoning. Two effective profiles: anticipative reasoners and principle-based explainers.

**Product translation:** Coach UX should elicit *those* moves — “What rule licenses this step?” / “What is this step trying to achieve?” / “What happens next?” — not “Write anything about your feelings about math” and not “Hear the AI explain for 90 seconds.”

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## XL.4 Scaffolded prompts + fading (coach design spine)

**FACT (fading + principle prompts):** Atkinson, Renkl & Merrill (2003, *Journal of Educational Psychology*, doi:10.1037/0022-0663.95.4.774) — successively fading worked-out steps helps near transfer vs example–problem pairs, but far transfer is unreliable from fading alone. Combining fading with prompts to identify the **underlying principle** for each worked step produced medium-to-large effects on **near and far transfer without extra time on task**.

**FACT (Cognitive Tutor classroom):** Aleven & Koedinger (2002, *Cognitive Science*, doi:10.1016/S0364-0213(02)00061-7) — geometry Cognitive Tutor; students who explained steps by citing geometric rules (glossary) learned with greater understanding and better transfer than problem-solving-only peers. Interpretation: better-integrated declarative knowledge, less shallow procedural knowledge. Scales in classroom software — not only lab talk-alouds.

**HYPOTHESIS for MindCraft:** Ingredient card\_templates + bridge cards are the worked-example substrate; coach prompts should (a) attach principle/ingredient labels to faded steps, (b) require a brief generation or structured pick *before* revealing the next full AI paragraph, (c) fade prompts as mastery rises (expertise reversal risk if we nag forever).

**Tie to SAFE-CRAFT (XXVI):** Micro-attempt / completion blank inside an example *plus* a principle prompt beats cold multi-step solve *or* passive read-only cards.

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## XL.5 Math meta-analysis: what we may claim (and must not)

**FACT (math SE meta-analysis):** Rittle-Johnson, Loehr & Durkin (2017, *ZDM*, doi:10.1007/s11858-017-0834-z) — prompted self-explanation in mathematics: small-to-moderate immediate gains — procedural knowledge ES ≈ .28, conceptual ES ≈ .33, procedural transfer ES ≈ .46. Effects **stronger when high-quality explanations are scaffolded** (training or structured responses). Time-on-task control did **not** erase the benefit (moderator). **Caveats they stress:** evidence for reliable classroom (esp. primary/secondary non-ITS) effects and for **retention over delay** is much thinner; delayed procedural/conceptual effects often weak/null while delayed transfer ES ≈ .32 in the subset that measured it; substantial heterogeneity (some null/negative studies).

**Instructional recommendations from that paper (treat as design doctrine, not slogans):**

1. Scaffold high-quality explanations (structure or train).
2. Design prompts so they do **not** steal attention from other critical content.
3. Prompt learners to explain **correct** information.

4. Prompt why **common misconceptions** are incorrect.

**Kill:** Marketing “self-explanation always raises grades” / absolute classroom-retention claims from lab immediate ES alone. Claim ladder: L1 “structured why-this-step prompts aim at transfer”; L2+ only after SE-\* experiments with delayed mixed transfer\_pass.

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## XL.6 Constraints: when prompts backfire

**FACT (constraint review):** Rittle-Johnson & Loehr (2017, *Psychonomic Bulletin & Review*, doi:10.3758/s13423-016-1079-5) — four constraints: (1) SE favors principle-guided domains and transfer, can hurt detail/exception learning; (2) explaining **one’s own** (often wrong) ideas can entrench them — better to explain known-correct / known-incorrect content; (3) prompts focus attention — mis-aimed prompts tax the wrong content; (4) alternatives (instructional explanation, retrieval, unfamiliar problems) can match or beat SE in some conditions.

**HYPOTHESIS for soft-wrong:** Order matters. After a wrong MC choice: (A) brief correct-principle or contrastive “why B fails” prompt with scaffold → help; (B) “Explain why you picked C” *before* any correctness signal → risk of justifying the misconception (constraint 2). Prefer: soft-wrong frame → show contrastive target → generate/select why the foil fails → then optional deeper AI wrap (XXXIII: calibration > fluency).

**Contradiction to hold:** Open generation is the romantic “thinker” move; structured glossary/ingredient picks (Aleven) often deliver the measurable classroom gain. MindCraft should not shame structured picks as “not real thinking” — they are the scalable SE scaffold.

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## XL.7 Competitive / product audit (explanation lens)

| Surface                        | Typical explanation pattern                    | MindCraft stance                                                                                               |
|--------------------------------|------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Khan / video</b>            | Instructional explanation; optional reflection | Do not clone lecture-first as identity product                                                                 |
| <b>Duo</b>                     | Rare deep SE; habit + bits                     | Habit ≠ principle generation                                                                                   |
| <b>Brilliant</b>               | Puzzle insight; light justify                  | Insight ≠ scaffolded SE curriculum                                                                             |
| <b>ChatGPT Solver</b>          | High-fluency instructional explanation         | <b>Not</b> self-explanation; overtrust risk (XXXIII)                                                           |
| <b>Cognitive Tutor lineage</b> | Step + rule citation                           | Closest evidence-backed software pattern                                                                       |
| <b>MindCraft today</b>         | Cards + Solver + coach drafts                  | Need <b>student generation gates</b> before AI walls of text; wire misconception bank into contrastive prompts |

**FOUNDER BELIEF:** Solver that answers *for* Maya without a generation gate competes on Layer-1 (answers get cheap) and loses the identity thesis. Coach that forces a principle pick / misconception contrast competes on Layer 2–6.



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## XL.8 Commercial implications (copy, product, growth)

### Marketing language that survives

- **Allowed:** “We don’t just show the steps — we ask you to say *why* the step works.”
- **Allowed:** “Wrong answers become contrast: why that trap fails, not a shame spiral.”
- **Allowed (parent):** “We train explaining the rule, not only picking the letter — that’s what transfers.”
- **Banned:** “AI explains everything so tutoring is free.”
- **Banned:** “Any chat box = self-explanation science.”
- **Banned:** Absolute meta-analytic grade claims; 2-sigma; streak-as-understanding.
- **Banned:** Forcing students to justify unchecked wrong guesses as the default pedagogy.

### Feature claim ladder (Part XXXIV hygiene)

| Feature                                                  | Claim max without SE-* data                                 |
|----------------------------------------------------------|-------------------------------------------------------------|
| AI worked-example narration                              | L0 instructional help; <b>not</b> SE claim                  |
| Structured principle / ingredient pick on faded step     | L1 SE scaffold (Aleven/Atkinson-aligned intent)             |
| Contrastive “why this distractor fails” after soft-wrong | L1 misconception-SE intent                                  |
| Open essay “explain your answer” every item              | L0 until quality+affect validated; risk of load + avoidance |
| “SE raises ACT / identity” hero                          | L0 until SE-1/SE-2 delayed transfer                         |

### Growth / North Star hygiene

Chat turns, explanation word count, and Solver dwell time are **vanity** if uncoupled from transfer\_pass and strategy-class errors. Prefer: rate of principle-consistent explanations (rubric or structured correct pick), misconception-contrast success, delayed mixed accuracy, retry\_120s after contrastive miss — not tokens generated by the model.

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## XL.9 Experiments spawned

| ID   | Question                                                                  | Design                     | Primary                                              | Kill condition                                                             |
|------|---------------------------------------------------------------------------|----------------------------|------------------------------------------------------|----------------------------------------------------------------------------|
| SE-1 | Principle-prompt on faded card steps vs same cards with AI narration only | 2-arm; matched time budget | 48h near + far / mixed transfer; shallow-copy errors | Narration-only $\geq$ prompted on transfer <b>and</b> lower load/avoidance |

|         |                                                                                          |            |                                                               |                                                                           |
|---------|------------------------------------------------------------------------------------------|------------|---------------------------------------------------------------|---------------------------------------------------------------------------|
| SE-2    | Soft-wrong → contrastive “why foil fails” vs soft-wrong → explain-own-choice first       | 2-arm      | Misconception recurrence; retry_120s; delayed item isomorphic | Explain-own-first wins transfer → constraint-2 doctrine wounded           |
| SE-3    | Structured ingredient/glossary pick vs open text SE vs no SE                             | 3-arm      | Transfer; time; completion rate (Maya anxiety segment)        | Open text wins transfer but kills return → ship structured default        |
| SE-4    | Prompt every step vs adaptive fade of SE prompts with mastery                            | 2-arm      | Transfer; prompt skip/nag irritation                          | Always-on prompts equal fade on transfer but raise churn → fade mandatory |
| SE-QUAL | 10 Maya interviews: “When did an explanation feel like <i>yours</i> vs the app talking?” | Appendix B | coded ownership / passivity moments                           | Nobody distinguishes → UX isn’t creating generation                       |

**Pre-reg (XXXIV):** SE-\* identify **prompt regime** → **transfer** / **misconception recurrence**. They do **not** identify long-term identity from one session of explain boxes. Densifies Experiment A: coach content should be SE-constrained, not generic pep talk.

**Supersedes / densifies:** XXVI worked-example fading notes; Experiment D (explanation timing × anxiety) → prefer SE-2/SE-3 as operational pairs; XXXIII “pedagogy wrap” → wrap = student generation + calibration, not longer AI prose.

## XL.10 Confidence table

| Claim                                                                          | Label                              | Confidence |
|--------------------------------------------------------------------------------|------------------------------------|------------|
| Higher-quality spontaneous SE while studying examples predicts better learning | FACT (Chi et al., 1989)            | High       |
| Prompting SE can improve understanding vs reread / no prompt                   | FACT (Chi et al., 1994; math meta) | High       |
| Qualitative SE features predict gains beyond time-on-task                      | FACT (Renkl, 1997)                 | High       |
| Fading + principle prompts aid near & far transfer without extra time          | FACT (Atkinson et al., 2003)       | High       |

|                                                                                                     |                                        |               |
|-----------------------------------------------------------------------------------------------------|----------------------------------------|---------------|
| Step explanation via rule citation helps Cognitive Tutor geometry transfer                          | FACT (Aleven & Koedinger, 2002)        | High          |
| Math prompted-SE immediate effects small–moderate; scaffold helps                                   | FACT (Rittle-Johnson et al., 2017 ZDM) | High          |
| Classroom + delayed retention evidence thinner than immediate lab/ITS                               | FACT (same meta)                       | High          |
| Explaining own wrong ideas can entrench; prefer correct/incorrect targets                           | FACT (constraint review)               | High          |
| AI instructional explanation $\equiv$ self-explanation benefit                                      | SPECULATION / false                    | Low (against) |
| Default MindCraft path = faded examples + structured principle/misconception prompts before AI wrap | FOUNDER BELIEF / HYPOTHESIS            | Medium–High   |

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## XL.11 What this chapter kills

1. **Kill:** Equating Solver/AI monologue with self-explanation pedagogy.
2. **Kill:** “Explain why you chose that” *before* correctness/contrast as the default after every wrong answer.
3. **Kill:** Absolute “SE always raises scores / always works in class / always sticks for months” marketing.
4. **Wound:** Open-ended explain boxes as the only SE UX — survives as optional depth, not as the scalable default.
5. **Survive (constrained):** Scaffolded prompts on **correct steps and known misconceptions**, paired with worked-example fading, measured by transfer — the coach spine for identity-relevant competence evidence.

**Doctrine until data:** Build and market **student-generated why** (structured first) on faded examples and contrastive wrongs. AI explains *after* or *around* that generation — never as a substitute for it. Fluency of the model is not mastery of the student.

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## Part XLI — Desirable Difficulties $\times$ Anxiety: When Struggle Helps vs When It Becomes Threat

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30

**Primary question:** How should MindCraft reconcile Bjork-style desirable difficulties (spacing, interleaving, retrieval, generation) with Ashcraft-style math-anxiety / working-memory threat — so product + copy do not sell either “grind harder” or “never make it hard”?

**Owners:** Learning Science · Affect / Anxiety · Product (Practice sequencing + evaluation climate) · Growth / Positioning · Red Team

**Commercial job:** Resolve the I.4 bottleneck: ship faded desirable difficulties **without** converting anxious Maya’s session into shame theater — and market the distinction between *productive struggle* and *threat* in language parents and students can trust.

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## XLI.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), and XL (self-explanation) all lean on **desirable difficulties**: conditions that slow short-term performance but improve durable access and transfer. Part XXIV (affect/anxiety) and Ashcraft’s line show that math anxiety **already** loads working memory like a dual task. Part XXXI (challenge–skill) and XXXVIII (goal climate) warn that harder-feeling practice can cue avoidance under performance threat.

v1.6 left an explicit bottleneck: “*Part XLI must resolve desirable difficulties × anxiety so interleaving/SE/spacing do not ship as shame.*”

**FOUNDER BELIEF under audit:** Identity transformation requires competence evidence that survives delay and mix — which *requires* some desirable difficulty — but for threatened learners, unsequenced difficulty is not “elite training”; it is **threat + WM collapse + exit**.

**Claim we refuse as doctrine:** “Struggle is always good” / “always crank desirable difficulties” *and* its twin “anxious students should only get easy fluency drills forever.” Both are commercially tempting; both fail Maya.

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## XLI.2 Constructs (product language)

| Construct                          | Plain definition                                                                 | Product signal                                                       | Failure mode if misused                                     |
|------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Desirable difficulty (DD)</b>   | Challenge that forces encoding/retrieval processes that support later access     | Mixed review, spaced retries, generation before reveal               | Feels worse today → students & parents quit if misframed    |
| <b>Undesirable difficulty</b>      | Challenge the learner cannot productively meet (knowledge, cues, or affect load) | Early full interleave; open SE under panic; high-stakes quiz theater | Errors without learning; avoidance; “I’m not a math person” |
| <b>Math anxiety load</b>           | Worry/rumination consuming WM resources needed for multi-step math               | Freeze, hint binge, rapid exit after first hard miss                 | Interpreted as “lazy” or “needs more struggle”              |
| <b>Stakes / evaluation climate</b> | Whether the hard moment is framed as judgment vs training                        | Soft-wrong, mastery climate, no streak-as-worth                      | High stakes turn retrieval into threat (Hinze & Rapp)       |

|                         |                                                                       |                                                         |                                                   |
|-------------------------|-----------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|
| <b>Equipped success</b> | Prior knowledge + cues sufficient to <i>respond</i> to the difficulty | Block-lite → fade → mix; worked examples; structured SE | DD without equipment = cruelty dressed as science |
|-------------------------|-----------------------------------------------------------------------|---------------------------------------------------------|---------------------------------------------------|

**Operational definition (HYPOTHESIS):** A MindCraft difficulty is *desirable* only when (a) the student has enough schema/cues to respond with non-chance success or productive partial success, **and** (b) evaluation climate keeps the moment in mastery/training frame, **and** (c) affect load is not already saturating WM. Else it is undesirable — even if the *same* schedule would be desirable for a calm, prepared peer.

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### XLI.3 Bjork line: difficulty that builds durable access

**FACT (origin):** Bjork (1994, in Metcalfe & Shimamura, *Metacognition: Knowing about knowing*, MIT Press, pp. 185–205) — conditions that slow acquisition (spacing, variation, testing/generation as learning events, contextual interference) can improve long-term retention and transfer relative to conditions that make performance look strong during practice.

**FACT (desirable ≠ any difficulty):** Bjork & Bjork (2011/2014 reviews; JARMAC 2020 comment, doi:10.1016/j.jarmac.2020.09.003) — many difficulties are undesirable forever; a difficulty is desirable only when the learner can respond successfully with the encoding/retrieval processes the difficulty is meant to trigger. *If the learner is not equipped, the difficulty becomes undesirable.*

**FACT (learning vs performance illusion):** Instructors and students systematically misread current fluency as learning (Bjork lineage; see also XXIX / XXXIX). Blocked / massed / reread conditions feel productive; spaced / interleaved / tested conditions feel worse *and often teach more*.

**Product translation:** MindCraft cannot sell “we make practice feel easy” as proof of learning, nor “we make practice feel hard” as proof of rigor. The commercial claim must be **durable choosing and retrieving under exam-like demand** — measured by delayed mix / transfer\_pass / solo\_transfer\_pass — not by session comfort or session pain.

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### XLI.4 Ashcraft line: anxiety as a competing load

**FACT (WM disruption):** Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — high math-anxiety individuals show reduced WM spans (especially computation-based spans); dual-task math + memory load amplifies RT/errors. Anxiety acts as a **transitory disruption of working memory**, not merely as a stable “low ability” trait.

**FACT (review / education bridge):** Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, doi:10.3758/BF03194059) — math beyond simple retrieval depends on WM; high math anxiety functions like a resource-demanding secondary task (preoccupation with fear). Global consequence: avoidance of elective math, majors, and careers.

**FACT (carry from XXIV):** Association of math anxiety with performance is real but modest on average (meta  $\sim r \approx -0.17$ ); mechanism candidates include WM competition + avoidance. Anxiety is **one bottleneck among many**, not destiny.

**Conflict statement (HYPOTHESIS):** Desirable difficulties often *increase* momentary retrieval/selection demand — exactly the demand anxiety already taxes. Unbuffered DD therefore risks converting a would-be desirable difficulty into an undesirable one for HMA students: same schedule, different learner state → different pedagogy.

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## XLI.5 Evidence that stakes and anxiety change whether “hard practice” helps

**FACT (pressure can kill retrieval benefits):** Hinze & Rapp (2014, *Applied Cognitive Psychology*, doi:10.1002/acp.3032) — high-stakes vs low-stakes quizzes: quiz performance similar, but **final test better after low-stakes**; only low-stakes quizzes beat reread. Authors propose **retrieval disruption**: pressure can impair the memory-formation benefits of retrieval even when online quiz scores look fine.

**FACT (classroom retrieval often lowers reported test anxiety):** Agarwal, D’Antonio, Roediger, McDermott & McDaniel (2014, *JARMAC*, doi:10.1016/j.jarmac.2014.07.002) — survey of 1,408 middle/high-school students in classrooms using retrieval practice; 72% reported feeling **less nervous** for unit tests/exams when retrieval practice was used; 92% said it helped learning. Self-report, not RCT — but directionally opposite to “tests always raise anxiety.”

**FACT (meta: practice tests tend to reduce test anxiety):** Yang, Li, Zhao, Luo & Shanks (2023, *Educational Psychology Review*, doi:10.1007/s10648-023-09801-w) — 24 studies / 25 effects /  $N \approx 3,374$ ; practice tests reduce test anxiety with Hedges’  $g \approx -0.52$ ; easy practice tests more mitigating than difficult ones. **Caveat**: mostly classroom/quiz anxiety outcomes — not a free pass to ship brutal mixed ACT sims as day-one “practice tests.”

**FACT (anxiety can reshape what is learned, not only performance):** Fioriti, Pizzie, Evans, Green & Lyons (2025, *JEP:LMC*, doi:10.1037/xlm0001453) — adults learning unfamiliar multiplication across days: high math anxiety associated with **impaired procedural learning** on unrepeated problems but **accelerated retrieval learning** on heavily repeated problems — consistent with effort allocation toward declarative retrieval routes and away from procedural efficiency gains.

**HYPOTHESIS for MindCraft:** Anxious Maya may *prefer* and even accelerate on repeated retrieval of the same items (looks like mastery, feels safer) while under-building procedural flexibility and strategy selection — exactly the exam skills interleaving/SE target. Product that only rewards repeated-item fluency can **serve anxiety’s preference** while failing transfer.

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## XLI.6 Resolution doctrine (not a slogan)

**FOUNDER BELIEF / HYPOTHESIS — SAFE-DD stack (ties FEI / mastery climate / IL / SE):**

1. **Equip before you strain.** Short block-lite + worked-example / card fade until execution errors are mostly slips (XXVI, XXXIX). DD without schema is undesirable.
2. **Lower stakes while raising retrieval.** Soft-wrong, mastery climate, no public rank, no streak-as-worth (XXXVIII; Hinze & Rapp). Retrieval as *training*, not judgment.
3. **Dose difficulty to remaining WM.** Prefer structured SE picks over open essays under high affect (XL SE-3); prefer faded interleave over day-one full mix (IL-2); prefer spaced short sets over marathon mixed threat.
4. **Frame the feeling.** Copy: “This mix trains *choosing* — feeling slower is the workout, not proof you’re behind.” Never: “hard mode for elites” / “grind through the pain.”

5. **Measure the right win.** Session accuracy under block is not the North Star. Prefer `retry_120s`, `delayed mix / transfer_pass`, `solo_transfer_pass`, and anxiety-segment retention — not streaks or hint-free speed alone.

6. **Do not abandon DD for the anxious segment.** Permanent easy fluency is identity colonization by avoidance (XXV). Graded exposure to equipped difficulty is the path — therapy-adjacent but not a therapy claim (queue id 58).

**Kill the false dichotomy:** Bjork vs Ashcraft is not “science wars.” Bjork already says unequipped difficulty is undesirable; Ashcraft explains *why anxiety unequips* (WM). The product job is **re-equip + destake + sequence**, then apply DD.

## XLI.7 Competitive / marketing implications

| Competitor pattern                  | What it often does under anxiety                         | MindCraft wedge                                                  |
|-------------------------------------|----------------------------------------------------------|------------------------------------------------------------------|
| <b>Duolingo-like streaks</b>        | Turns miss into identity failure; pressures retrieval    | Ban streak-as-learning; train without shame timers               |
| <b>Blocked textbook / many apps</b> | Fluency comfort; weak strategy choice                    | Honest: early block OK; claim exam-ready only after mix          |
| <b>Brutal “exam sim” day one</b>    | High stakes + hard mix → Hinze-style disruption + exit   | Soft sim → rise stakes after <code>retry_120s</code> healthy     |
| <b>ChatGPT tutor monologue</b>      | Soothes without generation; may reduce threat short-term | Student-generated <i>why before</i> AI wrap (XL); calm ≠ mastery |
| <b>“Growth mindset” posters</b>     | Empty struggle praise without equipment                  | Struggle + scaffolds + mastery climate + transfer metrics        |

### Marketing language that survives

- **Allowed:** “Hard practice only after you’re equipped — then we mix so exams don’t surprise you.”
- **Allowed:** “Feeling slower on mixed sets is the training signal, not a verdict.”
- **Allowed (parent):** “We separate *training retrieval* from *graded judgment* so practice builds memory instead of panic.”
- **Banned:** “Struggle always equals learning.”
- **Banned:** “We never make anxious students struggle.”
- **Banned:** Streak / blocked fluency / AI comfort as proof of readiness.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free.

### Feature claim ladder (Part XXXIV hygiene)

| Feature                                        | Claim max without DD×ANX data          |
|------------------------------------------------|----------------------------------------|
| Soft-wrong + mastery copy on hard items        | L1 affect-safe practice climate        |
| Block → fade → interleave schedule             | L1 sequenced DD (intent aligned XXXIX) |
| Low-stakes spaced retrieval before “exam mode” | L1 Hinze-aligned retrieval hygiene     |

|                                                         |                                    |
|---------------------------------------------------------|------------------------------------|
| “Our struggle pedagogy raises ACT for anxious students” | L0 until DD-1/DD-2                 |
| Always-on max mix / open SE under threat                | L0 — likely undesirable difficulty |

## XLI.8 Experiments spawned

| ID      | Question                                                                   | Design                                           | Primary                                                             | Kill condition                                                                     |
|---------|----------------------------------------------------------------------------|--------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------|
| DD-1    | Faded DD (block→mix + soft-wrong) vs early full mix same items             | 2-arm; stratify high vs low math-anxiety pretest | 7d transfer_pass; retry_120s; exit after first miss                 | Early-mix wins transfer <b>and</b> retry in HMA segment → fade doctrine wounded    |
| DD-2    | Low-stakes retrieval sets vs same items framed as scored “exam drill”      | 2-arm (Hinze-style stakes)                       | Delayed retention; state anxiety; return D+1                        | High-stakes ≥ low-stakes on delayed + no anxiety cost → stakes doctrine overstated |
| DD-3    | Structured SE under mix vs open SE under mix for HMA segment               | 2-arm (densifies SE-3)                           | Completion; transfer; hint binge                                    | Open SE wins without churn → structured-default less necessary                     |
| DD-4    | Repeated-item fluency path vs varied procedural path (Fioriti-inspired)    | 2-arm within HMA                                 | Procedural transfer / strategy-class errors vs item-repeat accuracy | Repeat-fluency equals varied on transfer → anxiety-shaped preference harmless      |
| DD-QUAL | 10 Maya interviews: “When did hard practice feel like training vs attack?” | Appendix B                                       | coded stake/equipment moments                                       | Nobody separates → UX framing invisible                                            |

**Pre-reg (XXXIV):** DD-\* identify **schedule** × **stakes** × **anxiety** → **transfer/return**. They do **not** identify long-term identity from one hard session. Densifies Experiment D and IL-2/SE-3 as the anxiety-gated versions of desirable difficulty.

**Supersedes / densifies:** XXIX.4 “never introduce DD as shame”; XXXIX IL-2; XL SE-3; I.4 bottleneck → now has an operational SAFE-DD stack pending DD-1/2.



## XLI.9 Confidence table

| Claim                                                                                                    | Label                                                 | Confidence                          |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------|-------------------------------------|
| Spacing/interleaving/retrieval/generation can improve long-term learning vs fluent short-term conditions | FACT (Bjork lineage)                                  | High                                |
| Difficulties are desirable only if the learner can respond successfully                                  | FACT (Bjork & Bjork)                                  | High                                |
| Math anxiety disrupts WM / acts like dual-task load                                                      | FACT (Ashcraft & Kirk, 2001; Ashcraft & Krause, 2007) | High                                |
| High stakes can erase retrieval-practice benefits on delayed tests                                       | FACT (Hinze & Rapp, 2014)                             | High                                |
| Classroom retrieval programs often associate with <i>lower</i> reported exam nerves                      | FACT (Agarwal et al., 2014; self-report)              | Medium–High                         |
| Practice testing meta-analytically reduces test anxiety (esp. easier quizzes)                            | FACT (Yang et al., 2023)                              | High                                |
| HMA may accelerate repeated-item retrieval learning while impairing procedural learning                  | FACT (Fioriti et al., 2025)                           | Medium–High (adult lab; arithmetic) |
| SAFE-DD (equip → destake → dose → frame → measure) is the right MindCraft default                        | FOUNDER BELIEF / HYPOTHESIS                           | Medium–High                         |
| “Struggle always” or “never struggle anxious kids” is good marketing                                     | SPECULATION / false                                   | Low (against)                       |

## XLI.10 What this chapter kills

1. **Kill:** “Desirable difficulties always help — turn difficulty up for everyone.” (Ignores Bjork’s own *equipped* clause + Ashcraft WM.)
2. **Kill:** “Anxious students should only get easy, blocked fluency forever.” (Serves avoidance; fails transfer; Fioriti-shaped trap.)
3. **Kill:** High-stakes “exam mode” as the default vehicle for retrieval practice. (Hinze & Rapp.)
4. **Wound:** Self-report that “quizzes reduce anxiety” as license for brutal day-one mixed sims — survives as *low-stakes, well-sequenced* retrieval, not as cruelty.
5. **Survive (constrained):** Faded, low-stakes, mastery-framed desirable difficulties that produce delayed mix evidence — the commercial spine for “exam-ready without panic theater.”

**Doctrine until data:** Ship **SAFE-DD:** equip → lower stakes → dose to WM → frame as training → measure delayed transfer and `retry_120s` in the anxiety segment. Market productive struggle only when those guards are visible. Never sell pain or comfort as mastery.

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## Part XLII — Social Comparison & Leaderboards: When Ranks Help vs Hurt Novices

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 3)

**Primary question:** When do social-comparison surfaces (especially leaderboards) motivate learning versus threaten identity, and what should MindCraft default for Maya — the anxious / novice / identity-fragile segment?

**Owners:** Motivation · Affect / Anxiety · Product (Practice / Community UX) · Growth / Positioning · Red Team

**Commercial job:** Kill “ranks = engagement” as a North Star; harden the I.4 TARGET rule (default-ban appearance UX) into a shippable comparison policy competitors can be audited against.

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### XLII.1 Why this chapter exists

Part XXXVIII (goal orientation) already bans **appearance** goal structures and streak-as-North-Star. Part XXIV / XLI show social-evaluative threat taxes working memory and can turn desirable difficulty into threat. Competitive products (Duolingo-like ranks, classroom Top-N boards, public classroom streaks) still sell comparison as “motivation.”

**FOUNDER BELIEF under audit:** Identity transformation needs competence evidence — not a public ladder that tells bottom-half Maya she is behind *people* rather than behind a *skill criterion*.

**Claim we refuse as doctrine:** “Leaderboards motivate learners” (full stop) — and its twin “if we anonymize, any infinite rank is fine.”

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### XLII.2 Constructs (product language)

| Construct                          | Plain definition                                 | Product signal                                                   | Failure mode if misused                                                        |
|------------------------------------|--------------------------------------------------|------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Social comparison</b>           | Evaluating self via others’ standing (Festinger) | Any peer rank, percentile, “class average,” public tutor callout | Turns mastery climate into appearance climate                                  |
| <b>Upward / downward / lateral</b> | Compare to better / worse / similar others       | Top-N board; “you beat 80%”; neighbor band                       | Extreme upward gap → give-up; downward → complacent ego boost without learning |

|                                        |                                                    |                                                     |                                                        |
|----------------------------------------|----------------------------------------------------|-----------------------------------------------------|--------------------------------------------------------|
| <b>Absolute vs relative board</b>      | All players ranked vs local neighbors / Top-N only | Infinite classroom board vs “near you” strip        | Absolute + real names = max face threat for low ranks  |
| <b>Criterion / mastery feedback</b>    | Progress vs a task standard, not vs peers          | Path progress, concept mastery bars, personal bests | Still can be gamed if criterion ≠ transfer             |
| <b>Normative / appearance feedback</b> | Standing vs others; looking smart/fast             | Public leaderboard, streak leagues                  | Identity threat; WM load; exit after first public miss |

**Operational definition (HYPOTHESIS):** A MindCraft comparison surface is *safe* only when (a) the default is **self/criterion**, (b) peer comparison is **opt-in**, (c) comparison targets are **attainable** (local band, not unreachable Top-1), (d) metrics ranked are **effort/process-compatible** with mastery climate (not speed-only or streak-only), and (e) low standing is never **publicly named** for the anxiety / novice segment.

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### XLII.3 Festinger → competition: the comparison engine

**FACT (origin):** Festinger (1954, *Human Relations*, “A Theory of Social Comparison Processes”) — when objective standards are ambiguous, people evaluate abilities/opinions by comparing with others; ability comparison carries a unidirectional drive upward (improve toward better standards).

**FACT (selection + reaction meta):** Gerber, Wheeler & Suls (2018, *Psychological Bulletin*, doi:10.1037/bul0000127) — across 60+ years: when choosing up vs down, strong preference for **upward** comparison under no threat; **contrast** (self-evaluation moves *away* from the target) dominates reactions — ability estimates and affect often decline after upward contact. People often choose upward comparisons *even though* the typical reaction is self-deflating contrast.

**FACT (competition model):** Garcia, Tor & Schiff (2013, *Perspectives on Psychological Science*, doi:10.1177/1745691613504114) — social comparison is a major source of competitive attitudes/behavior; situational amplifiers include **proximity to a standard** (near #1 vs far away) and **N** (few vs many competitors). Ranking UIs are engineered comparison machines.

**Product translation:** Shipping a leaderboard is not a neutral “engagement widget.” It is a deliberate injection of comparison → contrast risk → competitive framing. For Maya, that often means **appearance climate** (XXXVIII), not mastery.

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### XLII.4 Leaderboards can move behavior — under narrow conditions

**FACT (HE experiment):** Landers & Landers (2014, *Simulation & Gaming*, doi:10.1177/1046878114563662) — random assignment to a course wiki **leaderboard** (not tied to grades) increased project interactions (~+29.6) vs control; time-on-task mediated academic performance. Supports: leaderboards *can* raise engagement via goal/conflict attributes **in that setting**.

**Caveats (FACT / scope):** Undergraduate / course project; not math-anxious high-school Maya; not public infinite classroom shame; achievement metric was interaction volume, not durable transfer. Landers’ theory of gamified

learning (2014, doi:10.1177/1046878114563660) treats game attributes as mediators/moderators of *instructional* quality — gamification does not replace pedagogy.

**FACT (HE systematic review):** Li, Liang, Fryer & Shum (2024, *Journal of Computer Assisted Learning*, doi:10.1111/jcal.13077) — 20 articles / 22 studies / 29 interventions (2014–2023), **all higher education**; ~half ≤1 hour. Leaderboards *can* benefit motivation/engagement/performance, but **effects hinge on design**. Authors recommend (among others): absolute boards for motivation *with* anonymity especially for low ranks; mix performance + engagement criteria; caution that HE short-duration findings do **not** generalize cleanly to K–12.

**HYPOTHESIS:** Landers-style gains are most likely when learners already have task competence + low identity threat + voluntary/low-stakes ranking. Minecraft’s default user is the opposite segment on day one.

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## XLII.5 Math-specific: anxiety × infinite ranks

**FACT (primary maths game, large N):** Almo, Rocha, Brennan & Dondio (2024, *International Journal of Serious Games*, doi:10.17083/ijsg.v11i4.794) — N=1,389 Irish primary students, 6-week *Seven Spells* programme with an **infinite leaderboard**. Leaderboard enjoyment predicted by rank position **and** maths anxiety: **maths-anxious players disliked the leaderboard more than non-anxious peers even after controlling for position**. Preference for playing against others also predicted enjoyment. Small correlation between liking the board and liking the game.

**Implication (HYPOTHESIS):** For anxious learners, public infinite rank is not a free engagement boost — it is a trait-moderated threat surface. “They’re low because they’re bad at the game” does not exhaust the mechanism; anxiety predicts dislike **net of rank**.

**FACT (negative feedback climate):** Fong, Patall, Vasquez & Stautberg (2019, *Educational Psychology Review*, doi:10.1007/s10648-018-9446-6) — meta-analysis (78 studies): negative feedback reduces intrinsic motivation vs positive feedback; harm lessens when feedback includes instructional how-to-improve detail and uses **criterion-based** standards (vs purely normative). Public low rank is often pure normative negative feedback without instructional content.

**Bridge to XLI / XXIV:** Social-evaluative threat (public miss, public bottom third) is exactly the stakes Hinze & Rapp warn can disrupt retrieval benefits — and Ashcraft-style WM load. Infinite boards amplify that climate at scale.

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## XLII.6 Resolution doctrine — SAFE-COMPARE

**FOUNDER BELIEF / HYPOTHESIS — SAFE-COMPARE stack (ties TARGET / SAFE-DD / FEI):**

1. **Default = criterion, not peers.** Show progress vs path / mastery / personal best. Peers off by default.
2. **Opt-in competition only.** Never force infinite boards onto gap-scan novices or high-anxiety segment.
3. **If peers: local + attainable.** Prefer neighbor bands / similar-ability cohorts over Top-1 theater (Garcia proximity + Gerber contrast).
4. **Anonymize low standing.** No named public bottom ranks (Li et al. Rec. 2; Bai et al. cited therein).
5. **Rank what mastery climate allows.** Prefer process-compatible signals (retries completed, transfer attempts, concepts unlocked) over speed leagues / streak leagues. Never rank “hint-free seconds” as identity.

6. **Instructional wrap on any relative miss.** If relative feedback ships, attach *how to improve next* (Fong moderators) — not only “you’re #47.”

7. **Measure the right win.** `retry_120s`, `challenge_accept` (mastery-coded), `transfer_pass` / `solo_transfer_pass` — not `leaderboard_views` or rank delta as learning.

**Kill the false dichotomy:** “Competition is toxic” vs “competition is always motivating.” Competition can raise time-on-task for some prepared learners (Landers). For Maya default, **unbuffered infinite peer rank is appearance UX** — banned until SC experiments say otherwise.

## XLII.7 Competitive / marketing implications

| Competitor pattern                   | Comparison mechanism                                           | MindCraft wedge                                                                  |
|--------------------------------------|----------------------------------------------------------------|----------------------------------------------------------------------------------|
| <b>Duolingo-like leagues / ranks</b> | Normative weekly standing; streak adjacent                     | Mastery path progress; ban streak/rank as learning proof                         |
| <b>Classroom Top-N boards</b>        | Extreme upward targets; face threat                            | Criterion bars; optional local bands                                             |
| <b>Khan / practice dashboards</b>    | Often self-paced; weaker public rank                           | Keep self/criterion; don’t “gamify catch-up” with shame ranks                    |
| <b>Brilliant prestige signals</b>    | Ability-appearance among peers                                 | Sell <i>solo transfer</i> and <i>challenge-seeking why</i> , not percentile flex |
| <b>ChatGPT tutor alone</b>           | No peer board — but social comparison still via school/parents | Own the <i>safe</i> comparison story: progress vs yesterday’s self               |

### Marketing language that survives

- **Allowed:** “Progress against *your* path — not a public ladder.”
- **Allowed:** “Competition is optional; mastery climate is the default.”
- **Allowed (parent):** “We don’t put anxious kids on a named scoreboard to ‘motivate’ them.”
- **Banned:** “Leaderboards keep students hooked” as a learning claim.
- **Banned:** Rank / league / streak as North Star or as proof of identity change.
- **Banned:** Bloom 2-sigma; absolute tutoring-is-free; empty mindset posters.

### Feature claim ladder (Part XXXIV hygiene)

| Feature                                                  | Claim max without SC data              |
|----------------------------------------------------------|----------------------------------------|
| Criterion progress / personal bests default              | L1 mastery-climate alignment (XXXVIII) |
| Opt-in anonymized local band                             | L1 SAFE-COMPARE intent                 |
| “Our leaderboard raises ACT / identity for anxious Maya” | L0 until SC-1/SC-2                     |

|                                       |                                         |
|---------------------------------------|-----------------------------------------|
| Forced infinite named classroom board | L0 — appearance UX; likely harm for HMA |
|---------------------------------------|-----------------------------------------|

## XLII.8 Experiments spawned

| ID      | Question                                                                    | Design                                   | Primary                                                             | Kill condition                                                                                 |
|---------|-----------------------------------------------------------------------------|------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------|
| SC-1    | Criterion-only progress vs infinite named leaderboard (same practice items) | 2-arm; stratify high vs low math-anxiety | retry_120s; exit after public miss; 7d transfer_pass                | Infinite board wins transfer <b>and</b> retry in HMA → default-ban wounded                     |
| SC-2    | Opt-in local anonymized band vs forced absolute board                       | 2–3 arm                                  | Enjoyment; state anxiety; return D+1; challenge_accept motive codes | Forced absolute ≥ opt-in local on retention without anxiety cost → anonymity/opt-in overstated |
| SC-3    | Rank process metrics (retries, mix attempts) vs rank speed/streak           | 2-arm within volunteers                  | Strategy-class errors; transfer_pass vs session speed               | Speed rank equals process rank on transfer → metric choice moot                                |
| SC-4    | Criterion negative feedback + how-to-improve vs rank-only “you’re #N”       | 2-arm (Fong-inspired)                    | Intrinsic motivation proxy; retry                                   | Rank-only ≥ instructional criterion → Fong moderator fails in product                          |
| SC-QUAL | 10 Maya interviews: “When did seeing others’ scores help vs freeze you?”    | Appendix B                               | coded comparison threat moments                                     | Nobody distinguishes → copy invisible                                                          |

**Pre-reg (XXXIV):** SC-\* identify **comparison surface** × **anxiety** → **return/transfer**. They do **not** identify that “healthy competition builds character” from one league season.

**Densifies:** XXXVIII appearance ban; I.4 TARGET “leaderboard kill for Maya default”; XXIV social-evaluative threat hypothesis.

## XLII.9 Confidence table

| Claim                                                                                                  | Label                            | Confidence           |
|--------------------------------------------------------------------------------------------------------|----------------------------------|----------------------|
| People use others to evaluate abilities when standards are ambiguous                                   | FACT (Festinger, 1954)           | High                 |
| Upward comparison often preferred; contrast often dominates self-eval/affect                           | FACT (Gerber et al., 2018)       | High                 |
| Ranking contexts amplify competitive social comparison                                                 | FACT (Garcia et al., 2013)       | High                 |
| Leaderboards can increase time-on-task / mediate performance in some HE settings                       | FACT (Landers & Landers, 2014)   | High (scope-limited) |
| Leaderboard effects in education are design-dependent; HE evidence $\neq$ K–12                         | FACT (Li et al., 2024)           | High                 |
| Maths-anxious players dislike infinite boards more, net of rank                                        | FACT (Almo et al., 2024)         | High                 |
| Normative negative feedback tends to hurt intrinsic motivation vs positive; criterion + how-to softens | FACT (Fong et al., 2019)         | High                 |
| SAFE-COMPARE default (criterion, opt-in, local, anonymize low, process metrics) is right for MindCraft | FOUNDER BELIEF / HYPOTHESIS      | Medium–High          |
| “Leaderboards motivate everyone / raise learning for Maya”                                             | SPECULATION / false as universal | Low (against)        |

## XLII.10 What this chapter kills

1. **Kill:** Universal “leaderboards motivate students” as product doctrine or marketing. (Design- and trait-dependent; HE  $\neq$  Maya.)
2. **Kill:** Infinite named public ranks as default engagement for novices / high math-anxiety. (Almo et al.; TARGET / appearance.)
3. **Kill:** Rank delta / league standing / streak leagues as North Star or as proof of identity transformation.
4. **Wound:** Absolute-board motivation gains from HE reviews as license for K–12 public boards — survives only as *design hypothesis*, not as Maya default.
5. **Survive (constrained):** Opt-in, anonymized, attainable, criterion-wrapped comparison for learners who already seek competition — never as the onboarding spine.

**Doctrine until data:** Ship **SAFE-COMPARE:** criterion default → peer opt-in → local/attainable → anonymize low standing → rank mastery-compatible process → instructional wrap → measure transfer/retry, not rank views. Market

“progress without public ladders.” Never sell ranks as belonging.

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## Part XLIII — Habit Formation Science (Wood/Clear): Cue Design Without Identity Colonization

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 6 ≡ Red Team slot; ch43 never written → prefer Researcher per rotation)

**Primary question:** How should MindCraft use habit science (context cues, repetition, automaticity) to raise *return to practice* without making streaks / daily check-ins the student’s identity — or mistaking habit for learning?

**Owners:** Motivation · Product (Practice / notifications / recognition) · Growth / Positioning · Red Team

**Commercial job:** Harden Insight 5 / HID into a shippable **SAFE-HABIT** stack competitors (especially Duo-like streak products) can be audited against; kill “habit = mastery” and “21 days” folklore in copy and KPIs.

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### XLIII.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance / streak North Star ban), XLI (SAFE-DD), and XLII (SAFE-COMPARE) already treat streaks and ranks as threat surfaces when they become *worth*. Market gravity still pulls edtech toward Duolingo-shaped habit loops: push cue → micro-session → streak reward → loss aversion.

**FOUNDER BELIEF under audit:** Identity transformation needs *reliable return to difficulty* — so cue design is load-bearing — but the habit that forms must be “open practice and try the next hard thing,” not “protect a number that proves I am consistent.”

**Claims we refuse as doctrine:**

1. “Streaks = learning” / streak length as North Star.
  2. “Habits take 21 days” as science.
  3. Treating popular packaging (*Atomic Habits*) as peer-reviewed habit theory without Wood/Gardner/Lally grounding.
  4. Habit loops that **colonize identity** — the student becomes “a streak-keeper” rather than “someone who can recover from wrong.”
- 

### XLIII.2 Constructs (product language)

| Construct | Plain definition | Product signal | Failure mode if misused |
|-----------|------------------|----------------|-------------------------|
|-----------|------------------|----------------|-------------------------|



|                                              |                                                                                                                      |                                                                          |                                                                     |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------|
| <b>Habit (scientific)</b>                    | Learned context→response association in procedural memory; cue activates response with limited goal mediation (Wood) | Stable time/place/preceding-action cue for practice open                 | Shipping “motivation” when the real lever is <i>cue stability</i>   |
| <b>Automaticity</b>                          | Behavior feels efficient / less deliberative                                                                         | Self-report automaticity; shorter time-to-first-problem                  | Confusing app-open automaticity with <i>math skill</i> automaticity |
| <b>Context cue</b>                           | Physical setting, time, people, prior action in a sequence                                                           | After-school phone dock; tutor session end → 10-min solo; calendar block | Push spam as fake context; unstable cues never habitize             |
| <b>Implementation intention</b>              | If-then plan: when Y, do X (Gollwitzer)                                                                              | Onboarding “After dinner, I open Minecraft Map”                          | Vague goals (“practice more”) with no situational trigger           |
| <b>Identity colonization (product sense)</b> | Habit metric becomes who the student <i>is</i> / fears losing                                                        | Streak as self-worth; freeze culture; XP-as-self                         | HID failure — return without identity change; break → exit          |
| <b>SAFE-HABIT</b>                            | Cue + small start + miss-forgiveness + learning metrics above habit metrics                                          | See XLIII.6                                                              | Habit theater that still ranks streak / DAU as proof                |

**Operational definition (HYPOTHESIS):** A Minecraft habit surface is *safe* only when (a) cues target **opening equipped practice**, not protecting a counter, (b) miss recovery is cheap (Lally: one miss ≠ wipe), (c) recognition prefers process/identity markers over streak length, and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / identity items — never streak days.

### XLIII.3 Wood: habits are context–response, not vibes

**FACT (review):** Wood & Rünger (2016, *Annual Review of Psychology*, doi:10.1146/annurev-psych-122414-033417) — habits are learned associations between recurring **context cues** and responses, strengthened gradually via associative / reward learning; once formed, perception of the cue activates the response representation with limited need for executive control. Defining features of habit automaticity include cue activation and **relative insensitivity to short-term goal/outcome changes**.

**FACT (habit–goal interface):** Wood & Neal (2007, *Psychological Review*, “A New Look at Habits and the Habit–Goal Interface”) — goals help *start* repetition and exposure to cues; once habits form, contexts can trigger responses without a mediating goal. Habit change therefore often requires changing cues / environments, not only “trying harder.”

**FACT (systems view):** Wood, Mazar & Neal (2021, *Perspectives on Psychological Science*, habits vs goals as separate but interacting systems) — habitual responses can activate regardless of current goals; people may still inhibit

them, but cue-driven activation is real. Product translation: a streak notification is a **Pavlovian/context cue** engineered by the app — not proof the student wants math.

**Product translation:** If MindCraft wants return, design **stable contexts** (same after-school window; end-of-tutor-session stack; device home-screen tile) more than pep talks. If MindCraft wants identity, do **not** make the strongest cue “don’t break the number.”

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## XLIII.4 Clear / Fogg packaging vs the evidence spine

**FACT (popularization, not primary science):** Clear (*Atomic Habits*, 2018) packages cue / craving / response / reward loops and “habit stacking” (“After [current habit], I will [new habit]”) for a mass audience. Fogg (*Tiny Habits*) popularized tiny start + anchor habits. Neither replaces Wood’s procedural-memory account.

**FACT (what stacking actually is):** Habit stacking’s evidence spine is largely **implementation intentions** — Gollwitzer & Sheeran (2006, *Advances in Experimental Social Psychology*, doi:10.1016/S0065-2601(06)38002-1): across 94 tests, if-then plans show a medium-to-large effect on goal attainment ( $d \approx 0.65$ ), via heightened cue accessibility and automated initiation.

**HYPOTHESIS:** Clear is usable *copy scaffolding* for onboarding (“After tutoring, open Map for one mission”) if MindCraft cites the Gollwitzer mechanism and measures initiation → equipped practice, not Clear’s compound-growth metaphors as science.

**Kill:** Marketing that says “built on Atomic Habits science” as if the book were a methods paper. Prefer: “cue-based practice plans (implementation intentions; habit research).”

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## XLIII.5 How long, how fragile, what to measure

**FACT (timeline myth):** Lally, van Jaarsveld, Potts & Wardle (2010, *European Journal of Social Psychology*, doi:10.1002/ejsp.674) — daily repetition in a consistent context; automaticity rises on an asymptotic curve; among good-fit participants, median time to ~95% of asymptote  $\approx$  **66 days** (range **18–254**). Missing **one** opportunity did not materially derail formation. ~Half of motivated participants still failed to perform consistently enough to reach habit status in the study window.

**Kill:** “21 days to form a habit” (Maltz folklore, not this literature). Also kill product copy that implies a week-long streak is a habit.

**FACT (definition hygiene):** Gardner (2015, *Health Psychology Review*, doi:10.1080/17437199.2013.876238) — habit as a *process* whereby a stimulus generates an impulse to act via learned S–R association; impulse need not equal behavior. Most health-habit studies are correlational / self-report-heavy — confidence ceiling for edtech transfer claims.

**FACT (measurement trap):** Verplanken & Orbell (2003, *Journal of Applied Social Psychology*, doi:10.1111/j.1559-1816.2003.tb01951.x) — SRHI includes history of repetition, automaticity, **and expressing identity**. Self-report habit strength can smuggle identity language into a frequency metric.

**Product implication (HYPOTHESIS):** Do not treat SRHI-like “this is typically me” items as proof of *math identity transformation* (Part XXV / id 61 queue). Separate: (1) practice-open automaticity, (2) math self-efficacy / narrative identity scales, (3) transfer metrics.

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## XLIII.6 Duolingo-shaped products: habit works — and colonizes

**FACT (industry, self-report):** Duolingo publicly designs streaks using habit research framing; company blog claims learners who reach short streaks are far more likely to continue (e.g., 7-day streak correlational lifts — *not* causal RCTs published as independent learning science). Streak freezes / separation of streak vs daily goal are retention experiments on **engagement**, not transfer.

**FACT (misuse qualitative):** Hadi Mogavi et al. (2022, *Learning @ Scale*, doi:10.1145/3491140.3528274) — forum analysis + interviews on Duolingo: **gamification misuse** when users fixate on XP/streaks/leaderboards and get distracted from learning; reasons include competitiveness, playfulness overlearning, dark nudges, compulsion, herding. Ramifications touch learning efficiency, well-being, and ethics.

**Bridge to XLII / XXXVIII:** Streak leagues + infinite ranks are the same family as appearance climates — consistency theater. Wood predicts cues will keep firing even when the *learning* goal cooled; Hadi Mogavi shows the product can train the wrong response (protect streak / farm XP).

**FOUNDER BELIEF:** MindCraft may borrow **cue stability + small start + miss forgiveness**. MindCraft must not borrow **streak-as-self** or **loss-aversion-as-pedagogy**.

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## XLIII.7 Resolution doctrine — SAFE-HABIT

**FOUNDER BELIEF / HYPOTHESIS — SAFE-HABIT stack (ties HID / TARGET / SAFE-DD / SAFE-COMPARE):**

1. **Cue the practice, not the counter.** Default cues: calendar/time, tutor-session end stack, story-world return point — not “your streak dies at midnight” as the primary cue.
2. **Implementation intention onboarding.** One written if-then in gap-scan aftermath or first session (“After [stable cue], I open one mission”). Measure initiation, not slogan recall.
3. **Tiny equipped start.** First action  $\leq$  friction of one problem with SAFE-DD equipment — not a multi-screen habit tutorial.
4. **Miss  $\neq$  identity failure.** One miss does not reset belonging (Lally). Prefer gentle resume copy over shame / “you broke it.” Streak freezes are a *retention patch*, not a learning feature — if shipped, never as North Star.
5. **Habit metrics subordinate.** Track D1/D7 return, cue-honoring rate, time-to-first-problem as *diagnostics*. Learning North Stars remain retry / transfer / solo transfer / challenge\_accept (mastery-coded).
6. **Identity markers elsewhere.** Map fill, story unlocks, tutor recognition of *recovery*, parent “they tried the hard one again” — not streak length as “who they are.”
7. **No identity colonization.** Ban copy that equates worth with unbroken days. Ban ranking streaks (XLII). Ban streak leagues as proof of transformation.

**Kill the false dichotomy:** “We refuse all habit design” vs “we must out-Duo Duo.” Refuse *colonizing* habit design; keep *cue science*.

---

## XLIII.8 Competitive / marketing implications

| Competitor pattern                    | Habit mechanism                            | MindCraft wedge                                               |
|---------------------------------------|--------------------------------------------|---------------------------------------------------------------|
| <b>Duolingo streaks / freezes</b>     | Loss aversion + daily cue; retention-first | Cue practice return; miss-forgiving; never streak-as-learning |
| <b>Duo leagues + XP</b>               | Habit + appearance compound                | SAFE-COMPARE + SAFE-HABIT; sell solo transfer                 |
| <b>Khan session streaks / energy</b>  | Mild habit / progress bars                 | Keep criterion progress; don't add shame timers               |
| <b>Brilliant daily puzzles</b>        | Delight + return habit                     | Return <i>into</i> recoverable struggle, not only delight     |
| <b>ChatGPT “study every day” tips</b> | Advice without cue architecture            | Own <i>if-then</i> + <i>equipped mission</i> as the plan      |

### Marketing language that survives

- **Allowed:** “Same time, same cue — open one hard problem and recover.”
- **Allowed:** “Habits get you to the desk; identity is what you do when it's wrong.”
- **Allowed (parent):** “We build a practice cue without making a broken streak mean failure.”
- **Banned:** “Streaks create math identity.”
- **Banned:** “21-day habit challenge = mastery.”
- **Banned:** Streak / DAU as North Star or as proof of transformation.
- **Banned:** Bloom 2-sigma; tutoring-is-free; empty mindset posters.

### Feature claim ladder (Part XXXIV hygiene)

| Feature                                       | Claim max without HAB data                |
|-----------------------------------------------|-------------------------------------------|
| If-then practice cue + tiny equipped start    | L1 initiation design (Gollwitzer-aligned) |
| Miss-forgiving resume without streak shame    | L1 HID hygiene                            |
| “Our streak raises ACT / math identity”       | L0 until HAB-1/HAB-2                      |
| Forced streak + public streak rank as default | L0 — colonization UX                      |

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## XLIII.9 Experiments spawned

| ID | Question | Design | Primary | Kill condition |
|----|----------|--------|---------|----------------|
|----|----------|--------|---------|----------------|

|          |                                                                                      |                         |                                                                 |                                                                                   |
|----------|--------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------|
| HAB-1    | If-then cue plan vs streak-primary onboarding (same missions)                        | 2-arm; new users        | D7 return; time-to-first-problem; 14d transfer_pass             | Streak arm wins transfer <b>and</b> identity item → on-ramp-only doctrine wounded |
| HAB-2    | Miss-shame copy (“you broke your streak”) vs miss-forgiving resume                   | 2-arm after forced miss | Return D+1; state anxiety; retry_120s                           | Shame ≥ forgive on return without anxiety cost → forgive overstated               |
| HAB-3    | Stable calendar cue vs push-only reminders                                           | 2-arm                   | Cue-honoring rate; 4-week automaticity proxy                    | Push-only equals calendar on return → context-stability claim weak                |
| HAB-4    | Recognition: process/recovery badge vs streak badge vs none                          | 3-arm (ties GO-2)       | challenge_accept motives; identity item; streak obsession items | Streak badge wins engagement <b>and</b> transfer → HID weaker than feared         |
| HAB-QUAL | 10 Maya interviews: “What does your streak (elsewhere) make you feel when you miss?” | Appendix B              | coded colonization vs helpful cue                               | Nobody reports threat → copy over-indexed                                         |

**Pre-reg (XXXIV):** HAB-\* identify **cue/recognition** × **return/transfer**. They do **not** identify that “daily habit builds mathematicians” from DAU alone.

**Densifies:** Insight 5 / HID; I.4 TARGET streak ban; XXXVIII appearance; XLII streak-league kill.

### XLIII.10 Confidence table

| Claim                                                                                            | Label                                         | Confidence |
|--------------------------------------------------------------------------------------------------|-----------------------------------------------|------------|
| Habits are context–response associations; cue-triggered; relatively goal-insensitive once strong | FACT (Wood & Rünger, 2016; Wood & Neal, 2007) | High       |
| Goals help form habits via repetition/exposure; changing cues aids habit change                  | FACT (Wood line)                              | High       |

|                                                                                                                   |                                       |                    |
|-------------------------------------------------------------------------------------------------------------------|---------------------------------------|--------------------|
| Implementation intentions raise goal attainment (medium–large)                                                    | FACT (Gollwitzer & Sheeran, 2006)     | High               |
| Automaticity plateaus on the order of weeks–months; median ~66d in Lally sample; high variance; one miss often OK | FACT (Lally et al., 2010)             | High               |
| “21 days” is not the Lally result                                                                                 | FACT                                  | High               |
| Clear/Fogg are popular packaging; stacking ≈ if-then planning                                                     | HYPOTHESIS / FACT (mechanism mapping) | Medium–High        |
| Gamification can distract from learning (Duo case)                                                                | FACT (Hadi Mogavi et al., 2022)       | High (qualitative) |
| Streaks can raise return without identity / transfer                                                              | HYPOTHESIS (HID; industry)            | Medium–High        |
| SAFE-HABIT is right default for MindCraft                                                                         | FOUNDER BELIEF / HYPOTHESIS           | Medium–High        |
| “Streaks = learning / identity”                                                                                   | SPECULATION / false as universal      | Low (against)      |

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### XLIII.11 What this chapter kills

1. **Kill:** Streak length / DAU / “habit streak” as North Star or as proof of identity transformation. (HID; standing ban densified with Wood + Duo misuse.)
2. **Kill:** “21-day habit” marketing as science. (Lally variance + asymptote.)
3. **Kill:** Equating *Atomic Habits* branding with peer-reviewed habit doctrine. (Use Wood/Gardner/Gollwitzer; Clear as optional UX copy only.)
4. **Kill:** Shame-on-miss and public streak ranks as default engagement for Maya. (Colonization; XLII.)
5. **Wound:** “Any daily reminder is habit science” — survives only if tied to **stable context + repeated equipped response**, not push spam.
6. **Survive (constrained):** Cue design, tiny starts, implementation intentions, and miss forgiveness as **on-ramps** to FEI / transfer — never as the product thesis.

**Doctrine until data:** Ship **SAFE-HABIT**: cue practice (not the counter) → if-then onboarding → tiny equipped start → miss ≠ identity failure → habit metrics subordinate → identity markers elsewhere → ban streak colonization. Market “we get you back to the hard problem — without making a broken streak mean you’re not a math person.”

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## Part XLIV — Intrinsic Motivation Killers (Deci): Controlling Rewards and the Caveats

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 9; hour%6≠0; 3 researcher entries since synthesizer v1.6 → Researcher, not Synthesizer)

**Primary question:** When do MindCraft rewards, XP, badges, streaks, and “do this for the prize” UX *undermine* the intrinsic interest and autonomy we claim to build — and when are they informational rather than controlling?

**Owners:** Motivation · Product (Practice / recognition / parent copy) · Growth / Positioning · Red Team

**Commercial job:** Turn the vague “extrinsic bad / Deci says so” slogan into a shippable **SAFE-REWARD** stack competitors can be audited against; kill both “all rewards kill love of math” absolutism and “gamify everything” as identity strategy.

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## XLIV.1 Why this chapter exists

Parts XXV (habit ≠ identity), XXXVIII (appearance climates), XLII (SAFE-COMPARE), and XLIII (SAFE-HABIT) already treat streaks, ranks, and consistency theater as threat surfaces. Market gravity still pulls edtech toward *expected tangible / token rewards for doing the task*: XP for opening the app, coins for finishing any problem, badges for login streaks, parent dashboards that score “minutes practiced.”

**FOUNDER BELIEF under audit:** Identity transformation needs *felt autonomy* and *competence information* — not the absence of all rewards. The killer is **controlling functional significance** (pressure to behave for the inducement), not the existence of feedback or unexpected acknowledgment.

**Claims we refuse as doctrine:**

1. “All rewards kill intrinsic motivation” (Cameron–Pierce counter-literature exists; CET is differentiated).
  2. “Gamification is always fine if engagement rises” (engagement ≠ free-choice interest; Hadi Mogavi / Duo misuse).
  3. “Verbal praise always helps kids” (Deci: controlling praise / “you should keep it up” can undermine).
  4. Treating SDT as a poster (“autonomy / competence / relatedness”) without contingency design.
- 

## XLIV.2 Constructs (product language)

| Construct                              | Plain definition                                                                                               | Product signal                                                                            | Failure mode if misused                         |
|----------------------------------------|----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------|
| <b>Intrinsic motivation</b>            | Doing the activity for inherent interest / satisfaction                                                        | Free-choice return to hard problems; curiosity asks; challenge_accept for mastery reasons | Confusing “opens app for XP” with interest      |
| <b>Extrinsic motivation (spectrum)</b> | Behavior for separable outcome; can be more/less autonomous (external → introjected → identified → integrated) | Parent-pressure vs “I chose ACT path because...”                                          | Treating all extrinsic as poison or all as fine |

|                                     |                                                                                                         |                                                           |                                                              |
|-------------------------------------|---------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| <b>Controlling vs informational</b> | Event felt as pressure to behave vs competence feedback (CET)                                           | “Finish 10 for a coin” vs “you used the bridge correctly” | Same badge can be either — design of <i>how</i> it is framed |
| <b>Engagement-contingent reward</b> | Expected reward for <i>doing</i> the task (no quality bar)                                              | XP for any attempt; login coins                           | Classic undermine path (Deci 1999)                           |
| <b>Completion-contingent</b>        | Expected reward for finishing                                                                           | “Complete mission → loot” regardless of strategy quality  | Similar undermine; weak competence signal                    |
| <b>Performance-contingent</b>       | Reward for meeting a standard                                                                           | Badge for transfer pass; tutor note for recovery          | Can inform competence <i>or</i> control — climate decides    |
| <b>SAFE-REWARD</b>                  | Prefer informational feedback; avoid expected tangible for mere engagement; autonomy-supportive framing | See XLIV.7                                                | Reward theater that still trains external PLOC               |

**Operational definition (HYPOTHESIS):** A MindCraft reward surface is *safe* only when (a) it primarily conveys **competence / strategy information**, (b) expected tangible / token rewards are **not** the reason to open practice, (c) framing is autonomy-supportive (choice, rationale, non-pressuring language), and (d) North Stars remain `retry_120s` / `transfer_pass` / `solo_transfer_pass` / `identity` items — never XP velocity or badge count.

### XLIV.3 SDT spine (needs, not posters)

**FACT (overview):** Ryan & Deci (2000, *American Psychologist*, 55(1), 68–78, doi:10.1037/0003-066X.55.1.68) — Self-Determination Theory posits innate needs for **competence, autonomy, and relatedness**; social contexts that support these needs facilitate intrinsic motivation and internalization; contexts that thwart them yield alienation and depleted motivation.

**FACT (CET):** Cognitive Evaluation Theory (Deci & Ryan line; summarized in Deci, Koestner & Ryan, 1999) — rewards and external events have **functional significance**: controlling aspects shift perceived locus of causality outward (undermine intrinsic motivation); informational aspects affirm competence (can enhance). Competence feedback without autonomy does not reliably enhance intrinsic motivation.

**Product translation:** MindCraft’s brand voice (“you’re becoming someone who recovers”) must not be paired with UX that says “do this *or* lose the prize.” Relatedness (tutor, story world, parent) is not a license for controlling rewards.

### XLIV.4 The classic finding — and the differentiated meta



**FACT (origin):** Deci (1971, *Journal of Personality and Social Psychology*, 18(1), 105–115, doi:10.1037/h0030644) — expected extrinsic rewards for an interesting activity can reduce subsequent free-choice engagement (the “undermining” / overjustification pattern that launched the literature).

**FACT (Deci–Koestner–Ryan 1999 meta):** Deci, Koestner & Ryan (1999, *Psychological Bulletin*, 125(6), 627–668, doi:10.1037/0033-2909.125.6.627) — 128 studies. As predicted by CET:

- **Engagement-contingent** rewards undermine free-choice intrinsic motivation ( $d \approx -0.40$ ).
- **Completion-contingent** ( $d \approx -0.36$ ) and **performance-contingent** ( $d \approx -0.28$ ) also undermine free-choice behavior on average.
- Engagement- and completion-contingent also hurt self-reported interest ( $d \approx -0.15 / -0.17$ ).
- **Unexpected** tangible rewards: no average undermine on free-choice.
- **Positive feedback / verbal rewards:** enhance free-choice ( $d \approx 0.33$ ) and interest ( $d \approx 0.31$ ) on average.
- Tangible rewards tended to be **more detrimental for children** than college students; verbal rewards less enhancing for children than for college students.

**FACT (education restatement):** Deci, Koestner & Ryan (2001, *Review of Educational Research*, 71(1), 1–27, doi:10.3102/00346543071001001) — argue Cameron & Pierce (1994) understated undermining; conclude tangible rewards have a substantial undermining effect when CET contingencies are respected; defend keeping CET for educational practice.

**Product implication (HYPOTHESIS):** Expected XP/coins for *mere engagement* (open app, any answer, any completion) is the highest-risk MindCraft pattern for Maya-aged users. Informational soft-wrong + strategy feedback is the low-risk pattern. Performance-contingent badges *can* work if they mark real competence (transfer) and are not administered as pressure.

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## XLIV.5 Caveats — Cameron/Pierce, climate, and “not all rewards”

**FACT (counter-meta):** Cameron & Pierce (1994, *Review of Educational Research*, 64(3), 363–423, doi:10.3102/00346543064003363) — across 96 between-groups experiments, *overall* reward did not decrease intrinsic motivation; verbal praise increased it; the clearest negative effect was expected tangible rewards for *simply doing* a task (free-choice time after reward removal). They argued negative effects are circumscribed and avoidable.

**FACT (debate):** Ryan & Deci (1996, *RER*), Lepper et al. (1996), Kohn (1996) vs Cameron & Pierce (1996, *RER*) — methodological dispute over aggregation, moderators, and what “overall” means. **Not settled as “rewards are harmless.”**

**FOUNDER BELIEF / resolution for MindCraft:** Do **not** cite “Deci proved all gamification is evil.” Do **not** cite Cameron to greenlight engagement-contingent loot. Cite the **contingency**: expected tangible for mere doing is the kill zone; informational verbal/competence feedback is the build zone; performance-contingent is climate-sensitive.

**FACT (controlling praise):** Ryan (1982) and Ryan, Mims & Koestner (1983) — positive feedback administered with controlling language (e.g., “you *should* keep up the good work”) reduces intrinsic motivation relative to informational administration. Deci 1999 summarizes interpersonal climate as a moderator of performance-contingent rewards.

**Kill:** “We praise a lot, so SDT is satisfied.” Praise can be controlling.

**FACT (choice as autonomy support):** Patall, Cooper & Robinson (2008, *Psychological Bulletin*, 134(2), 270–300, doi:10.1037/0033-2909.134.2.270) — meta of 41 studies: providing choice enhances intrinsic motivation, effort, task performance, and perceived competence. Effects stronger for instructionally *irrelevant* choices, when ~2–4 successive choices are offered, when rewards are **not** stacked after the choice manipulation, and vs highly controlling controls; stronger for children than adults.

**Product implication:** Meaningful micro-choice (which story frame, which representation after mastery structure is set) can support autonomy — but **choice + immediately dangling reward** weakens the choice benefit. Do not “choose your path → earn 50 coins.”

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## XLIV.6 Competitive patterns under CET audit

| Competitor pattern                             | CET reading                                                                             | MindCraft wedge                                                                 |
|------------------------------------------------|-----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| <b>Duolingo XP / streaks / leagues</b>         | Expected engagement- and completion-contingent tokens + appearance climate (XLII/XLIII) | Informational recovery feedback; no XP-as-reason-to-open; SAFE-HABIT cues       |
| <b>Khan energy / points-ish progress</b>       | Mild completion signals; mostly mastery bars                                            | Keep criterion progress; avoid coinization of minutes                           |
| <b>Brilliant delight unlocks</b>               | Intrinsic interest scaffold; some completion loot                                       | Keep delight <i>inside</i> the problem; don't pay for opening                   |
| <b>ChatGPT “I’ll quiz you if you want”</b>     | Low token reward; high fluency risk (XXXIII)                                            | Autonomy of asking ≠ competence evidence; measure solo transfer                 |
| <b>Parent dashboards: minutes / day streak</b> | Parent-facing controlling contingency spills to student                                 | Report retry_120s, transfer, challenge_accept — not “minutes bought with coins” |

**Bridge:** XLIII says cue the practice, not the counter. XLIV says: if you must reward, reward **informative competence**, not engagement.

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## XLIV.7 Resolution doctrine — SAFE-REWARD

**FOUNDER BELIEF / HYPOTHESIS — SAFE-REWARD stack (ties TARGET / SAFE-COMPARE / SAFE-HABIT / FEI):**

1. **No expected tangible / token for mere engagement.** Ban default XP-for-open, coins-for-any-answer, loot-for-login. (Deci 1999 engagement-contingent.)
2. **Feedback > inducement.** Soft-wrong, strategy labels, bridge-gap naming, and “you recovered” tutor notes are the primary reinforcer. Prefer unexpected acknowledgment over announced prizes.
3. **If performance-contingent, mark real competence.** Badges/unlocks tied to transfer\_pass / solo\_transfer\_pass / mastery-coded challenge\_accept — not to streak days or raw volume. Administer without “you should” pressure.

4. **Autonomy-supportive framing.** Offer rationale; allow meaningful micro-choice (Patall); never “do this or lose the reward.”
5. **Don’t stack choice with dangling rewards.** Choice then immediate coin undermines the autonomy gain.
6. **Child-risk premium.** Maya-aged users are in the higher tangible-harm band of Deci 1999 — stricter default than adult SaaS gamification norms.
7. **Metrics hygiene.** Track XP/badge counts as *diagnostics of system behavior*, never as North Stars. Learning NS remain retry / transfer / solo transfer / identity.
8. **Parent copy:** Sell “she chose the harder one again” and competence evidence — not “she earned 400 XP.”

**Kill the false dichotomy:** “No rewards ever” vs “gamify everything.” Refuse *controlling engagement loot*; keep *informational competence signals* and story unlocks that are earned by transfer, not by attendance.

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## XLIV.8 Marketing language that survives

- **Allowed:** “We don’t pay kids to open the app — we show them what they figured out.”
- **Allowed:** “Unlocks follow real transfer, not login streaks.”
- **Allowed (parent):** “Progress you can trust: recovery and transfer — not points theater.”
- **Banned:** “Rewards create love of math.”
- **Banned:** “XP builds identity.”
- **Banned:** Absolute “all extrinsic motivation is bad” / “tutoring is free so motivation is free.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streak/XP as North Star.

### Feature claim ladder (Part XXXIV hygiene)

| Feature                                                           | Claim max without IM data            |
|-------------------------------------------------------------------|--------------------------------------|
| Informational soft-wrong + strategy feedback (no engagement loot) | L1 CET-aligned hygiene               |
| Unexpected tutor/parent acknowledgment after recovery             | L1 (unexpected tangible / social)    |
| “Our XP system raises intrinsic math interest”                    | L0 until IM-1/IM-2                   |
| Expected coins for any completion as default                      | L0 — engagement-contingent kill zone |

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## XLIV.9 Experiments spawned

| ID | Question | Design | Primary | Kill condition |
|----|----------|--------|---------|----------------|
|----|----------|--------|---------|----------------|

|         |                                                                                                                         |                          |                                                                                                |                                                                                               |
|---------|-------------------------------------------------------------------------------------------------------------------------|--------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| IM-1    | Engagement-loot (XP for any attempt) vs no-loot informational feedback (same missions)                                  | 2-arm; new users         | Free-choice analog: voluntary hard-problem rate post-session; 14d interest item; transfer_pass | Loot arm wins free-choice <b>and</b> transfer → engagement-loot ban overstated                |
| IM-2    | Performance badge on transfer_pass (informational copy) vs same badge with controlling copy (“you should keep earning”) | 2-arm                    | challenge_accept motives; interest; return                                                     | Controlling ≥ informational on interest → climate claim weak                                  |
| IM-3    | Micro-choice without reward vs choice + dangling coins                                                                  | 2-arm (Patall moderator) | Intrinsic motivation proxy; effort; autonomy need satisfaction                                 | Choice+coins ≥ choice-only → “don’t stack” overstated                                         |
| IM-4    | Unexpected recovery acknowledgment vs announced prize for recovery                                                      | 2-arm                    | Free-choice return; attribution coding                                                         | Announced ≥ unexpected on free-choice without control feelings → unexpected preference weaker |
| IM-QUAL | 10 Maya interviews: “When a learning app gives points, what does that make you feel about the math?”                    | Appendix B               | coded controlling vs informational                                                             | Nobody reports control → copy over-indexed                                                    |

**Pre-reg (XXXIV):** IM-\* identify **reward contingency** × **free-choice/interest/transfer**. They do **not** identify “gamification builds mathematicians” from XP velocity.

**Densifies:** TARGET appearance ban; SAFE-HABIT; SAFE-COMPARE; Insight on extrinsic traps; standing streak NS ban.

## XLIV.10 Confidence table

| Claim | Label | Confidence |
|-------|-------|------------|
|-------|-------|------------|

|                                                                                                 |                                       |                  |
|-------------------------------------------------------------------------------------------------|---------------------------------------|------------------|
| Autonomy, competence, relatedness support intrinsic motivation / wellness                       | FACT (Ryan & Deci, 2000)              | High             |
| Expected engagement-/completion-contingent tangible rewards undermine free-choice IM on average | FACT (Deci, Koestner & Ryan, 1999)    | High             |
| Unexpected tangible rewards do not show average undermine                                       | FACT (Deci et al., 1999)              | High             |
| Positive feedback enhances IM on average; controlling administration can reverse                | FACT (Deci et al., 1999; Ryan line)   | High             |
| Tangible rewards more detrimental for children than college samples                             | FACT (Deci et al., 1999)              | High             |
| Cameron & Pierce find overall null with circumscribed negatives                                 | FACT (1994); interpretation contested | Medium (dispute) |
| Choice enhances IM / effort / competence perceptions                                            | FACT (Patall et al., 2008)            | High             |
| Stacking rewards after choice weakens choice benefit                                            | FACT (Patall moderator)               | Medium–High      |
| SAFE-REWARD is right default for MindCraft                                                      | FOUNDER BELIEF / HYPOTHESIS           | Medium–High      |
| “All rewards kill intrinsic motivation” / “XP creates math love”                                | SPECULATION / false as universals     | Low (against)    |

## XLIV.11 What this chapter kills

1. **Kill:** Expected engagement-contingent XP/coins/loot as default motivation architecture for Maya. (Deci 1999  $d \approx -0.40$  free-choice.)
2. **Kill:** Marketing that “rewards create love of math” or “XP = identity.” (External PLOC; HID.)
3. **Kill:** Absolute slogan “all extrinsic rewards are bad” used to ban informational feedback and earned transfer unlocks. (CET differentiation; Cameron caveat acknowledged.)
4. **Kill:** Controlling praise / “you should keep earning” as the voice of competence support.
5. **Wound:** “Performance badges are always safe” — survives only with informational climate + real competence criterion.
6. **Survive (constrained):** Unexpected acknowledgment, strategy-informative feedback, autonomy-supportive micro-choice, and transfer-gated unlocks under **SAFE-REWARD**.

**Doctrine until data:** Ship **SAFE-REWARD**: no expected tokens for mere engagement → feedback over inducement → performance marks only for real transfer/recovery → autonomy framing → don’t stack choice with dangling coins

→ child-risk premium → XP never North Star. Market “we don’t pay kids to practice — we make the competence visible.”

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## Part XLV — Mathematical Resilience: Growth Under Adversity Without Grit Theater

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 12 ≡ Red Team slot, but ch45 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.6 → not yet Synthesizer)

**Primary question:** What is *mathematical resilience* as an actionable construct (Johnston-Wilder & Lee), how does it differ from grit posters and empty “bounce back” copy, and what product stack keeps Maya in the growth zone without dumping her into the danger zone?

**Owners:** Motivation / Affect · Product (Practice / coach / tutor briefing) · Positioning · Red Team

**Commercial job:** Replace “build grit” / “resilience mindset” marketing with a shippable **SAFE-RESILIENCE** stack competitors can be audited against — growth-zone design, support recruitment, struggle-as-normal, value without coercion — tied to FEI / SAFE-DD / SAFE-REWARD.

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### XLV.1 Why this chapter exists

Parts XXIV (anxiety), XXV (self-efficacy / habit≠identity), XXX (attribution / helplessness), XXXI (flow / challenge–skill), XLI (SAFE-DD), and XLIV (SAFE-REWARD) already treat threat, recovery, and controlling incentives. Market and school language still collapses everything into **grit** or generic **resilience** slogans: “just push through,” “bounce back harder,” streak-as-character, leaderboard-as-grit-proof.

**FOUNDER BELIEF under audit:** Identity transformation under difficulty needs *domain-specific* resilience — knowing struggle is normal in math, valuing the work, knowing how to work at it, and **recruiting support** before anxiety flips the lid — not a personality transplant labeled grit.

**Claims we refuse as doctrine:**

1. Duckworth-style grit as MindCraft’s North Star or primary product claim (Credé et al. wound the construct).
  2. “Desirable difficulty” = keep students in the danger/anxiety zone (conflicts SAFE-DD / Ashcraft).
  3. Empty resilience posters without support architecture (tutor cards, peers, ontology scaffolds).
  4. Treating mathematical resilience as synonymous with growth-mindset wall art.
- 

### XLV.2 Constructs (product language)

| Construct | Plain definition | Product signal | Failure mode if misused |
|-----------|------------------|----------------|-------------------------|
|-----------|------------------|----------------|-------------------------|

|                                          |                                                                                                                                                         |                                                                             |                                                             |
|------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-------------------------------------------------------------|
| <b>Developmental resilience (Masten)</b> | Ordinary adaptive systems enabling competence despite adversity — not rare “super kids”                                                                 | Design that protects basic systems (safety, belonging, competence feedback) | Selling “extraordinary character” as the product            |
| <b>Mathematical resilience (JW/Lee)</b>  | Positive stance that lets learners engage despite affective barriers: growth belief, value, knowing how to work at math, knowing how to recruit support | Stay in growth zone; ask for help; recover after stuck; continue study path | Posterizing the four aspects without UX                     |
| <b>Growth Zone Model</b>                 | Comfort (easy / little growth) → Growth (challenge + support) → Danger/Anxiety (threat / thinking collapses)                                            | Challenge calibrated + support on tap; exit danger fast                     | Framing red-zone panic as “grit training”                   |
| <b>Grit (Duckworth)</b>                  | Perseverance + consistency of interest toward long-term goals                                                                                           | Tempting NS copy                                                            | Near-conscientiousness ; weak intervention leverage (Credé) |
| <b>SAFE-RESILIENCE</b>                   | Equip → name struggle → recruit support → destake danger → measure recovery/transfer — never grit KPI                                                   | See XLV.7                                                                   | Resilience theater that still shames help-seeking           |

**Operational definition (HYPOTHESIS):** A MindCraft session builds mathematical resilience only when the student (a) experiences struggle as *expected* not as personal defect, (b) has a clear path to **support** (cards, tutor, peer, worked scaffold) before threat spikes, (c) returns to challenge after recovery (retry\_120s / challenge\_accept), and (d) can transfer without the model finishing the thought — not when streak or “grit score” rises.

### XLV.3 Ordinary magic — resilience is not a unicorn trait

**FACT:** Masten (2001, *American Psychologist*, 56(3), 227–238, doi:10.1037/0003-066X.56.3.227) — resilience in development is often **ordinary magic**: competence despite adversity typically arises from normative adaptational systems (attachment, learning systems, self-regulation, community supports). Greatest threats are those that *compromise* these protective systems. Policy implication: protect and restore ordinary systems rather than hunt rare resilient personalities.

**Product translation:** MindCraft should not market “we forge resilient geniuses.” Market “we keep the ordinary protective systems online while math gets hard” — safety, relatedness (tutor/world), competence information, recoverable miss. That is Masten-aligned and FEI-aligned.

**Kill:** “Resilience is a rare character trait we select for.”

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## XLV.4 Mathematical resilience as a pragmatic construct

**FACT (origin framing):** Johnston-Wilder & Lee (2010) introduced **mathematical resilience** as a pragmatic positive construct opposite avoidance, anxiety, and helplessness — enabling learners to engage with mathematics despite affective barriers (cited as foundational in later scale and handbook work; e.g. Lee & Johnston-Wilder construct summaries; Johnston-Wilder & Lee handbook chapter, doi:10.4324/9781315740713-34).

**FACT (four aspects — construct literature):** Lee & Johnston-Wilder summarize mathematical resilience as four pragmatic aspects:

1. **Growth** — belief that mathematical capability can develop with effort and appropriate support (growth/incremental theory of learning).
2. **Value** — mathematics has personal / world value and the learner is valued in the learning community.
3. **Knowing how to work at mathematics** — struggle is expected; strategies exist for stuckness.
4. **Knowing how to recruit support** — peers, adults, resources; stuck is not permanent solitude.

**FACT (Growth Zone Model):** Johnston-Wilder et al. (2013 onward; e.g. pre-service teacher applications, Warwick WRAP) — three zones:

- **Comfort (green):** routine; little growth.
- **Growth (amber/yellow):** challenge requiring support, persistence, agency; productive struggle.
- **Danger / anxiety (red):** challenge perceived beyond capability even with support; survival response; repeated exposure → persistent anxiety / avoidance.

**HYPOTHESIS (product):** MindCraft’s “soft-wrong + cards + tutor” stack is literally a **support-recruitment machine** for staying in the growth zone. Competitors that only escalate difficulty (or shame help) shove Maya into red.

**Bridge to SAFE-DD (XLI):** Desirable difficulties belong in the *growth* zone with equipment and destaking — not in the danger zone. “Struggle” ≠ “panic.”

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## XLV.5 Measurement — MRS is attitudes, not magic scores

**FACT:** Kooken, Welsh, McCoach, Johnston-Wilder & Lee (2016, *Measurement and Evaluation in Counseling and Development*, 49(3), 217–242, doi:10.1177/0748175615596782) — developed and validated the **Mathematical Resilience Scale (MRS)** via EFA/CFA across three samples. Three correlated factors: **Value**, **Struggle**, and **Growth**. Positioned as a way to gauge likelihood of participation and persistence in mathematics.

**Product implication (HYPOTHESIS):** MRS-style items (value / struggle-as-normal / growth belief) can inform parent research and longitudinal identity panels — but they are **not** a North Star and not a gamified in-app grit meter. Self-report attitudes ≠ transfer. Pair with `retry_120s`, `challenge_accept`, `transfer_pass`, help-recruitment events (opened card / asked tutor / peer).

**Kill:** Shipping a “Resilience Score™” badge as proof of identity change (L0 → L3 leap; Part XXXIV hygiene).

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## XLV.6 Grit theater — what we refuse to sell

**FACT (origin):** Duckworth, Peterson, Matthews & Kelly (2007, *Journal of Personality and Social Psychology*, 92(6), 1087–1101, doi:10.1037/0022-3514.92.6.1087) — grit as perseverance and passion for long-term goals; predictive claims in selective samples fueled popular education discourse.

**FACT (meta wound):** Credé, Tynan & Harms (2017, *Journal of Personality and Social Psychology*, 113(3), 492–511, doi:10.1037/pspp0000102) — 584 effects, 88 samples,  $N \approx 66,807$ : higher-order grit structure not confirmed; grit **very strongly** overlaps conscientiousness; overall grit–performance/retention relations **modest**; perseverance-of-effort facet carries more criterion validity than consistency-of-interest; **interventions to enhance grit may have only weak effects**; construct validity questioned; primary utility may lie in perseverance facet alone.

**FOUNDER BELIEF / resolution:** Perseverance *after equipped struggle* remains commercially meaningful (retry\_120s). Packaging that as Duckworth “grit” invites overclaim, personality fatalism, and equity harm (blame the child’s character when curriculum/support failed). Prefer **mathematical resilience** language: growth zone + support recruitment + struggle-as-normal.

**Standing Constitution already warned** (core OS): grit/resilience hype is contested; structural inequality and curriculum quality matter more than character slogans. This chapter densifies that kill with Credé + JW/Lee specificity.

**Kill:** “MindCraft builds grit.” **Survive (constrained):** “MindCraft keeps you in the growth zone — challenge with support, recovery without shame.”

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## XLV.7 Competitive audit — who shoves students into the red zone?

| Pattern                                                        | Zone reading                                                          | MindCraft wedge                                                |
|----------------------------------------------------------------|-----------------------------------------------------------------------|----------------------------------------------------------------|
| <b>Duolingo streak shame / league pressure</b>                 | Danger + appearance climate (XLII/XLIII); “grit” via fear of break    | Miss ≠ identity failure; cue practice not counter; no grit KPI |
| <b>Khan mastery lock without affect design</b>                 | Can strand in stuck without emotional exit tools                      | Soft-wrong + ingredient cards as support recruitment           |
| <b>Brilliant hard delight</b>                                  | Often growth-zone when curiosity high; weak for high-anxiety Maya     | SAFE-DD equip/destake before delight escalation                |
| <b>Unguarded ChatGPT “just try harder / here’s the answer”</b> | Either skips struggle (comfort via answer) or fluent wrong confidence | PWC + solo transfer; support = pedagogy wrap not answer dump   |
| <b>School “resilience assembly”</b>                            | Poster without recruit-support architecture                           | Operationalize support buttons in-session                      |

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## XLV.8 Resolution doctrine — SAFE-RESILIENCE

FOUNDER BELIEF / HYPOTHESIS — SAFE-RESILIENCE stack (ties FEI / SAFE-DD / TARGET / SAFE-REWARD / EVT value):

- 1. **Name the zones in product language.** Comfort / growth / danger — coach and tutor briefs use this, not “be gritty.”
  - 2. **Equip before escalate.** SAFE-DD first: representations, faded examples, destaked attempts — then raise challenge.
  - 3. **Struggle is expected, not character failure.** Attribution copy (XXX): strategy and effort-with-strategy; never “you’re not resilient enough.”
  - 4. **Support recruitment is a first-class action.** Cards, ask-tutor, peer, worked scaffold — log as positive resilience behavior, not as “cheating the streak.”
  - 5. **Exit the danger zone fast.** Anxiety spike → reduce stakes, shorten item, switch representation, offer support — do not double dose “desirable difficulty.”
  - 6. **Value without coercion.** Eccles utility/attainment (XXXVII) and story worlds — not “suffer because grit.”
  - 7. **Measure recovery and transfer, not grit scores.** retry\_120s, help-then-solo sequences, challenge\_accept, transfer\_pass / solo\_transfer\_pass; optional MRS-like survey offline.
  - 8. **Parent copy:** Sell “she got stuck, recruited help, and came back” — not “she’s a gritty kid now.”
- Kill the false dichotomy:** “Coddle forever in comfort” vs “forge character in the red zone.” Refuse both. Ship growth-zone challenge with support on tap.

XLV.9 Marketing language that survives

- **Allowed:** “Hard math with a way out — support when you’re stuck, challenge when you’re ready.”
- **Allowed:** “Stuck isn’t failure. Stuck is when you recruit the next tool.”
- **Allowed (parent):** “We train recovery and transfer — not grit slogans.”
- **Banned:** “We build grit.” / “Resilience score.” / “Push through the panic.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as proof of resilience; absolute tutoring-is-free.

Feature claim ladder (Part XXXIV hygiene)

| Feature                                                   | Claim max without RES data     |
|-----------------------------------------------------------|--------------------------------|
| Growth-zone framing + support CTAs in coach               | L1 hygiene aligned with JW/Lee |
| Soft-wrong + card recruitment → retry                     | L1 mechanism claim             |
| “Raises mathematical resilience (MRS)” as product outcome | L0 until RES-1                 |
| “Builds grit / character”                                 | L0 — banned packaging          |

XLV.10 Experiments spawned

| ID | Question | Design | Primary | Kill condition |
|----|----------|--------|---------|----------------|
|----|----------|--------|---------|----------------|

|          |                                                                                           |                           |                                                             |                                                                                                                    |
|----------|-------------------------------------------------------------------------------------------|---------------------------|-------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------|
| RES-1    | Growth-zone coach copy + explicit support CTA vs grit/perseverance copy (same difficulty) | 2-arm; anxiety-stratified | retry_120s; help-recruit rate; post affect; transfer_pass   | Grit copy $\geq$ growth-zone on retry+transfer without raising danger affect $\rightarrow$ zone framing overstated |
| RES-2    | Support-first after miss (card/tutor offer) vs no offer (same soft-wrong)                 | 2-arm                     | Time-to-retry; danger-zone self-report; solo transfer later | No-offer $\geq$ support-first on retry without harm $\rightarrow$ recruitment less load-bearing                    |
| RES-3    | Escalate difficulty after equip vs escalate without equip (SAFE-DD $\times$ RES)          | 2 $\times$ 2              | Challenge_accept ; WM/anxiety proxy; transfer               | Unequipped escalate wins transfer in high-anxiety $\rightarrow$ SAFE-DD sequencing weak                            |
| RES-4    | MRS Value/Struggle/Growth $\Delta$ over 4 weeks of SAFE-RESILIE NCE UX vs control         | Survey + behavior         | MRS factors + behavioral NS                                 | MRS rises, behavior flat $\rightarrow$ attitude theater                                                            |
| RES-QUAL | 10 Maya interviews: “When you’re stuck in math, what does ‘being resilient’ mean?”        | Appendix B                | coded grit-vs-support meanings                              | Everyone already means support recruitment $\rightarrow$ copy shift lower priority                                 |

**Pre-reg (XXXIV):** RES-\* identify **zone framing**  $\times$  **support recruitment**  $\times$  **recovery/transfer**. They do **not** identify “we create resilient mathematicians” from MRS alone.

**Densifies:** SAFE-DD; FEI; TARGET; standing grit-hype ban; help-seeking as virtue not streak cheat.

## XLV.11 Confidence table

| Claim                                                                  | Label               | Confidence |
|------------------------------------------------------------------------|---------------------|------------|
| Developmental resilience often arises from ordinary protective systems | FACT (Masten, 2001) | High       |

|                                                                                                             |                                      |                                                             |
|-------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------------------------------------------------------|
| Mathematical resilience = growth + value + how-to-work + recruit support                                    | FACT (JW/Lee construct line)         | High (construct); Medium (intervention effect sizes sparse) |
| Growth Zone Model: comfort / growth / danger                                                                | FACT (JW et al. model literature)    | High as pedagogy model; Medium as RCT-proven UX             |
| MRS measures Value, Struggle, Growth attitudes                                                              | FACT (Kookken et al., 2016)          | High                                                        |
| Grit higher-order structure weak; overlaps conscientiousness; modest prediction; weak intervention leverage | FACT (Credé et al., 2017)            | High                                                        |
| Danger-zone exposure as “training” harms anxious learners                                                   | HYPOTHESIS (Ashcraft line + SAFE-DD) | Medium–High                                                 |
| SAFE-RESILIENCE is right default for MindCraft                                                              | FOUNDER BELIEF / HYPOTHESIS          | Medium–High                                                 |
| “Grit posters raise transfer” / “Resilience Score™ = identity”                                              | SPECULATION / false as doctrine      | Low (against)                                               |

## XLV.12 What this chapter kills

1. **Kill:** Grit / “build character” as MindCraft marketing North Star. (Credé 2017; Constitution grit warning.)
2. **Kill:** Treating panic / danger-zone as desirable difficulty. (Growth Zone Model + SAFE-DD.)
3. **Kill:** Resilience as rare trait branding (“extraordinary kids”). (Masten ordinary magic.)
4. **Kill:** In-app Resilience Score™ as proof of identity transformation. (MRS ≠ transfer; claim ladder.)
5. **Wound:** Growth-mindset language alone — survives only inside growth + value + struggle know-how + **support recruitment**, not as wall art.
6. **Survive (constrained):** Growth-zone challenge with support on tap; struggle-as-normal; recovery metrics; parent story of “stuck → help → return.”

**Doctrine until data:** Ship **SAFE-RESILIENCE**: name zones → equip before escalate → struggle expected → support recruitment first-class → exit danger fast → value without coercion → measure recovery/transfer not grit. Market “hard math with a way out.”

## Part XLVI — Teacher/Tutor Expectancy: Pygmalion Without Magic, Golem Without Labels

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 15; hour%6≠0 → Researcher; 5 researcher entries since synthesizer v1.6 → not yet Synthesizer)

**Primary question:** What do teacher/tutor expectancy effects actually justify for MindCraft — and what briefing ethics keep college tutors from installing a Golem on Maya?

**Owners:** Tutor ops / Product (briefing cards) · Equity · Positioning · Red Team

**Commercial job:** Replace “we believe in every student” poster copy and “Pygmalion IQ” folklore with a shippable **SAFE-EXPECTANCY** stack — evidence-based briefs, class-level high challenge + support, ban stigmatizing labels, measure mediation behaviors not vibes.

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## XLVI.1 Why this chapter exists

Parts XXV (self-efficacy), XXX (attribution), XXXVI (equity), XXXVII (EVT expectancy), and XLV (support recruitment) already treat *student-side* “can I?” and belonging. They do not fully answer: **what beliefs do human tutors carry into the session, how do those beliefs leak into climate/input/output/feedback, and how should MindCraft’s pre-session briefing be designed so gap-scan weakness does not become a self-fulfilling low label?**

**FOUNDER BELIEF under audit:** A knowledge-graph + tutor hybrid only compounds identity if the *human* in the loop treats Maya as capable of the next hard step with support — not as a fixed “weak student” whose low mastery scores license colder climate, thinner input, and fewer chances to respond.

**Claims we refuse as doctrine:**

1. Marketing the classic Pygmalion IQ story as MindCraft’s differentiator (“our tutors raise IQ by believing harder”).
  2. Briefing tutors with trait labels (“low ability,” “slow,” “not STEM material”) drawn from gap-scan or weakness ranking.
  3. Treating teacher-expectancy literature as proof that positive vibes alone move transfer.
  4. Ignoring Golem / differential-treatment risks for stigmatized or “weakness-tagged” students.
- 

## XLVI.2 Constructs (product language)

| Construct                                      | Plain definition                                                                 | Product signal                                          | Failure mode if misused                         |
|------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------|-------------------------------------------------|
| <b>Induced expectancy (Pygmalion paradigm)</b> | Authority given false “bloomers” info → treats them differently → outcomes shift | Early-session / cold-start briefing window              | Selling IQ magic; ignoring replication limits   |
| <b>Accuracy vs prophecy</b>                    | Expectations often predict because they track real prior performance             | Gap-scan / graph as <i>task</i> evidence, not character | Confusing prediction with causation             |
| <b>Golem effect</b>                            | Low expectations → colder / more dogmatic treatment → worse performance          | Ban negative trait labels in tutor cards                | “Honest” weakness dump that licenses withdrawal |

|                                               |                                                                                                       |                                                   |                                        |
|-----------------------------------------------|-------------------------------------------------------------------------------------------------------|---------------------------------------------------|----------------------------------------|
| <b>Four-factor mediation</b>                  | Climate, input, output opportunity, feedback carry expectancy into behavior                           | Audit tutor/coach on those four channels          | Soft climate without hard input/output |
| <b>High-expectation teacher (class-level)</b> | High challenge + high support for <i>all</i> , not only nominated stars                               | Tutor playbook: everyone gets hard Qs + scaffolds | Favorites / “bloomers only” tracks     |
| <b>SAFE-EXPECTANCY</b>                        | Evidence brief → high challenge-with-support → no trait labels → mediate via CIOF → measure behaviors | See XLVI.7                                        | Belief posters; Golem briefs           |

**Operational definition (HYPOTHESIS):** A MindCraft tutor session is expectancy-*safe* only when (a) the brief states **next playable tasks and scaffolds**, not trait judgments, (b) the tutor offers **hard-enough** work with support on tap (XLV growth zone), (c) climate stays warm under struggle, and (d) the student gets **output opportunity** (speak/solve before tutor finishes) — not when the tutor “believes harder” while finishing the math.

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### XLVI.3 Origin story — real, contested, not marketing copy

**FACT:** Rosenthal & Jacobson (1968, *Pygmalion in the Classroom*; also *Urban Review* summary, doi:10.1007/BF02322211) — teachers told that randomly selected elementary students were poised for intellectual blooming; those students later showed larger IQ gains on average in the reported analyses, especially in early grades. The study launched classroom expectancy research and intense methodological critique (e.g. Thorndike and later replication debates).

**FACT (meta, induction credibility):** Raudenbush (1984, *Journal of Educational Psychology*, 76(1), 85–97, doi:10.1037/0022-0663.76.1.85) — synthesis of 18 Pygmalion-style experiments: expectancy effects on pupil IQ are larger when teachers have had **little prior contact** with pupils at induction ( $\leq 2$  weeks); effects shrink when teachers already know students well. Grade pattern: stronger early-grade effects in the corpus; design features of induction matter.

**Product translation:** MindCraft’s **first tutor sessions and cold-start briefs** are the high-leverage expectancy window — analogous to early-year induction. Once a tutor has rich behavioral evidence, “random bloomers” folklore is the wrong metaphor; **accurate, task-level** expectations dominate.

**Kill:** “Pygmalion proves we raise IQ by believing.” Sell mediation design in the first hours of contact, not IQ alchemy.

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### XLVI.4 What 35 years actually say (Jussim & Harber)

**FACT (review):** Jussim & Harber (2005, *Personality and Social Psychology Review*, 9(2), 131–155, doi:10.1207/s15327957pspr0902\_3) — after ~35 years:

1. Self-fulfilling prophecies in classrooms **do occur**, but effects are **typically small**, do not accumulate greatly across perceivers/time, and may dissipate more than snowball.

2. **Powerful** effects may selectively hit students from **stigmatized** social groups.
3. Whether expectancy effects reliably change *intelligence*, and whether they do more harm than good in general, remains unclear.
4. Teacher expectations often predict outcomes **because they are accurate**, not only because they are self-fulfilling.

**HYPOTHESIS for MindCraft:** Gap-scan / mastery graphs will *accurately* predict struggle. Accuracy is not permission to install a Golem. The commercial risk is **translating accurate weakness into low-challenge, low-warmth treatment** — exactly the path that can still produce small-but-real prophecy costs, amplified for already-stereotyped students (ties Part XXXVI equity).

**Wound:** “Expectancy effects are huge and ubiquitous.” Survives only as: small average effects; context/stigma moderators; accuracy dominates prediction.

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## XLVI.5 Mediation — climate, input, output, feedback

**FACT (behavioral mediation):** Brophy & Good (1970, *Journal of Educational Psychology*, doi:10.1037/h0029908) — observational study: teachers demanded better performance and praised it more for high-expectation students; accepted poor performance more readily from lows and were less likely to praise good performance from lows when it occurred.

**FACT (four-factor meta):** Harris & Rosenthal (1985, *Psychological Bulletin*, 97(3), 363–386, doi:10.1037/0033-2909.97.3.363) — 31 meta-analyses of mediating behaviors; Rosenthal’s four-factor frame (**climate, feedback, input, output/opportunity**) organizes how expectancies become treatment. Expectancy→behavior and behavior→outcome links support multiple mediators (e.g. warmer climate, longer/richer interaction, questioning). Later summaries often note **feedback** as the weaker of the four average links, pushing affect/effort emphasis.

**Product checklist (HYPOTHESIS — CIOF audit):**

| Channel         | Tutor/coach do                                        | Anti-pattern                                      |
|-----------------|-------------------------------------------------------|---------------------------------------------------|
| <b>Climate</b>  | Warmth under struggle; curiosity about wrong paths    | Cold distance after “weak” brief                  |
| <b>Input</b>    | Richer explanations, multiple representations (cards) | Thin “just try” for lows; lectures only for highs |
| <b>Output</b>   | Wait time; let Maya finish; SE prompts (Part XL)      | Tutor finishes every thought (Solver crutch)      |
| <b>Feedback</b> | Criterion + strategy (XXX); soft-wrong                | Praise only highs; criticism without how-to       |

**Kill:** Climate-only positivity (“you’ve got this!”) without input/output redesign — that is poster Pygmalion.

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## XLVI.6 Golem risk — negative expectancy is the ops hazard

**FACT:** Babad, Inbar & Rosenthal (1982, *Journal of Educational Psychology*, 74(4), 459–474, doi:10.1037/0022-0663.74.4.459) — distinguished positive expectancy outcomes (“Galatea”) from **negative** ones (“Golem”). Teachers high in susceptibility to biasing information treated low-potential nominees more negatively (notably dogmatic behavior); unbiased teachers treated groups more equitably. Differential negative treatment showed up in teacher behavior **and** student task performance.

**FACT (student-perceived differentiation):** Brattesani, Weinstein & Marshall (1984, *Journal of Educational Psychology*, 76(2), 236–247, doi:10.1037/0022-0663.76.2.236) — in classrooms where students perceive **high differential treatment**, teacher expectations contribute more to achievement than in low-PDT classrooms. Students can see the track.

**Product translation:** A tutor card that says “Maya is weak on linear equations / low confidence / hard gap-scan” is biasing information. High-bias tutors will leak Golem through climate and dogmatism. Low-PDT design = same high standards + differentiated *scaffolds*, not differentiated *respect*.

**Kill:** “Honest labels help tutors calibrate.” Honest **task evidence** helps; honest **trait verdicts** harm.

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## XLVI.7 High-expectation teachers — class-level, not favorites

**FACT:** Rubie-Davies (2007, *British Journal of Educational Psychology*, doi:10.1348/000709906X101601) — contrasts high- vs low-expectation *teachers* (expectations for the class as a whole). High-expectation teachers spent more time teaching, oriented lessons, linked prior knowledge, explained carefully, asked mixed open/closed questions including **high-level questions of all students**, and managed more positively; lows were more dismissive of incorrect answers and relied on closed questioning.

**HYPOTHESIS:** MindCraft’s competitive wedge vs “AI that flatters then finishes” and vs “tutor who only drills the labeled weak topic gently” is **Rubie-Davies-shaped ops**: every student gets high-level questioning + scaffolds; incorrect answers get learning support, not dismissal.

**Bridge to SAFE-RESILIENCE / SAFE-DD:** High expectation ≠ danger zone. High expectation = growth-zone challenge with support recruitment — not grit theater and not thin input for “lows.”

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## XLVI.8 Competitive audit (expectancy lens)

| Competitor pattern | Expectancy mechanism                                                                 | MindCraft counter                              |
|--------------------|--------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Duo</b>         | Habit/streak as “you’re the type who shows up”; little academic expectancy mediation | Transfer + CIOF; no streak-as-belief           |
| <b>Khan</b>        | Self-serve library; weak human expectancy channel                                    | Human tutor brief + graph without Golem labels |
| <b>Brilliant</b>   | Prestige challenge; can cue performance-approach for novices                         | Mastery structures + expectancy-managed gates  |



|                                |                                                                                          |                                                   |
|--------------------------------|------------------------------------------------------------------------------------------|---------------------------------------------------|
| <b>ChatGPT Base</b>            | Sycophantic high-warmth climate + answer dump (fake high expectancy, zero output demand) | Warm climate <b>plus</b> forced solo output / PWC |
| <b>Unbriefed college tutor</b> | Forms expectancies from accent, race, “seems lost,” or raw weakness dump                 | SAFE-EXPECTANCY brief + playbook                  |

## XLVI.9 Resolution doctrine — SAFE-EXPECTANCY

### FOUNDER BELIEF / HYPOTHESIS — SAFE-EXPECTANCY stack:

- 1. Brief tasks, not traits.** Tutor card: current concept, recommended level, formats struggled, successful scaffolds last time, next playable question IDs — **never** “low ability / not a math person.”
- 2. Protect the cold-start window.** First sessions: explicit high-challenge-with-support script (Raudenbush induction logic).
- 3. Class-level high expectation.** Rubie-Davies: hard questions for everyone; wrong answers → teach, don’t dismiss.
- 4. Mediate via CIOF.** Climate warmth + rich input (cards) + output opportunity (wait / SE) + strategy feedback — audit these, not “belief intensity.”
- 5. Accuracy without Golem.** Graph weakness drives *scaffold choice*, not *respect withdrawal*.
- 6. Stigma premium.** Extra review when briefs could activate stereotype (XXXVI); ban demographic proxies in tutor UI.
- 7. AI coach inherits the same rules.** No “you’re behind your peers” cold open; no finishing her thought under “support.”
- 8. Measure mediation + FEI, not IQ folklore.** Tutor wait-time, student talk ratio, help-then-solo, `retry_120s`, `transfer_pass` / `solo_transfer_pass`.

**Kill the false dichotomy:** “Ignore data, just believe” vs “dump the weakness label so tutors are realistic.” Refuse both. Ship **evidence-based high challenge with equitable treatment**.

## XLVI.10 Marketing language that survives

- **Allowed:** “Tutors see what to teach next — not who she is forever.”
- **Allowed:** “High challenge for everyone, support on tap — no tracks of respect.”
- **Allowed (parent):** “We brief tutors on the next skill, not on a fixed ability story.”
- **Banned:** “Pygmalion IQ gains.” / “Believe and they bloom.” / “We tell tutors who’s weak so they know.”
- **Banned:** Bloom 2-sigma; empty mindset posters; streaks as expectancy proof; tutoring-is-free.

### Feature claim ladder (Part XXXIV hygiene)

|         |                            |
|---------|----------------------------|
| Feature | Claim max without EXP data |
|---------|----------------------------|

|                                             |                                                 |
|---------------------------------------------|-------------------------------------------------|
| Trait-label ban + task-only tutor brief     | L1 hygiene aligned with Golem literature        |
| CIOF playbook / wait-time coaching          | L1 mechanism claim                              |
| “Raises IQ via tutor expectancy”            | L0 — banned                                     |
| “Eliminates achievement gaps via Pygmalion” | L0 until EXP-* + equity design (Jussim caveats) |

## XLVI.11 Experiments spawned

| ID       | Question                                                                                     | Design                       | Primary                                                   | Kill condition                                                                                    |
|----------|----------------------------------------------------------------------------------------------|------------------------------|-----------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| EXP-1    | Task-only brief vs trait/weakness-label brief (same content otherwise)                       | 2-arm tutors; record session | Student talk ratio; wait-time; post affect; transfer_pass | Label brief ≥ task-only on transfer without colder climate → Golem risk overstated for our tutors |
| EXP-2    | High-expectation playbook (hard Q + scaffold) vs “remediate gently only” after same gap-scan | 2-arm                        | challenge_accept; solo transfer; help-recruit             | Gentle-only wins transfer+return in high-anxiety → Rubie-Davies transfer fails for Maya           |
| EXP-3    | Cold-start script (first 2 sessions) vs unstructured tutor open                              | 2-arm new matches            | FEI events week 1; retention to session 3                 | Unstructured ≥ scripted → induction window less load-bearing                                      |
| EXP-4    | AI coach: warm+finish vs warm+output-demand (SE/wait)                                        | 2-arm                        | solo_transfer_pass; PWC                                   | Warm+finish ≥ warm+output on solo transfer → climate-only expectancy myth revived                 |
| EXP-QUAL | 10 tutor + 10 Maya: “What did the brief make you assume?”                                    | Interview                    | coded trait vs task inferences                            | Tutors already ignore labels → UI priority shifts                                                 |

**Pre-reg (XXXIV):** EXP-\* identify **brief content** × **CIOF mediation** × **transfer**. They do **not** identify “we create bloomers” or IQ change.

**Densifies:** Equity (XXXVI); EVT expectancy; SAFE-RESILIENCE support; FEI; tutor ops queue item 68.

## XLVI.12 Confidence table

| Claim                                                                                                               | Label                           | Confidence                                                          |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------------------------------------------------------------|
| Rosenthal & Jacobson launched expectancy research; original IQ story is contested                                   | FACT                            | High                                                                |
| Induced expectancy effects larger when teacher knows student less (Raudenbush, 1984)                                | FACT                            | High                                                                |
| Average self-fulfilling effects small; accuracy often drives prediction; stigma can amplify (Jussim & Harber, 2005) | FACT                            | High                                                                |
| Differential treatment observable; highs get more demand/praise (Brophy & Good, 1970)                               | FACT                            | High                                                                |
| Four-factor mediation frame useful (Harris & Rosenthal, 1985)                                                       | FACT                            | High (framework); Medium (which channel dominates in tutoring apps) |
| Golem / high-bias negative treatment real (Babad et al., 1982)                                                      | FACT                            | High                                                                |
| High-expectation teachers differ in practice for whole class (Rubie-Davies, 2007)                                   | FACT                            | High (observational/descriptive line); Medium (causal ops transfer) |
| SAFE-EXPECTANCY is right default for MindCraft tutor briefing                                                       | FOUNDER BELIEF / HYPOTHESIS     | Medium–High                                                         |
| “Pygmalion IQ” / belief posters as product claim                                                                    | SPECULATION / false as doctrine | Low (against)                                                       |

## XLVI.13 What this chapter kills

1. **Kill:** Marketing classic Pygmalion IQ blooming as MindCraft proof. (Raudenbush limits; Jussim accuracy; claim ladder.)
2. **Kill:** Tutor briefs that install trait/weakness identity labels. (Babad Golem; Weinstein PDT.)
3. **Kill:** Climate-only positivity without input/output demand. (Harris & Rosenthal mediation; Bastani/PWC line.)
4. **Kill:** “Just be realistic — tell tutors she’s weak” as ops hygiene.
5. **Wound:** Expectancy as a large, cumulative social-problem engine — survives only as small average effects with stigma/PDT moderators.
6. **Survive (constrained):** Cold-start high-challenge-with-support; task-only briefs; CIOF audit; Rubie-Davies whole-class high expectation; measure mediation + transfer.

**Doctrine until data:** Ship **SAFE-EXPECTANCY**: brief the next skill, not the fixed self; high challenge for everyone with scaffolds; warm climate under struggle; force student output; never weaponize gap-scan as a Golem tag. Market “tutors see what to teach next — not who she is forever.”

## Part XLVII — Sleep, Stress, and Learning: Timing Practice Without All-Nighter Theater

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 18 ≡ Red Team slot, but ch47 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.6 → not yet Synthesizer)

**Primary question:** How should MindCraft time practice and exam-week product behavior given sleep-dependent consolidation and stress/pressure drain on working memory — without becoming a sleep-coaching app or glorifying late-night grind?

**Owners:** Product (session timing / exam modes) · Parent messaging · Affect pipeline · Red Team

**Commercial job:** Ship a **SAFE-TIMING** stack — overnight spacing as consolidation infrastructure, exam-week load softening, ban streak/all-nighter heroism, measure delayed transfer not midnight minutes.

### XLVII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXIV/XLI (anxiety × desirable difficulty), XXXI (flow/challenge–skill), and XLV (growth vs danger zone) already treat *within-session* difficulty and affect. They do not fully answer: **when** should Maya practice relative to sleep, and what should the product do in high-stakes stress weeks when encoding, attention, and WM are compromised?

**FOUNDER BELIEF under audit:** Identity transformation requires recoverable struggle *and* biological conditions that let struggle stick. A product that pushes “one more mission” at 1 a.m. during exam week can look engaged while sabotaging consolidation and installing danger-zone affect.

**Claims we refuse as doctrine:**

- 1. Marketing MindCraft as a sleep-hygiene / wellness brand (“we fix teen sleep”).
- 2. Celebrating all-nighters, midnight streaks, or “grind hours” as learning proof.
- 3. Treating acute exam pressure as desirable difficulty.
- 4. Absolute claims that “sleep always consolidates math” independent of encoding quality, task type, and sleep opportunity.

### XLVII.2 Constructs (product language)

| Construct | Plain definition | Product signal | Failure mode if misused |
|-----------|------------------|----------------|-------------------------|
|-----------|------------------|----------------|-------------------------|

|                                      |                                                                     |                                                          |                                                  |
|--------------------------------------|---------------------------------------------------------------------|----------------------------------------------------------|--------------------------------------------------|
| <b>Sleep-dependent consolidation</b> | Offline reprocessing after encoding that stabilizes/reshapes memory | Space practice across nights; delay transfer checks      | Sell “sleep = magic” without encoding first      |
| <b>Encoding under restriction</b>    | Multi-night short sleep impairs <i>new</i> learning capacity        | Soften new-concept loads after chronic short sleep flags | Blame student “laziness” for restriction biology |
| <b>Pressure / choke</b>              | High-stakes appraisal consumes WM needed for hard math              | Destake practice; hide peer ranks in exam week           | Treat choke as low ability                       |
| <b>Math-anxiety dual-task</b>        | Worry acts like a secondary load on WM                              | SAFE-DD before hard items                                | Add timers/leaderboards under anxiety            |
| <b>SAFE-TIMING</b>                   | Time practice for consolidation + protect WM under stress           | See XLVII.8                                              | Sleep-app feature creep; grind theater           |

**Operational definition (HYPOTHESIS):** A MindCraft practice plan is *timing-safe* when (a) hard encoding sessions are followed by a real sleep opportunity before high-stakes transfer claims, (b) multi-night sleep debt / high stress triggers **load softening** (fewer new concepts, more retrieval of known, support on tap), and (c) engagement metrics never reward post-midnight volume as mastery.

### XLVII.3 Sleep consolidates — and evolves — memory

**FACT (systems review):** Diekelmann & Born (2010, *Nature Reviews Neuroscience*, 11, 114–126, doi:10.1038/nrn2762) — sleep optimizes consolidation depending on learning conditions and sleep timing; slow-wave sleep (SWS) and REM support complementary system- vs synaptic-consolidation roles; consolidation can change representations quantitatively *and* qualitatively.

**FACT (broader offline processing):** Stickgold (2005, *Nature*, 437, 1272–1278, doi:10.1038/nature04286) — converging evidence that offline memory reprocessing during sleep is a major component of how memories are formed and shaped (not a single all-or-none event).

**FACT (integrative/schema angle):** Walker & Stickgold (2010, *Nature Reviews Neuroscience*, 11, 218, doi:10.1038/nrn2762-c1) — beyond veridical preservation, sleep (often REM-linked in their framing) supports unitization, assimilation into remote networks, and abstraction of rules/schemas — relevant to MindCraft’s transfer goal, not only verbatim item memory.

**FACT (mechanism exemplar):** Gais & Born (2004, *PNAS*, 101(7), 2140–2144, doi:10.1073/pnas.0305404101) — elevating acetylcholine during SWS-rich sleep blocked declarative word-pair consolidation while sparing a nondeclarative skill task — evidence that the *neurochemical milieu* of SWS matters for declarative consolidation, not “rest” generically.

**Product translation:** Spacing missions across nights (Part XXIX) is not only spacing theory — it is a **consolidation window**. Claiming “exam-ready” from a same-day blocked accuracy binge without overnight delay is scientifically lazy (ties XXXIX readiness signal).

**Kill:** “Sleep replaces practice.” Consolidation operates on what was encoded. No encoding → nothing to consolidate.

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## XLVII.4 Adolescent sleep restriction is causal for learning damage

Maya is a high-schooler. Population sleep debt is not a vibe; experimental restriction shows classroom-relevant harm.

**FACT (simulated classroom):** Beebe, Rose & Amin (2010, *Journal of Adolescent Health*, 47(5), 523–525, doi:10.1016/j.jadohealth.2010.03.005) — within-subject multi-night restriction vs healthy duration: sleep-restricted adolescents showed poorer learning, more inattentive behaviors, and lower arousal in a simulated classroom (medium-to-large effects; small *n*, preliminary but experimental).

**FACT (review):** Beebe (2011, *Pediatric Clinics of North America*, doi:10.1016/j.pcl.2011.03.002) — synthesizes cognitive, behavioral, and functional consequences of inadequate sleep in children/adolescents; chronic restriction is common and not benign.

**FACT (long-term factual retention):** Cousins, Wong & Chee (2019, *Journal of Adolescent Health*, 65(4), 549–557, doi:10.1016/j.jadohealth.2019.04.030) — adolescents (15–18) with 5 h sleep opportunity for 5 nights vs 9 h controls: after four restriction nights, learning detailed factual material showed **more forgetting** at 30 min (~26%), Day 3 (~34%), and Day 42 (~65%) relative to controls. Vigilance also fell; memory impairment did not simply track vigilance correlation.

**HYPOTHESIS for MindCraft:** A “productive” ACT cram week of late missions on short sleep can **look** like engagement while destroying the encoding that gap-scan and practice are supposed to build. Parent-visible “minutes practiced” without sleep context is a vanity metric.

**Wound:** “Weekend catch-up fixes the week.” Literature flags incomplete restitution and cumulative cost (see Beebe 2019-line summaries); do not market weekend binge as full repair.

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## XLVII.5 Stress and pressure — WM theft, not character failure

Sleep loss and evaluative stress often co-occur in exam weeks. Separate the mechanisms.

**FACT (math anxiety × WM):** Ashcraft & Krause (2007, *Psychonomic Bulletin & Review*, 14, 243–248, doi:10.3758/BF03194059) — math performance depends on WM for beyond-retrieval arithmetic; high math anxiety behaves like a dual-task (worry consumes capacity); anxious students also avoid math pathways. Complements Ashcraft & Kirk (2001) line already in Part XXIV.

**FACT (choking under pressure):** Beilock & Carr (2005, *Psychological Science*, 16(2), 101–105, doi:10.1111/j.0956-7976.2005.00789.x) — performance pressure harmed **high**-WM individuals on the most WM-demanding math problems — pressure consumes the capacity their superior strategies rely on. Not “dumb under stress”; often “capable under load.”

**Bridge to SAFE-DD / FEI:** Exam-week timers, public ranks (XLII), and “prove you’re ready tonight” framing are pressure injectors. Desirable difficulty assumes encoding capacity; pressure + anxiety can erase the dose.

**HYPOTHESIS:** Affective check-in stress > 0.7 already softens target\_mastery in /recommend — that is the correct direction. Extend it: under high stress / self-reported short sleep, prefer retrieval + support recruitment over new-concept flooding and appearance UX.

**Kill:** “Pressure makes diamonds” as practice design. Pressure makes WM poorer for the problems that need it most.

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## XLVII.6 Competitive audit (timing lens)

| Competitor pattern  | Timing / stress mechanism                                    | MindCraft counter                                                          |
|---------------------|--------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Duo</b>          | Streaks + late-night nagging; habit over consolidation       | SAFE-HABIT + SAFE-TIMING: miss≠identity; no midnight streak heroism        |
| <b>Khan</b>         | Infinite library; learner can cram to exhaustion             | Spaced missions + delayed transfer as readiness                            |
| <b>Brilliant</b>    | Delight / prestige sessions; can invite performance pressure | Mastery climate; destake under stress                                      |
| <b>ChatGPT Base</b> | Always-on 2 a.m. answers; no consolidation pedagogy          | Human/AI wrap that schedules <i>struggle then sleep</i> , not answer dumps |
| <b>Tutor shops</b>  | Pack hours before exam Saturday                              | Brief tutors: encode early in week; retrieval near exam; never shame sleep |

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## XLVII.7 Exam weeks — special product mode, not harder grind

**SPECULATION → product hypothesis:** Default “more missions” during ACT week maximizes anxiety and restriction. Safer mode:

1. **Front-load encoding** earlier in the week when sleep opportunity is better.
2. **Shift to retrieval + mixed review** 48–72 h pre-exam (XXIX/XXXIX).
3. **Lower appearance cues** (no leaderboards, no streak threats).
4. **Keep growth-zone items** but exit danger fast (XLV); prefer scaffolds over new ontology leaps.
5. **Parent copy:** sleep is part of the study plan — not “rest is for quitters.”

**FOUNDER BELIEF:** Selling calm competence under exam load beats selling heroic sleep debt.

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## XLVII.8 Resolution doctrine — SAFE-TIMING

**FOUNDER BELIEF / HYPOTHESIS — SAFE-TIMING stack:**

1. **Encode → sleep → transfer claim.** Treat overnight delay as part of readiness (with XXIX spacing), not optional wellness.
2. **Never reward midnight volume.** Ban streak/XP incentives that spike after local quiet hours; engagement ≠ consolidation.
3. **Restriction-aware load.** If multi-night short sleep or high stress is signaled (check-in, parent note, exam-week flag): fewer new concepts, more retrieval, CIOF support (XLVI), SAFE-DD destake.
4. **Pressure ≠ DD.** Timers, ranks, and “prove it tonight” are kill switches under high anxiety/stress.
5. **Exam-week mode.** Soften novelty; maximize recoverable retrieval; protect sleep opportunity messaging.
6. **Do not become a sleep clinic.** Nudge timing and load — do not diagnose disorders or sell melatonin theater.
7. **Tutor briefs include timing.** “Practice early evening when possible; protect sleep before transfer checks” — task ops, not character judgment.
8. **Measure delayed** transfer\_pass / solo\_transfer\_pass, not minutes-after-midnight.

**Kill the false dichotomy:** “Ignore biology, just grind” vs “pivot the company into sleep coaching.” Refuse both. Ship **timing-aware learning load.**

### XLVII.9 Marketing language that survives

- **Allowed:** “Hard practice sticks better when she sleeps on it.”
- **Allowed:** “We don’t measure mastery at 1 a.m.”
- **Allowed (parent):** “Exam week ≠ all-nighter week — we shift to review and protect sleep.”
- **Banned:** “Our app fixes teen sleep.” / “Grind while they sleep.” / Streak-as-study-ethic.
- **Banned:** Bloom 2-sigma; empty mindset; tutoring-is-free; pressure-as-pedagogy.

#### Feature claim ladder (Part XXXIV hygiene)

| Feature                                  | Claim max without TIM-* data                    |
|------------------------------------------|-------------------------------------------------|
| Overnight spacing before transfer checks | L1 mechanism aligned with consolidation reviews |
| Exam-week load softening + destake       | L1 product hygiene under Ashcraft/Beilock       |
| “Sleep mode raises ACT scores”           | L0–L1 only with pre-reg; not L3 identity        |
| Sleep-clinic / wellness brand claims     | L0 — banned                                     |

### XLVII.10 Experiments spawned

| ID | Question | Design | Primary | Kill condition |
|----|----------|--------|---------|----------------|
|----|----------|--------|---------|----------------|



|          |                                                                                                             |             |                                                     |                                                                                                |
|----------|-------------------------------------------------------------------------------------------------------------|-------------|-----------------------------------------------------|------------------------------------------------------------------------------------------------|
| TIM-1    | Same content: practice ending $\geq 1$ h before habitual bedtime vs late-night block; transfer next evening | 2-arm       | next-day / 48h transfer_pass                        | Late $\geq$ early $\rightarrow$ consolidation timing overstated for our items                  |
| TIM-2    | Exam-week mode (retrieval+ destake) vs “more new missions” in last 5 days pre-mock                          | 2-arm       | mock score; affect; sleep opportunity self-report   | More-new wins mock+return $\rightarrow$ softening wrong default                                |
| TIM-3    | High-stress check-in $\rightarrow$ auto soft path vs ignore stress                                          | 2-arm       | retry_120s; WM-demand item accuracy; return         | Ignore $\geq$ soft on transfer without affect cost $\rightarrow$ modifier useless              |
| TIM-4    | Parent message: “sleep is study” vs “push one more hour”                                                    | message A/B | next-night session timing; parent WTP/trust         | Push-hour increases volume <i>and</i> next-day transfer $\rightarrow$ grind narrative survives |
| TIM-QUAL | 10 Maya exam-week diaries: when practice helped vs harmed                                                   | interview   | coded sleep debt $\times$ danger zone $\times$ cram | Students already self-time well $\rightarrow$ UI priority drops                                |

**Pre-reg (XXXIV):** TIM-\* identify **timing  $\times$  load  $\times$  delayed transfer**. They do **not** identify clinical sleep treatment or ACT score miracles from a bedtime tip.

**Densifies:** XXIX spacing; XLI SAFE-DD; XLII compare ban; XLIII habit without colonization; XLV danger exit; affect check-in; Part LXIII exam-pressure queue.

## XLVII.11 Confidence table

| Claim                                                                                                                          | Label | Confidence |
|--------------------------------------------------------------------------------------------------------------------------------|-------|------------|
| Sleep supports declarative/procedural consolidation with stage-sensitive mechanisms (Diekelmann & Born, 2010; Stickgold, 2005) | FACT  | High       |

|                                                                                                           |                                 |                                                            |
|-----------------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------------------------------|
| Sleep can support schema/abstraction beyond verbatim storage (Walker & Stickgold, 2010)                   | FACT                            | Medium–High (active research debate on REM vs SWS mapping) |
| Low ACh during SWS matters for some declarative consolidation (Gais & Born, 2004)                         | FACT                            | High (specific manipulation); Medium (product generality)  |
| Multi-night adolescent restriction impairs classroom learning/attention (Beebe et al., 2010; Beebe, 2011) | FACT                            | High (direction); Medium (effect sizes / sample limits)    |
| Restriction impairs long-term factual retention in teens (Cousins et al., 2019)                           | FACT                            | High                                                       |
| Math anxiety consumes WM (Ashcraft & Krause, 2007)                                                        | FACT                            | High                                                       |
| Pressure can choke high-WM math performance (Beilock & Carr, 2005)                                        | FACT                            | High                                                       |
| SAFE-TIMING is right default for MindCraft scheduling/exam mode                                           | FOUNDER BELIEF / HYPOTHESIS     | Medium–High                                                |
| MindCraft-as-sleep-app is a good growth wedge                                                             | SPECULATION / false as doctrine | Low (against)                                              |

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## XLVII.12 What this chapter kills

1. **Kill:** All-nighter / midnight-streak heroism as learning culture. (Cousins retention; Beebe classroom; SAFE-HABIT.)
2. **Kill:** Treating exam pressure and timers as desirable difficulty. (Beilock choke; Ashcraft dual-task; SAFE-DD.)
3. **Kill:** Same-day cram accuracy as exam-ready proof without overnight/delayed transfer. (Diekelmann/Born; XXIX/XXXIX.)
4. **Kill:** Pivoting brand into clinical sleep wellness.
5. **Wound:** “Sleep always helps every math skill equally.” Survives only as: condition- and task-dependent consolidation; encoding first.
6. **Survive (constrained):** Encode→sleep→claim; restriction/stress-aware load; exam-week retrieval mode; parent “sleep is study”; measure delayed transfer.

**Doctrine until data:** Ship **SAFE-TIMING:** practice hard enough to encode, sleep enough to consolidate, soften novelty under debt and pressure, never sell grind-at-1-a.m. as identity. Market “hard practice sticks better when she sleeps on it.”

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# Part XLVIII — Transfer & Variation Theory: Far Transfer Design in the Ontology

**Chapter status:** Living evidence brief — Researcher tick 2026-07-30 (UTC hour 21; hour%6≠0 → Researcher; 7 researcher entries since synthesizer v1.6 → not yet Synthesizer; this tick becomes the 8th)

**Primary question:** How should MindCraft design question sequences and “transfer” claims so Maya discerns *critical mathematical structure* — without selling story-world immersion or blocked accuracy as far transfer?

**Owners:** Product (questionBank / ontology / pathfinder) · Content ops · Marketing claim ladder · Red Team

**Commercial job:** Ship a **SAFE-TRANSFER** stack — deliberate variation/invariance for discernment, hugged near transfer toward ACT formats, bridged high-road abstraction only where designed, measure delayed/format/solo transfer — kill “narrative = transfer for free.”

## XLVIII.1 Why this chapter exists

Parts XXIX (spacing/retrieval), XXXIX (interleaving), XL (self-explanation), XLI (SAFE-DD), and XLVII (encode→sleep→claim) already constrain *when* and *how hard* practice should feel. They do not fully answer: **what pattern of examples** makes a concept transferable, and what marketing language about “transfer” survives evidence?

MindCraft’s ontology already encodes ingredients, bridges, combinations, formats, and archetypes. The risk is product language that treats any format change, story skin, or mixed set as “far transfer” — while competitors sell fluency theater as readiness.

**FOUNDER BELIEF under audit:** Identity as a mathematical thinker requires *flexible discernment* of structure across contexts — not one beautiful tutorial problem solved once. That flexibility is **designed**, not an automatic side-effect of lore.

**Claims we refuse as doctrine:**

- 1. Story worlds automatically produce far transfer.
- 2. Blocked same-day accuracy ≡ exam-ready / transferable skill.
- 3. “We teach for transfer” without specifying content×context dimensions (Barnett & Ceci).
- 4. Random interleaving or random story skins as a substitute for *principled* variation of critical aspects.

## XLVIII.2 Constructs (product language)

| Construct           | Plain definition                                                       | Product signal                                                | Failure mode if misused        |
|---------------------|------------------------------------------------------------------------|---------------------------------------------------------------|--------------------------------|
| Near / far transfer | Application of learning across some distance in content and/or context | Label every transfer claim with <i>which</i> dimensions moved | Vague “far transfer” marketing |

|                               |                                                      |                                                                          |                                                                  |
|-------------------------------|------------------------------------------------------|--------------------------------------------------------------------------|------------------------------------------------------------------|
| <b>Low-road / high-road</b>   | Automatic similarity trigger vs mindful abstraction  | Hugging (ACT-like items) vs bridging (explicit compare/abstract prompts) | Expect far transfer from low-road drill alone                    |
| <b>Object of learning</b>     | What is to be discerned (content + critical aspects) | Ingredient / bridge as the intended discernment target                   | Teach “one thing” without opening critical dimensions            |
| <b>Variation / invariance</b> | What changes vs what stays fixed across a sequence   | Example sets that separate critical aspects                              | Vary everything at once → noise; vary nothing → surface fixation |
| <b>SAFE-TRANSFER</b>          | Transfer claims + sequences that survive evidence    | See XLVIII.8                                                             | Story-skin theater; blocked-accuracy readiness                   |

**Operational definition (HYPOTHESIS):** A MindCraft mission is transfer-*safe* when (a) the sequence makes at least one critical aspect *discernable* via planned contrast/generalization/separation/fusion, (b) “transfer” checks change a declared Barnett–Ceci dimension (format, surface story, temporal delay, or solo vs scaffolded), and (c) marketing never claims far transfer from immersion alone.

### XLVIII.3 Far transfer is rare without design — name the dimensions

**FACT (taxonomy):** Barnett & Ceci (2002, *Psychological Bulletin*, 128(4), 612–637, doi:10.1037/0033-2909.128.4.612) — a century of “does far transfer occur?” debates compared apples to oranges; they propose a nine-dimension content×context framework. Estimating a *single* far-transfer effect size is misguided; evidence is substantial under some conditions and untested under others.

**FACT (prosecution):** Detterman (1993, in Detterman & Sternberg, Eds., *Transfer on Trial*, Ablex) — argues spontaneous far/general transfer is rare; apparent transfer often reflects cues/hints or other mechanisms; humans more often learn specific facts and patterns than free-floating abstractions.

**FACT (analogical noticing failure):** Gick & Holyoak (1980, *Cognitive Psychology*, 12, 306–355, doi:10.1016/0010-0285(80)90013-4) — subjects who read an analogous military “fortress” story often failed to spontaneously apply the convergence principle to Duncker’s radiation problem; **hints** dramatically raised solution rates. Structural isomorphism ≠ noticed transfer.

**Product translation:** MindCraft must not market “she’ll just see it on the ACT” after a kitchen-world linear-equations arc. Spontaneous far transfer is the *exception*. Design either **hugs** the exam context or **bridges** with explicit abstraction (compare two instances; name the invariant).

**Kill:** “Our stories create far transfer.” Stories can raise utility/attainment (XXXVII) and belonging (XXXVI); they do not substitute for noticing structure.

## XLVIII.4 Mechanisms that *do* teach for transfer — low road, high road, hug, bridge

**FACT (mechanisms):** Salomon & Perkins (1989, *Educational Psychologist*, 24(2), 113–142, doi:10.1207/s15326985ep2402\_1) — **low-road** transfer: extensive, varied practice → automatic triggering in similar stimulus conditions; **high-road** transfer: mindful abstraction, forward- or backward-reaching search for connections. Findings of transfer/nontransfer reflect whether those conditions were met.

**FACT (instructional strategies):** Perkins & Salomon (1992, “Transfer of Learning,” *International Encyclopedia of Education*, 2nd ed.; building on Perkins & Salomon, 1988, *Educational Leadership*, 46(1), 22–32) — **hugging** (practice that approximates target performance) vs **bridging** (prompt abstraction, metacognition, planned connections). Education gets transfer when designed for it — not by hoping.

### HYPOTHESIS for MindCraft:

- **Hug:** format-tagged ACT/GCSE-like items, timed retrieval near exam, archetype-matched instances (Layer 2/3) after ingredient mastery.
- **Bridge:** coach prompts that force comparison (“what stayed the same?”), self-explanation of the invariant (XL), and bridge-gap cards when two concepts are known but unconnected.
- **Low-road alone** (Duo-like drill skins) will not deliver ACT problem-type novelty. **High-road alone** (chatty abstraction without varied instances) will not stick.

**Wound:** Absolute “transfer can’t be taught” (hard Detterman reading). Survives only as: spontaneous far transfer is rare; *cued and designed* transfer is the product job.

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## XLVIII.5 Variation theory — discernment requires planned difference

**FACT (foundations):** Marton & Booth (1997, *Learning and Awareness*, Lawrence Erlbaum) — learning is experiencing an object of learning in a new way; discerning critical aspects requires experiencing patterns of **variation and invariance**.

**FACT (math classroom design):** Kullberg, Runesson Kempe & Marton (2017, *ZDM Mathematics Education*, 49, 559–569, doi:10.1007/s11858-017-0858-4) — variation theory as a design principle changes *what is made possible to learn*; teachers using VT systematically open critical aspects (e.g., relating concepts rather than teaching one-at-a-time isolation); example sets matter, and teaching must **draw attention** to the patterns (presence of variation ≠ experienced variation).

**FACT (dimensions of possible variation):** Watson & Mason (line; e.g., Watson & Mason, 2005/2006 math-education elaborations of Marton) — to learn a mathematical concept is to become aware of which aspects can vary while remaining an example, and over what **range of permissible change**. Teacher and student may stress different dimensions.

**FACT (Chinese bianshi tradition):** Gu, Huang & Marton (2004, in Fan et al., Eds., *How Chinese Learn Mathematics*, World Scientific, pp. 309–347, doi:10.1142/9789812562241\_0012) — Chinese “teaching with variation” systematically juxtaposes conceptual and procedural variants so students deepen structure; indigenous practice, not a Western bolt-on.

**FACT (curriculum variation problems):** Sun (2011a, *Educational Studies in Mathematics*, 76(1), 65–85, doi:10.1007/s10649-010-9263-4) — Chinese textbook “variation problems” organize connectedness (one problem,

multiple changes/solutions) to highlight invariants; contrasts with curricula that treat problems as isolated context decorations.

#### Product translation for the ontology:

1. Pick the **object of learning** (ingredient or bridge), not “finish five questions.”
2. Hold the critical structure **invariant**; vary one dimension at a time (numbers → representation → story skin → exam archetype).
3. After separation, **fuse** (vary jointly) only when WM/SAFE-DD allows (XXVI/XLI).
4. Prompt noticing (“what changed / what stayed the same?”) — ties XL self-explanation.
5. Interleaving (XXXIX) is related but not identical: IL trains discrimination among *categories*; VT trains discernment of *critical aspects within* an object of learning. Use both deliberately.

**Kill:** “More random story skins = better transfer.” Random surface change without invariant focus is entertainment variance, not Martonian variation.

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### XLVIII.6 Competitive audit (transfer / variation lens)

| Competitor pattern  | Transfer mechanism                                                              | MindCraft counter                                                          |
|---------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| <b>Khan</b>         | Huge item pool; learner may block by topic; transfer left to chance             | VT sequences + delayed mixed transfer checks                               |
| <b>Duo</b>          | Low-road habit; skin-deep variety; streaks as proof                             | SAFE-HABIT + SAFE-TRANSFER: variety serves discernment, not XP             |
| <b>Brilliant</b>    | Elegant interactive instances; may under-specify exam hug                       | Hug ACT formats after insight; measure transfer_pass                       |
| <b>ChatGPT Base</b> | Fluent explanations; weak structured example contrast; sycophancy risk (XXXIII) | Deterministic card/path spine; SE + contrast prompts; no monologue⇒mastery |
| <b>Tutor shops</b>  | Ad-hoc worksheets; expert tutors often vary well by instinct                    | Codify variation patterns in bank + tutor briefs (CIOF-compatible, XLVI)   |

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### XLVIII.7 Ontology / bank design implications (MindCraft-specific)

#### HYPOTHESIS — variation layers already latent in the stack:

- **Ingredient card styles** (geometric / algebraic / procedural) = representation variation with concept held.
- **Format axis** (word\_problem, diagram, symbolic\_expression, ...) = context/content-adjacent transfer dimensions — prefer matched then force a declared hop.
- **Bridges** = high-road objects of learning (connection itself is the critical aspect).
- **Archetypes (Layer 2)** = exam-pattern hug targets after near mastery.

- **Combinations** = fusion stage (multiple ingredients co-activated) — do not dump at cold start.

**SPECULATION** → **product rule**: Generation and bank curation should tag each practice *bundle* with: `invariant_aspect`, `varied_dimension`, `transfer_check_dimension`. Untagged bundles default to “drill,” not “transfer mission.”

**FOUNDER BELIEF**: Pathfinder recommendations that only maximize weakness severity without sequencing variation will grind the same surface until boredom — and still fail the ACT’s novel wrap.

## XLVIII.8 Resolution doctrine — SAFE-TRANSFER

**FOUNDER BELIEF / HYPOTHESIS — SAFE-TRANSFER stack:**

1. **Name the hop.** Every transfer claim states which Barnett–Ceci-style dimension moved (format, story surface, delay, scaffold removal, domain).
2. **Design variation, don’t sprinkle it.** Sequences: contrast → separate critical aspect → generalize range → fuse when ready.
3. **Hug the exam; bridge the abstraction.** ACT-like items after structure is discernable; explicit compare/SE prompts for high-road.
4. **Never equate story immersion with transfer.** Narrative is motivation/belonging infrastructure (WORLD\_VISION / XXXVI–XXXVII), not a transfer engine by itself.
5. **Blocked accuracy ≠ transferable skill.** Require delayed / mixed / format-shifted / solo checks (`transfer_pass`, `solo_transfer_pass`).
6. **Draw attention to the pattern.** Coach/tutor line: “What stayed the same?” is load-bearing UX, not flavor text.
7. **Respect load and anxiety.** Fusion + far hops only under SAFE-DD; exam week prefers hugged retrieval (XLVII).
8. **Measure what you sell.** Instrument transfer hops; kill missions that raise same-surface accuracy without hop gains.

**Kill the false dichotomy**: “Transfer is a myth, only drill” vs “Our world magically creates genius generalizers.” Refuse both. Ship **designed discernment**.

## XLVIII.9 Marketing language that survives

- **Allowed**: “We change one thing at a time until the structure is obvious — then we change the wrapping.”
- **Allowed**: “Ready means she can do it in a new format tomorrow, not five identical items tonight.”
- **Allowed (parent)**: “Practice looks varied on purpose — same math backbone, different disguises.”
- **Banned**: “Immersive stories create far transfer.” / “AI explanations transfer automatically.” / Blocked-streak-as-mastery.
- **Banned**: Bloom 2-sigma; empty mindset; tutoring-is-free; vague far-transfer heroics.

### Feature claim ladder (Part XXXIV hygiene)

| Feature | Claim max without TR-* data |
|---------|-----------------------------|
|---------|-----------------------------|

|                                                   |                                                         |
|---------------------------------------------------|---------------------------------------------------------|
| VT-sequenced bundles + noticing prompts           | L1 mechanism aligned with Marton/Kullberg/Gu            |
| Format-hop / delayed transfer checks as readiness | L1 product metric hygiene (Barnett–Ceci + Gick/Holyoak) |
| “Raises ACT by teaching far transfer”             | L0–L1 only with pre-reg estimand; not L3 identity       |
| Story-world $\equiv$ far transfer                 | L0 — banned                                             |

## XLVIII.10 Experiments spawned

| ID      | Question                                                                     | Design    | Primary                                      | Kill condition                                                         |
|---------|------------------------------------------------------------------------------|-----------|----------------------------------------------|------------------------------------------------------------------------|
| TR-1    | Same concept: VT sequence (one varied dimension) vs random surface skins     | 2-arm     | next-day format-hop transfer_pass            | Random $\geq$ VT $\rightarrow$ principled variation overstated         |
| TR-2    | Bridge object: explicit compare/SE vs monologue explanation then items       | 2-arm     | bridge-gap item accuracy; solo_transfer_pass | Monologue $\geq$ compare $\rightarrow$ XL/bridge UX useless            |
| TR-3    | Hug-then-hop: ACT-archetype items after VT vs archetype-first cold           | 2-arm     | archetype accuracy; affect; time             | Cold $\geq$ VT-prep $\rightarrow$ hug sequencing wrong                 |
| TR-4    | Transfer check: delayed mixed vs same-day blocked as “mission complete” gate | 2-arm     | 48h transfer; return                         | Same-day gate predicts 48h as well $\rightarrow$ delay rule overstated |
| TR-QUAL | 10 Maya think-alouds on “what changed?” prompts                              | interview | coded discernment of invariant               | Students already name invariants $\rightarrow$ prompt priority drops   |

**Pre-reg (XXXIV):** TR-\* identify **sequence design**  $\times$  **declared transfer hop**. They do **not** identify “stories cause identity” or ACT score miracles from a single varied set.

**Densifies:** XXIX delayed readiness; XXXIX IL discrimination; XL SE; XLI SAFE-DD; format-weakness workstream; Layer 2/3 archetype bank; Part XLIX misconceptions queue.



## XLVIII.11 Confidence table

| Claim                                                                                                               | Label                           | Confidence    |
|---------------------------------------------------------------------------------------------------------------------|---------------------------------|---------------|
| Far-transfer debates need multi-dimension taxonomy (Barnett & Ceci, 2002)                                           | FACT                            | High          |
| Spontaneous far/general transfer is rare; cues matter (Detterman, 1993; Gick & Holyoak, 1980)                       | FACT                            | High          |
| Low-road vs high-road + hug/bridge are usable design levers (Salomon & Perkins, 1989; Perkins & Salomon, 1992/1988) | FACT                            | High          |
| Discernment requires variation/invariance patterns (Marton & Booth, 1997; Kullberg et al., 2017)                    | FACT                            | High          |
| Chinese variation / bianshi is a systematic task-design tradition (Gu et al., 2004; Sun, 2011a)                     | FACT                            | High          |
| SAFE-TRANSFER is right default for MindCraft bank/path/copy                                                         | FOUNDER BELIEF / HYPOTHESIS     | Medium–High   |
| Story immersion alone produces far transfer                                                                         | SPECULATION / false as doctrine | Low (against) |

## XLVIII.12 What this chapter kills

1. **Kill:** Marketing story worlds as automatic far-transfer engines. (Gick/Holyoak noticing; Detterman rarity.)
2. **Kill:** Blocked same-surface accuracy as transferable / exam-ready proof. (Barnett–Ceci; XXIX/XXXIX readiness.)
3. **Kill:** Random cosmetic variety as “variation theory.” (Kullberg: what is made possible to learn; attention to patterns.)
4. **Kill:** AI monologue explanations as sufficient for transfer. (Hint/schema need; XL SE; XXXIII trust.)
5. **Wound:** “Transfer cannot be taught.” Survives only as: spontaneous far transfer is rare; designed hug/bridge/VT can raise hop success.
6. **Survive (constrained):** Named transfer hops; VT sequences; hug ACT + bridge abstraction; noticing prompts; delayed/format/solo metrics.

**Doctrine until data:** Ship **SAFE-TRANSFER**: vary on purpose so the backbone becomes obvious, hug the exam when claiming readiness, bridge with explicit abstraction, never sell lore or identical-item streaks as transfer. Market “same math backbone, different disguises — then prove it in a new wrapping tomorrow.”

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# Part XLIX — Misconceptions as Productive: Diagnostic Wrong Answers & Soft-Wrong Science

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 00 ≡ Red Team slot, but ch49 never written → prefer Researcher per rotation; 8 researcher entries since synthesizer v1.6 → Synthesizer deferred to next eligible non-RT hour)

**Primary question:** How should MindCraft treat wrong answers — as shame tokens to erase, or as *diagnostic resources* that drive soft-wrong coaching, misconception routing, and honest mastery updates?

**Owners:** Product (Diagnostic / practice soft-wrong / Layer-4 misconception memory) · Engine (ingredient/misconception links) · Marketing claim ladder · Red Team

**Commercial job:** Ship a **SAFE-MISCON** stack — elicit → classify (lack vs confident misconception) → springboard without public shame → route remediation → measure retry + recalibration — kill “instant red-X pedagogy” and “all wrongs are equal.”

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## XLIX.1 Why this chapter exists

MindCraft already ships pieces of this thesis: soft-wrong UX, hide-correctness diagnostic (C4), Eedi misconception bank, Layer-1 canonical\_misconception\_family fields, and Experiment A (wizard coach on soft-wrong). What the Constitution lacked was an evidence spine that says *why* those features are load-bearing — and which competitor postures they kill.

Parts XXX (attribution), XXXVIII (mastery climate), XL (self-explanation), XLI (SAFE-DD), and XLVIII (SAFE-TRANSFER) constrain how misses feel and how sequences vary. They do not fully answer: **what is a productive wrong answer**, and when does revealing correctness help vs corrupt learning and the graph?

**FOUNDER BELIEF under audit:** Identity as a mathematical thinker grows when a miss becomes *information about structure* — not a grade, not a personality verdict, and not a fluent AI monologue that skips the student’s idea.

**Claims we refuse as doctrine:**

- 1. Misconceptions are unitary “bugs” that instruction must confront-and-replace wholesale.
  - 2. Instant correctness feedback = better learning for every miss.
  - 3. Soft-wrong tone alone equals diagnostic teaching.
  - 4. All incorrect selections update mastery the same way (guess ≡ confident wrong ≡ slip).
  - 5. Marketing “we celebrate mistakes” without routing, classification, or retry instrumentation.
- 

## XLIX.2 Constructs (product language)

| Construct | Plain definition | Product signal | Failure mode if misused |
|-----------|------------------|----------------|-------------------------|
|-----------|------------------|----------------|-------------------------|

|                                        |                                                                                 |                                                      |                                                                                            |
|----------------------------------------|---------------------------------------------------------------------------------|------------------------------------------------------|--------------------------------------------------------------------------------------------|
| <b>Misconception (classic)</b>         | Stable, theory-like wrong idea that interferes with learning                    | Named misconception_id; distractors that diagnose it | Treat every wrong as a deep theory                                                         |
| <b>Knowledge-in-pieces / resources</b> | Fragmentary, context-sensitive knowledge elements; flawed <i>and</i> productive | Route to refine/reorganize, not only “delete”        | Romanticize every wrong as deep wisdom                                                     |
| <b>Diagnostic wrong</b>                | Incorrect response that reveals <i>which</i> resource/misconception fired       | Option → misconception map; confidence tier          | Beauty distractors with no diagnostic map                                                  |
| <b>Soft-wrong</b>                      | Low-shame evaluation structure: miss is safe, informative, retry-inviting       | Tone + no public rank + coach invite                 | Soft tone with no next move = empty kindness                                               |
| <b>Hide-correctness (C4)</b>           | Serve real items; record outcome; do <b>not</b> reveal key during diagnostic    | Gap-scan / FEI probes                                | Revealing a wrong key as “gotcha” corrupts trust + can seed wrong keys into memory theater |
| <b>SAFE-MISCON</b>                     | Productive-error doctrine that survives evidence                                | See XLIX.8                                           | Mistake-celebration posters; red-X gamification                                            |

**Operational definition (HYPOTHESIS):** A MindCraft miss is *productive* when (a) the response maps to a known misconception family or resource pattern, (b) the UX destakes identity threat (soft-wrong / mastery climate), (c) the next move uses the wrong as a springboard (compare, conflict, SE, bridge card) rather than only stating the key, and (d) telemetry separates lack-of-knowledge, slip, and high-confidence misconception for graph updates.

### XLIX.3 Misconceptions reconceived — flawed *and* productive

**FACT (constructivist critique):** Smith, diSessa & Roschelle (1993/1994, *Journal of the Learning Sciences*, 3(2), 115–163, doi:10.1207/s15327809jls0302\_1) — acknowledge empirical findings of misconceptions research but reject the accompanying view that students hold strongly held flawed ideas that instruction must *confront and replace*. That view overemphasizes discontinuity with experts and conflicts with constructivism: advanced knowledge is built from prior understandings. They sketch learning as refinement and reorganization within a complex-systems view; misconceptions are **both flawed and productive**.

**FACT (knowledge in pieces):** diSessa (1993, *Cognition and Instruction*, 10(2–3), 105–225, doi:10.1207/s1532690xci1002&3\_2) — intuitive knowledge as weakly organized elements (p-prims / sense of mechanism), not a coherent wrong theory ready for wholesale replacement. Expertise densifies and reorganizes the system.

**FACT (instructional lens):** Hammer (1996, *Journal of the Learning Sciences*, 5(2), 97–127, doi:10.1207/s15327809jls0502\_1) — “misconceptions” vs “p-prims” change what a teacher sees as the instructional task (replace a theory vs refine activation conditions of resources).

**Product translation:** MindCraft’s Layer-1 *failure\_mode* / *canonical\_misconception\_family* and Eedi *mis\_* IDs should drive **refinement routes** (when does this resource work? when does it fail?) — not only a shameful “you have Misconception X” badge. Coach copy: “That move works for this case — here’s where it breaks” beats “Wrong. Correct answer is B.”

**Kill:** Product language that treats Maya as carrying a disease-like misconception to be purged. Survive: diagnostic naming for *engine routing*, student-facing language as productive refinement.

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## XLIX.4 Errors as springboards — inquiry, not erasure

**FACT (teaching experiment):** Borasi (1994, *Journal for Research in Mathematics Education*, 25(2), 166–208, doi:10.5951/jresmetheduc.25.2.0166) — secondary students can capitalize on errors as **springboards for inquiry**; variations of the strategy exist; the point is stimulating mathematical inquiry from the error, not only teacher-side diagnosis for remediation.

**FACT (error analysis tradition):** Radatz (1979, *Journal for Research in Mathematics Education*, 10(3), 163–172, doi:10.5951/jresmetheduc.10.3.0163) — systematic error analysis has long diagnostic value for teachers; Borasi’s contribution is turning that value toward *student* inquiry.

**FACT (diagnostic teaching / conflict):** Bell (1993a, *Educational Studies in Mathematics*, 24, 5–34; 1993b, *Educational Studies in Mathematics*, 24, 115–137) and Swan’s diagnostic-teaching line (e.g., Swan, 2001, “Dealing with misconceptions in mathematics,” in *Issues in Mathematics Teaching*, Routledge) — long-term learning improves when lessons **expose** existing thinking, generate cognitive conflict across methods/answers, and resolve via reflective discussion — vs curricula that avoid analyzing errors or front-load explanations then graded practice.

**HYPOTHESIS for MindCraft:** Soft-wrong + wizard coach should implement a *compressed* springboard cycle: expose idea → contrast with invariant/conflict item → student-generated why (XL) → optional AI wrap under PWC — not Duo-style instant red/green with explanation dump.

**Wound:** Classroom conflict discussion does not transfer 1:1 to solo app UX. Survive as: design *micro-conflict* (two worked paths; “which claim survives?”) rather than claiming Socratic seminar in a phone UI.

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## XLIX.5 Soft-wrong science — safety is necessary, not sufficient

**FACT (affect × attempt):** Prior Constitution densification (XXIV anxiety/WM; XXXVIII mastery climate; XLI SAFE-DD destake) — threat appraisal suppresses attempt quality; evaluation structure that makes misses identity-safe raises willingness to engage difficulty.

**FOUNDER BELIEF:** Soft-wrong is mastery *evaluation structure*, not brand voice. Without diagnostic mapping + next move, soft-wrong is empty kindness.

**HYPOTHESIS:** Soft-wrong’s commercial wedge vs Khan/Duo is not “nicer feedback” — it is **recoverable struggle with competence evidence** (retry\_120s, challenge\_accept, solo\_transfer\_pass). Competitors can copy warmer copy; they rarely ship miss → misconception route → bridge/ingredient card → delayed transfer check.

**Contradicting pressure:** Parents often want instant correctness theater (“show the answer”). Claim ladder: L0–L1 session metrics for soft-wrong; do not sell L3 identity from tone alone (XXXIV).

**Kill:** Soft-wrong as marketing slogan without Experiment A instrumentation. Survive: soft-wrong as FEI stage (informative miss) gated by SAFE-MISCON routing.

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## XLIX.6 Confidence, calibration, and hide-correctness

**FACT (hypercorrection):** Butterfield & Metcalfe (2001, *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27(6), 1491–1494, doi:10.1037//0278-7393.27.6.1491) — contrary to interference predictions, **high-confidence errors** were *more* likely to be corrected after feedback than low-confidence errors (hypercorrection).

**FACT (mechanism sketch):** Butterfield & Metcalfe (2006, *Memory & Cognition* / follow-on attentional work; see also Metcalfe & Finn line) — corrective feedback after surprising high-confidence errors captures more attention; later work notes latent knowledge / “knew it all along” contributions. Product implication: confidence on wrongs is diagnostically precious, not just a UI flourish.

**FACT (authentic math diagnostics online):** Foster, Woodhead, Barton & Clark-Wilson (2022 advance of 2021 online-first; *Educational Studies in Mathematics*, doi:10.1007/s10649-021-10084-7) — large-scale diagnostic quiz data: higher confidence on an incorrect response associated with higher later correction probability on analogous items (hypercorrection signal in authentic math learning), after controls discussed in-paper; confidence assessment can prompt reflection.

**FACT (calibration hygiene):** Confidence–accuracy work in diagnostics (e.g., overconfidence on alternative conceptions; gender patterns in under/overconfidence across studies) — poor calibration is common; separating *misinformed* (confident wrong) from *uninformed* (low confidence / blank) changes instruction.

**Product translation — C4 hide-correctness:** During gap-scan / identity-sensitive diagnostics, revealing keys mid-flow can (a) spike threat, (b) turn the session into answer-hunting, (c) risk encoding wrong “gotcha” narratives. Record outcomes + confidence; defer key reveal to a designed springboard moment (or never in pure diagnostic). **Do not** market hide-correctness as “we never teach the answer” — that is false and banned under tutoring-is-free style slogans.

**HYPOTHESIS:** Practice mode should often *use* hypercorrection: after a high-confidence miss, prioritize attended corrective feedback + isomorphic retest; after low-confidence miss, prioritize scaffold/hug before confrontation (SAFE-DD equip).

**Kill:** Treating every wrong as equal graph evidence. Survive: confidence-tiered update policy (aligns with Layer-4 evidence\_update\_policy intent: don’t over-update from one item; separate can-do from can-recognize).

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## XLIX.7 What this means for the bank, ontology, and Solver

**Bank / C5:** Distractors should be designed misconceptions, not random wrongs. Eedi ingestion already mints mis\_ IDs — enrichment linking misconceptions → ingredients remains the annotation bottleneck (CLAUDE.md). Without links, soft-wrong cannot route.

**Ontology:** Prefer nested ingredient failure\_mode + Layer-2 common\_misconception\_ids as the join surface for coach cards. Variation sequences (XLVIII) that juxtapose the misconception-triggering surface vs the invariant are VT done right.

**Solver / AI:** Under PWC (XXXIII), AI wrap after student springboard — not “fluent correct answer” that ignores the selected distractor. Sycophancy risk: agreeing with a wrong path. Guard: name the student’s move, state where it applies, show the break.

**Competitive wedge (XXXV):** Duo celebrates streaks and instant binary feedback; ChatGPT delivers fluent keys; Khan explains. Minecraft sells **diagnostic wrong** → **recoverable path** with evidence the parent can see (retry, transfer) without leaderboard shame (SAFE-COMPARE).

### XLIX.8 SAFE-MISCON stack (ship rule)

- 1. **Elicit** — Real items; prefer misconception-tagged distractors.
- 2. **Classify** — Lack / slip / high-confidence misconception (confidence + item family).
- 3. **Destake** — Soft-wrong mastery climate; no public shame; no streak-as-punishment for miss.
- 4. **Springboard** — Micro-conflict, compare, SE prompt, or Borasi-style inquiry move *using the wrong*.
- 5. **Route** — Ingredient / bridge / VT contrast card from misconception map.
- 6. **Reveal with purpose** — Feedback timing by confidence tier; hide-correctness in diagnostic; attended feedback after high-confidence miss.
- 7. **Measure** — retry\_120s, recalibration Δ, misconception recurrence, delayed isomorphic accuracy — not “mistakes celebrated” vanity.

**Marketing language that survives:** “Wrong answers that teach,” “find the pattern in the miss,” “confidence that matches skill.”

**Marketing language that dies:** “We never say you’re wrong,” “mistakes are magic,” instant red-X as rigor, Soft-wrong™ as identity transformation (L3 from tone).

### XLIX.9 Claim ladder

| Claim                                                             | Max ladder without new data |
|-------------------------------------------------------------------|-----------------------------|
| Soft-wrong raises attempt/retry vs harsh red-X                    | L1 (Experiment A)           |
| Misconception-tagged distractors improve routing vs random wrongs | L1                          |
| Confidence-tiered feedback beats uniform instant key              | L1                          |

|                                                             |                                          |
|-------------------------------------------------------------|------------------------------------------|
| Hide-correctness diagnostic yields cleaner weakness ranking | L1 (C1/C4 product bet)                   |
| Productive-error UX → identity as mathematical thinker      | L3 — ban until transfer + identity items |

### XLIX.10 Experiments spawned

| ID       | Question                                                              | Design    | Primary                                           | Kill condition                                              |
|----------|-----------------------------------------------------------------------|-----------|---------------------------------------------------|-------------------------------------------------------------|
| MIS-1    | Soft-wrong + springboard coach vs soft-wrong + key-only               | 2-arm     | retry_120s; next-item accuracy                    | Key-only ≥ springboard → inquiry UX overstated              |
| MIS-2    | Misconception-mapped distractors vs random wrongs                     | 2-arm     | route precision; recurrence of same mis-ID        | Random ≥ mapped → bank annotation ROI fails                 |
| MIS-3    | Confidence-tiered feedback (hypercorrection-aware) vs uniform instant | 2-arm     | high-conf miss correction @ 48h                   | Uniform ≥ tiered → tiering complexity unjustified           |
| MIS-4    | Hide-correctness diagnostic vs reveal-as-you-go for gap-scan          | 2-arm     | completion; affect; later worstweakness stability | Reveal ≥ hide on weakness quality + affect → C4 overstated  |
| MIS-QUAL | 10 Maya think-alouds after soft-wrong                                 | interview | coded: shame vs curiosity; names own idea?        | Students already springboard unaided → coach priority drops |

**Pre-reg (XXXIV):** MIS-1...4 identify **error-handling design × classification**. They do **not** identify “celebrating mistakes causes identity.”

**Densifies:** Experiment A; C4 diagnostic; Layer-4 misconception memory; XL SE; XLVIII VT conflict pairs; Part L confidence calibration queue.

### XLIX.11 Confidence table

| Claim | Label | Confidence |
|-------|-------|------------|
|-------|-------|------------|

|                                                                                                                         |                                 |                                                     |
|-------------------------------------------------------------------------------------------------------------------------|---------------------------------|-----------------------------------------------------|
| Misconceptions are both flawed and productive resources (Smith et al., 1993/94)                                         | FACT                            | High                                                |
| Knowledge-in-pieces / p-prims reframes instructional task (diSessa, 1993; Hammer, 1996)                                 | FACT                            | High                                                |
| Errors can be springboards for inquiry (Borasi, 1994)                                                                   | FACT                            | High                                                |
| Diagnostic teaching / conflict discussion beats error-avoidant exposition for long-term learning (Bell/Swan line)       | FACT                            | Medium–High (classroom evidence; app transfer open) |
| High-confidence errors are hypercorrected after feedback (Butterfield & Metcalfe, 2001; Foster et al. math diagnostics) | FACT                            | High                                                |
| SAFE-MISCON is right default for MindCraft soft-wrong / diagnostic / bank                                               | FOUNDER BELIEF / HYPOTHESIS     | Medium–High                                         |
| Soft tone alone transforms identity                                                                                     | SPECULATION / false as doctrine | Low (against)                                       |

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## XLIX.12 What this chapter kills

1. **Kill:** Misconceptions-as-disease / confront-and-replace-only pedagogy as company doctrine. (Smith/diSessa/Roschelle; Hammer.)
2. **Kill:** Instant red-X + answer dump as the default “rigorous” product. (Borasi springboard; Bell/Swan diagnostic teaching.)
3. **Kill:** Soft-wrong as empty brand kindness without diagnostic route + retry metrics.
4. **Kill:** Equal graph updates for guess, slip, and confident misconception. (Hypercorrection / calibration literature.)
5. **Kill:** Hide-correctness marketed as “we never teach answers.”
6. **Wound:** Classroom conflict discussion  $\equiv$  mobile UX. Survive as designed micro-conflict + SE, not seminar cosplay.
7. **Survive (constrained):** Misconception-mapped distractors; confidence-tiered feedback; soft-wrong as FEI informative miss; springboard-before-monologue; MIS-1...4.

**Doctrine until data:** Ship **SAFE-MISCON:** treat wrong answers as diagnostic fuel, destake identity threat, springboard from the student’s idea, route to structure, measure retry and recalibration — never sell mistake-celebration posters or red-X theater as learning science.

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# Part L — Confidence Calibration: Over/Underconfidence, Hide-Correctness, and Honest Self-Knowledge

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 06 ≡ Red Team slot, but ch50 never written → prefer Researcher per rotation; 0 researcher entries since synthesizer v1.7)

**Primary question:** How should MindCraft treat student *confidence* — as a vibe to inflate, a score to gamify, or a *calibrated signal* that drives feedback, graph updates, and identity-safe diagnostics?

**Owners:** Product (Diagnostic C4 / soft-wrong / confidence UX) · Engine (evidence\_update\_policy / confidence tiers) · Marketing claim ladder · Red Team

**Commercial job:** Ship a **SAFE-CALIB** stack — elicit confidence → classify miss type → destake → recalibrate without inflation theater → tier feedback → measure confidence\_miss\_tier — kill “raise confidence” as North Star and “how sure?” without routing.

## L.1 Why this chapter exists

The research question of this Constitution is how humans become *confident mathematical thinkers*. That phrase is dangerous if “confident” means louder, happier, or more certain. The commercial and scientific load-bearing meaning is **calibrated competence**: knowing when you know, knowing when you do not, and updating both skill and self-appraisal under informative misses.

Parts XXV (self-efficacy), XXX (attribution), XXXIII (AI overtrust), XLIX (SAFE-MISCON / hypercorrection), and the v1.7 metric confidence\_miss\_tier already point here. What was missing was a chapter that (a) defines calibration rigorously, (b) separates over- from underconfidence product responses, (c) ties C4 hide-correctness to metacognitive hygiene, and (d) refuses confidence theater as marketing.

**FOUNDER BELIEF under audit:** Identity as a mathematical thinker includes *honest self-knowledge* — not perpetual certainty, and not chronic self-doubt dressed as humility.

### Claims we refuse as doctrine:

- 1. Raise confidence = improve learning (always).
- 2. Confidence meters / “Belief Score™” as identity North Stars.
- 3. Asking “how sure?” without changing feedback, routing, or graph updates.
- 4. Hide-correctness = never teaching answers.
- 5. Dunning–Kruger as a meme that justifies shaming low performers.
- 6. Instant correctness reveal as the only path to calibration.

## L.2 Constructs (product language)

| Construct | Plain definition | Product signal | Failure mode if misused |
|-----------|------------------|----------------|-------------------------|
|-----------|------------------|----------------|-------------------------|

|                               |                                                                     |                                                                   |                                                            |
|-------------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------|------------------------------------------------------------|
| <b>Calibration</b>            | Match between stated confidence and empirical accuracy across items | Confidence × correctness curves; bias & absolute accuracy indices | One-item “I feel sure” = calibrated                        |
| <b>Overconfidence</b>         | Confidence systematically exceeds accuracy                          | High-conf wrongs; skipped review; false mastery                   | Shame the student; ignore hypercorrection opportunity      |
| <b>Underconfidence</b>        | Accuracy exceeds confidence                                         | Correct + low conf; stuck on secure content                       | Force “believe harder” posters; skip challenge progression |
| <b>Confidence of response</b> | Certainty that <i>this</i> answer is correct (item-level)           | 0–10 or low/med/high next to answer                               | Global “math confidence” survey only                       |
| <b>Hide-correctness (C4)</b>  | Record outcome; defer or withhold key reveal in diagnostic          | Gap-scan / FEI probes                                             | Never-teach-answers slogan; corrupt key theater            |
| <b>SAFE-CALIB</b>             | Calibration doctrine that survives evidence                         | See L.8                                                           | Confidence gamification; inflation theater                 |

**Operational definition (HYPOTHESIS):** MindCraft treats calibration as a *session and graph* property when (a) item-level confidence is elicited on diagnostic and soft-wrong paths, (b) high-conf misses trigger attended corrective / springboard sequences, (c) low-conf corrects unlock challenge progression rather than endless drill, (d) mastery updates weight miss type (guess ≠ confident misconception), and (e) marketing claims stay at L0–L1 until CAL experiments report.

### L.3 Calibration science — what “well calibrated” means

**FACT (definition):** Lichtenstein & Fischhoff (1977, *Organizational Behavior and Human Performance*, 20, 159–183; classic line) and Fischhoff, Slovic & Lichtenstein (1977, *Journal of Experimental Psychology: Human Perception and Performance*, 3(4), 552–564, doi:10.1037/0096-1523.3.4.552) — a judge is well calibrated if, over the long run, for propositions assigned probability  $p$ , the proportion true  $\approx p$ . Extreme certainty is routinely *inappropriate*: people are wrong too often when they feel certain.

**FACT (bias pattern):** Across general-knowledge and many educational tasks, the dominant bias is **overconfidence**, especially when accuracy is near chance; as accuracy rises, overconfidence often shrinks and underconfidence can appear at high accuracy (Lichtenstein & Fischhoff, 1977 summary pattern; replicated in educational monitoring work).

**FACT (measurement plurality):** Lingel, Lenhart & Schneider (2019, *ZDM*, metacognitive monitoring in mathematics — ERIC EJ1222655) — absolute accuracy, relative accuracy (e.g. Gamma), and diagnostic sensitivity/specificity are *not interchangeable*. Method choices change conclusions. Seventh-graders showed **pervasive overconfidence**; retrospective judgments and some scale formats improved monitoring precision vs prospective.

**Product translation:** MindCraft should log item-level confidence + correctness and report **bias** (over/under) and **discrimination** (does higher confidence predict correctness?) separately. Do not ship a single “Calibration %” vanity

metric parents can misunderstand.

**Kill:** One glamorous calibration number as growth KPI. Survive: confidence\_miss\_tier + bias curves for product learning, not for public leaderboards (SAFE-COMPARE).

---

## L.4 Overconfidence, underconfidence, and the dual commercial problem

**FACT (school math overconfidence):** García, Rodríguez, González-Castro, González-Pienda & Torrance (2016, *Metacognition and Learning*; ERIC EJ1105998) — elementary students solving math problems were poorly calibrated and tended toward **overconfidence**; inaccurate calibrators differed in reported metacognitive process use.

**FACT (ability × miscalibration):** Kruger & Dunning (1999, *Journal of Personality and Social Psychology*, 77(6), 1121–1134, doi:10.1037/0022-3514.77.6.1121) — bottom-quartile performers on humor/grammar/logic tasks grossly overestimated ability; incompetence can impair the metacognitive skill needed to recognize incompetence. Improving skill can improve self-appraisal.

**Wound (popular DK):** The meme version (“stupid people are always overconfident”) overclaims, ignores regression-to-the-mean critiques, and invites product cruelty. Use as **FACT** about *metacognitive dual burden in low skill*, not as a brand insult toward Maya.

**FACT (underconfidence is also costly):** Foster’s CA program (Foster, 2016, *Educational Studies in Mathematics*, 91(2), 271–288, doi:10.1007/s10649-015-9660-9; Foster, 2016, *Mathematics Teaching*, 251, 11–13; Centre for Mathematical Cognition summary) — underconfidence can reduce satisfaction, block progression to harder material, and trap students in repetitive secure practice; overconfidence can embed uncorrected errors and complacency. Pedagogical aim is **appropriate** confidence, not maximal confidence.

**HYPOTHESIS for MindCraft:** Overconfident Maya needs destaked confrontation + springboard (SAFE-MISCON + hypercorrection). Underconfident Maya needs earned-win evidence and challenge unlocks (FEI), not pep talks. One “confidence booster” feature cannot serve both.

**Kill:** “We build confidence” as undifferentiated slogan. Survive: “confidence that matches skill” / “know what you know.”

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## L.5 Confidence assessment in classrooms — promise and null on attainment

**FACT (CA mechanism):** Confidence assessment asks students to rate certainty alongside each answer; scoring that rewards correct high-confidence and penalizes incorrect high-confidence (and also punishes under-reporting when correct) incentivizes *truthful* ratings (Foster, 2016; Gardner-Medwin line in HE). Aim includes discouraging guess-and-hope.

**FACT (student acceptability):** Foster (2016 ESM) — secondary students can make usable confidence judgments on procedures; generally positive about CA in short trials.

**FACT (null on summative attainment — load-bearing contradiction):** Foster (2021/2022, *International Journal of Science and Mathematics Education*, 20, 1411–1429, doi:10.1007/s10763-021-10207-9) — quasi-experimental CA across four secondary schools (N = 475), meta-analytic Cohen’s  $d \approx -0.02$  [95% CI  $-0.22, 0.19$ ], Bayes factor favoring

null on attainment gains vs controls. Conclusion: incorporating CA into low-stakes formative assessments does **not** appear detrimental — and also does **not** show a clear positive attainment effect from CA alone.

**Commercial implication (critical):** Shipping a “how sure?” widget is cheap and marketable. Evidence says **CA alone is not an attainment engine**. MindCraft’s wedge is not novelty CA — it is CA **wired** to soft-wrong routing, misconception maps, confidence-tiered feedback, and graph policy. Competitors can copy a confidence slider; they rarely ship the spine.

**HYPOTHESIS:** MindCraft CA improves *process metrics* (retry, misconception route hit, delayed isomorphic correction after high-conf miss) before it moves ACT scores. Claim ladder: process L1 first; attainment L2+ only after CAL + MIS experiments.

**Kill:** Marketing “confidence assessment raises grades.” Survive: CA as diagnostic fuel + metacognitive prompt, non-inferior on attainment per Foster 2021, valuable when coupled to FEI/SAFE-MISCON.

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## L.6 Hypercorrection, contradicting reasons, and feedback design

**FACT (hypercorrection):** Butterfield & Metcalfe (2001, *JEP:LMC*, 27(6), 1491–1494, doi:10.1037//0278-7393.27.6.1491) — high-confidence errors more often corrected after feedback than low-confidence errors.

**FACT (math diagnostics):** Foster, Woodhead, Barton & Clark-Wilson (*Educational Studies in Mathematics*, doi:10.1007/s10649-021-10084-7) — authentic online math diagnostic data show hypercorrection-consistent patterns; confidence assessment can prompt reflection.

**FACT (debiasing via contradicting reasons):** Koriat, Lichtenstein & Fischhoff (1980, *Journal of Experimental Psychology: Human Learning and Memory*, 6(2), 107–118) — listing reasons *against* the chosen answer improved appropriateness of confidence; supporting reasons alone did not. Confidence tracks strength of supporting evidence; people neglect disconfirming evidence unless prompted.

### Product translation:

1. After high-conf miss → attended feedback + micro-conflict / “why might this be wrong?” (SE + Koriat) before fluent AI monologue (PWC / XXXIII).
2. After low-conf miss → scaffold/hug first (SAFE-DD), then lighter confrontation.
3. After low-conf correct → show competence evidence; invite slightly harder item (underconfidence unlock).
4. C4 hide-correctness during gap-scan: elicit confidence + answer; **defer** key reveal so the session stays diagnostic, not answer-hunting — then use purposeful reveal in practice springboards (XLIX).

**Kill:** Uniform instant red/green for every miss. Survive: confidence-tiered reveal + contradicting-reason prompt (aligns MIS-3).

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## L.7 Hide-correctness, AI trust, and graph integrity

**FOUNDER BELIEF:** C4 exists because revealing keys mid-diagnostic can spike threat, turn FEI into answer theater, and (if keys are wrong) poison trust. Recording outcomes without reveal preserves graph honesty for weakness ranking

(C1).

**HYPOTHESIS:** Calibration *improves* when students later reconcile confidence with delayed feedback in a destaked springboard — not when every item is a public gotcha.

**Link to XXXIII:** Fluent AI tutors create **overtrust** (calibration failure toward the *system*). Student self-appraisal and AI-skepticism are sibling problems.

**Layer-4 alignment:** Confidence tiers operationalize `evidence_update_policy` — high-conf wrong → stronger misconception evidence; low-conf wrong → weaker / “uninformed” path.

**Kill:** Equal mastery deltas for all wrongs; “we never teach the answer.” Survive: purposeful reveal; confidence-weighted updates; AI wrap after student generation.

---

## L.8 SAFE-CALIB stack (ship rule)

1. **Elicit** — Item-level confidence on diagnostic and on soft-wrong / challenge items (not only global surveys).
2. **Classify** — Correct/incorrect × low/med/high → miss classes (guess, slip, confident misconception, underconfident correct).
3. **Destake** — Soft-wrong / mastery climate; never public-shame high-conf wrongs; no streak punishment for honest low confidence.
4. **Recalibrate** — Show mismatch privately (“sure + wrong” or “unsure + right”) as information, not identity verdict.
5. **Tier feedback** — Hypercorrection path for high-conf miss; hug-then-dose for low-conf miss; unlock challenge after underconfident correct.
6. **Prompt against** — One contradicting reason / “where does this break?” before answer dump (Koriat).
7. **Measure** — `confidence_miss_tier`, calibration bias  $\Delta$ , high-conf miss correction @ delay, `underconf→challenge_accept` — not Belief Score™ or confidence streaks.

**Marketing language that survives:** “Know what you know,” “confidence that matches skill,” “wrong answers that teach,” “honest diagnostics.”

**Marketing language that dies:** “We raise confidence,” “believe in yourself” as product, Confidence League ranks, “never say you’re wrong,” CA-raises-grades claims without data.

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## L.9 Competitive wedge (brief)

Duo (binary + streaks), Khan (explain-first), Brilliant (scaffold dopamine), and ChatGPT tutors (fluent certainty → overtrust) all fail *calibration* differently. MindCraft’s L1-safe line: not louder confidence — **earned, calibrated** confidence that survives a format hop (solo transfer + confidence on unaided items + PWC).

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## L.10 Claim ladder

| Claim                                                                              | Max ladder without new data                              |
|------------------------------------------------------------------------------------|----------------------------------------------------------|
| Item-level CA is usable with teens and non-inferior on attainment vs BAU           | L1 (Foster 2016; 2021 null)                              |
| High-conf misses deserve different feedback than low-conf misses                   | L1 (hypercorrection literature)                          |
| Contradicting-reason prompts improve confidence appropriateness                    | L1 (Koriat et al., 1980)                                 |
| C4 hide-correctness improves weakness ranking quality + affect vs reveal-as-you-go | L1 product bet (MIS-4 / CAL-4)                           |
| Confidence-tiered graph updates beat equal-wrong updates                           | L1                                                       |
| CA alone raises summative math attainment                                          | <b>Banned</b> pending contrary RCT (Foster 2021 against) |
| Calibrated confidence → identity as mathematical thinker                           | L3 — ban until transfer + identity instruments           |

## L.11 Experiments spawned

| ID    | Question                                                            | Design                       | Primary                                              | Kill condition                                            |
|-------|---------------------------------------------------------------------|------------------------------|------------------------------------------------------|-----------------------------------------------------------|
| CAL-1 | Confidence-tiered feedback vs uniform instant key                   | 2-arm                        | high-conf miss correction @ 48h; retry_120s          | Uniform $\geq$ tiered → tiering unjustified               |
| CAL-2 | Contradicting-reason prompt vs explain-correct-only after miss      | 2-arm                        | calibration bias $\Delta$ ; next isomorphic accuracy | Explain-only $\geq$ prompt → Koriat UX overstated for app |
| CAL-3 | Underconfident-correct → challenge unlock vs more secure drill      | 2-arm                        | challenge_accept; later accuracy                     | Drill $\geq$ unlock on growth → underconf policy wrong    |
| CAL-4 | Hide-correctness diagnostic vs reveal-as-you-go (shared with MIS-4) | 2-arm                        | completion; affect; worstweakness stability          | Reveal wins on weakness + affect → C4 overstated          |
| CAL-5 | Confidence-weighted mastery updates vs equal-wrong updates          | 2-arm offline / sim + online | prediction of next-session accuracy                  | Equal $\geq$ weighted → engine complexity unjustified     |

|          |                                                                                |           |                                                      |                                                       |
|----------|--------------------------------------------------------------------------------|-----------|------------------------------------------------------|-------------------------------------------------------|
| CAL-QUAL | 10 Maya interviews:<br>meaning of<br>“sure” / shame<br>after high-conf<br>miss | interview | coded: threat vs<br>curiosity; wants<br>reveal when? | Students refuse<br>confidence UX<br>→ redesign elicit |
|----------|--------------------------------------------------------------------------------|-----------|------------------------------------------------------|-------------------------------------------------------|

**Pre-reg (XXXIV):** CAL-\* identify **confidence** × **feedback** / **update design**. They do **not** identify “raising confidence causes identity.”

**Densifies:** MIS-3/4; C4; Layer-4 evidence policy; XXV self-efficacy; XXXIII AI overtrust; metric confidence\_miss\_tier.

## L.12 Confidence table

| Claim                                                                                          | Label                           | Confidence    |
|------------------------------------------------------------------------------------------------|---------------------------------|---------------|
| Calibration = long-run match of confidence to accuracy (Fischhoff/Lichtenstein line)           | FACT                            | High          |
| Overconfidence is common; extreme certainty often unwarranted                                  | FACT                            | High          |
| School math students often poorly calibrated / overconfident (García et al.; Lingel et al.)    | FACT                            | High          |
| DK dual burden for low skill is real; meme uses are overstated                                 | FACT / WOUND                    | Medium–High   |
| CA usable + non-inferior attainment; not a proven attainment booster alone (Foster 2016; 2021) | FACT                            | High          |
| Hypercorrection after high-conf errors (Butterfield & Metcalfe; Foster et al.)                 | FACT                            | High          |
| Contradicting reasons improve confidence appropriateness (Koriat et al., 1980)                 | FACT                            | High          |
| SAFE-CALIB is right default for MindCraft diagnostic / soft-wrong / graph                      | FOUNDER BELIEF / HYPOTHESIS     | Medium–High   |
| Raise-confidence marketing transforms identity                                                 | SPECULATION / false as doctrine | Low (against) |

## L.13 What this chapter kills

1. **Kill:** “Raise confidence” as undifferentiated North Star or brand promise. (Foster: appropriate confidence; under- and over- both hurt.)
2. **Kill:** Confidence meters, Belief Score™, or confidence streaks as identity engines. (SAFE-COMPARE / SAFE-HABIT alignment.)
3. **Kill:** “How sure?” UI without tiered feedback, routing, or update policy. (Foster 2021 null on CA-alone attainment.)
4. **Kill:** CA / confidence assessment marketed as raising grades without new evidence.
5. **Kill:** Equal graph updates for guess vs high-confidence misconception.
6. **Kill:** Hide-correctness as “we never teach answers.”
7. **Kill:** Dunning–Kruger shaming as brand voice.
8. **Wound:** Classroom CA  $\equiv$  app CA; HE medical CA  $\equiv$  HS math. Survive as: elicit + truthful incentive design + wired spine, not worksheet cosplay.
9. **Survive (constrained):** Item-level confidence; SAFE-CALIB stack; hypercorrection-aware feedback; underconfidence unlocks; CAL-1...5; coupling to SAFE-MISCON.

**Doctrine until data:** Ship **SAFE-CALIB**: treat confidence as a diagnostic signal for honest self-knowledge and differential pedagogy — never as a vibe to inflate, a league to rank, or a substitute for FEI evidence.

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## Part LI — Deliberate Practice in Tutoring Sessions: Ericsson vs Hambrick, and the 45-Minute Design

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 09; hour%6≠0; 1 researcher entry since synthesizer v1.7 → Researcher)

**Primary question:** What should a MindCraft tutoring / practice session *do* if “practice” is not volume, not streak minutes, and not answer delivery — and how far can Ericsson’s deliberate-practice (DP) frame travel before Macnamara–Hambrick evidence kills overclaim?

**Owners:** Product (session architecture / tutor briefing) · Engine (weakness targeting / feedback loops) · Marketing claim ladder · Red Team

**Commercial job:** Ship a **SAFE-DP** session stack — diagnose → isolate edge skill → goal → effortful attempt → informative feedback → revise → brief rest → solo check — kill 10,000-hours theater, grind-volume North Stars, and “we do deliberate practice” as brand without a coach-designed edge task.

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### LI.1 Why this chapter exists

MindCraft sells sessions: college tutors, practice missions, Solver, Map. The Constitution’s research question is identity transformation under difficulty. Deliberate practice is the popular science answer to “how experts get good.” It is also a marketing trap — competitors and edtech decks cite Ericsson while shipping naive drill (more of the same),



gamified volume (streaks/XP), or fluent AI monologue (pseudo-coaching).

Parts XXVI (CLT / fading), XXIX (spacing/retrieval), XXXI (challenge–skill), XXXIX (interleave), XL (self-explanation), XLI (SAFE-DD), and L (SAFE-CALIB) already constrain *what kind* of difficulty and feedback are safe. What was missing was a chapter that (a) defines DP vs purposeful vs naive practice, (b) faces the Macnamara–Hambrick variance numbers honestly, (c) translates math-education DP (Lehtinen line) into a **45-minute** tutor/product design, and (d) refuses talent-denial and grind-as-virtue as doctrine.

**FOUNDER BELIEF under audit:** A high-quality MindCraft session is closer to *coach-designed practice at the edge of ability with feedback* than to homework help, content library, or engagement loops.

**Claims we refuse as doctrine:**

1. 10,000 hours / “practice makes perfect” as causal story.
2. Accumulated minutes / streaks as proof of DP.
3. DP explains nearly all expert variance (popular overread).
4. Any hard drill = DP; AI chat without diagnosis = DP coach; talent-denial as brand.

## LI.2 Constructs (product language)

| Construct                                  | Plain definition                                                                                                                                                                          | Product signal                                                         | Failure mode if misused               |
|--------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------|
| <b>Naive practice</b>                      | Repeating what you can already do; play / homework slog                                                                                                                                   | High accuracy, low struggle, no edge target                            | “Completed 40 problems” vanity        |
| <b>Purposeful practice</b>                 | Goal + focus + feedback, possibly self-directed (Ericsson & Pool, <i>Peak</i> , 2016)                                                                                                     | Student sets a micro-goal; app gives feedback                          | Duo-like streaks dressed as purpose   |
| <b>Deliberate practice (strict)</b>        | Coach/teacher designs tasks beyond current reliable performance; immediate informative feedback; refinement; builds mental representations (Ericsson et al., 1993; Ericsson & Pool, 2016) | Tutor/engine picks weakness; task just beyond; feedback revises method | Labeling all structured homework “DP” |
| <b>Structured practice (meta-analytic)</b> | Broad “practice designed to improve” used in Macnamara et al. (2014) — often looser than Ericsson’s DP                                                                                    | Hours logged in studies                                                | Equating study-hours with DP          |

|                              |                                                                                                       |                                      |                                |
|------------------------------|-------------------------------------------------------------------------------------------------------|--------------------------------------|--------------------------------|
| <b>Mental representation</b> | Organized internal model that lets experts monitor and improve (Ericsson line; Lehtinen et al., 2017) | Transfer / format hop / self-monitor | Fluency without representation |
| <b>SAFE-DP</b>               | DP doctrine that survives Hambrick wound + anxiety/CLT gates                                          | See LI.8                             | Grind theater; 10k-hours copy  |

**Operational definition (HYPOTHESIS):** A MindCraft block counts as *deliberate-practice-shaped* when (a) target is a diagnosed edge (concept, ingredient, bridge, format, or miss class), (b) task is beyond reliable solo performance but solvable with brief support, (c) feedback changes the *next attempt's method*, not only the answer key, (d) student generates at least one why / check before AI wrap (PWC / XL), and (e) session ends with a short unaided probe (solo\_transfer\_pass or isomorphic check) — not with “minutes practiced.”

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### LI.3 Ericsson’s core claim — what is actually said

**FACT (foundational paper):** Ericsson, Krampe & Tesch-Römer (1993, *Psychological Review*, 100(3), 363–406, doi:10.1037/0033-295X.100.3.363) — expert performance is strongly associated with accumulated *deliberate practice*: highly structured activities designed to improve specific aspects of performance, typically with teacher/coach guidance, clear goals, feedback, and effortful concentration. Mere experience / play / mindless repetition do not substitute.

**FACT (effort constraint):** Same paper and later summaries — effective daily DP duration is limited; early practice often ~1 hour or less; laboratory extended-practice protocols commonly ~1 hr sessions; pushing past optimal duration yields diminishing returns and fatigue. Rest is part of the training system, not laziness (aligns SAFE-TIMING).

**FACT (Peak distinction):** Ericsson & Pool (2016, *Peak*) — **purposeful practice** = well-defined goal, focus, feedback, get-out-of-comfort-zone; **deliberate practice** = purposeful practice *plus* a field with known training methods and a teacher who designs/adapts exercises. Popular “10,000 hours” readings flatten this distinction and often cite Gladwell’s popularization rather than Ericsson’s constraints.

**Product translation:** MindCraft’s tutor + ontology spine can *aspire* to the coach role (diagnose → design edge task → feedback). Solo app loops without diagnosis are at best purposeful practice — and often naive practice with a nicer UI.

**Kill:** “10,000 hours of MindCraft.” Survive: “coach-designed practice at your edge, in a session you can sustain.”

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### LI.4 The Hambrick / Macnamara wound — variance is not destiny theater

**FACT (meta-analysis):** Macnamara, Hambrick & Oswald (2014, *Psychological Science*, 25(8), 1608–1618, doi:10.1177/0956797614535810) — across domains, deliberate-practice measures explained roughly **26%** of variance in games, **21%** music, **18%** sports, **4%** education, and **<1%** professions. Conclusion: DP is important, but **not as important as popular arguments claim**.

**FACT (sports follow-up):** Macnamara, Moreau & Hambrick (2016, *Perspectives on Psychological Science*, 11(3), 333–350, doi:10.1177/1745691616635591) — in sports, substantial unexplained variance remains; elite vs sub-elite comparisons weaken simple “more DP → elite” stories.

**FACT (reply chain):** Ericsson (2016, *Perspectives on Psychological Science*, doi:10.1177/1745691616635600) argues meta-analyses sum *heterogeneous practice hours* and dilute the construct; Macnamara & Hambrick (2016, doi:10.1177/1745691616635614) reply that Ericsson’s criteria shift and that existing evidence does not support “largely accounted for by DP.”

**FACT (definition rescue attempt):** Ericsson & Harwell (2019, *Frontiers in Psychology*, doi:10.3389/fpsyg.2019.02396) — re-emphasize original DP criteria; claim Macnamara et al. often measured **structured practice**, not strict DP; reanalyses with stricter inclusion raise explained variance. Independent methodologists still warn against treating any reanalysis as settling talent, opportunity, and starting-skill confounds.

**Wound for MindCraft marketing:** Education’s ~4% figure is a Red Team grenade against “science says deliberate practice is everything.” Even if Ericsson is right that many education studies measured weak proxies, MindCraft cannot sell DP as a near-sufficient cause of identity or ACT gains.

**HYPOTHESIS:** MindCraft should treat DP as a **session design grammar** (how to spend 45 minutes), not as a **variance-explained brand claim**. Commercial strength is *individualized edge practice with feedback* — the part both camps agree matters more than mindless hours — without promising that practice volume explains Maya’s future.

**Kill:** “Experts are made of deliberate practice — so we made you an expert factory.” Survive: “Most homework isn’t practice that changes you; we design the next hard thing on purpose.”

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## LI.5 Mathematics education — drill vs deliberate practice

**FACT (math DP review):** Lehtinen, Hannula-Sormunen, McMullen & Gruber (2017, *ZDM Mathematics Education*, 49, 625–636, doi:10.1007/s11858-017-0856-6) — school math “practice” is often **drill-and-practice** that automatizes procedures but can yield *inert* routines rather than adaptive number knowledge; DP in expertise research involves thinking, problem solving, reflection, and dynamically retargeted training — not mechanical fluency alone. They summarize Ericsson (2016) features: coach-known skills, beyond-current-ability effort, specific goals, full attention, feedback + modification, building mental representations, improving prior skills with correct fundamentals.

**FACT (study time ≠ improvement):** Plant, Ericsson, Hill & Asberg (2005, *Contemporary Educational Psychology*, 30(1), 96–116) — in higher education, amount of study time did not significantly predict GPA once quality/concentration/goals entered; improvement tied to concentrated, goal-directed learning (cited in Lehtinen et al. as caution against raw hours).

**HYPOTHESIS for MindCraft:** Ontology ingredients + bridges are the *right decomposition* for DP-style targeting (isolate the failing subprocess, then rebuild), but only if sessions refuse “do 20 similar ACT items” as the default. Blocked accuracy is acquisition theater (XXXIX); DP needs the edge miss class (SAFE-MISCON / SAFE-CALIB), then varied probes (SAFE-TRANSFER).

**Link to CLT / anxiety:** Beyond-ability tasks without destake/equip violate SAFE-DD. DP’s “near-maximal effort” is **not** permission for threat flooding. For Maya, the coach’s first job is often making the edge task *informative*, not merely hard.

**Kill:** Back-to-basics grind marketed as Ericsson. Survive: adaptive practice that builds representations — Lehtinen’s contrast as competitive copy vs worksheet mills.

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## LI.6 What a 45-minute MindCraft session should look like

**FOUNDER BELIEF:** Human tutor sessions (~45–60 min) and digital missions should share one DP-shaped spine so marketing demos, tutor briefs, and app UX do not diverge into “warm homework help” vs “streak game.”

**Proposed spine (HYPOTHESIS — SAFE-DP):**

| Minute block | Move                                                                                                 | Maps to                                                  |
|--------------|------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| 0–5          | Affect + goal: name <i>one</i> edge target (weakness / bridge / format / miss class)                 | FEI safety; E×V; SAFE-EXPECTANCY task-only language      |
| 5–12         | Quick diagnose / warm isomorphic (hide-correctness OK if FEI)                                        | C4; SAFE-CALIB elicit if used                            |
| 12–28        | Edge attempts: 1–3 items beyond reliable solo; student generation before explain                     | DP effort; PWC; SE (XL); SAFE-DD dose                    |
| 28–36        | Feedback that revises method: contradicting reason / springboard / ingredient card — not answer dump | Koriat; SAFE-MISCON; ingredient runtime                  |
| 36–42        | Near transfer / format hop / interleaved foil                                                        | SAFE-TRANSFER; XXXIX                                     |
| 42–45        | Solo check + attribution (“what changed in your method?”) + next cue                                 | solo_transfer_pass; SAFE-HABIT if-then; no streak sermon |

**Effort constraint product rule:** Prefer one sharp DP cycle over stuffing eight topics. Marathon “grind nights” violate SAFE-TIMING and Ericsson’s own duration cautions.

**Tutor ethics:** Brief tutors on *task* targets from the graph — never trait labels (SAFE-EXPECTANCY). “Today we strengthen the functions→rate bridge” beats “she’s weak and unmotivated.”

**AI role:** Engine proposes the edge target and card order; LLM wraps language *after* student attempt (PWC). Unguarded Solver-as-session is not DP.

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## LI.7 Competitive wedge (brief)

Khan often explain-then-block; Duo sells streak volume; Brilliant scaffolds delight; ChatGPT tutors monologue without diagnosis; traditional tutoring delivers answers. MindCraft’s L1-safe line: not “more practice” —

**better-designed practice** (one edge, method-changing feedback, solo proof). Minutes, XP, and fluent help are not DP.

---

## LI.8 SAFE-DP stack (ship rule)

1. **Diagnose** — Pick one edge from graph / gap / tutor brief (concept, ingredient, bridge, format, or confidence-miss class).
2. **Isolate** — Decompose to the failing subprocess (ontology), not a random worksheet.
3. **Goal** — State a performance micro-goal for the block (“solve rate problems when the story hides the function”).
4. **Stretch** — Tasks beyond reliable solo, inside SAFE-DD capacity (equip/destake first if anxious).
5. **Attempt** — Student generates; no AI monologue-first (PWC / XL).
6. **Feedback** — Informative, method-revising; confidence-tiered when elicited (SAFE-CALIB / MISCON).
7. **Revise** — Immediate retry or isomorphic variant that uses the new method.
8. **Prove** — Short unaided / format-hop check; log `solo_transfer_pass`, not minutes.
9. **Stop** — Respect effort constraint; schedule sleep/spacing (SAFE-TIMING / XXIX).
10. **Measure** — Edge-target hit rate, method-change coding, transfer — never 10k-hours progress bars.

**Marketing language that survives:** “Practice at your edge,” “coach-designed sessions,” “feedback that changes your next try,” “show what you can do alone.”

**Marketing language that dies:** “10,000 hours,” “grind mindset,” “practice makes perfect,” “XP = mastery,” “our AI is your deliberate-practice coach” without diagnosis spine, talent-denial (“anyone can be elite if they DP”).

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## LI.9 Claim ladder

| Claim                                                                                               | Max ladder without new data                                |
|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Strict DP ≠ mere study hours / drill fluency                                                        | L1 (Ericsson 1993; Lehtinen 2017; Plant et al. 2005)       |
| Popular “DP explains almost all expertise” overclaims; education variance small in Macnamara et al. | L1 (2014 meta; reply chain)                                |
| Coach/teacher-designed edge tasks + feedback beat naive repetition for skill change                 | L1 (expertise literature consensus on quality > raw hours) |
| MindCraft 45-min SAFE-DP spine improves solo transfer vs homework-help control                      | L1 product bet — needs DP experiments                      |
| DP volume alone creates identity as mathematical thinker                                            | <b>Banned</b> (L3 without instruments; Hambrick wound)     |
| Talent/opportunity irrelevant                                                                       | <b>Banned</b> as brand                                     |

---

## LI.10 Experiments spawned

| ID      | Question                                                                     | Design    | Primary                                              | Kill condition                                                                        |
|---------|------------------------------------------------------------------------------|-----------|------------------------------------------------------|---------------------------------------------------------------------------------------|
| DP-1    | SAFE-DP spine vs homework-help / explain-first session (same tutor time)     | 2-arm     | solo_transfer_pass; method-change code               | Help $\geq$ DP spine $\rightarrow$ spine overstated                                   |
| DP-2    | One edge target vs multi-topic coverage in 45 min                            | 2-arm     | delayed isomorphic accuracy; cognitive load / affect | Coverage $\geq$ focus on learning $\rightarrow$ “one edge” wrong for ACT cram segment |
| DP-3    | Engine-proposed ingredient target vs tutor-chosen topic without graph        | 2-arm     | edge-hit; transfer                                   | Tutor-only $\geq$ engine $\rightarrow$ graph targeting weak                           |
| DP-4    | Feedback revises method (SE + springboard) vs answer-key-only after miss     | 2-arm     | next-attempt success; retry_120s                     | Key-only $\geq$ method feedback $\rightarrow$ DP feedback claim weak                  |
| DP-5    | Session cap (~45–60 focused min) vs encouraged grind (+30 min similar items) | 2-arm     | delayed accuracy; sleep/affect                       | Grind $\geq$ cap $\rightarrow$ effort-constraint messaging wrong for this population  |
| DP-QUAL | 10 Maya + 5 tutor interviews: “real practice” vs help                        | interview | coded: edge / threat / agency                        | Spine feels like cruelty $\rightarrow$ redesign destake                               |

**Pre-reg (XXXIV):** DP-\* identify **session architecture / targeting / feedback quality** — not “DP causes identity” or “explains elite careers.” Densifies FEI, PWC, SAFE-DD/TRANSFER/MISCON/CALIB/TIMING; prefer transfer metrics over minutes.

## LI.11 Confidence table

| Claim                                                                                                   | Label | Confidence |
|---------------------------------------------------------------------------------------------------------|-------|------------|
| Ericsson et al. (1993): DP = coach-guided, feedback-rich, effortful practice beyond current performance | FACT  | High       |

|                                                                                                    |                                 |               |
|----------------------------------------------------------------------------------------------------|---------------------------------|---------------|
| Practice duration constrained; rest matters; purposeful ≠ deliberate without coach ( <i>Peak</i> ) | FACT                            | High          |
| Macnamara et al. (2014): modest DP variance; ~4% education; definition dispute unsettled           | FACT                            | High          |
| Lehtinen et al.: math drill ≠ DP; Plant et al.: hours ≠ achievement when quality ignored           | FACT                            | High          |
| SAFE-DP is right default session grammar for MindCraft                                             | FOUNDER BELIEF / HYPOTHESIS     | Medium–High   |
| 10k-hours / talent-denial / grind NS transform learners                                            | SPECULATION / false as doctrine | Low (against) |

## LI.12 What this chapter kills

1. **Kill:** 10,000-hours / “practice makes perfect” as causal brand.
2. **Kill:** Streak minutes, problem counts, or XP as DP proof.
3. **Kill:** “DP explains expertise — therefore our app creates experts” (education ~4% wound).
4. **Kill:** Hard drill ≡ DP; AI monologue ≡ DP coach; talent-denial; multi-topic coverage theater as default rigor.
5. **Wound:** Violin-lab DP ≠ anxious ACT novice tutoring — survive as quality grammar + SAFE-DD, not music cosplay.
6. **Survive:** SAFE-DP spine; ontology edge targeting; method-revising feedback; effort caps; DP-1...5; “better-designed practice” copy.

**Doctrine until data:** Ship **SAFE-DP:** one diagnosed edge, effortful attempt, feedback that changes the next try, short solo proof, stop before grind — never sell hours, streaks, or talent-denial as the learning engine.

## Part LII — Community of Practice (Lave/Wenger): Legitimate Peripheral Participation — Careful Transfer

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 12 ≡ Red Team slot, but ch52 never written → prefer Researcher per rotation; 2 researcher entries since synthesizer v1.7 → Researcher)

**Primary question:** What does *community of practice* / *legitimate peripheral participation* (LPP) actually claim — and how far can MindCraft transfer those ideas into product, tutoring, story worlds, and “Community” UX before the transfer becomes cosplay?

**Owners:** Product (social / tutor / world design) · Marketing claim ladder · Equity (whose community?) · Red Team

**Commercial job:** Ship a **SAFE-CoP** stack — participation trajectories with real old-timers and real practices, not Discord-as-community, leaderboard-as-belonging, or “we’re a CoP” brand — while killing apprenticeship theater that

ignores Boylan/Fuller wounds.

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## LII.1 Why this chapter exists

The Constitution’s thesis is identity transformation under difficulty. Lave & Wenger’s situated-learning line is the classic social account of how newcomers become members of a practice — midwives, tailors, quartermasters — not how students complete problem sets. Edtech decks love the phrase *community of practice* because it sounds like belonging, mentors, and identity without specifying *what practice* students are joining.

MindCraft already has social-adjacent surfaces: college tutors, story worlds, Community nav, parent/tutor roles, Map as shared ontology. Without this chapter, those surfaces drift toward (a) Duo-style leagues as belonging, (b) forum/Discord theater as CoP, or (c) AI chat as “apprenticeship” with no old-timer practice. Parts XXV (identity), XXXVI (equity of worlds), XLII (SAFE-COMPARE), XLVI (tutor expectancy), and LI (SAFE-DP coach role) constrain pieces; what was missing was an honest transfer of LPP into product law.

**FOUNDER BELIEF under audit:** Becoming a confident mathematical thinker is partly *joining a practice* (ways of posing, checking, arguing, recovering from wrong), not only acquiring items — but school/ACT prep is not a tailor shop, and naming the product a CoP does not make LPP true.

### Claims we refuse as doctrine:

1. “MindCraft is a community of practice” as marketing without naming the practice and the peripheral→central trajectory.
2. Forums, streaks, or chat volume as proof of LPP.
3. AI monologue as old-timer apprenticeship.
4. School/ACT classrooms (or solo app sessions) as interchangeable with ethnographic apprenticeship sites.

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## LII.2 Constructs (product language)

| Construct                                        | Plain definition                                                                                                         | Product signal                                                                       | Failure mode if misused                     |
|--------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Community of practice (CoP)</b>               | People engaged over time in a joint enterprise with shared repertoire (Wenger, 1998)                                     | Tutors + peers + world characters sharing <i>how math is done here</i>               | Calling any group chat a CoP                |
| <b>Legitimate peripheral participation (LPP)</b> | Newcomers participate in real practice at the edge; access and legitimacy grow toward fuller roles (Lave & Wenger, 1991) | Student contributes partial reasoning / checks; tutor/old-timer holds the full craft | Watching a video ≠ peripheral participation |



|                               |                                                                            |                                                         |                                              |
|-------------------------------|----------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------|
| <b>Old-timer / newcomer</b>   | Asymmetric competence + responsibility in the practice                     | College tutor, veteran Maya peer, coach spine           | Peer leaderboard as fake expertise hierarchy |
| <b>Repertoire</b>             | Shared ways of talking, tools, stories, criteria of “good work”            | Soft-wrong language; Map ingredients; story check-moves | Lore dump without practice criteria          |
| <b>Acquisition metaphor</b>   | Learning as gaining knowledge/skills as possessions (Sfard, 1998)          | Mastery graph, item bank, transfer probes               | Only metaphor → identity ignored             |
| <b>Participation metaphor</b> | Learning as becoming a member of a practice (Sfard, 1998)                  | “How we recover from wrong here”                        | Only metaphor → contentless belonging        |
| <b>SAFE-CoP</b>               | CoP doctrine that survives Boylan/Fuller transfer wounds + Sfard pluralism | See LII.8                                               | Apprenticeship cosplay; Discord theater      |

**Operational definition (HYPOTHESIS):** A MindCraft interaction counts as *LPP-shaped* when (a) the student does a real piece of the practice (attempt, explain, check, revise — not only consume), (b) an old-timer (human tutor or SE-constrained coach) models criteria of competence, (c) legitimacy is task/role based (you may try this edge), not trait based (you are a weak student — SAFE-EXPECTANCY), and (d) progress is visible as *changing participation* (more solo responsibility, wider repertoire) — not as XP.

---

### LII.3 Lave & Wenger (1991) — what LPP actually says

**FACT (foundational book):** Lave & Wenger (1991, *Situated Learning: Legitimate Peripheral Participation*, Cambridge University Press, ISBN 9780521423748; doi:10.1017/CBO9780511815355) — learning viewed as situated activity has as its central defining characteristic **legitimate peripheral participation**: learners participate in communities of practitioners and move toward fuller participation in the sociocultural practices of the community. LPP names relations among newcomers and old-timers, activities, identities, artifacts, and communities of knowledge and practice — not a classroom seating chart.

**FACT (cases):** Their analytic cases are midwives, Yucatec midwives’ apprentices, Vai and Gola tailors, navy quartermasters, butchers, and Alcoholics Anonymous — craft and mutual-aid settings with durable joint enterprises, not standardized exam prep.

**FACT (learning subsumed):** They argue that becoming a participant *includes* skill learning; identity and membership are not add-ons after content delivery.

**Product translation:** MindCraft’s research question (“confident mathematical thinkers”) rhymes with *identity in practice*. The danger is importing the *label* while shipping acquisition-only UX (explain → pick A–D → streak).

**Kill:** “Situated learning = our story world backdrop.” Survive: “Participation means doing pieces of mathematical practice with growing responsibility.”

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## LII.4 Wenger (1998) — meaning, practice, community, identity

**FACT:** Wenger (1998, *Communities of Practice: Learning, Meaning, and Identity*, Cambridge University Press; doi:10.1017/CBO9780511803932) — social theory of learning with four interlocking components: **meaning** (experiencing the world as meaningful), **practice** (shared historical/social resources for mutual engagement), **community** (social configurations where competence is recognized), **identity** (learning changes who we are). CoPs are informal groupings that form as people pursue shared enterprises over time — primary unit of analysis is neither lone individual nor formal institution alone.

**HYPOTHESIS for MindCraft:** The product must specify the *enterprise* (e.g., recovering from productive wrongs on ACT-path concepts; telling-and-checking in a story world) and the *repertoire* (soft-wrong, SE prompts, Map language). Without enterprise + repertoire, “Community” nav is a social tab, not a CoP.

**Wound:** Corporate/edtech CoP literature often flattens Wenger into “knowledge-sharing groups.” MindCraft must not sell community features as identity transformation (claim ladder L3) from engagement metrics (L0/L1).

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## LII.5 The transfer wound — school math is not a tailor shop

**FACT (school mathematics critique):** Boylan (2010, *Teaching and Teacher Education*, 26(1), 61–70, doi:10.1016/j.tate.2009.08.005) — LPP was developed from informal apprenticeship contexts; applicability to school classrooms is **problematic**, especially where teacher-centred, decontextualised procedural practices dominate “usual” school mathematics. He critiques LPP transfer on social practice, learning relationships, power, agency, and identity; proposes **ecologies of participation** to capture multidimensional participation (imposed/voluntary, engagement/disengagement, identity risk/maintain, marginal/peripheral, etc.). He notes that if LPP is not found in usual school math, treating it as a *universal* feature of situated learning is undermined — while still valuing Lave & Wenger as a contrastive model and intervention heuristic.

**FACT (workplace reassessment):** Fuller, Hodkinson, Hodkinson & Unwin (2005, *British Educational Research Journal*, 31(1), 49–68, doi:10.1080/0141192052000310029) — LPP remains insightful for workplace learning research, but has **significant limitations** in complex institutional settings (manufacturing, secondary schools): diverse patterns of participation; institutional environments configure opportunities and barriers that ethnographic craft cases under-specify.

**FACT (metaphor pluralism):** Sfard (1998, *Educational Researcher*, 27(2), 4–13, doi:10.3102/0013189X027002004) — acquisition and participation metaphors each entail desirable and harmful practices; **choosing just one** risks theoretical distortion and bad pedagogy. MindCraft’s graph/mastery engine is acquisition-shaped; identity thesis is participation-shaped — doctrine needs both, not a CoP slogan that deletes assessment.

**FACT (math classroom situated work — careful ally):** Boaler (1999 AERA paper / related classroom studies; see also Boaler & Greeno identity/agency line summarized in math-education CoP literature) — classroom communities’ constraints and affordances shape what students learn to *do*; formalized classrooms attune students to practices that may not transfer. Useful as evidence that *participation forms matter*, not as proof that MindCraft already is an apprenticeship CoP.

**HYPOTHESIS:** Treat LPP as a **design aspiration for tutor+practice sessions** (peripheral contribution → fuller solo responsibility) and a **diagnostic for fake community**, not as a claim that the app is a CoP.

**Kill:** Universal “students learn by LPP here.” Survive: “Usual school math often blocks LPP; we design for legitimate attempts at the edge with old-timer criteria.”

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## LII.6 What “practice” means for MindCraft (not vibes)

Align with Constitution practices already named:

1. **Diagnose** → **attempt** → **informative miss** → **revise** → **solo check** (FEI / SAFE-DP / SAFE-MISCON).
2. **Student-generated why before AI wrap** (PWC / XL).
3. **Named transfer hops** (SAFE-TRANSFER).
4. **Calibrated confidence talk** (SAFE-CALIB) — saying “I’m sure” as part of the repertoire, not a Belief Score™.
5. **Story worlds as wrappers of frozen math** (equity + meaning) — characters can model *peripheral roles* (watcher → helper → solver) without claiming ethnographic authenticity.

**SPECULATION (to test):** Peer “Community” features help only when they circulate the repertoire above (how we check; how we destake wrongs). Social chat about school stress without practice criteria may raise belonging feelings without changing mathematical participation (wound for growth loops).

---

## LII.7 Competitive positioning

| Competitor pattern          | CoP-shaped risk                                      | MindCraft wedge                                 |
|-----------------------------|------------------------------------------------------|-------------------------------------------------|
| Duo leagues / streaks       | Belonging as rank theater (SAFE-COMPARE)             | Criterion competence, not named infinite ranks  |
| Khan comments / help forums | Peripheral <i>talk about</i> problems ≠ LPP in craft | Tutor/old-timer + attempt-first                 |
| Brilliant delight social    | Shared puzzles without recovery repertoire           | Soft-wrong + solo transfer as membership proof  |
| ChatGPT “study buddy”       | Fluency without legitimacy criteria                  | PWC + ontology spine; AI ≠ old-timer by default |
| Discord “study community”   | CoP brand, acquisition optional                      | Enterprise + repertoire required for CoP copy   |

**Commercial implication:** Do not out-community Discord. Sell *becoming someone who participates differently in hard math* — with evidence (solo\_transfer\_pass, method-change after miss), not member counts.

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## LII.8 SAFE-CoP stack (product law)

1. **Name the practice** — joint enterprise in copy and UX (“how we recover and prove on this edge”), not “join our community.”
  2. **Peripheral must be real work** — attempt, partial solution, check, SE prompt; watching/scrolling is not LPP.
  3. **Old-timer required for CoP claims** — human tutor or SE-constrained coach with criteria; ungarded AI chat ≠ apprenticeship.
  4. **Legitimacy without trait labels** — access to roles by task readiness; ban Golem scripts (SAFE-EXPECTANCY).
  5. **Trajectory > tab** — measure changing participation (solo share up; help faded) not Community page visits.
  6. **Plural metaphors** — keep acquisition instruments (graph, bank); never replace them with belonging KPIs (Sfard).
  7. **Equity of membership** — whose history/stories define the repertoire (Part XXXVI); token “community” without belonging audit is a kill.
  8. **Claim ladder** — CoP/LPP language max L1 product bet until CoP-\* experiments; ban L3 “identity via community features” without instruments.
- 

## LII.9 Claim ladder

| Claim                                                                 | Max ladder without new data            |
|-----------------------------------------------------------------------|----------------------------------------|
| LPP describes apprenticeship-style learning in craft/mutual-aid cases | L1 (Lave & Wenger, 1991)               |
| Usual school math often fails LPP conditions; transfer is non-trivial | L1 (Boylan, 2010; Fuller et al., 2005) |
| Acquisition + participation metaphors both needed; one alone distorts | L1 (Sfard, 1998)                       |
| MindCraft tutor sessions can implement LPP-shaped trajectories        | L1 product bet — needs CoP experiments |
| Forums / streaks / AI chat = CoP / LPP                                | <b>Banned</b>                          |
| Shipping Community features creates mathematical identity             | <b>Banned</b> L3 without instruments   |

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## LII.10 Experiments spawned

| ID    | Question                                                                                                        | Design | Primary                                       | Kill condition                            |
|-------|-----------------------------------------------------------------------------------------------------------------|--------|-----------------------------------------------|-------------------------------------------|
| CoP-1 | LPP-shaped session (peripheral contribution + old-timer criteria + fade to solo) vs explain-first homework help | 2-arm  | solo_transfer_pass; coded participation moves | Help ≥ LPP spine → CoP framing overstated |

|          |                                                                     |           |                                              |                                                             |
|----------|---------------------------------------------------------------------|-----------|----------------------------------------------|-------------------------------------------------------------|
| CoP-2    | Named repertoire onboarding (“how we check here”) vs feature tour   | 2-arm     | next-session method language; retry_120s     | Tour ≥ repertoire → practice-naming useless                 |
| CoP-3    | Human tutor old-timer vs ungarded AI buddy (same time)              | 2-arm     | solo transfer; overtrust flags (XXXIII)      | AI ≥ tutor on solo → old-timer claim weak for this segment  |
| CoP-4    | Community chat with repertoire prompts vs open social chat          | 2-arm     | practice-talk rate; delayed accuracy         | Open ≥ prompted → repertoire constraint wrong               |
| CoP-5    | Story peripheral roles (watcher→helper→solver) vs flat problem list | 2-arm     | challenge_accept ; belonging items; transfer | Flat ≥ roles on learning → role trajectory theater          |
| CoP-QUAL | 10 Maya + 5 tutor: “who is in your math people?”                    | interview | coded: old-timer / repertoire / legitimacy   | “Community” = friends only, no practice → social tab demote |

**Pre-reg (XXXIV):** CoP-\* identify **participation design** / **old-timer necessity** / **fake-community kill** — not “CoP causes identity” or “LPP is universal in apps.”

## LII.11 Confidence table

| Claim                                                                                        | Label | Confidence |
|----------------------------------------------------------------------------------------------|-------|------------|
| Lave & Wenger (1991): LPP as central descriptor of situated learning in apprenticeship cases | FACT  | High       |
| Wenger (1998): CoP as meaning –practice–community–identity                                   | FACT  | High       |
| Boylan (2010): LPP transfer to usual school math is problematic; ecologies of participation  | FACT  | High       |
| Fuller et al. (2005): LPP limited in complex institutions                                    | FACT  | High       |
| Sfard (1998): do not choose only acquisition or only participation                           | FACT  | High       |

|                                                             |                                 |               |
|-------------------------------------------------------------|---------------------------------|---------------|
| SAFE-CoP is right default for MindCraft social/tutor design | FOUNDER BELIEF / HYPOTHESIS     | Medium        |
| Community features alone transform math identity            | SPECULATION / false as doctrine | Low (against) |

## LII.12 What this chapter kills

1. **Kill:** “We are a community of practice” as brand without enterprise + repertoire + old-timer.
2. **Kill:** Forums, Discord, streaks, leagues, or chat volume as LPP proof.
3. **Kill:** AI monologue  $\equiv$  apprenticeship / old-timer.
4. **Kill:** Treating school/ACT/app sessions as ethnographically equivalent to Lave & Wenger’s craft cases.
5. **Wound:** Participation-only pedagogy that deletes mastery/transfer instruments (Sfard).
6. **Survive:** SAFE-CoP; LPP-shaped tutor spine; peripheral = real work; trajectory metrics; CoP-1...5; “join the practice of recovering and proving” copy — not “join the community.”

**Doctrine until data:** Ship **SAFE-CoP** — name the practice, require real peripheral work and old-timer criteria, measure participation trajectories under acquisition+participation pluralism — never sell social tabs or CoP labels as identity engines.

## Part LIII — Religion / Ritual Light-Touch Design: Meaning Without Cult Dynamics

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 15; hour%6≠0; 3 researcher entries since synthesizer v1.7 → Researcher)

**Primary question:** How can MindCraft design *light-touch rituals* that mark transitions, regulate affect, and carry meaning — without drifting into cult dynamics, sacred-science branding, or high-arousal initiation theater?

**Owners:** Product (session / world / onboarding arcs) · Brand & copy · Equity / trust · Red Team

**Commercial job:** Ship a **SAFE-RITUAL** stack — brief, opt-in, secular, reversible meaning-markers tied to real practice — while killing “join the order,” streak liturgies, confession milieus, and unretracted ritual→grade overclaims.

### LIII.1 Why this chapter exists

The Constitution’s thesis is identity transformation under difficulty. Humans mark identity shifts with *rituals*: greetings before hard work, closing circles after miss→revise, naming a first solo transfer. Edtech and consumer apps already smuggle ritual language — streaks as devotion, leagues as congregations, onboarding as initiation, founder myth as sacred science — usually without naming the load-bearing psychology or the failure modes.

MindCraft has natural ritual-shaped moments: gap-scan completion, first soft-wrong recovery, tutor session open/close, story-world threshold crossings, Map unlocks. Parts XXIV (affect), XXV (identity), XXXI (flow), XLIII (SAFE-HABIT), XLIV (SAFE-REWARD), XLV (resilience), LII (SAFE-CoP) constrain pieces. What was missing was an honest doctrine that (a) uses ritual science for *light* meaning, (b) borrows transition structure without religion cosplay, and (c) installs a cult-dynamics kill list so “meaning” never becomes totalism.

**FOUNDER BELIEF under audit:** Becoming a confident mathematical thinker needs *marked moments* (not only item completion) — but the company must never sell belonging-through-devotion, milieus that punish exit, or doctrine that overrides the student’s evidence.

#### Claims we refuse as doctrine:

1. Rituals (or “sacred practice”) as proven grade engines from unretracted lab theater.
2. Streak / league / XP liturgy as identity North Star.
3. High-arousal initiation, confession milieus, or “you’re not one of us until...” gates.
4. Founder/product doctrine as sacred science that cannot be falsified (contradicts the lab itself).

## LIII.2 Constructs (product language)

| Construct              | Plain definition                                                                                                      | Product signal                                                        | Failure mode if misused                       |
|------------------------|-----------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------|-----------------------------------------------|
| <b>Ritual (psych)</b>  | Predefined, symbolic, often repetitive sequence that is <i>more than</i> its instrumental steps (Hobson et al., 2018) | Brief open/close; naming a miss class; candle-free “we begin”         | Calling every UX animation a ritual           |
| <b>Routine / habit</b> | Repeated behavior cue→response without required symbolic meaning (SAFE-HABIT)                                         | If-then practice cue                                                  | Relabeling habit as sacred                    |
| <b>Rite of passage</b> | Transition marked by separation → liminal → reaggregation (van Gennep; Turner)                                        | Gap-scan → liminal “not yet rated” → diagnosticCompleted + first path | Initiation hazing; permanent limbo            |
| <b>Imagistic mode</b>  | Infrequent, high-arousal rituals → vivid episodic memory; small exclusive groups (Whitehouse, 2004)                   | Rare world “boss” moments                                             | Trauma theater; faction bonding over learning |

|                                 |                                                                                                                                     |                                                      |                                             |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------|
| <b>Doctrinal mode</b>           | Frequent, low-arousal repetition → semantic/procedural memory; larger inclusive scale (Whitehouse, 2004)                            | Soft-wrong language; Map repertoire; session grammar | Indoctrination scripts; empty liturgy       |
| <b>Totalism / cult dynamics</b> | Thought-reform pattern: milieu control, confession cult, sacred science, loaded language, doctrine-over-person, exit costs (Lifton) | Anti-patterns in copy, retention, community          | “Family,” purity tests, unfalsifiable brand |
| <b>SAFE-RITUAL</b>              | Light-touch ritual doctrine that survives cult kills + retraction wounds                                                            | See LIII.8                                           | Sacred streak; initiation cosplay           |

**Operational definition (HYPOTHESIS):** A MindCraft moment counts as *SAFE-RITUAL* when (a) it is brief and optional or easily skippable, (b) symbolism points at *practice repertoire* (recover, prove, calibrate) not brand devotion, (c) exit/skip has no status punishment, (d) no confession or purity demand, (e) claims stay on the ladder (affect/transition aid ≠ causal identity engine).

### LIII.3 Psychology of ritual — regulatory functions (not magic grades)

**FACT (integrative review):** Hobson, Schroeder, Risen, Xygalatas & Inzlicht (2018, *Personality and Social Psychology Review*, 22(3), 260–284, doi:10.1177/1088868317734944) — organize empirical ritual work around three regulatory functions: **(a) emotion regulation, (b) performance goal states, (c) social connection**. Mechanisms include bottom-up features (repetition, formality, synchrony) and top-down meaning (belief that the sequence *is* a ritual / carries symbolic significance).

**FACT (grief / control):** Norton & Gino (2014, *Journal of Experimental Psychology: General*, 143(1), 266–272, doi:10.1037/a0031772) — rituals after losses (loved ones, relationships, lotteries) can reduce grief feelings and increase felt control relative to no-ritual controls. Useful as evidence that symbolic sequences can regulate emotion and agency — **not** as a math-learning RCT.

**FACT (consumption / meaning frame):** Vohs, Wang, Gino & Norton (2013, *Psychological Science*, 24(9), 1714–1721, doi:10.1177/0956797613478949) — ritualized actions can enhance consumption experience via greater involvement/savoring. Ally for “mark the moment” design; kill if used to justify gamified consumption of XP as learning.

**WOUND (retraction — do not market):** Brooks, Schroeder, Risen, Gino, Galinsky, Norton & Schweitzer (2016, *Organizational Behavior and Human Decision Processes*, 137, 71–85, doi:10.1016/j.obhdp.2016.07.004) — claimed pre-performance rituals reduce anxiety and improve performance (including high-stakes math-like tasks). **This article has been retracted.** MindCraft must **not** cite it as surviving evidence or sell “rituals raise exam scores.” Treat as a Red Team kill on ritual→grade overclaim.



**HYPOTHESIS for MindCraft:** Prefer Hobson’s *emotion regulation* and *goal-state* functions for session openers/closers and soft-wrong recovery markers; treat performance/grade claims as **banned** until unretracted, pre-registered evidence on MindCraft metrics (retry\_120s, solo\_transfer\_pass, affect check-ins).

**Kill:** “Science proves rituals improve math performance.” Survive: “Light symbolic sequences can regulate affect and mark transitions; grade claims need new evidence.”

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## LIII.4 Modes of religiosity — why high-arousal initiation is the wrong default

**FACT:** Whitehouse (2004, *Modes of Religiosity: A Cognitive Theory of Religious Transmission*, AltaMira Press / Rowman & Littlefield, ISBN 978-0-7591-0615-4) — religions tend to coalesce around **imagistic** vs **doctrinal** poles. Imagistic: infrequent, high-arousal rituals → flashbulb-like episodic memory; small, exclusive, often intense group bonds. Doctrinal: frequent, low-arousal, routinized teaching/ritual → semantic and procedural memory; larger, more inclusive, centrally scaffolded communities.

**FACT (extreme ritual ally — careful):** Xygalatas, Mitkidis, Fischer, Reddish, Skewes, Geertz et al. (2013, *Psychological Science*, 24(8), 1602–1605, doi:10.1177/0956797612472910) — extreme rituals can increase prosociality in field settings. This is **not** a product recommendation for teens in ACT prep; it shows imagistic arousal can bond groups — which is exactly the risk surface for cult-adjacent design.

**HYPOTHESIS:** MindCraft’s default should be **doctrinal-light**: frequent, low-arousal repertoire (session grammar, soft-wrong language, Map names) — not imagistic initiation (ordeal unlocks, shame fires, exclusive survivor factions). Rare story “threshold” moments may use mild imagistic *flavor* only if destaked, optional, and never required for belonging (SAFE-DD / equity).

**Product translation:** Gap-scan and first recovery are transitions, not hazing. World chapters can feel special without pain-as-proof.

**Kill:** Bootcamp/ordeal branding; “you haven’t really suffered until...” Survive: Frequent low-arousal practice markers + rare optional celebration.

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## LIII.5 Rites of passage — structure without cult costume

**FACT:** van Gennep (1909; English trans. 1960, *The Rites of Passage*, University of Chicago Press) — life transitions commonly share three phases: **separation**, **margin/limen**, **aggregation** (reincorporation).

**FACT:** Turner (1969, *The Ritual Process*; see also “Liminality and Communitas”) — elaborates the liminal phase as *betwixt and between*: ambiguous status, stripped markers, intense potential for *communitas* (anti-structure bonding). Powerful analytically; dangerous if product designers chase liminality as engagement hack (keep students permanently “in between,” dependent on the brand for status).

**HYPOTHESIS for MindCraft transitions:**

| Phase | Product move | Guardrail |
|-------|--------------|-----------|
|-------|--------------|-----------|

|             |                                                                           |                                                             |
|-------------|---------------------------------------------------------------------------|-------------------------------------------------------------|
| Separation  | Leave “I don’t know where I stand” → enter gap-scan / session             | No shame script; task language only (SAFE-EXPECTANCY)       |
| Liminal     | Hide-correctness diagnostic; soft-wrong; “not yet aggregated”             | Time-bounded; no status mockery; skippable ceremony         |
| Aggregation | diagnosticCompleted; first path; named repertoire (“how we recover here”) | Grant competence access, not trait identity; no purity test |

**Link to SAFE-CoP:** Aggregation should admit the student to *practice roles* (peripheral → fuller), not to a brand congregation.

**Kill:** Permanent liminality as retention. Survive: Clear exits into rights/obligations of the practice (you may attempt this edge solo).

## LIII.6 Cult dynamics — the kill list (Lifton as anti-pattern checklist)

**FACT (classic totalitarian criteria):** Lifton (1961, *Thought Reform and the Psychology of Totalism*; later editions restate cult criteria) — thought-reform / totalistic environments characteristically include patterns such as: **milieu control, mystical manipulation, demand for purity, cult of confession, sacred science, loading the language, doctrine over person, dispensing of existence** (who counts as real/valid). Cult-like groups often combine charismatic leader-focus, thought-reform patterns, and exploitation of idealism.

**FOUNDER BELIEF:** MindCraft’s research lab and product ethics require the *opposite* of sacred science: every doctrine chapter is falsifiable; Red Team is a feature; students may leave without metaphysical damnation.

**Product anti-patterns (SPECULATION → treat as design law until wounded):**

| Lifton pattern        | Edtech failure mode                                     | MindCraft ban                                        |
|-----------------------|---------------------------------------------------------|------------------------------------------------------|
| Milieu control        | App as only legitimate math space; shame offline help   | Allow tutors, school, parents; no “only here” purity |
| Mystical manipulation | Streak miracles; “the algorithm chose you” destiny copy | Transparent recommend reasons; no fate talk          |
| Demand for purity     | Perfect streaks; zero-miss identity                     | Soft-wrong; SAFE-MISCON; miss as data                |
| Cult of confession    | Forced public failure theater / oversharing             | Private by default; opt-in sharing                   |
| Sacred science        | Unfalsifiable brand claims; ban critique                | Claim ladder; lab kills; no 2-sigma marketing        |
| Loading the language  | In-group jargon that replaces thinking                  | Repertoire words OK; ban thought-stopping slogans    |
| Doctrine over person  | Override student evidence with brand story              | Graph + outcomes over vibes; SAFE-CALIB              |

|                         |                             |                           |
|-------------------------|-----------------------------|---------------------------|
| Dispensing of existence | Rank/league as who “counts” | SAFE-COMPARE; no exile UX |
|-------------------------|-----------------------------|---------------------------|

**Kill:** “We’re a movement / family / order.” Survive: “We’re a practice with optional meaning-markers and exit dignity.”

### LIII.7 Competitive audit — ritual theater in the market

Duo streaks/leagues = doctrinal liturgy + mild imagistic competition (habit colonization; cue habits without sacred streak NS). Khan mastery energy = completion ritual (keep instruments; add recovery markers). Brilliant “aha” = imagistic spark (celebrate then probe transfer). GPT chat = pseudo-confession / endless liminality (SE-constrained close). Studygram hustle = purity + confession (ban grind-as-virtue).

**Commercial implication:** Position as **meaning without membership taxes** — mark *mathematical practice*, not brand devotion. Copy: “mark the recovery,” “open like a craft,” “close with what changed” — never “join the faithful.”

### LIII.8 SAFE-RITUAL stack (product law)

1. **Light and short** — seconds to a minute; never a sermon that displaces attempts (SAFE-DP time budget).
2. **Optional / skippable** — no status loss for skip; no streak-as-morality.
3. **Point at practice** — symbolize recover / prove / calibrate / transfer — not founder, brand, or league.
4. **Doctrinal default** — frequent low-arousal repertoire; rare celebration only, never ordeal.
5. **Bounded liminality** — transitions resolve into clear rights (you may try X); no permanent betwixt for retention.
6. **Anti-totalism** — ban milieu monopoly, confession theater, sacred science, purity, exit shame.
7. **Affect ≠ grade claim** — may aim at calm / agency / belonging-to-practice; **no** unretracted ritual→score marketing.
8. **Equity** — no borrowed sacred forms that mock students’ real religions; secular by default (Part XXXVI).
9. **Claim ladder** — ritual language max L1 product bet until RIT-\* experiments; ban L3 identity-via-ceremony without instruments.

### LIII.9 Claim ladder

| Claim                                                                      | Max ladder without new data |
|----------------------------------------------------------------------------|-----------------------------|
| Rituals can regulate emotion / goal states / connection (mechanism review) | L1 (Hobson et al., 2018)    |
| Symbolic sequences can increase felt control after loss                    | L1 (Norton & Gino, 2014)    |
| Imagistic vs doctrinal modes describe divergent ritual ecologies           | L1 (Whitehouse, 2004)       |

|                                                                                  |                                                               |
|----------------------------------------------------------------------------------|---------------------------------------------------------------|
| Rites of passage often share separation–liminal–aggregation                      | L1 (van Gennep; Turner)                                       |
| Lifton patterns are useful anti-design checks for edtech                         | L1 design heuristic (not a clinical diagnosis of competitors) |
| MindCraft light rituals improve <code>retry_120s</code> / affect / solo transfer | L1 product bet — needs RIT experiments                        |
| Rituals raise ACT scores / prove identity transformation                         | <b>Banned</b> without unretracted evidence                    |
| Streak liturgy / ordeal initiation / “family” milieus as identity engines        | <b>Banned</b>                                                 |

### LIII.10 Experiments spawned

| ID       | Question                                                                     | Design    | Primary                                          | Kill condition                                                           |
|----------|------------------------------------------------------------------------------|-----------|--------------------------------------------------|--------------------------------------------------------------------------|
| RIT-1    | 30s session open ritual (name edge + breath/pose optional) vs silent start   | 2-arm     | pre-task affect; <code>challenge_accept</code>   | Silent $\geq$ ritual on start+transfer → opener useless                  |
| RIT-2    | Soft-wrong recovery marker (“name the miss class”) vs correct/incorrect only | 2-arm     | <code>retry_120s</code> ; method-change code     | Binary $\geq$ marker → recovery ritual theater                           |
| RIT-3    | Gap-scan aggregation ceremony (brief) vs auto-redirect no mark               | 2-arm     | next-session return; belonging -to-practice item | Auto $\geq$ ceremony → skip ceremony permanently                         |
| RIT-4    | Optional celebration on first <code>solo_transfer_pass</code> vs none        | 2-arm     | delayed transfer; skip rate                      | Forced > optional engagement but worse affect → force banned             |
| RIT-5    | Streak-framed cue vs practice-repertoire cue (same schedule)                 | 2-arm     | 4-week return; identity items; anxiety           | Streak $\geq$ repertoire on learning → still ban streak as NS; copy only |
| RIT-QUAL | 10 Maya: which moments felt meaningful vs creepy/cultish?                    | interview | coded: light / pressure / exit fear              | “Family/order” language → kill that copy                                 |

**Pre-reg (XXXIV):** RIT-\* identify **transition/affect design and anti-totalism** — not “ritual causes identity” or “rituals raise scores.”

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### LIII.11 Confidence table

| Claim                                                             | Label                                                               | Confidence         |
|-------------------------------------------------------------------|---------------------------------------------------------------------|--------------------|
| Hobson et al. (2018): three regulatory functions of ritual        | FACT                                                                | High               |
| Norton & Gino (2014): rituals can ease grief / raise felt control | FACT                                                                | High               |
| Brooks et al. (2016) ritual→performance paper                     | FACT it was published;<br><b>retracted</b> — do not use as evidence | High (against use) |
| Whitehouse (2004): imagistic vs doctrinal modes                   | FACT                                                                | High               |
| van Gennep / Turner: passage / liminality structure               | FACT                                                                | High               |
| Lifton totalism criteria as edtech anti-patterns                  | HYPOTHESIS / design heuristic                                       | Medium–High        |
| SAFE-RITUAL is right default for MindCraft meaning UX             | FOUNDER BELIEF / HYPOTHESIS                                         | Medium             |
| Ceremony alone transforms math identity                           | SPECULATION / false as doctrine                                     | Low (against)      |

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### LIII.12 What this chapter kills

1. **Kill:** Ritual→exam-score marketing (especially via retracted Brooks et al., 2016).
2. **Kill:** Streak / league / XP as sacred liturgy or North Star.
3. **Kill:** High-arousal initiation, confession milieus, purity tests, exit shame.
4. **Kill:** “Movement / family / order” copy that loads Lifton patterns.
5. **Wound:** Permanent liminality-as-retention; imagistic faction bonding.
6. **Survive:** SAFE-RITUAL; doctrinal-light repertoire; bounded passages; anti-totalism checklist; RIT-1...5; “mark the recovery” copy — not “join the faithful.”

**Doctrine until data:** Ship **SAFE-RITUAL** — brief, skippable, practice-pointing meaning-markers under an explicit cult-dynamics ban — never sell devotion, ordeal, or unretracted ritual-performance claims as the identity engine.

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# Part LIV — Military / Aviation Brief–Debrief: After-Action Reviews for Math Sessions

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 18 ≡ Red Team slot, but ch54 never written → prefer Researcher per rotation; 4 researcher entries since synthesizer v1.7 → Researcher)

**Primary question:** What transferable structure do military after-action reviews and aviation brief/debrief cycles offer for MindCraft tutoring and practice sessions — without militaristic cosplay, blame culture, or overclaiming effect sizes as ACT marketing?

**Owners:** Product (session spine / Notes close) · Tutor ops · Coach UX · Brand & copy · Red Team

**Commercial job:** Ship a **SAFE-AAR** stack — brief → attempt → structured debrief with objective traces — while killing bootcamp branding, lecture-as-debrief, trauma-debrief confusion, and “debriefs improve performance 25%” as an undifferentiated growth claim.

## LIV.1 Why this chapter exists

Parts LI (SAFE-DP) and LIII (SAFE-RITUAL) already constrain *time budgets* and *light meaning markers*. What was missing is the **operational cycle** high-reliability domains use to convert experience into the next attempt: a short **brief** that sets intent and roles, an event under observation, and a **debrief / after-action review (AAR)** that compares expected vs actual performance without collapsing into critique-as-humiliation.

MindCraft already has the raw materials: session open (affect check-in), attempt traces, soft-wrong codes, outcomes posted to the graph, Notes/session summaries, and coach cards. Competitors either skip structured reflection (answer-delivery chat), fake it with streak liturgy (Duo), or replace it with instructor monologue (many tutoring products and AI wrap text). The commercial wedge is not “we’re like the military.” It is: **every hard attempt leaves a learning artifact the student helps interpret.**

**FOUNDER BELIEF under audit:** Identity transformation under difficulty requires *reviewed attempts*, not only more problems — but review must be non-punitive, student-participatory, and tied to the next edge (SAFE-DP), not to brand devotion (SAFE-RITUAL) or trait shame (SAFE-EXPECTANCY).

**Claims we refuse as doctrine:**

- 1. Militaristic / “warrior math” branding as belonging technology.
- 2. AAR ≡ critical-incident stress debriefing (trauma processing) — different animal.
- 3. Marketing “+25% performance” from debrief meta-analyses as if it were an ACT score guarantee.
- 4. Instructor/AI lecture that *calls itself* a debrief while blocking student self-analysis.

## LIV.2 Constructs (product language)

| Construct | Plain definition | Product signal | Failure mode if misused |
|-----------|------------------|----------------|-------------------------|
|-----------|------------------|----------------|-------------------------|

|                                    |                                                                                                                  |                                                                                 |                                        |
|------------------------------------|------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|----------------------------------------|
| <b>Brief</b>                       | Short pre-event alignment on intent, success criteria, roles, likely traps (CRM / FAA markers)                   | Session open: target edge, level, “what counts as success,” optional coach role | 10-min sermon; fate talk; trait labels |
| <b>After-action review (AAR)</b>   | Systematic technique turning a recent event into learning via feedback, reflection, discussion (Keiser & Arthur) | Post-mission 4-question close; tutor end-of-session AAR                         | Critique lecture; blame; vibes-only    |
| <b>Debrief (training)</b>          | Structured reflection after simulation/performance; often facilitator-guided                                     | Coach/tutor facilitation; self-led when traces exist                            | Endless chat liminality; AI monologue  |
| <b>Objective performance media</b> | Records of what happened (video, logs, instruments) — not memory alone                                           | Attempt list, soft-wrong codes, timing, outcomes JSON                           | Relying on “how did it feel?” only     |
| <b>Facilitation (vs lecture)</b>   | Elicit crew/student self-analysis; deepen participation (Dismukes et al.)                                        | Advocacy–inquiry; student names frame first                                     | Tutor/AI tells the moral of the story  |
| <b>Good judgment stance</b>        | Share evaluative judgment + curiosity about learner’s frames (Rudolph et al.)                                    | Soft-wrong + “what were you seeing?”                                            | Fake nonjudgmental praise; or blame    |
| <b>Alignment</b>                   | Match participants ↔ focus ↔ measurement (team vs individual)                                                    | Solo practice AAR on solo metrics; tutor+student AAR on shared plan             | Team vibes measured by individual XP   |
| <b>SAFE-AAR</b>                    | Brief–attempt–debrief doctrine that survives militarism / blame / overclaim kills                                | See LIV.8                                                                       | Bootcamp cosplay; 25%-as-marketing     |

**Operational definition (HYPOTHESIS):** A MindCraft close counts as *SAFE-AAR* when (a) a pre-brief named the intended edge and success criterion, (b) the debrief answers expected / actual / sustain / improve using **at least one objective trace**, (c) the student produces  $\geq 1$  self-analytic statement before any AI wrap, (d) the output is a next-attempt commitment (not a grade sermon), (e) no trait labels, no punishment for miss, no militaristic belonging tax.

### LIV.3 Army AAR doctrine — four questions, not a roast

**FACT (doctrine lineage):** U.S. Army training circular **TC 25-20** (*A Leader's Guide to After Action Reviews*, 30 Sep 1993) and subsequent training doctrine institutionalize the AAR as a professional discussion focused on training objectives — not a humiliation ritual. Contemporary restatements (e.g., Center for Army Lessons Learned handbooks; USAID AAR guidance adapted from TC 25-20) center four questions:

1. What was **expected** to happen?
2. What **actually** happened?
3. What went **well**, and why?
4. What can be **improved**, and how?

**FACT (process intent):** Formal Army AAR guidance emphasizes open-ended questioning so participants discover gaps vs standards; facilitators must avoid critique-as-lecture (FM 25-101 Appendix G / CALL lineage). Focus: **results vs objectives**, sustain strengths, actionable fixes.

**HYPOTHESIS for MindCraft:** Map onto a 60–120s practice close and a 3–5 min tutor close — brief names expected edge; traces show actual; student names one sustain; one improve-how (SAFE-HABIT if-then or method change). Kill “AAR = public roast.” Survive: standards-referenced discovery.

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## LIV.4 Meta-analytic evidence — real gains, careful transfer

**FACT:** Tannenbaum & Cerasoli (2013, *Human Factors*, 55(1), 231–245, doi:10.1177/0018720812448394) — meta-analysis of 46 samples ( $N = 2,136$ ). Debriefs / AARs improve effectiveness vs control by about **25% on average** ( $d \approx 0.67$ ). Effects similar for teams and individuals, simulated and real settings, medical and nonmedical samples. **Alignment** (participants, intent, measurement level) bolsters effects; facilitated and structured debriefs trend stronger (facilitation estimate based on fewer nonfacilitated studies).

**FACT:** Keiser & Arthur (2021, *Journal of Applied Psychology*, 106(7), 1007–1032; advance online 2020, doi:10.1037/apl0000821) — bare-bones meta-analysis of 61 studies (107 *ds*; 915 teams and 3,499 individuals). Overall AAR effect  $d \approx 0.79$  across multiple training criteria — larger than Tannenbaum & Cerasoli's task-performance-focused estimate. Two characteristics consistently help: **(a) alignment** to individual vs team, **(b) objective performance review media**. Facilitation interacts: self-led + team-aligned and self-led + objective media among stronger combinations; structure matters more in military than healthcare contexts.

**HYPOTHESIS (transfer wound):** These samples are largely military, healthcare simulation, and organizational training — **not** ACT prep for anxious Maya. Treat  $d \approx 0.67$ – $0.79$  as evidence that structured post-event reflection is a high-leverage training design pattern — **not** license to claim “MindCraft AARs raise SAT/ACT by 25%.”

### Commercial claim ladder:

- L1: Debriefs/AARs improve training outcomes in reviewed domains (cite metas).
- L1 product bet: SAFE-AAR improves MindCraft instruments (retry\_120s, method-change codes, solo\_transfer\_pass).
- **Banned:** Undifferentiated “science says +25%” on landing pages without domain caveat and MindCraft pre-reg.

**Kill:** 2-sigma-style overclaim via borrowed meta  $d$ . Survive: “Structured brief–debrief with traces beats vibes-only close.”

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## LIV.5 Aviation — brief as error management; debrief as facilitated self-analysis

**FACT (CRM briefing as performance marker):** FAA Advisory Circular **AC 120-51E** (*Crew Resource Management Training*) lists effective **briefings** among CRM behavioral markers: thorough, address coordination/planning/problems, establish open communication, set expectations for SOP deviations and automation duties. Briefings are error-management tools — they align the crew *before* the event, not after the crash.

**FACT (CRM evolution):** Helmreich, Merritt & Wilhelm (1999, *International Journal of Aviation Psychology*, 9(1), 19–32; see also Helmreich line of NASA/UT work) — CRM evolved toward error management: avoid, trap, mitigate. Briefing strategies and situation awareness modules are framed as countermeasures, not soft skills theater.

**FACT (LOFT debrief facilitation):** Dismukes, Jobe & McDonnell (1997, NASA TM-110442, *LOFT Debriefings: An Analysis of Instructor Techniques and Crew Participation*) — analyzed 36 LOFT debriefings at five U.S. airlines; developed the Debriefing Assessment Battery. **Facilitation** (vs instructor lecture) increases depth of crew participation and self-analysis of CRM performance; instructor skill varied dramatically; crews were responsive but rarely fully led their own debriefs. Companion: McDonnell, Jobe & Dismukes (1997, NASA TM-112192, *Facilitating LOS Debriefings: A Training Manual*).

### HYPOTHESIS for MindCraft:

- **Brief** = gap-scan / mission card / tutor open: name the edge, success criterion, known misconception trap, role of coach (hints vs silence).
- **Debrief** = student-led analysis with tutor/coach as facilitator; AI may *prompt* advocacy–inquiry, must not deliver the sermon first (aligns with Part XL SE constraints and Part XXXIII AI trust).

**Product translation:** Tutor briefing ethics from SAFE-EXPECTANCY (task-only language) + CRM brief markers (coordination, traps, automation = “when does the coach speak?”).

**Kill:** Captain-monologue onboarding. Survive: short interactive brief + facilitated close.

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## LIV.6 Healthcare simulation — good judgment, not fake nonjudgment

**FACT:** Rudolph, Simon, Dufresne & Raemer (2006, *Simulation in Healthcare*, 1(1), 49–55, doi:10.1097/01266021-200600110-00006; PubMed 19088574) — “There’s no such thing as ‘nonjudgmental’ debriefing.” **Debriefing with good judgment:** (1) frames drive actions → results; (2) stance of genuine curiosity (mistakes as puzzles); (3) **advocacy–inquiry** pairs (state observation + judgment, then ask what the learner’s frame was). Helps instructors share evaluative judgment without destroying trust.

**Link to SAFE-MISCON / SAFE-CALIB:** Soft-wrong is not empty kindness; it is destaked judgment plus frame inquiry. Hide-correctness diagnostic is a special brief (no outcome reveal) — its “debrief” must not smuggle keys that corrupt the graph (C4).

**HYPOTHESIS:** MindCraft coach cards should default to advocacy–inquiry templates (“I notice you treated the inequality like an equation when you multiplied by  $-1$  — what were you assuming?”) rather than praise-only or scold-only.

**Kill:** “Nonjudgmental forever” (hides standards) and “brutal honesty” (hides curiosity). Survive: good-judgment debrief.

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## LIV.7 Distinctions and kill list (scope control)

Do not confuse training AARs with **critical-incident stress debriefing** (trauma processing — different ethics). Ban militaristic belonging (“warrior,” bootcamp shame), streak-as-AAR, AI monologue labeled as debrief, vibes-only hot-wash when traces exist, public leaderboard “reviews” (SAFE-COMPARE), and “+25% smarter” ads from borrowed meta *d*.

**Wound:** Hot-wash alone under-captures learning — useful as phase 1, insufficient as the whole product. MindCraft default: immediate 60s close + optional next-day Notes deepening.

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## LIV.8 SAFE-AAR stack (product law)

1. **Brief before attempt** — name edge, success criterion, trap, coach role (CRM brief markers).
  2. **Attempt under observation** — log objective media (outcomes, codes, timing).
  3. **Four-question close** — expected / actual / sustain / improve (TC 25-20 lineage).
  4. **Student speaks first** —  $\geq 1$  self-analytic statement before tutor/AI wrap (Dismukes facilitation).
  5. **Good judgment** — advocacy-inquiry; standards visible; curiosity about frames (Rudolph).
  6. **Align level** — solo metrics for solo practice; shared plan metrics for tutor sessions (Tannenbaum; Keiser).
  7. **Next-attempt output** — one concrete improve-how tied to SAFE-DP time budget.
  8. **Non-punitive climate** — miss = data; no trait labels (SAFE-EXPECTANCY); no militaristic shame.
  9. **Claim ladder** — meta *d* stays L1 domain evidence; MindCraft ACT claims need AAR-\* experiments.
  10. **Anti-cosplay** — borrow the *cycle*, not the costume; secular practice language (SAFE-RITUAL).
- 

## LIV.9 Competitive audit — who actually closes the loop?

Khan tells correctness without student-led expected/actual. Duo substitutes streak liturgy for AAR. Brilliant celebrates aha but rarely closes sustain/improve. ChatGPT tutors lecture after chatty “goals.” Human tutoring quality varies with facilitator skill (Dismukes variance). MindCraft already logs graph outcomes — formalize SAFE-AAR in practice close + Notes.

**Commercial implication:** Position as **reviewed struggle** — brief the edge, attempt with traces, close expected/actual — never “military-grade” or borrowed “+25%” miracles. Parent copy: expected vs happened → what changes next. Student copy: “name what you thought would happen.”

---

## LIV.10 Claim ladder

|       |                             |
|-------|-----------------------------|
| Claim | Max ladder without new data |
|-------|-----------------------------|

|                                                                                                    |                                                         |
|----------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| AARs/debriefs improve training outcomes in meta-analyzed domains ( $d \approx 0.67\text{--}0.79$ ) | L1 (Tannenbaum & Cerasoli, 2013; Keiser & Arthur, 2021) |
| Four-question AAR structure is institutionalized Army training practice                            | L1 (TC 25-20 lineage / CALL restatements)               |
| Facilitation deepens self-analysis vs lecture in LOFT debriefs                                     | L1 (Dismukes et al., 1997)                              |
| CRM briefings are performance markers for coordination / error management                          | L1 (FAA AC 120-51E; Helmreich line)                     |
| Good-judgment advocacy–inquiry is a validated simulation debrief stance                            | L1 method (Rudolph et al., 2006)                        |
| SAFE-AAR improves MindCraft <code>retry_120s</code> / method-change / solo transfer                | L1 product bet — needs AAR experiments                  |
| “+25%” / meta $d$ as ACT or identity guarantee                                                     | <b>Banned</b>                                           |
| Bootcamp / warrior branding as belonging engine                                                    | <b>Banned</b>                                           |
| AAR $\equiv$ trauma CISM                                                                           | <b>Banned</b> conflation                                |

## LIV.11 Experiments spawned

| ID    | Question                                                     | Design        | Primary                                                       | Kill condition                                                                      |
|-------|--------------------------------------------------------------|---------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------|
| AAR-1 | 4-question practice close vs “nice job / try again” toast    | 2-arm         | <code>retry_120s</code> ; method-change code rate             | Toast $\geq$ AAR on retry+transfer $\rightarrow$ close theater                      |
| AAR-2 | Brief (edge+success+trap) vs no brief, same items            | 2-arm         | <code>challenge_accept</code> ; first-attempt process quality | No-brief $\geq$ brief $\rightarrow$ brief bloat; shorten or cut                     |
| AAR-3 | Objective-trace AAR vs memory-only AAR                       | 2-arm         | accuracy of self-report; next-session transfer                | Memory $\geq$ trace $\rightarrow$ still keep traces for graph; copy only            |
| AAR-4 | Student-first facilitation vs AI-monologue wrap              | 2-arm         | SE quality rubric; <code>solo_transfer_pass</code>            | Monologue $\geq$ student-first $\rightarrow$ contradicts XL; kill monologue default |
| AAR-5 | Tutor trained on advocacy–inquiry vs business-as-usual close | 2-arm cluster | session notes quality; student affect; 2-week transfer        | BAU $\geq$ trained $\rightarrow$ training content fail                              |

|          |                                                                  |           |                                 |                                      |
|----------|------------------------------------------------------------------|-----------|---------------------------------|--------------------------------------|
| AAR-QUAL | 10 Maya + 5 tutors: which closes felt useful vs military/shamey? | interview | coded: useful / cosplay / blame | “Bootcamp” language → kill that copy |
|----------|------------------------------------------------------------------|-----------|---------------------------------|--------------------------------------|

**Pre-reg (XXXIV):** AAR-\* identify **session architecture** (brief–debrief design) — not “military methods raise ACT” or identity-from-discipline myths.

## LIV.12 Confidence table

| Claim                                                                             | Label                           | Confidence    |
|-----------------------------------------------------------------------------------|---------------------------------|---------------|
| Tannenbaum & Cerasoli (2013): debriefs $d \approx 0.67$                           | FACT                            | High          |
| Keiser & Arthur (2021): AAR $d \approx 0.79$ ; alignment + objective media matter | FACT                            | High          |
| TC 25-20 / Army AAR four-question core                                            | FACT                            | High          |
| Dismukes et al. (1997): facilitation → deeper crew self-analysis                  | FACT                            | High          |
| Rudolph et al. (2006): good-judgment / advocacy–inquiry                           | FACT (method paper)             | High          |
| FAA CRM briefing markers                                                          | FACT (advisory doctrine)        | High          |
| Meta effects transfer cleanly to ACT prep for anxious teens                       | HYPOTHESIS / wounded            | Low–Medium    |
| SAFE-AAR is right default session spine for MindCraft                             | FOUNDER BELIEF / HYPOTHESIS     | Medium        |
| Military cosplay improves belonging or scores                                     | SPECULATION / false as doctrine | Low (against) |

## LIV.13 What this chapter kills

1. **Kill:** “Military-grade” / bootcamp / warrior branding as identity or efficacy claim.
2. **Kill:** Marketing meta-analytic  $d$  / “+25%” as undifferentiated ACT or IQ promise.
3. **Kill:** Conflating training AARs with trauma CISM.
4. **Kill:** Lecture or AI monologue that labels itself a debrief while blocking student self-analysis.
5. **Kill:** Vibes-only closes without objective traces when traces exist.

6. **Wound:** Hot-wash-only products; facilitator skill variance (train tutors; constrain AI).

7. **Survive:** SAFE-AAR; four-question close; CRM-style brief; facilitation + good judgment; AAR-1...5; copy as **reviewed struggle** — not costume cosplay.

**Doctrine until data:** Ship **SAFE-AAR** — brief the edge, attempt with traces, close with expected/actual/sustain/improve under non-punitive facilitation — and never sell discipline theater or borrowed percentage miracles as the identity engine.

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## Part LV — Sports Film-Study Pedagogy: Error Clips → Coach Cards

**Chapter status:** Living evidence brief — Researcher tick 2026-07-31 (UTC hour 21; hour%6≠0; 5 researcher entries since synthesizer v1.7 → Researcher)

**Primary question:** What transferable design does sports film-study / video-based feedback (VBF) offer for MindCraft practice review — turning attempt traces into sparse, student-controlled “clips” and coach cards — without sports-cosplay branding, feedback floods, or highlight-reel shame?

**Owners:** Product (Solver / Notes / coach cards) · Tutor ops · Coach UX · Brand & copy · Red Team

**Commercial job:** Ship a **SAFE-FILM** stack — objective attempt clips + less-is-more cueing + learner-timed review + self-model pairing — while killing “we’re like a sports academy” marketing, monologue film rooms, and flood-of-tips as pedagogy.

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### LV.1 Why this chapter exists

Part LIV (SAFE-AAR) established the *cycle*: brief → attempt with traces → structured close. Sports film study is the **media layer** of that cycle — high-performance domains do not rely on memory alone; they rewatch selected moments, cue attention, and commit the next play. MindCraft already logs the analogue of game film: attempt lists, soft-wrong codes (mis\_ IDs), timing, format tags, outcomes JSON. Competitors either never replay process (answer-key chat), replay only correctness (Khan-style), or flood the learner with AI commentary that no one remembers a week later.

**FOUNDER BELIEF under audit:** Identity as a mathematical thinker grows when the learner can *see their own decision frames* on a specific attempt — not when a brand dresses practice as a locker room. Film-study transfer is about **selective objective media + facilitation**, not jersey cosplay.

**Claims we refuse as doctrine:**

1. Sports-academy / “elite athlete” branding as belonging or efficacy technology.
  2. Coach monologue over 30 tips per “film room” (recall collapses).
  3. Highlight-reel of only successes *or* only failures as identity theater.
  4. Treating declarative recall of tips as if it were transfer to next-attempt performance.
  5. Equating any video UI with film-study pedagogy.
-

## LV.2 Constructs (product language)

| Construct                          | Plain definition                                                         | Product signal                                                      | Failure mode if misused               |
|------------------------------------|--------------------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------|
| <b>Film study / VBF</b>            | Structured review of performance via recorded moments + guided attention | Replay attempt; soft-wrong clip; coach card on <i>that</i> step     | Passive highlight reel; shame montage |
| <b>Performance analysis (PA)</b>   | Evaluate/describe performance with media (Pearson et al.)                | Graph outcomes + process codes as “tape”                            | Stats without learner agency          |
| <b>Self-observation</b>            | Learner watches <i>own</i> prior attempt                                 | “Replay your steps” before AI wrap                                  | Narcissistic doom-scroll of errors    |
| <b>Skilled / expert model</b>      | Observation of a competent demonstration                                 | Worked example / correct pathway card paired to same archetype      | Model-only → “I can’t do that” gap    |
| <b>Self-modeling / feedforward</b> | Edited view of self succeeding at a near-future challenge (Dowrick)      | Show student’s own prior <i>successful</i> transfer on related edge | Fake montages that invent competence  |
| <b>Attentional cueing</b>          | Direct gaze/attention to the decision that mattered                      | Coach card highlights inequality sign flip / bridge gap             | Cue everything = cue nothing          |
| <b>Self-controlled feedback</b>    | Learner chooses <i>when</i> to request video/KP review                   | “Review this attempt?” on demand vs forced after every item         | Autonomy theater with no cue quality  |
| <b>SAFE-FILM</b>                   | Film-study doctrine that survives sports cosplay / flood / shame kills   | See LV.8                                                            | Academy branding; tip dumps           |

**Operational definition (HYPOTHESIS):** A MindCraft review counts as *SAFE-FILM* when (a) the clip is an **objective attempt trace** (not vibes), (b)  $\leq 3$  attentional cues / coaching points are attached, (c) the learner can request or skip replay (self-control), (d) when a miss is shown, a **paired skilled model or student’s own prior success** on a related edge is available (self-model), (e) the output is a next-attempt commitment (SAFE-AAR improve-how), (f) no public shame reel, no sports-cosplay belonging tax.

## LV.3 Applied models of observation — who / what / when / how

**FACT:** Ste-Marie, Law, Rymal, Jenny, Hall & McCullagh (2012, *International Review of Sport and Exercise Psychology*, 5(2), 145–176, doi:10.1080/1750984X.2012.665076) advance an **Applied Model for the Use of Observation (AMUO)**: practitioners should assess observer and task characteristics, then vary **who** is observed, **what** is observed (and instructional features), **when** observation occurs, and **how** information is delivered — not treat “show

video” as a single intervention.

**FACT:** Ste-Marie et al. (2020, *Research Quarterly for Exercise and Sport*, doi:10.1080/02701367.2019.1693489) revisit AMUO across 2011–2018 applied-task articles; observation remains effective across a wider task base, but **guidelines still require intentional Who/What/When/How design** — effectiveness is not automatic from “video exists.”

**HYPOTHESIS for MindCraft:** Map AMUO onto practice:

- **Who:** self (attempt replay) ± skilled model (worked pathway for same archetype) ± peer only when SAFE-COMPARE allows.
- **What:** the decision frame that produced the soft-wrong — not the whole worksheet.
- **When:** after attempt, before next isomorphic item; optional delayed Notes deepen (aligns SAFE-AAR hot-wash + later deepen).
- **How:** sparse cueing + student-first naming (facilitation), not AI monologue.

**Kill:** “We added a replay button” as film-study. Survive: designed observation with AMUO knobs.

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## LV.4 Video feedback as performance analysis — learning ≠ tip retention

**FACT:** Pearson, Webb, Milligan & Dicks (2023/2025 print, *International Review of Sport and Exercise Psychology*, 18(1), 417–439, doi:10.1080/1750984X.2023.2235700) — integrative review of video feedback as a facet of PA (16 studies from 5,838 screened; OSF registration noted). Delivery mode (coach-led classroom vs self-led), environment, and **scheduling** shape learning; scheduling often aids **declarative knowledge retention** more clearly than transfer to on-field performance. Collaborative coach–athlete practices may enhance engagement. Authors call for better frameworks and experimental variety — PA is common; *learning-optimized* PA is under-specified.

**Wound (transfer):** Elite sport PA classrooms ≠ Maya’s 12-minute practice session. Treat Pearson et al. as a warning against equating “we reviewed the tape” with “performance transferred.”

**Commercial implication:** MindCraft must instrument **next-attempt behavior** (retry\_120s, method-change codes, solo\_transfer\_pass) — not tip-quiz scores after coach cards.

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## LV.5 Feedback floods kill recall — less is more

**FACT:** Mason, Farrow & Hattie (2020, *International Journal of Sports Science & Coaching*; doi:10.1177/1747954120951080) — exploratory study of one-to-one post-match video feedback in an AFL club (6 coach–player dyads). Coaches delivered on the order of ~**30 feedback messages** per meeting. One week later, players recalled about **50% of summarised themes** but only about **6% of fine-grained feedback idea units**. Authors encourage a “**less is more**” approach.

**HYPOTHESIS:** AI tutors and “helpful” coach UIs recreate the AFL flood at machine speed — worse for anxious novices (SAFE-DD × anxiety; CLT). Cap coach-card points per clip at **1–3**. Prefer one bridge/misconception handle over a laundry list.

**Product translation:** Soft-wrong → single primary mis\_ / ingredient cue → one advocacy–inquiry prompt (Rudolph/SAFE-AAR) → one next-attempt tweak. Kill “here are eight tips from your film.”

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## LV.6 Self-controlled video feedback — autonomy of *when*, not absence of coaching

**FACT:** van der Meer, van den Hoven, van der Kamp & Savelsbergh (2024, *Research Quarterly for Exercise and Sport*, 95(2), 537–545, doi:10.1080/02701367.2023.2275801; online 2023) — intermediate tennis players; self-controlled video feedback (request) vs yoked externally controlled schedule; coach provided video with attentional cueing and transitional tactical statements. Self-controlled group showed **larger tactical performance gains** vs pretest at posttest and **one-week retention**. Self-efficacy did not differ by group; gains were not predicted by self-efficacy / self-regulative questionnaire scores in this sample.

**FACT (supporting line):** Aiken, Fairbrother & Post (2012, *Frontiers in Psychology*, 3:338, doi:10.3389/fpsyg.2012.00338) — self-controlled video feedback benefits basketball set-shot learning relative to yoked schedules (motor-learning line often cited in later VBF work).

**Wound:** Meta-analytic debate exists on the strength of self-controlled feedback advantages (van der Meer et al. note McKay et al., 2022 vs Jimenez-Diaz et al., 2021). Do **not** market “science proves self-paced video raises ACT.” Treat as design prior: **learner-timed review + quality cueing** beats forced tip dumps *or* no review.

**HYPOTHESIS for MindCraft:** After a miss (or optional after a hard success), offer “Review this attempt?” rather than auto-playing a 90s AI lecture. Forced review every item → fatigue and controlling climate (SAFE-REWARD / SAFE-DD).

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## LV.7 Self-model pairing and feedforward — identity-safe media

**FACT (pairing):** Robertson, St. Germain & Ste-Marie (2018, *Journal of Motor Learning and Development*, 6(1), 18–34, doi:10.1123/jmld.2016-0027) — gymnasts learning two skills in a within-subject design: self-observation **paired with a skilled model** outperformed self-observation alone later in acquisition; error-identification **response sensitivity** was also higher for paired skills.

**FACT (self-modeling / feedforward):** Dowrick (1999, *Applied and Preventive Psychology*, 8(1), 23–39, doi:10.1016/S0962-1849(99)80009-2) reviews self-modeling; Dowrick (2012, *WIREs Cognitive Science*, 3(2), 215–230, doi:10.1002/wcs.1156) theorizes “learning from the future” via reconfigured component skills. Dowrick, Kim-Rupnow & Power (2006, *Journal of Special Education*, 39(4), 194–207, doi:10.1177/00224669060390040101) — video **feedforward** plus tutoring improved reading fluency; in 9/10 cases improvement rate was greatest during feedforward (small *N* — transfer carefully).

**HYPOTHESIS:** A miss clip should not stand alone. Pair with (1) a skilled worked pathway for the same archetype and/or (2) the student’s own prior success on a related edge. Feedforward spirit without fake editing: **show competence already owned, then the gap**.

**Kill:** Failure-only shame reels; success-only dopamine montages that hide productive misconceptions (SAFE-MISCON). Survive: paired clip design.

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## LV.8 SAFE-FILM stack (product law)

1. **Clip = objective attempt media** — steps, codes, timing; not “how did it feel?” alone (extends Keiser objective-media finding via LIV).
  2. **Less is more** —  $\leq 3$  cues per clip; prefer one primary soft-wrong handle (Mason et al.).
  3. **AMUO design** — decide who/what/when/how before shipping replay UI (Ste-Marie 2012/2020).
  4. **Self-controlled timing** — learner can request review; default is offer, not force-flood (van der Meer et al.; Aiken et al.).
  5. **Self-model pairing** — miss clip + skilled pathway and/or own prior success (Robertson/Ste-Marie line; Dowrick feedforward spirit).
  6. **Student names the frame first** — then advocacy–inquiry coach card (SAFE-AAR / Rudolph).
  7. **Transfer metric, not tip quiz** — instrument next attempt (Pearson wound).
  8. **Anti-cosplay** — borrow the *media pedagogy*, not locker-room brand (SAFE-RITUAL / SAFE-AAR).
  9. **No public error theater** — film study is private or tutor-dyad unless SAFE-COMPARE + consent (SAFE-CoP caution).
  10. **Claim ladder** — sport VBF evidence stays L1 domain evidence; MindCraft ACT/identity claims need FILM-\* experiments.
- 

## LV.9 Competitive audit — who runs film study?

ChatGPT-style tutors rarely replay *your* attempt structure and risk tip floods. Khan shows correctness with weak process pairing. Duo streak liturgy  $\neq$  film study. Brilliant ahas rarely close a self-model loop. Human “sports-style” tutoring inherits facilitator variance (Pearson). MindCraft already has attempt traces + soft-wrong + cards + Notes — **formalize SAFE-FILM**.

**Commercial implication:** Position as **reviewed decisions** — cue the one hinge, try the next play — never “train like the pros.” Parent copy: replay *how* they thought. Student copy: “pick a clip to review.”

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## LV.10 Claim ladder

| Claim                                                                                                    | Max ladder without new data                   |
|----------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| Observation interventions should vary who/what/when/how (AMUO)                                           | L1 (Ste-Marie et al., 2012; 2020)             |
| VBF-as-PA often improves declarative retention more clearly than transfer; design/delivery matter        | L1 (Pearson et al., 2023)                     |
| Dense coach tip floods $\rightarrow$ poor week-later fine-grained recall (~6%); less-is-more recommended | L1 exploratory (Mason, Farrow & Hattie, 2020) |
| Self-controlled video feedback can outperform yoked schedules on tactical learning (tennis)              | L1 single study (van der Meer et al., 2024)   |

|                                                                                |                                                           |
|--------------------------------------------------------------------------------|-----------------------------------------------------------|
| Self-skilled-model pairing can beat self-observation alone (gymnastics)        | L1 (Robertson, St. Germain & Ste-Marie, 2018)             |
| Video feedforward can accelerate skill acquisition in some educational samples | L1 small-N (Dowrick et al., 2006); theory (Dowrick, 2012) |
| SAFE-FILM improves MindCraft retry_120s / method-change / solo_transfer_pass   | L1 product bet — needs FILM experiments                   |
| “Sports academy methods raise ACT / identity” undifferentiated                 | <b>Banned</b>                                             |
| Highlight-shame reels / tip floods as pedagogy                                 | <b>Banned</b>                                             |

## LV.11 Experiments spawned

| ID        | Question                                                               | Design    | Primary                                      | Kill condition                                                                      |
|-----------|------------------------------------------------------------------------|-----------|----------------------------------------------|-------------------------------------------------------------------------------------|
| FILM-1    | ≤3-cue clip card vs uncapped AI tip flood after miss                   | 2-arm     | tip recall @24h; retry_120s; method-change   | Flood ≥ sparse on transfer → revisit cue quality only                               |
| FILM-2    | Self-controlled “Review?” vs forced replay every miss                  | 2-arm     | completion; affect; next-item transfer       | Forced ≥ choice on transfer <i>and</i> better affect → keep forced for novices only |
| FILM-3    | Miss-only clip vs miss+skilled model vs miss+own prior success         | 3-arm     | error ID; solo_transfer_pass                 | Miss-only ≥ paired → pairing theater                                                |
| FILM-4    | AMUO-scripted coach card (who/what/when/how) vs generic praise/explain | 2-arm     | SE quality; transfer                         | Generic ≥ AMUO → card content fail                                                  |
| FILM-5    | Film close + SAFE-AAR four questions vs film without AAR questions     | 2-arm     | improve-how specificity; week-later transfer | No-AAR ≥ AAR → film without close still ok; keep AAR optional                       |
| FILM-QUAL | 10 Maya: which clip UX felt useful vs shamey/sportsy?                  | interview | coded: useful / cosplay / flood / shame      | “Athlete” language → kill that copy                                                 |

**Pre-reg (XXXIV):** FILM-\* identify **review media architecture** — not “sports methods raise scores” or identity-from-athleticism myths.

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## LV.12 Confidence table

| Claim                                                              | Label                               | Confidence    |
|--------------------------------------------------------------------|-------------------------------------|---------------|
| Ste-Marie et al. AMUO who/what/when/how                            | FACT                                | High          |
| Pearson et al. VBF-as-PA: delivery/scheduling; transfer caution    | FACT                                | High          |
| Mason et al.: ~30 tips; ~6% fine-grained week recall; less-is-more | FACT (exploratory, small <i>N</i> ) | Medium–High   |
| van der Meer et al.: self-controlled VBF > yoked on tennis tactics | FACT (single study)                 | Medium        |
| Robertson, St. Germain & Ste-Marie: self∪model > self alone        | FACT                                | Medium–High   |
| Dowrick feedforward reading gains                                  | FACT (small <i>N</i> )              | Medium        |
| Sport VBF transfers cleanly to ACT prep for anxious teens          | HYPOTHESIS / wounded                | Low–Medium    |
| SAFE-FILM is right default review layer for Minecraft              | FOUNDER BELIEF / HYPOTHESIS         | Medium        |
| Sports-cosplay branding improves belonging or scores               | SPECULATION / false as doctrine     | Low (against) |

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## LV.13 What this chapter kills

1. **Kill:** Sports-academy / pro-athlete / locker-room branding as efficacy or belonging claim.
2. **Kill:** Feedback floods (30 tips / uncapped AI wrap) labeled as film study.
3. **Kill:** Equating tip recall or “we watched the replay” with transfer / exam readiness.
4. **Kill:** Failure-only shame reels and success-only dopamine montages as identity engines.
5. **Kill:** Replay UI without AMUO who/what/when/how design.
6. **Wound:** Classroom PA and elite dyads under-transfer to short teen sessions; facilitator variance.
7. **Survive:** SAFE-FILM; less-is-more clips; self-controlled review timing; self∪model pairing; feedforward spirit via own prior success; FILM-1...5; copy as **reviewed decisions** — not sports cosplay.

**Doctrine until data:** Ship **SAFE-FILM** — objective attempt clips, sparse cues, learner-timed review, paired models, transfer metrics — stacked on SAFE-AAR’s brief–debrief spine, and never sell athlete mythology or tip floods as the transformation engine.

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## Part LVI — Music Pedagogy Ladders: Scales → Repertoire → Recital Identity

**Chapter status:** Living evidence brief — Researcher tick 2026-08-01 (UTC hour 00 ≡ Red Team slot, but ch56 never written → prefer Researcher per rotation; 6 researcher entries since synthesizer v1.7 → Researcher)

**Primary question:** What transferable design does instrumental music pedagogy offer for MindCraft’s learning ladder — technique drills → meaningful repertoire → public competence signal — without conservatory cosplay, minutes-practiced vanity, or recital theater as identity?

**Owners:** Product (Practice path / story quests / exam signal) · Coach UX · Brand & copy · Red Team

**Commercial job:** Ship a **SAFE-MUSIC** stack — scales-as-ingredient drills, repertoire-as-story/exam pieces with learner choice, recital-as-transfer proof not stage cosplay — while killing “practice like a virtuoso,” hours-logged as mastery, and pass-floor exam identity.

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### LVI.1 Why this chapter exists

The core OS already sketches a music analogy (drill → quest → “recital” signal). Parts LI (SAFE-DP) and LV (SAFE-FILM) borrowed *session architecture* and *review media* from high-performance domains. Music pedagogy is the clearest **ladder** domain: beginners do not start with concertos; they build technique, then play repertoire they care about, then face graded exams or recitals that mark identity — *if* practice quality, motivation, and self-belief cohere.

MindCraft’s product already has the three rungs in embryo: ingredient/concept drills (scales), story-wrapped and exam-tagged questions (repertoire), ACT/diagnostic public signals (recital). Competitors collapse the ladder: Duo gamifies minutes; Khan floods explanations; ChatGPT skips technique for fluent answers; Brilliant jumps to ahas without graded public proof.

**FOUNDER BELIEF under audit:** Mathematical identity grows when learners climb a **named ladder** — controlled technique → chosen meaningful pieces → evidence they can perform under evaluation — not when a brand dresses homework as a conservatory.

**Claims we refuse as doctrine:**

1. Conservatory / “virtuoso / prodigy” branding as belonging or efficacy.
  2. Practice **minutes** or streaks as achievement (quantity without formal practice quality).
  3. Scales-forever without repertoire *or* repertoire without technique scaffolds.
  4. Recital/stage theater as identity without solo transfer evidence.
  5. Treating a bare “pass” exam outcome as motivational success.
-

## LVI.2 Constructs (product language)

| Construct                     | Music meaning                                                    | MindCraft analogue                                             | Failure mode if misused                                          |
|-------------------------------|------------------------------------------------------------------|----------------------------------------------------------------|------------------------------------------------------------------|
| <b>Scales / technique</b>     | Isolated skill building with immediate feedback                  | Ingredient drills; format reps; bridge micro-skills            | Endless drill without meaning → boredom / controlling motivation |
| <b>Repertoire</b>             | Whole pieces worth playing; often student-chosen                 | Story quests; archetype instances; exam-tagged sets            | Pretty narrative with no technique → theater                     |
| <b>Recital / grade exam</b>   | Public or adjudicated performance identity marker                | Diagnostic; timed mixed set; solo_transfer_pass                | Stage cosplay; pass-floor as “done”                              |
| <b>Formal practice</b>        | Goal-directed time with SRL + deliberate strategies              | SAFE-DP session spine with goals + feedback                    | Naive play-through of worksheets                                 |
| <b>Ineffective strategies</b> | Error-chasing without analysis; mindless repetition              | Immediate correction spam; tip floods; blocked vanity accuracy | Hallam “ineffective strategies” factor                           |
| <b>SAFE-MUSIC</b>             | Ladder doctrine that survives cosplay / quantity / theater kills | See LVI.8                                                      | Conservatory brand; minutes KPI                                  |

**Operational definition (HYPOTHESIS):** A MindCraft path counts as *SAFE-MUSIC* when (a) technique drills are **goal-tagged ingredients** with feedback (not infinite XP), (b) repertoire includes **learner-chosen or interest-aligned** problem sets when feasible, (c) public signals require **mixed / transfer** performance not blocked accuracy, (d) minutes practiced never outrank formal-practice quality metrics, (e) branding names the ladder (skills → pieces → proof) without virtuoso mythology.

## LVI.3 Deliberate practice’s home domain — music, not Gladwell hours

**FACT:** Ericsson, Krampe & Tesch-Römer (1993, *Psychological Review*, 100(3), 363–406, doi:10.1037/0033-295X.100.3.363) grounded deliberate practice in the training of instrumental musicians: effortful, teacher-guided activities with goals, immediate informative feedback, and repetition after reflection — distinct from playful performance or mere hours logged. Music academy students’ structured solitary practice was the empirical prototype.

**Wound (already in SAFE-DP / Part LI):** Hambrick-line critiques and domain limits mean “10,000 hours” and undifferentiated practice time are **banned** marketing. Music supplies the *definition*, not a guarantee that ACT scores rise from branded “deliberate practice.”

**Product implication:** Scales = short, goalful ingredient reps with feedback; never sell “log more hours this week” as the transformation engine (aligns Bonneville-Roussy below; SAFE-HABIT / SAFE-REWARD).

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## LVI.4 Strategy quality rises with expertise — not all practice develops

**FACT:** Hallam, Rinta, Varvarigou, Creech, Papageorgi, Gomes & Lanipekun (2012, *Psychology of Music*, 40(5), 652–680, doi:10.1177/0305735612443868) surveyed **N = 3,325** young instrumentalists across nine expertise/grade levels. Factor analysis yielded seven practice-related factors, including: systematic practice strategies; organization of practice; use of recordings/metronome for feedback; analytic strategies; **ineffective strategies**; concentration; immediate correction of errors. Statistically significant linear relationships with grade level appeared for systematic strategies, recordings/metronome use, ineffective strategies, and immediate error correction — but **not** for organization, analytic strategies, or concentration. Educational implication: expertise tracks *which* strategies learners adopt, not merely “they practiced.”

**HYPOTHESIS for MindCraft:** Instrument strategy\_quality proxies (sectioning hard items, requesting SAFE-FILM clip, naming soft-wrong before retry) — not minutes\_in\_app. Decline of enjoyment of practice with advancement in related Hallam work is a wound: advanced ladders need repertoire meaning, not more grind UI.

**Kill:** Progress bars powered only by time-on-task. Survive: strategy factors as product instrumentation.

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## LVI.5 Quantity is not enough — formal practice mediates time → achievement

**FACT:** Bonneville-Roussy & Bouffard (2015, *Psychology of Music*, 43(5), 686–704, doi:10.1177/0305735614534910; online 2014) propose an integrative framework: **formal practice** = goal-directed, focused practice that includes both self-regulation and deliberate practice strategies. In a 4-month prospective SEM study, their integrative model predicted musical achievement better than traditional time-only approaches; practice time predicted achievement **when associated with formal practice**, not as raw quantity.

**Commercial implication:** Parent dashboards that lead with “hours practiced” recreate the failed music predictor. Lead with formal-practice signals: goals set, feedback used, transfer passed. Copy: “focused reps on the weak bridge,” never “she practiced 4 hours.”

**Aligns:** SAFE-DP (goals/feedback/rep), SAFE-HABIT (cue ≠ colonization), streak ban as North Star.

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## LVI.6 Repertoire choice — interest upgrades practice behavior

**FACT:** Renwick & McPherson (2002, *British Journal of Music Education*, 19(2), 173–181, doi:10.1017/S0265051702000256) — case study of a beginning clarinetist (“Clarissa”). When practising **self-selected** repertoire vs teacher-assigned pieces, she engaged strategies typical of more advanced stages (silent fingering, silent thinking, singing), spent more time, and persevered through difficulty. Authors link choice/interest to higher-level cognitive engagement (consistent with broader motivation findings cited therein).

**FACT / FOUNDER-adjacent:** Evans (2015) applies Self-Determination Theory to music education: scales are often **not** intrinsically enjoyable; motivation quality depends on whether learners internalize the *value* of technique (identified regulation) vs pure external control (teacher said so / praise). External control of scale drills risks the Deci overjustification pattern already killed in SAFE-REWARD (Part XLIV).

**HYPOTHESIS:** MindCraft “repertoire” should mix (1) teacher/engine-assigned technique (scales) framed with **why this ingredient unlocks the next piece**, and (2) **choice among story arcs / problem sets** when mastery allows — choice upgrades strategy quality, not just delight metrics.

**Kill:** Forced syllabus-only grind with no choice forever. Kill: choice-only playground with no technique ladder. Survive: scales with identified value + repertoire choice windows.

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## LVI.7 Recital / grade exams — identity markers that can demotivate

**FACT:** Hallam, Papageorgi, Varvarigou & Creech (2020 print / 2019 online, *Psychology of Music*, doi:10.1177/0305735618816168) — N = 2,131 musicians aged 6–19 across expertise and exam outcomes (fail → highly commended). Those awarded **merely a pass** tended to undertake the **least** practice and responded least positively on organization, recordings/metronome use, analytic strategies, social life, and self-belief in musical ability. Those who **failed** were most likely to adopt ineffective strategies and less likely to enjoy performing, playing, lessons, and practice. Motivation models in the paper foreground musical **identity**, self-belief, goals, attributions, and supportive environment — not exam theater alone.

**Wound:** Graded exams are identity technology *and* stress technology (parent/teacher reputation dynamics noted in related literature the authors cite). A MindCraft “recital” that is only a score reveal can create pass-floor demotivation or fail-shame — opposite of FEI.

**HYPOTHESIS:** Public signals must be (a) criterion-referenced transfer proofs (solo\_transfer\_pass, mixed formats), (b) mastery-climate framed (SAFE TARGET / Part XXXVIII), (c) never sold as “your recital proves you’re a genius,” (d) paired with SAFE-AAR debrief + SAFE-FILM clip on the weak bar, not curtain-call XP.

**Kill:** Recital cosplay / standing ovation UI as identity. Kill: celebrating bare pass as transformative. Survive: adjudicated *competence evidence* with recovery path.

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## LVI.8 SAFE-MUSIC stack (product law)

1. **Name the three rungs** — technique (scales) → repertoire (pieces) → proof (recital/transfer).
2. **Scales = goalful ingredient drills** with immediate feedback (Ericsson definition; SAFE-DP).
3. **Repertoire includes choice** when possible — interest upgrades strategy quality (Renwick & McPherson).
4. **Frame technique value** (identified regulation) — why this drill unlocks the piece (Evans / SDT); avoid pure controlling rewards (SAFE-REWARD).
5. **Formal practice > minutes** — instrument strategy quality; never North-Star hours (Bonneville-Roussy & Bouffard; Hallam 2012).
6. **Public signal = mixed/transfer proof** — not blocked accuracy or stage theater (Hallam 2019 wound + SAFE-TRANSFER).
7. **Pass-floor ≠ win** — design recovery for bare-pass and fail states; protect self-belief (Hallam exam outcomes).
8. **Anti-cosplay** — borrow the *ladder*, not conservatory/prodigy brand (SAFE-RITUAL / SAFE-AAR pattern).
9. **Stack with prior SAFE-\*** — DP session spine, FILM review of the hard bar, AAR close, CALIB confidence tiers.

10. **Claim ladder** — music evidence stays L1 domain evidence; MindCraft ACT/identity claims need MUSIC-\* experiments.

## LVI.9 Competitive audit — who runs the ladder?

| Competitor              | Scales          | Repertoire            | Recital                      | Ladder failure                         |
|-------------------------|-----------------|-----------------------|------------------------------|----------------------------------------|
| Khan                    | Skill practice  | Weak story meaning    | Unit mastery quizzes         | Explanation flood; weak identity proof |
| Duo                     | Micro-drills    | Limited choice depth  | Leagues/streaks as “recital” | Quantity/streak liturgy (banned NS)    |
| Brilliant               | Insight puzzles | Delight-as-repertoire | Weak graded public ACT proof | Aha without adjudicated transfer       |
| ChatGPT tutor           | Skips scales    | Answers as repertoire | No solo proof                | Fluent wrongness; no ladder            |
| Human music-style tutor | Often strong    | Often assigned-only   | Real juries                  | Variance; stress; syllabus boredom     |

**Commercial implication:** Position MindCraft as **skills** → **pieces** → **proof** — “drill the hinge, play the piece, show you can transfer” — never “train like Juilliard” or “practice minutes that compound.” Parent copy: quality of practice and transfer proof. Student copy: “unlock the next piece” / “show what you can play cold.”

## LVI.10 Claim ladder

| Claim                                                                                          | Max ladder without new data                 |
|------------------------------------------------------------------------------------------------|---------------------------------------------|
| Deliberate practice prototype is music training with goals, feedback, repetition               | L1 (Ericsson et al., 1993)                  |
| Strategy factors shift with grade level in large youth sample; time alone insufficient framing | L1 (Hallam et al., 2012)                    |
| Practice time predicts achievement when tied to formal (SRL+DP) practice                       | L1 (Bonneville-Roussy & Bouffard, 2015)     |
| Student-selected repertoire can upgrade practice strategies vs assigned pieces                 | L1 case study (Renwick & McPherson, 2002)   |
| Scales often need internalized value, not pure intrinsic fun                                   | L1 theoretical/applied SDT (Evans, 2015)    |
| Bare-pass exam outcomes associate with weaker practice/motivation profiles                     | L1 observational (Hallam et al., 2019/2020) |



|                                                                             |                                          |
|-----------------------------------------------------------------------------|------------------------------------------|
| SAFE-MUSIC improves MindCraft transfer / identity metrics vs minutes-led UX | L1 product bet — needs MUSIC experiments |
| “Conservatory methods raise ACT / identity” undifferentiated                | <b>Banned</b>                            |
| Hours practiced / streak as mastery or recital                              | <b>Banned</b>                            |

## LVI.11 Experiments spawned

| ID         | Question                                                                   | Design    | Primary                                                | Kill condition                                                                    |
|------------|----------------------------------------------------------------------------|-----------|--------------------------------------------------------|-----------------------------------------------------------------------------------|
| MUSIC-1    | Named ladder UX (scales→piece→proof) vs flat question feed                 | 2-arm     | completion; solo _transfer_pass; perceived progress    | Flat ≥ ladder on transfer → ladder is copy theater                                |
| MUSIC-2    | Choice among 2–3 repertoire sets vs assigned-only                          | 2-arm     | strategy proxies; time-on-hard-items; affect           | Assigned ≥ choice on transfer <i>and</i> affect → choice optional                 |
| MUSIC-3    | Technique framed with “unlocks piece X” vs controlling “do your scales”    | 2-arm     | persistence; identified reasons; next-piece start rate | Controlling ≥ identified → SDT frame fail                                         |
| MUSIC-4    | Parent dashboard: formal-practice metrics vs minutes-first                 | 2-arm     | parent WTP/comprehension; student pressure talk        | Minutes-first better WTP without pressure → still wound educationally; don’t ship |
| MUSIC-5    | Recital = mixed transfer set + AAR vs score-reveal only                    | 2-arm     | self-belief; retry; week-later transfer                | Reveal-only ≥ AAR on belief <i>and</i> transfer → simplify                        |
| MUSIC-QUAL | 10 Maya: does “scales/pieces/recital” language feel clarifying or elitist? | interview | coded: clarifying / cosplay / pressure                 | Conservatory words → kill that copy                                               |

**Pre-reg (XXXIV):** MUSIC-\* identify **ladder architecture** — not “music methods raise scores” or prodigy-identity myths.

## LVI.12 Confidence table

| Claim                                                                   | Label                           | Confidence    |
|-------------------------------------------------------------------------|---------------------------------|---------------|
| Ericsson et al.: DP rooted in music training structure                  | FACT                            | High          |
| Hallam et al. 2012: large-N strategy factors × grade                    | FACT                            | High          |
| Bonneville-Roussy & Bouffard: formal practice mediates time→achievement | FACT                            | High          |
| Renwick & McPherson: choice upgrades strategies (case)                  | FACT (small <i>N</i> )          | Medium        |
| Evans SDT: scales need internalized value                               | FACT / applied theory           | Medium–High   |
| Hallam et al. 2019: pass-floor weakest practice/motivation profile      | FACT (observational)            | Medium–High   |
| Music ladder transfers cleanly to ACT prep for anxious teens            | HYPOTHESIS / wounded            | Low–Medium    |
| SAFE-MUSIC is right default path metaphor for MindCraft                 | FOUNDER BELIEF / HYPOTHESIS     | Medium        |
| Conservatory cosplay / hours KPI improve belonging or scores            | SPECULATION / false as doctrine | Low (against) |

## LVI.13 What this chapter kills

1. **Kill:** Conservatory / virtuoso / prodigy branding as efficacy or belonging claim.
2. **Kill:** Minutes practiced, streaks, or hours-logged as North Star mastery (quantity without formal practice).
3. **Kill:** Scales-only forever without repertoire meaning *or* repertoire theater without technique.
4. **Kill:** Recital/stage cosplay and bare-pass celebration as identity transformation.
5. **Kill:** Equating “we practiced a lot” with exam readiness or musical/math identity.
6. **Wound:** Graded public performance can demotivate (pass-floor, fail-shame, parent pressure); transfer carefully.
7. **Survive:** SAFE-MUSIC ladder; formal-practice metrics; choiceful repertoire; identified technique value; transfer-as-recital; MUSIC-1...5; copy as **skills** → **pieces** → **proof**.

**Doctrine until data:** Ship **SAFE-MUSIC** — goalful scales, meaningful chosen pieces, adjudicated transfer proof — stacked on SAFE-DP / SAFE-FILM / SAFE-AAR, and never sell conservatory mythology or practice minutes as the transformation engine.

# Part LVII — Chess Annotation & Metacognition: Postmortem UX Without Grandmaster Cosplay

**Chapter status:** Living evidence brief — Researcher tick 2026-08-01 (UTC hour 03; hour%≠0; 7 researcher entries since synthesizer v1.7 → Researcher)

**Primary question:** What transferable design does chess *annotation* / *postmortem* offer for MindCraft’s after-attempt metacognition — student-generated explanation of critical positions, error classification into next drills, analysis of wins as well as losses — without grandmaster cosplay, engine-eval floods, or rating-as-identity?

**Owners:** Product (Practice debrief / Solver review / Notes) · Coach UX · ML soft-wrong → drill loop · Brand & copy · Red Team

**Commercial job:** Ship a **SAFE-ANNOT** stack — annotate before reveal, self-explain critical moves, classify → next drill, analyze successes not only failures — while killing “think like a GM,” engine dump ≡ coaching, and games-played/rating as learning proof.

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## LVII.1 Why this chapter exists

Parts LIV–LVI borrowed *session close* (SAFE-AAR), *review media* (SAFE-FILM), and *ladder architecture* (SAFE-MUSIC) from adjacent high-performance domains. Chess is the classic cognitive-science laboratory for **move choice under uncertainty** and for **post-game annotation** as deliberate learning — the closest analogue to MindCraft’s “attempt → review → next weak bridge” loop.

Competitors already fake the chess metaphor poorly: ChatGPT tutors monologue the “best line”; puzzle apps spam tactics without annotation; Duo/Khan celebrate streaks and accuracy without a student-authored postmortem. MindCraft’s soft-wrong science (SAFE-MISCON), hide-correctness diagnostic (SAFE-CALIB), and coach self-explanation doctrine (Part XL / SAFE-SE) need a **postmortem UX law** that is evidence-shaped, not chess-club branding.

**FOUNDER BELIEF under audit:** Mathematical identity grows when learners *annotate their own critical attempts* — name what they saw, why they chose, what principle failed — then convert that into one next drill, not when an engine or LLM floods them with better answers.

### Claims we refuse as doctrine:

1. Grandmaster / “think like a master” branding as efficacy or belonging.
  2. Engine evaluation bars / tip floods labeled as “annotation.”
  3. Games played, rating, streaks, or puzzle counts as North Star mastery.
  4. Loss-only shame review (never annotate wins / successful strategies).
  5. Novice JOLs / vague “how well did you understand?” without forced restudy of critical moves.
  6. AI postmortem as automatically closing skill gaps (self-selection wound).
- 

## LVII.2 Constructs (product language)

| Construct                      | Chess meaning                                                    | MindCraft analogue                                | Failure mode if misused               |
|--------------------------------|------------------------------------------------------------------|---------------------------------------------------|---------------------------------------|
| <b>Critical position</b>       | Branch point where evaluation swings                             | Soft-wrong item; bridge fail; timed blunder       | Annotating every move → exhaustion    |
| <b>Annotation / postmortem</b> | Written/verbal account of thought at critical nodes              | Student-first debrief after attempt/session       | Engine dump before student speech     |
| <b>Candidate moves</b>         | Alternatives considered before choice                            | Alternative strategies / formats considered       | Fake multiple-choice theater          |
| <b>Self-explanation</b>        | Why this move / principle applies                                | Chi/Renkl-style why-prompts (Part XL)             | “Explain” that restates the tip       |
| <b>Engine / AI feedback</b>    | Stockfish-style eval after human thought                         | Coach card / model solution <i>after</i> annotate | Reveal-first collapses generation     |
| <b>Template / chunk</b>        | Pattern recognition structures in LTM                            | Ingredient / archetype recognition                | “Memorize more patterns” vanity       |
| <b>SAFE-ANNOT</b>              | Postmortem doctrine that survives cosplay / flood / rating kills | See LVII.9                                        | Chess branding; eval bars as pedagogy |

**Operational definition (HYPOTHESIS):** A MindCraft review counts as *SAFE-ANNOT* when (a) the learner generates explanation of **1–3 critical attempts before** correctness/model reveal, (b) annotation names a **principle or misconception family** (not vibes), (c) the loop ends in a **linked next drill** (motif → ingredient set), (d) wins/successes are eligible for annotation (not only failures), (e) engine/LLM text never substitutes for student generation, (f) branding borrows the *method*, not grandmaster mythology.

### LVII.3 de Groot — verbal protocols are the annotation prototype

**FACT:** Adriaan de Groot’s classic work (*Het denken van den Schaker*, 1946; English *Thought and Choice in Chess*, 1965/1978, Mouton / Amsterdam University Press lineage) collected **thinking-aloud protocols** from players of varying strength (including masters and world champions) who were asked to choose a move in unfamiliar tournament positions *as if in a real game*, verbalizing plans, calculations, and considerations. The object was the **choice-of-move problem**, not whole-game biography.

**FACT (structure):** de Groot described phases of chess thought commonly summarized as orientation → exploration → investigation → proof (see secondary summaries in the chess-programming / cognitive literature citing de Groot). Expertise showed up less as “search farther on every branch” in the original macrostructure story, and more as **better initial perception and selection of what to investigate**.

**Wound (citation hygiene):** Bilali■, McLeod & Gobet (2008, *Cognitive Science* line / “Sixty years of citing de Groot”) document that textbooks often **misreport** de Groot as “experts and novices search identically.” Later replications find additional search-structure differences. Product implication: do **not** market “masters don’t calculate more — they just see” as a slogan that excuses skipping hard work *or* as a claim that MindCraft’s annotation UI replicates grandmaster perception.

**Product implication:** Annotation UX = structured verbalization at critical nodes (what did you notice → candidates → why this choice → what would falsify it), not a vibes journal. Aligns SAFE-AAR question spine and Part XL self-explanation prompts.

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## LVII.4 Chunks and templates — recognition scaffolds search (not “genius”)

**FACT:** Chase & Simon (1973) chunking theory — experts recall meaningful game positions far better than novices; advantage collapses on random boards — grounded expertise in **pattern memory**, not mystical IQ (classic result; see also Gobet & Simon extensions).

**FACT:** Gobet & Simon’s **template theory** (e.g. Gobet & Simon, 1996; Gobet, 1998 *Cognition* comparison of expert-memory theories) proposes that recurring chunks evolve into richer templates with slots — unifying low-level chunks with higher-level schematic knowledge. Gobet (2005) notes deliberate practice matters because unsupervised “just play” can acquire **irrelevant** chunks that fail to improve (or even hinder) performance.

**Commercial implication:** MindCraft “annotation” should cue **ingredient/archetype recognition** (“which pattern is this?”) before deep calculation monologues. Kill copy that says students must “think like Kasparov.” Survive: pattern labels tied to ontology IDs after student attempt.

**Kill:** Genius branding. Kill: unstructured blitz-volume as chunk building. Survive: deliberate attention to the patterns that matter (SAFE-DP).

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## LVII.5 Self-explanation beats prediction-only for chess novices

**FACT:** de Bruin, Rikers & Schmidt (2007, *Contemporary Educational Psychology*, 32(2), 188–205, doi:10.1016/j.cedpsych.2006.01.001) — novices learning a King+Rook vs King endgame. Conditions: observe computer moves; predict next move; **predict + self-explain**. Self-explanation condition showed better principled understanding (predictions applying endgame principles) and more often checkmated in the test phase. Prediction alone did **not** beat observation.

**HYPOTHESIS for MindCraft:** Postmortem prompts must force **why/principle language**, not mere “what would you play next?” prediction clicks. This is the chess-domain replication of Chi/Renkl SE doctrine already in Part XL — use it to harden coach UX after attempts.

**Kill:** Silent engine scrubbing. Kill: “tap the best move” prediction theater without explanation. Survive: annotate-why before reveal.

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## LVII.6 Novice metacognition is fragile — JOLs can fail; forced restudy helps

**FACT:** de Bruin, Rikers & Schmidt (2005, *Applied Cognitive Psychology*, 19(2), 167–181, doi:10.1002/acp.1109) — novice chess endgame learning with judgments of learning (JOLs) and move selection for restudy. Forced selection of moves for restudy improved learning vs free selection (even controlling for number restudied). Providing JOLs showed

better *self-regulatory behavior* on some measures but **no or negative** performance benefit vs no-JOL — authors argue JOLs may place high ineffective load on novice working memory.

**FACT:** de Bruin, Rikers & Schmidt (2007, *European Journal of Cognitive Psychology* / related metacomprehension skill-acquisition paper, doi:10.1080/09541440701326204) — experienced chess players showed higher metacomprehension accuracy and self-regulation than novices when learning an endgame; novice absolute accuracy near zero. Expertise affects monitoring quality in skill domains.

**Product implication:** Do not ship bare confidence sliders as “metacognition” for Maya-novices (also SAFE-CALIB wound). Prefer **forced critical-move restudy** + principle annotation. Gate fancy monitoring UI by expertise / after scaffolding.

**Kill:** “Rate your understanding 1–5” as the whole postmortem. Survive: select 1–3 critical items → explain → restudy linked drill.

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## LVII.7 Analyzing games beats mere volume — and wins matter

**FACT (observational, large-N):** Yiannakoulis (2026, *Simulation & Gaming*, doi:10.1177/10468781261443352) — ~2M online blitz games / ~2.1k active Lichess players (1600–1800 band). Computer-assisted game analysis positively associated with rating improvement; **analyzing wins** showed a clearer association than analyzing losses in main models; total games played only weakly associated. Authors stress measurement noise (analysis flag is blunt; biases toward null) and exploratory observational status — not an RCT.

**HYPOTHESIS:** MindCraft should prompt annotation on **successful hard items** and efficient solutions (Dowrick-adjacent feedforward / SAFE-FILM success clips), not only on failures — “study your brilliancies” as well as soft-wrongs. Loss-only review risks ostrich/shame dynamics the paper discusses via adjacent feedback literature.

**Wound:** Associational evidence ≠ “annotation raises ACT.” Claim ladder stays L1 for chess rating deltas; MindCraft transfer needs ANNOT experiments.

**Kill:** “Play more puzzles” as improvement doctrine. Kill: failure-only shame reels. Survive: reflective analysis of critical successes and failures.

---

## LVII.8 AI / engine feedback — self-selection and fluency wounds

**FACT / high-relevance wound:** Riedl & Bogert (2024/2026 update, arXiv:2409.18660) — 5+ years, ~52k individuals on an online chess platform with optional AI game analysis. Motivated and higher-skilled players **self-select** into AI feedback and use it more productively; apparent learning gains can be an **illusion of AI effectiveness** once endogenous motivation is accounted for; AI access can **widen skill gaps**; centralized AI feedback can reduce intellectual diversity (supported via platform natural experiments in the paper).

**Aligns:** Part XXXIII AI trust/sycophancy — fluent eval bars are not pedagogy; overtrust risk. SAFE-FILM less-is-more; SAFE-AAR facilitation not lecture.

**Product implication:** Optional “engine” / model solution is a **second pass** after student annotation. Do not market auto-analysis as closing equity gaps. Instrument who seeks postmortem help (selection bias).

**Kill:** “AI reviews every game so everyone improves equally.” Kill: reveal-first Stockfish cosplay in Solver. Survive: human-first annotation → sparse model contrast → one drill.

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## LVII.9 SAFE-ANNOT stack (product law)

1. **Student annotates first** — 1–3 critical attempts before correctness / coach / model reveal.
  2. **Prompt for principles** — why/candidates/what would change your mind (de Groot structure + de Bruin SE).
  3. **Forced restudy selection** for novices — don’t rely on raw JOLs (de Bruin 2005).
  4. **Classify → next drill** — motif/misconception → ingredient set (close the loop; not insight theater).
  5. **Annotate wins too** — successful strategies and hard-earned corrects (Yiannakoulis).
  6. **Sparse AI second pass** — contrast, don’t replace; watch self-selection (Riedl & Bogert).
  7. **Pattern labels after attempt** — chunks/templates as ontology tags, not genius myths (Chase/Simon; Gobet).
  8. **Anti-cosplay** — borrow postmortem method, not “grandmaster academy” brand (SAFE-RITUAL / SAFE-MUSIC pattern).
  9. **Stack** — SAFE-AAR close, SAFE-FILM clip, SAFE-MISCON soft-wrong, SAFE-SE prompts, SAFE-CALIB tiers.
  10. **Claim ladder** — chess evidence = L1 domain; MindCraft ACT/identity claims need ANNOT-\* experiments.
- 

## LVII.10 Competitive audit — who runs a real postmortem?

| Competitor          | Annotation?                 | Student generation first?    | Loop to next drill?               | Failure mode                     |
|---------------------|-----------------------------|------------------------------|-----------------------------------|----------------------------------|
| Khan                | Hints/explanations          | Often reveal/explain-first   | Weak systematic soft-wrong→drill  | Explanation flood                |
| Duo                 | Minimal                     | No                           | Streak/XP loop                    | Quantity liturgy                 |
| Brilliant           | Aha rationales              | Partial                      | Weak exam transfer proof          | Insight without postmortem habit |
| ChatGPT tutor       | Fluent postmortem monologue | Usually model-first          | Rare durable drill link           | Sycophancy; fluent wrongness     |
| Chess.com / Lichess | Engine analysis UX          | Optional; often engine-first | Puzzle motifs if user closes loop | Eval bars; self-selection        |
| Human tutor         | Variable verbal postmortem  | Often strong                 | Variable                          | Unscalable; shame risk           |

**Commercial implication:** Position MindCraft as **annotate** → **classify** → **drill** — “replay the critical move in your own words, then train the hinge” — never “AI grandmaster reviews your session.” Parent copy: reflection quality and linked practice, not games played. Student copy: “name the mistake / name the win, then one next rep.”

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## LVII.11 Claim ladder

| Claim                                                                               | Max ladder without new data                             |
|-------------------------------------------------------------------------------------|---------------------------------------------------------|
| de Groot: verbal protocols reveal structured move-choice thought                    | L1 (classic methods)                                    |
| Chase/Simon + Gobet/Simon: expertise tied to chunks/templates, not mysticism        | L1                                                      |
| Self-explanation + prediction > prediction/observation for chess-novice principles  | L1 (de Bruin et al., 2007 CEP)                          |
| Forced restudy selection helps novices; JOLs can fail/overload                      | L1 (de Bruin et al., 2005 ACP)                          |
| Metacomprehension accuracy rises with chess expertise in skill learning             | L1 (de Bruin et al., 2007 EJCP)                         |
| Game analysis associated with rating gains; play volume weak; wins analysis salient | L1 observational (Yiannakoulias, 2026)                  |
| AI feedback gains partly selection; can widen gaps / reduce diversity               | L1 observational + natural experiments (Riedl & Bogert) |
| SAFE-ANNOT improves MindCraft transfer / identity vs reveal-first UX                | L1 product bet — needs ANNOT experiments                |
| “Grandmaster methods” / engine dumps raise ACT undifferentiated                     | <b>Banned</b>                                           |
| Rating / streak / games-played as mastery North Star                                | <b>Banned</b>                                           |

## LVII.12 Experiments spawned

| ID      | Question                                     | Design | Primary                                                             | Kill condition                                                                |
|---------|----------------------------------------------|--------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|
| ANNOT-1 | Annotate-before-reveal vs reveal-first coach | 2-arm  | retry_120s; principle-coded annotation quality; week-later transfer | Reveal-first $\geq$ annotate on transfer $\rightarrow$ annotation is friction |
| ANNOT-2 | SE-why prompts vs prediction-only taps       | 2-arm  | principled language rate; next-drill completion                     | Prediction $\geq$ SE on transfer $\rightarrow$ simplify                       |
| ANNOT-3 | Forced 2 critical restudies vs free JOL-only | 2-arm  | restudy adherence; transfer                                         | JOL-only $\geq$ forced on transfer <i>and</i> affect $\rightarrow$ revisit    |



|            |                                                                                  |           |                                              |                                                                                                    |
|------------|----------------------------------------------------------------------------------|-----------|----------------------------------------------|----------------------------------------------------------------------------------------------------|
| ANNOT-4    | Annotate wins+losses vs losses-only                                              | 2-arm     | return rate; self-belief; strategy reuse     | Losses-only $\geq$ mixed on learning without shame cost $\rightarrow$ still prefer mixed ethically |
| ANNOT-5    | AI second-pass on vs off after annotation                                        | 2-arm     | overtrust incidents; transfer; time-to-drill | AI-on hurts calibration or transfer $\rightarrow$ default off for novices                          |
| ANNOT-QUAL | 10 Maya: does “postmortem / annotate” language feel clarifying or chess-elitist? | interview | coded: clarifying / cosplay / shame          | Chess words $\rightarrow$ plain “replay the hard step”                                             |

**Pre-reg (XXXIV):** ANNOT-\* identify **postmortem architecture** — not “chess training raises ACT” or grandmaster-identity myths.

## LVII.13 Confidence table

| Claim                                                                              | Label           | Confidence  |
|------------------------------------------------------------------------------------|-----------------|-------------|
| de Groot protocols established structured verbal move-choice analysis              | FACT            | High        |
| Chunk/template accounts dominate expert chess memory explanations                  | FACT            | High        |
| de Bruin 2007: SE helps chess novices discover principles                          | FACT            | High        |
| de Bruin 2005: forced restudy > free; JOLs shaky for novices                       | FACT            | High        |
| Expertise improves metacomprehension in chess skill learning                       | FACT            | Medium–High |
| Yiannakoulias: analysis $\leftrightarrow$ rating; play volume weak (observational) | FACT (assoc.)   | Medium      |
| Riedl & Bogert: AI feedback self-selection / gap / diversity wounds                | FACT (platform) | Medium–High |

|                                                                      |                                 |               |
|----------------------------------------------------------------------|---------------------------------|---------------|
| Chess postmortem transfers cleanly to anxious ACT teens' identity    | HYPOTHESIS / wounded            | Low–Medium    |
| SAFE-ANNOT is right default review metaphor for MindCraft            | FOUNDER BELIEF / HYPOTHESIS     | Medium        |
| Grandmaster cosplay / engine-first dumps improve belonging or scores | SPECULATION / false as doctrine | Low (against) |

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### LVII.14 What this chapter kills

1. **Kill:** Grandmaster / “think like a master” branding as efficacy or belonging.
2. **Kill:** Engine-eval floods and LLM monologues labeled as annotation or coaching.
3. **Kill:** Games played, rating, streaks, or puzzle counts as North Star mastery.
4. **Kill:** Loss-only shame review; never studying successful critical decisions.
5. **Kill:** Bare JOLs / confidence sliders as sufficient metacognition for novices.
6. **Kill:** “AI analyzes every attempt so gaps close equally” (selection + diversity wound).
7. **Wound:** Textbook oversimplification of de Groot search findings; don’t sloganize perception.
8. **Survive:** SAFE-ANNOT; annotate-before-reveal; SE principles; forced critical restudy; classify→drill; annotate wins; sparse AI second pass; ANNOT-1...5.

**Doctrine until data:** Ship **SAFE-ANNOT** — student-generated postmortem on critical attempts, principle-tagged, linked to the next drill, stacked on SAFE-AAR / SAFE-FILM / SAFE-SE / SAFE-MISCON — and never sell chess mythology, engine bars, or volume metrics as the transformation engine.

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## Part LVIII — Therapy: Graded Exposure (Math Anxiety Ladders)

**Chapter status:** Living evidence brief — Researcher tick 2026-08-01

**Primary question:** How should MindCraft borrow *graded exposure* / *inhibitory learning* from anxiety treatment — without therapy cosplay, flooding, or “calm until zero” theater — so threatened learners approach math instead of avoiding it?

**Owners:** Affect / Anxiety · Learning Science · Product (Practice sequencing + stakes) · Growth / Positioning · Red Team

**Commercial job:** Turn math-anxiety avoidance into a **designed approach ladder** that parents can trust and students can finish — then market approach + expectancy violation, not “we cured anxiety” or day-one ACT sims as exposure therapy.

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### LVIII.1 Why this chapter exists

Parts XXIV (affect/anxiety), XLI (SAFE-DD), XLV (SAFE-RESILIENCE), and L (SAFE-CALIB) already say: anxiety taxes working memory, unbuffered desirable difficulty becomes threat, grit posters fail, and “raise confidence” is not a North Star. What they do **not** yet specify is the **approach schedule** when the threat *is the math itself* — timed items, word problems, graphs, exams, public board energy.

Competitors fake exposure poorly: Khan dumps harder items after a wrong answer; Duo streaks punish skip; Brilliant opens with delightful puzzles that never feel like the threat Maya avoids; ChatGPT tutors soothe (“you’ve got this”) while the student never approaches the feared format. Therapy apps and school worksheets sometimes cosplay CBT with breathing gifs and no expectancy violation.

**FOUNDER BELIEF under audit:** Identity transformation for high-math-anxiety (HMA) Maya requires **approaching feared math situations under destaked conditions until threat expectancies are violated** — not forever-easy fluency, not flooding, and not clinical cosplay.

#### Claims we refuse as doctrine:

1. Day-one full ACT / mixed timed exams labeled “exposure therapy.”
2. Habituation-only success (“stay until anxiety = 0”).
3. Clinical CBT / therapist branding without clinician oversight.
4. Permanent safety behaviors (hint binge, skip, parent rescue) as “support.”
5. Strict linear hierarchy forever (never vary order/context).
6. Anxiety meters / “Calm Score™” as North Star.
7. Exposure → ACT +2 $\sigma$  / Bloom marketing.

## LVIII.2 Constructs (product language)

| Construct                   | Therapy meaning                                                  | MindCraft analogue                               | Failure mode if misused         |
|-----------------------------|------------------------------------------------------------------|--------------------------------------------------|---------------------------------|
| <b>Fear hierarchy</b>       | Ranked approach steps from least → most feared                   | Format/stakes ladder (untimed soft → timed mix)  | Hierarchy as forever easy-mode  |
| <b>Graded exposure</b>      | Approach feared CS while learning safety                         | Planned approach to avoided item types           | Random harder = fake exposure   |
| <b>Expectancy violation</b> | Feared outcome does not occur (or is tolerable)                  | Soft-wrong + survive + retry without catastrophe | Calm breathing without approach |
| <b>Safety behavior</b>      | Acts that prevent learning (escape, ritual)                      | Instant full reveal; skip; streak protect        | “Support” that blocks learning  |
| <b>Inhibitory learning</b>  | New CS—no US memory competes with fear memory                    | Multiple contexts/formats after approach         | One safe session → exam panic   |
| <b>SAFE-EXPOSE</b>          | Exposure doctrine that survives flooding / cosplay / meter kills | See LVIII.9                                      | Therapy branding; flood sims    |

**Operational definition (HYPOTHESIS):** A MindCraft session counts as *SAFE-EXPOSE* when (a) the learner **approaches** a previously avoided or high-SUDS math situation, (b) stakes are **destaked** enough that WM can engage (SAFE-DD), (c) a named **threat expectancy** is checked (“I’ll freeze / blank / be stupid”), (d) outcome **violates or revises** that expectancy without catastrophe framing, (e) **safety behaviors are faded** (hints → cards → solo), (f) later practice **varies context/format** so learning is not locked to one cozy UI, (g) product never claims to *treat* clinical anxiety disorders.

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### LVIII.3 Why avoidance is the commercial enemy

**FACT (construct + avoidance):** Hembree (1990, *Journal for Research in Mathematics Education*, 21(1), 33–46, doi:10.2307/749455 / 10.5951/jresmetheduc.21.1.0033) — meta-analysis of 151 studies: mathematics anxiety relates to poorer achievement, poorer attitudes, and is **bound directly to avoidance** of math; valid treatments that reduce anxiety tend to accompany improved performance.

**FACT (WM disruption):** Ashcraft & Kirk (2001, *JEP: General*, doi:10.1037/0096-3445.130.2.224) — HMA individuals show reduced WM spans under math demand; anxiety acts as a **transitory dual-task load**, amplifying errors especially on carry/complex items (see also Ashcraft & Krause, 2007, *Psychonomic Bulletin & Review*).

**Commercial implication:** The product problem is not “Maya needs nicer explanations.” It is **avoidance of the situations that would create competence evidence**. Marketing that promises comfort without approach sells temporary relief and loses the identity loop (FEI). Marketing that floods her with timed mixed ACT on day one sells rigor theater and churn.

**Kill:** “More content solves anxiety.” Kill: “Just make it fun.” Survive: designed approach to avoided formats under recoverable stakes.

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### LVIII.4 Classical graded exposure and math desensitization lineage

**FACT (lineage):** Systematic desensitization (Wolpe tradition) pairs gradual hierarchy ascent with incompatible responses (often relaxation). Education/counseling literature adapted this to math anxiety via hierarchy construction + gradual approach (see secondary reviews; e.g. IntechOpen synthesis citing Wolpe 1958 and school applications — cite cautiously as review, not primary RCT).

**FACT (math-anxiety treatment comparison):** Zettle (2003, *The Psychological Record*, 53, 197–215, doi:10.1007/BF03395440) — Acceptance and Commitment Therapy (ACT) vs systematic desensitization for mathematics anxiety; both approaches target anxiety-related avoidance, with different mechanisms (acceptance/values vs desensitization). Product takeaway: **approach + new relationship to anxious thoughts** matters more than branding one school of therapy.

**FACT (case / protocol shape):** Open-access childhood MA CBT case work (e.g. Coping Cat–inspired hierarchy + graded exposure; PMC8283868) shows the clinical shape — least→most feared math situations. **Wound:** N=1 ≠ product efficacy claim.

**HYPOTHESIS for MindCraft:** Borrow *hierarchy + graded approach + destake*; do **not** ship therapist roleplay or “we treat anxiety disorders.”

**Kill:** Clinical cosplay. Survive: hierarchy of *math situations* (formats, timing, stakes), not feelings posters.

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## LVIII.5 Inhibitory learning upgrades habituation theater

**FACT (mechanism shift):** Craske, Treanor, Conway, Zbozinek & Vervliet (2014, *Behaviour Research and Therapy*, 58, 10–23, doi:10.1016/j.brat.2014.04.006; PMC4114726) — exposure optimized via **inhibitory learning**: goal is new inhibitory CS–no US associations and expectancy violation, **not** necessarily within-session fear reduction to zero. Strategies include expectancy violation, removal of safety signals, variability, multiple contexts, retrieval cues, affect labeling (among others).

**FACT (clinical overview):** Craske inhibitory-regulation line (doi:10.1159/000381574) — anxious individuals often show inhibitory learning/regulation deficits; strengthen inhibitory retrieval, not just “wait until calm.” Variability-in-exposure reviews (e.g. PMC6884337) argue varying hierarchy order/intensity can aid expectancy violation and cross-context retrieval — school-product transfer remains empirical.

**Product implication:** Success metric is **approach + violated expectancy + later approach in new context** (challenge\_accept, retry\_120s, format hops), **not** “anxiety slider hit zero.” Soft-wrong + survive is expectancy violation done right. Breathing-only without approach is cosplay.

**Kill:** Habituation-as-North-Star. Kill: “Stay in the hard problem until you feel calm.” Survive: name the feared outcome → approach → check what happened → vary next.

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## LVIII.6 Extinction is fragile — renewal kills one-session miracles

**FACT:** Bouton (2002, *Biological Psychiatry*, 52(10), 976–986, doi:10.1016/S0006-3223(02)01546-9; PubMed 12437938) — extinction does **not** erase the original fear association; it creates new, **context-dependent** learning. Return of fear via **renewal** (ABA / ABC patterns) is expected when context changes after extinction.

**FACT (learning & memory bridge):** Bouton (2004, *Learning & Memory*, 11, 485–494) — contexts set the occasion for CS–US vs CS–no US; mere removal from the extinction context can renew responding.

**Commercial implication:** One cozy Minecraft session where Maya survived word problems does **not** license “anxiety cured” copy — exam hall / parental watching / timed ACT is a **new context**. Product must plan **multi-context approach** (home app → timed → mixed → low-stakes quiz climate) and retrieve inhibitory learning with cues (SAFE-AAR “what was I afraid would happen?”).

**Kill:** One-and-done exposure marketing. Survive: spaced approach across formats/stakes (ties SAFE-TIMING / SAFE-TRANSFER).

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## LVIII.7 Brief cognitive tools that are *not* exposure — and must not replace it

**FACT:** Park, Ramirez & Beilock (2014, *Journal of Experimental Psychology: Applied*, doi:10.1037/xap0000013) — short **expressive writing** before a math exam reduced the HMA–LMA performance gap (N=80); among HMAs,

anxiety/cause/insight language related positively to math performance. Related classroom writing finding: Ramirez & Beilock (2011, *Science*, doi:10.1126/science.1199427).

**HYPOTHESIS:** Writing / affect labeling can free WM *before* approach (Craske lists affect labeling among optimizers) — but writing **without** subsequent approach is incomplete for avoidance. Product: optional 60–90s worry-dump *then* hierarchy step — not a journaling app substitute for practice.

**Kill:** Mindfulness theater as the whole product. Survive: brief unload → approach → expectancy check.

---

## LVIII.8 Crosswalk to SAFE-DD / FEI / soft-wrong

| Exposure principle          | MindCraft already-adjacent law | Product move                                             |
|-----------------------------|--------------------------------|----------------------------------------------------------|
| Equip before strain         | SAFE-DD equip                  | Hierarchy starts below panic; cards available then faded |
| Destake while approaching   | Soft-wrong; mastery climate    | No public rank; no streak-as-worth during expose steps   |
| Expectancy violation        | FEI fear→evidence              | Prompt: “What did you expect?” → “What happened?”        |
| Remove safety signals       | Hint fade / solo_transfer      | Cap hint depth; require attempt before full reveal       |
| Variability / multi-context | SAFE-TRANSFER / VT             | After approach succeeds, hop format/timing               |
| Do not flood                | SAFE-RESILIENCE growth-zone    | Ban day-one full timed mix for HMA segment               |

**FOUNDER BELIEF:** SAFE-EXPOSE is the **affective schedule** for SAFE-DD — difficulty dosing without approach design is incomplete for Maya who never opens the hard door.

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## LVIII.9 SAFE-EXPOSE stack (surviving product rules)

1. **Build a situation hierarchy, not a vibes hierarchy.** Steps = concrete math situations (untimed symbolic → word → diagram → timed soft → short mix). Prefer student-ranked SUDS-lite once, then product defaults.
2. **Approach before polish.** The session’s job is approach + expectancy check, not explanation volume.
3. **Destake the approach.** Soft-wrong, mastery climate, no league, no streak identity (SAFE-COMPARE / HABIT / REWARD).
4. **Name expectancy → violate → label.** Short prompt triad; optional expressive writing only as WM unload.
5. **Fade safety behaviors.** Hints → structured cards → annotate (SAFE-ANNOT) → solo transfer. Permanent hint binge = failed exposure.
6. **Vary after success.** Do not lock learning to one cozy context (Bouton renewal).
7. **Measure approach, not calm meters.** Primary: challenge\_accept, retry\_120s, avoided-format approach rate, later transfer\_pass. Anxiety self-report is covariate, not North Star.

8. **Anti-cosplay.** No “therapist mode,” no diagnosis, no clinical claims. Plain language: “practice the scary step under safe rules.”

**LVIII.10 Competitive audit — who actually runs exposure?**

| Competitor    | Hierarchy?                          | Destaked approach? | Expectancy check?             | Failure mode                                   |
|---------------|-------------------------------------|--------------------|-------------------------------|------------------------------------------------|
| Khan          | Weak (skill tree ≠ fear ladder)     | Partial            | Rare                          | Harder after miss ≈ flood/shame                |
| Duo           | No                                  | No (streak threat) | No                            | Avoidance via skip; streak colonization        |
| Brilliant     | Puzzle delight ≠ feared exam format | Low stakes yes     | Rare                          | Fun ≠ approach to avoided threat               |
| ChatGPT tutor | Ad hoc                              | Emotional soothe   | Often sycophantic reassurance | Soothe without approach; fluent wrongness      |
| Human tutor   | Often strong                        | Variable           | Variable                      | Unscalable; trait-label risk (SAFE-EXPECTANCY) |

**Commercial implication:** Position as **approach ladders for feared math situations** — scary step under safe rules, then prove in a new format — never “AI therapy” and never day-one elite timed drills.

**LVIII.11 Claim ladder**

| Claim                                                                            | Max ladder without new data                    |
|----------------------------------------------------------------------------------|------------------------------------------------|
| Math anxiety tied to avoidance; treatments that work often improve performance   | L1 (Hembree, 1990)                             |
| Anxiety disrupts WM during math                                                  | L1 (Ashcraft & Kirk, 2001)                     |
| Exposure benefits from inhibitory learning / expectancy violation > calm-to-zero | L1 clinical translation (Craske et al., 2014+) |
| Extinction is context-bound; renewal expected                                    | L1 (Bouton, 2002/2004)                         |
| Expressive writing can shrink HMA performance gap pre-test                       | L1 (Park/Ramirez/Beilock)                      |
| Zettle: SD and ACT both target math-anxiety avoidance pathways                   | L1 clinical trial lineage                      |

|                                                                             |                                          |
|-----------------------------------------------------------------------------|------------------------------------------|
| SAFE-EXPOSE raises MindCraft<br>challenge_accept / transfer for HMA segment | L1 product bet — needs EXP-O experiments |
| Flooding / day-one ACT sims / Calm Score™ /<br>therapy cosplay              | <b>Banned</b> as doctrine                |

## LVIII.12 Experiments spawned

| ID         | Question                                                                     | Design               | Primary                                      | Kill condition                                                                              |
|------------|------------------------------------------------------------------------------|----------------------|----------------------------------------------|---------------------------------------------------------------------------------------------|
| EXP-O1     | Hierarchy+destake<br>vs random<br>harder after miss                          | 2-arm HMA<br>segment | challenge_acce<br>pt; exit; week<br>transfer | Random ≥<br>hierarchy on<br>transfer without<br>churn cost →<br>revisit                     |
| EXP-O2     | Expectancy-viola<br>tion prompts on<br>vs off                                | 2-arm                | expectancy<br>revision rate;<br>retry_120s   | Prompts add<br>friction without<br>learning →<br>shorten/kill                               |
| EXP-O3     | Safety-behavior<br>fade vs<br>permanent hints                                | 2-arm                | solo_transfer_<br>pass; hint depth           | Permanent hints<br>≥ fade on transfer<br>→ still prefer<br>fade ethically;<br>redesign fade |
| EXP-O4     | Single-context<br>approach vs<br>multi-format<br>renewal plan                | 2-arm                | format-hop pass<br>after delay               | Single ≥ multi →<br>still keep multi<br>for exam ecology                                    |
| EXP-O5     | Worry-write then<br>approach vs<br>approach-only                             | 2-arm HMA            | accuracy under<br>load; completion           | Write hurts or<br>delays without<br>gain →<br>optional/off                                  |
| EXP-O-QUAL | 10 Maya:<br>“exposure<br>ladder” language<br>vs “practice the<br>scary step” | interview            | coded: clarifying<br>/ clinical / shame      | Clinical words →<br>plain language<br>only                                                  |

**Pre-reg (XXXIV):** EXP-O\* identify **approach architecture for anxiety** — not “we treat anxiety disorders” or ACT score guarantees.

## LVIII.13 Confidence table



| Claim                                                                 | Label                           | Confidence    |
|-----------------------------------------------------------------------|---------------------------------|---------------|
| Anxiety–avoidance–performance cluster is real at population level     | FACT                            | High          |
| WM dual-task account of online math-anxiety cost                      | FACT                            | High          |
| Inhibitory learning reframes exposure away from habituation-only      | FACT (clinical science)         | High          |
| Bouton renewal / context-bound extinction                             | FACT                            | High          |
| Expressive writing helps some HMA test performance                    | FACT                            | Medium–High   |
| Case/protocol SD for child MA generalizes to ACT teens in apps        | HYPOTHESIS / wounded            | Low–Medium    |
| SAFE-EXPOSE is right default for MindCraft HMA segment                | FOUNDER BELIEF / HYPOTHESIS     | Medium        |
| Therapy branding / flooding / calm meters improve belonging or scores | SPECULATION / false as doctrine | Low (against) |

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## LVIII.14 What this chapter kills

1. **Kill:** Day-one full timed mixed ACT / flooding labeled as exposure therapy.
2. **Kill:** Habituation-only success (“anxiety must hit zero”).
3. **Kill:** Clinical CBT / therapist / diagnosis cosplay in product or ads.
4. **Kill:** Permanent hint binge, skip, or parent-rescue as “support.”
5. **Kill:** Anxiety meters / Calm Score™ / “we cured math anxiety” as North Star.
6. **Kill:** One cozy session ≡ exam-ready (renewal wound).
7. **Kill:** Bloom 2-sigma / exposure→score miracle marketing.
8. **Wound:** N=1 clinical cases and review chapters as efficacy proof.
9. **Survive:** SAFE-EXPOSE; hierarchy of situations; destaked approach; expectancy violation; fade safety behaviors; multi-context renewal plan; EXP-O1...5.

**Doctrine until data:** Ship **SAFE-EXPOSE** — graded approach to feared math situations under recoverable stakes, with expectancy checks and faded safety behaviors, stacked on SAFE-DD / FEI / SAFE-CALIB — and never sell therapy cosplay, flooding, or calm meters as the transformation engine.

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## Part LIX — Parent WTP Experiments Design

**Chapter status:** Living evidence + experiment-design brief — Researcher tick 2026-08-01

**Primary question:** How should MindCraft *measure* what parents will pay for — and which message frames move purchase intent — without confusing stated preference for cash, or selling score theater / vibe identity?

**Owners:** Growth / Positioning · Behavioral Econ · Product (parent surfaces) · Red Team

**Commercial job:** Ship a **pre-registered WTP / conjoint / message stack** that tells founders which attributes to build and which claims to kill — before pricing pages and parent dashboards ossify around the wrong North Star.

### LIX.1 Why this chapter exists

Part XXVII mapped the post-AI WTP stack (answers → explanations → diagnosis → affect → accountability → trusted outcome signal) and spawned PAR-1 / COMP-1. Part XXXII flagged parent anxiety transmission; Part XXXVII’s EVT-4 already asked competence-evidence vs minutes vs AI-unlimited. What the Constitution still lacked is the **methods contract**: how to run conjoint / discrete-choice and message tests so results can raise claim-ladder level (Part XXXIV L4 market causal) instead of producing inspirational survey quotes.

**FOUNDER BELIEF under audit:** Families will pay for *trusted diagnosis + transfer evidence + human accountability* more than for unlimited AI chat — **if** we prove it with trade-off designs, not founder intuition.

**Claims we refuse as doctrine:**

- 1. “Parents only buy score spikes” as settled fact without trade-off data.
- 2. “Identity transformation” as a purchasable SKU without FEI evidence parents can parse.
- 3. Likert “how much would you pay?” as serious WTP.
- 4. Stated WTP ≡ revenue (hypothetical bias ignored).
- 5. Streak / minutes / “AI unlimited” as the parent value prop by default.
- 6. One national average price as the pricing strategy.

### LIX.2 Constructs (product language)

| Construct         | Research meaning                                                          | MindCraft analogue                               | Failure mode if misused           |
|-------------------|---------------------------------------------------------------------------|--------------------------------------------------|-----------------------------------|
| <b>WTP</b>        | Max price at which a package is chosen over alternatives / outside option | Monthly plan parents select in CBC               | Treat survey WTP as sticker price |
| <b>Attribute</b>  | Package feature with levels                                               | Diagnosis depth; tutor hours; report type; price | Too many attributes → noise       |
| <b>CBC / DCE</b>  | Choice among multi-attribute profiles                                     | Parent chooses Plan A vs B vs neither            | Rank-all-features surveys         |
| <b>Part-worth</b> | Utility of a level                                                        | Relative value of FEI report vs score-only       | Over-interpret small segments     |

|                     |                                                |                                  |                                      |
|---------------------|------------------------------------------------|----------------------------------|--------------------------------------|
| <b>Message test</b> | Framing → intent / comprehension               | Landing / email / dashboard copy | Confuse click with retention         |
| <b>SAFE-WTP</b>     | Experiment doctrine that survives vibe pricing | See LIX.10                       | Sell identity without transfer proof |

**Operational definition (HYPOTHESIS):** A MindCraft parent-pricing study counts as *SAFE-WTP* when (a) it uses **choice-based** trade-offs (or incentive-aligned follow-ups), not open “name a price,” (b) attributes map to **shippable product surfaces**, (c) a **price** or clear outside option is present, (d) **hypothetical-bias mitigations** are pre-registered, (e) primary outcomes separate **purchase intent**, **comprehension of child’s bottleneck**, and **homework-conflict risk**, (f) results update packaging — not vanity decks.

### LIX.3 Market facts — demand exists; composition is uneven

**FACT (US tutoring industry geography):** Kim, Goodman & West (2024, CESifo Working Paper No. 11251 / IZA DP 17178) — U.S. private tutoring *centers* more than tripled ~1997–2022 (~3,000 → ~10,000); locations concentrate in high-income, high-parent-education areas; >half in top income quintile; residual association with Asian American population share conditional on income/education. School-market structure (private/charter/magnet) is only marginally predictive.

**FACT (household spend snapshot, cited therein):** Using Consumer Expenditure Survey calculations, authors report ~6–7% of U.S. families with children 6–17 paid for tutoring in the past year (2022), averaging about **\$437 in months with a purchase**; top 10% / 5% of monthly spend near **\$1,200 / \$2,000**. (Treat as CES-derived descriptive, not MindCraft price ceiling.)

**FACT (shadow-education framing):** Bray (2010) and Bray & Lykins (2012, ADB) document private supplementary tutoring (“shadow education”) as a global, fee-based parallel to schooling — useful for competitive context, not a US price guide.

**Commercial implication:** There is a real paid market adjacent to MindCraft — but it is **skewed**. Pricing and positioning that assume median-America “all parents pay \$X for identity” will miss both the premium competitive segment and equity constraints. Segment conjoint (income × anxiety × exam stakes) is mandatory.

**Kill:** One national WTP number as strategy. Survive: segment-aware CBC + equity-aware packaging.

### LIX.4 Why Likert WTP fails — use stated choice

**FACT (method foundation):** Louviere, Hensher & Swait (2000, *Stated Choice Methods*, Cambridge) — stated preference via controlled choice experiments recovers relative attribute utilities under random-utility assumptions; preferable to unstructured preference talk when products are multi-attribute.

**FACT (RUM lineage):** McFadden (1974) random-utility discrete choice — choices reveal trade-offs; with a monetary attribute, researchers can express part-worths as WTP for levels.

**FACT (tutoring-adjacent DCE):** Gambi & De Witte (2024, *Higher Education*, doi:10.1007/s10734-024-01358-z) — DCE + information experiment on HE students’ tutoring preferences/consumption; high stated WTP coexists with low private uptake; SES demand vs ability-to-pay tension; behavioral frictions (status quo, image) matter. **Wound for K-12 parents:** student HE sample ≠ Maya’s parent — method transfers, coefficients do not. Broader HE DCE literature (UK/EU/Polish university-attribute studies building on Louviere et al.) shows the same tool recovers monetary trade-offs for staff expertise, flexibility, and mode.

**Product implication:** Stop asking “Would you pay for an identity-transforming math world?” Ask: *Between these two monthly packages, which do you choose — or neither?*

### LIX.5 Hypothetical bias — stated ≠ paid

**FACT:** Murphy, Allen, Stevens & Weatherhead (2005, *Environmental and Resource Economics*, 30(3), 313–325) — meta-analysis of hypothetical vs actual WTP; median hypothetical/actual ratio ~1.35 with severe positive skew; choice-based elicitation associated with lower bias than some contingent-valuation formats (model-sensitive).

**FACT (mitigation lineage):** Cummings & Taylor (1999, *American Economic Review*) — “cheap talk” scripts can reduce hypothetical overstatement (effects heterogeneous later). Schmidt & Bijmolt (2020, *JAMS*, doi:10.1007/s11747-019-00666-6) — consumer-goods meta: prefer indirect/choice over direct “what would you pay?”; incentive-aligned mechanisms help when feasible.

**HYPOTHESIS for MindCraft:** Parent CBC will still inflate absolute dollars; **rank-order of attributes** (FEI report vs streak dashboard vs AI chat hours vs human tutor minutes) is the decision-useful output. Validate top packages with a small **deposit** / **waitlist** / **paid pilot** before national pricing.

**Kill:** Survey WTP as P&L input. Survive: relative part-worths → packaging; cash tests → prices.

### LIX.6 Attribute set for MindCraft parent CBC (shippable)

**Design rule (FOUNDER BELIEF):** Every attribute level must map to a screen or ops cost we can actually ship in 90 days.

| Attribute            | Levels (draft)                                                                                      | Ties to doctrine                |
|----------------------|-----------------------------------------------------------------------------------------------------|---------------------------------|
| Diagnosis depth      | Vague “needs practice” / concept weakness list / bridge+format gap map                              | Ontology advantage; XXVII trust |
| Evidence report      | Score/minutes only / FEI pack (retry_120s, challenge-seeking, transfer) / FEI + honest plateau note | PAR-1; anti-streak              |
| Human accountability | None / async tutor note weekly / 2× live 25-min / 4× live                                           | Scarcity layer                  |

|                              |                                                                                |                             |
|------------------------------|--------------------------------------------------------------------------------|-----------------------------|
| <b>AI help policy</b>        | Unlimited chat / guarded pedagogy wrap / attempt-first then cards              | XXXIII; SE constraints      |
| <b>Practice architecture</b> | Streak gamification / mastery climate soft-wrong / SAFE-EXPOSE ladder language | XLII–XLIV; LVIII            |
| <b>Price</b>                 | e.g. \$49 / \$99 / \$179 / \$299 per month (calibrate to segment)              | Must include realistic span |
| <b>Outside option</b>        | “Neither — keep free Khan/GPT/status quo”                                      | Critical for WTP            |

**HYPOTHESIS:** Diagnosis depth + FEI report + modest human accountability will dominate unlimited AI and streak theater **in the anxious-middle segment**; score-chaser segment may overweight timed-test sims — measure, don’t assume.

**Red Team:** If FEI report wins stated choice but parents cannot *explain* the child’s bottleneck after viewing it, comprehension failed — packaging theater.

### LIX.7 Message tests are not conjoint (run both)

Conjoint answers *what package*; messages answer *which story opens the wallet without raising homework conflict*.

**Pre-registered message arms (builds EVT-4 / COMP-1):**

| Arm                      | Parent-facing frame                         | Risk                                              |
|--------------------------|---------------------------------------------|---------------------------------------------------|
| M1 Score relief          | “Raise ACT math”                            | Overclaim; retention crash if flat                |
| M2 Homework peace        | “End nightly fights”                        | Soothing without competence                       |
| M3 Diagnosis trust       | “See the exact bridge that’s stuck”         | Jargon; need plain language                       |
| M4 Identity / confidence | “Become someone who doesn’t freeze”         | Belief Score™ / raise-confidence kill (v1.7–v1.8) |
| M5 Hybrid                | Diagnosis + transfer check + honest plateau | Longer copy; may win retention                    |

**Primary outcomes:** click → trial start; 14-day activation; parent comprehension quiz (name bottleneck); reported homework conflict; **not** vanity “brand love.”

**Kill:** Identity-only ads that cannot point to a measurable FEI metric. Survive: hybrid that sells method, shows evidence.

### LIX.8 Qualitative gate before fielding CBC

**FACT / process:** Part XXVII and QUAL-P (XXXII) already require ~10 parent interviews. CBC attributes invented in a founder Slack will mis-specify levels.

**Protocol (HYPOTHESIS):** Semi-structured interviews → code “what I thought I bought / what would make me cancel / words I trust / words that feel scammy” → freeze attribute glossary → then CBC. Do not reverse the order.

**Segment quotas (SPECULATION until interviews):** (1) HMA-Maya parents (anxiety/avoidance), (2) score-chaser / selective-college, (3) homework-peace exhausted, (4) price-sensitive remedial. Run latent-class or split models; do not average them into one “parent.”

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## LIX.9 Equity and pricing ethics

**FACT (concentration):** Kim et al. (2024) — private center supply concentrates where income/education are high. Packaging only for that segment reproduces shadow-education inequality.

**FOUNDER BELIEF:** Charging for diagnosis+accountability is legitimate if transfer evidence is real (XXVII.7). Illegitimate if the paid tier is streak cosmetics + GPT wrap.

**Guardrails:**

- Paid claims ↔ measured FEI / learning metrics.
- No “guaranteed score jump” in any CBC level or ad arm.
- Consider a **honest free tier** that still shows diagnosis (trust acquisition) vs freemium that hides the bottleneck (dark pattern).
- Report segment WTP separately; do not market top-quintile WTP as universal.

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## LIX.10 SAFE-WTP stack (surviving product rules)

1. **Choice over chatter.** CBC/DCE or incentive-aligned choice; ban standalone “name your price.”
2. **Ship-mapped attributes only.** No fictional features.
3. **Always include price + neither.** Otherwise WTP is fantasy.
4. **Bias protocol.** Cheap talk / certainty / consequentiality / small cash validation — pre-register which.
5. **Three outcomes.** Intent ≠ comprehension ≠ conflict. Optimize packaging on all three.
6. **Segment, don’t average.** Latent class or pre-defined segments.
7. **Messages ≠ packages.** Run COMP/EVT message tests alongside CBC.
8. **Anti-theater.** FEI report that parents cannot interpret loses even if chosen.
9. **Anti-overclaim.** No 2σ, no guaranteed ACT points, no “AI tutor replaces human” as paid promise.
10. **Update doctrine with data.** Part-worths that kill founder favorites get logged as kills.

---

## LIX.11 Claim ladder

|       |                             |
|-------|-----------------------------|
| Claim | Max ladder without new data |
|-------|-----------------------------|

|                                                                             |                                                                      |
|-----------------------------------------------------------------------------|----------------------------------------------------------------------|
| US private tutoring centers grew sharply; high-income concentration         | L1 (Kim/Goodman/West 2024)                                           |
| Shadow education is a real global fee market                                | L1 (Bray line)                                                       |
| Stated choice recovers relative utilities better than unstructured WTP talk | L1 (Louviere/Hensher/Swait; McFadden)                                |
| Hypothetical WTP inflates; choice formats help but do not erase bias        | L1 (Murphy et al. 2005; Cummings & Taylor)                           |
| HE tutoring DCE: high WTP, low uptake, SES frictions                        | L1 adjacent (Gambi & De Witte 2024) — transfer wound to K-12 parents |
| MindCraft parents prefer FEI+diagnosis over unlimited AI / streaks          | <b>L0–L1 product bet</b> — needs WTP-* experiments                   |
| Survey dollars = revenue forecast                                           | <b>Banned</b>                                                        |

## LIX.12 Experiments spawned

| ID    | Question                                                  | Design               | Primary                              | Kill condition                                                                                       |
|-------|-----------------------------------------------------------|----------------------|--------------------------------------|------------------------------------------------------------------------------------------------------|
| WTP-1 | FEI report vs score/minutes vs streak dashboard in CBC    | CBC, price + neither | part-worths; segment classes         | Streak/minutes dominate FEI <i>and</i> later retention → revisit packaging (still educational wound) |
| WTP-2 | Diagnosis depth (vague / concept / bridge+format) × price | CBC                  | WTP for bridge+format; comprehension | Depth raises intent but kills comprehension → simplify language                                      |
| WTP-3 | Human minutes vs AI-unlimited vs guarded wrap             | CBC                  | part-worth human; churn proxy intent | AI-unlimited wins on intent <i>and</i> 30-day solo transfer → thesis wounded                         |
| WTP-4 | Message arms M1–M5                                        | RCT landing/email    | trial start; conflict; comprehension | Identity arm wins clicks + raises conflict without retention → kill identity-only                    |
| WTP-5 | Cheap-talk / certainty calibration on vs off              | 2× embedded in CBC   | calibration gap                      | No gap → still keep cash pilot                                                                       |

|          |                                              |           |                               |                                                      |
|----------|----------------------------------------------|-----------|-------------------------------|------------------------------------------------------|
| WTP-6    | Deposit / paid pilot of top CBC winner vs #2 | Field     | pay-through; 30-day retention | Stated winner loses cash → trust revealed preference |
| WTP-QUAL | 10 parents: cancel triggers + trusted words  | Interview | codebook → attribute freeze   | Founder glossary ≠ parent glossary → rewrite levels  |

**Pre-reg (XXXIV):** WTP-\* identify **parent package preference and message effects** — not “we proved identity sells” or national price optimality from N=200 survey.

**Ties:** EVT-4, PAR-1, COMP-1, MUSIC-4, TIM-4 become arms or covariates inside this family — do not fork duplicate surveys.

## LIX.13 Confidence table

| Claim                                                                               | Label                                     | Confidence    |
|-------------------------------------------------------------------------------------|-------------------------------------------|---------------|
| Paid tutoring demand exists and is income-skewed in the US center market            | FACT                                      | High          |
| CES spend snapshots describe <i>purchasers</i> , not all parents' WTP               | FACT                                      | High          |
| CBC/DCE is the right class of tool for multi-attribute tutoring packages            | FACT (methods)                            | High          |
| Absolute stated dollars will be biased upward                                       | FACT / HYPOTHESIS                         | High          |
| FEI+diagnosis beats unlimited AI for MindCraft's anxious-middle segment             | HYPOTHESIS                                | Medium        |
| Hybrid message (diagnosis + transfer + honest plateau) wins retention vs score-only | HYPOTHESIS                                | Medium        |
| Identity-only parent ads are net-positive                                           | SPECULATION / against doctrine until data | Low (against) |
| One price point fits all segments                                                   | SPECULATION / false as strategy           | Low (against) |

## LIX.14 What this chapter kills

1. **Kill:** Likert / open-ended “how much would you pay?” as pricing research.



2. **Kill:** Treating survey WTP as revenue or national sticker price.
3. **Kill:** “Parents only buy scores” or “parents buy identity” as settled without CBC.
4. **Kill:** Streak / minutes / AI-unlimited as default parent value props without losing on part-worths.
5. **Kill:** Guaranteed ACT points /  $2\sigma$  / Bloom in any package level or ad.
6. **Kill:** Averaging anxious-middle and score-chaser into one “parent persona.”
7. **Wound:** Importing HE or non-US tutoring conjoint coefficients as US ACT-parent truth.
8. **Survive:** SAFE-WTP; CBC with price+neither; ship-mapped attributes; bias protocol; message $\neq$ package; WTP-1...6 + QUAL; cash validation before scale.

**Doctrine until data:** Run **SAFE-WTP** — choice-based parent trade-offs on diagnosis, FEI evidence, human accountability, AI policy, and price — then package what wins *and* what parents can explain, not what flatters the identity thesis.